

21 May 2002

TO: Pre-Installation Tab 088

SUBJECT: **Direction 2304880, Signa TwinSpeed Upgrade Pre-Installation, Rev 1**

Enclosed is *Direction 2304880, Signa TwinSpeed Upgrade Pre-Installation, Rev 1* revision. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2304880 REVISION 1 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
1 SYSTEM CONFIGURATION	• Added Signa TwinSpeed with EXCITE Technology upgrade catalogs
2 ROOM LAYOUTS	• Added Signa TwinSpeed with EXCITE Technology upgrade equipment information • Update TSCC & TGWC for revised E&W information • Added Network connection and telephone lines option to system connectivity requirements • Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogens • Updated TAC Cabinet dimensions drawing
4 SITE ENVIRONMENT	• Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications • Updated Water Cooled TSCC & TGWC water cooling requirements • Added System Cabinet with EXCITE Technology alarm device information
5 POWER REQUIREMENTS	• Revised Voltage Transient specification • Added NEC 2002 Article references
6 INTERCONNECT DATA	• Added Signa TwinSpeed with EXCITE Technology information • Updated miscellaneous references
7 RF SHIELDED ROOM	• Corrected Table Of Contents typographical error
9 PREINSTALLATION CHECKLIST	• Updated SOI Request Form web address • Updated references to Network connection and telephone lines option to system connectivity • Revised checklist item to inform outside emergency personnel precautions to take in event of emergencies in advance of magnet ramp-up
10 TOOLS & TEST EQUIPMENT	• Added new & deleted obsolete test equipment & part numbers
APPENDIX B	• Added Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed" document.

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2294003 REV 1

Refer to Table 2 for instructions to properly update your Direction 2294003 Rev 0 with the enclosed Revision 1 pages.

TABLE 2
DIRECTION 2304880 REVISION 1 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
Front Matter	Title Page	Title Page
	A	A
	i to v	i to v
1 SYSTEM CONFIGURATION	1–1 to 1–33 (entire section)	1–1 to 1–58 (entire section)
2 ROOM LAYOUTS	2–1 to 2–62 (entire section)	2–1 to 2–68 (entire section)
4 SITE ENVIRONMENT	4–1 to 4–19 (entire section)	4–1 to 4–21 (entire section)
5 POWER REQUIREMENTS	5–1 to 5–17 (entire section)	5–1 to 5–17 (entire section)
6 INTERCONNECT DATA	6–1 to 6–50 (entire section)	6–1 to 6–69 (entire section)
7 RF SHIELDED ROOM	7–1/2	7–1/2
9 PREINSTALLATION CHECKLIST	9–1 to 9–7 (entire section)	9–1 to 9–7 (entire section)
10 TOOLS & TEST EQUIPMENT	10–1 to 1–10 (entire section)	10–1 to 1–10 (entire section)
APPENDIX B		Tab B Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed" document

Debra C. Balis
MR SERVICE ENGINEERING



GE Medical Systems

Technical Publications

Direction 2304880

Revision 1

Signa® 1.5T *TwinSpeed* Upgrade Pre-Installation

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Operating Documentation



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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
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- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
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0	Aug. 31, 2001	Initial document.
1	May 21, 2002	Added Signa TwinSpeed with EXCITE Technology system upgrade information throughout the document. Other major changes listed by section include the following Sec 2: Update TSCC & TGWC for revised E&W information; Added Network connection and telephone lines option to system connectivity requirements; Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogenics; Updated TAC Cabinet dimensions drawing. Sec 4: Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications; Updated Water Cooled TSCC & TGWC water cooling requirements. Sec 5: Revised Voltage Transient specification; Added NEC 2002 Article references. Sec 6: Updated miscellaneous references. Sec 7: Corrected Table Of Contents typographical error. Sec 9: Updated SOI Request Form web address; Updated references to Network connection and telephone lines option to system connectivity. Sec 10: Added new & deleted obsolete test equipment & part numbers; Revised checklist item to inform outside emergency personnel precautions to take in event of emergencies in advance of magnet ramp-up. Appendix B: Added Tab B and Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed" document.

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* This revision number/letter corresponds to the indicated document's revision control system.

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INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a Signa 1.5T TwinSpeed system upgrade. “Preinstallation work” is done to prepare the customer’s premises for the installation of the upgrade sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as facility input power line to the Main Disconnect Panel and Power Distribution Unit. If new wires are installed then the electrical contractor shall ring identify and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non-electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Site renovation.
- Installation of structural reinforcements as required.

All work **MUST** be in compliance with national and local building and safety codes.

All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Panel and Power Distribution Unit or the power lines required in the Magnet Room, the Main Disconnect Panel option, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, or the RF shielded room specified in this document.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Main Disconnect Panel and Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third-party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre-installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of the pre-installation information.

SECTION 1 – SYSTEM CONFIGURATION

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1-2-3	Signa LX 1.5T Without LCC Magnet Installed	1-14
1-2-4	Signa Horizon 1.5T	1-19
1-2-5	Signa Advantage 1.5T 5.X & 4.X	1-24
1-3	UPGRADES TO SIGNA TWINSPEED	1-31
1-3-1	Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet	1-31
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1-1 UPGRADE INTRODUCTION

Magnetic Resonance (MR) system upgrades to Signa® 1.5T *TwinSpeed* with EXCITE Technology or Signa® 1.5T *TwinSpeed* are documented throughout this publication. Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in SECTION 6 INTERCONNECT DATA.

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Pre-Installation document uses the technical document names.

SALES LITERATURE NAME	= TECHNICAL DOCUMENT NAME
Signa Infinity 1.5T <i>TwinSpeed</i> with EXCITE Technology	= Signa 1.5T <i>TwinSpeed</i> with EXCITE Technology
Signa Infinity 1.5T <i>TwinSpeed</i>	= Signa 1.5T <i>TwinSpeed</i>
CXK4 150 Magnet with K4 Technology	= 1.5T LCC (Active Shield) Magnet

Note

Throughout this document references to Signa *TwinSpeed* apply to both Signa 1.5T *TwinSpeed* with EXCITE Technology and Signa 1.5T *TwinSpeed* except where explicitly documented as different.

Note

Signa 1.5T *TwinSpeed* upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the *TwinSpeed* Upgrade.

WARNING!

UPGRADE TO AN LCC MAGNET HAS CUSTOMER SITE ENVIRONMENTAL FACTORS AND LOGISTICS THAT NEED TO BE UNDERSTOOD AND PLANNED FOR PRIOR TO THE MAGNET UPGRADE. DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION IS REFERENCED THROUGHOUT THIS DOCUMENT FOR TWINSPEED LCC MAGNET SPECIFICATIONS WHICH THE SITE MUST MEET.

1-1 UPGRADE INTRODUCTION (Continued)**WARNING!**

SIGNA 1.5T TWINSPEED SYSTEM REQUIRES THE ACGD/PDU CABINET. ALL SYSTEMS UPGRADING TO THIS SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

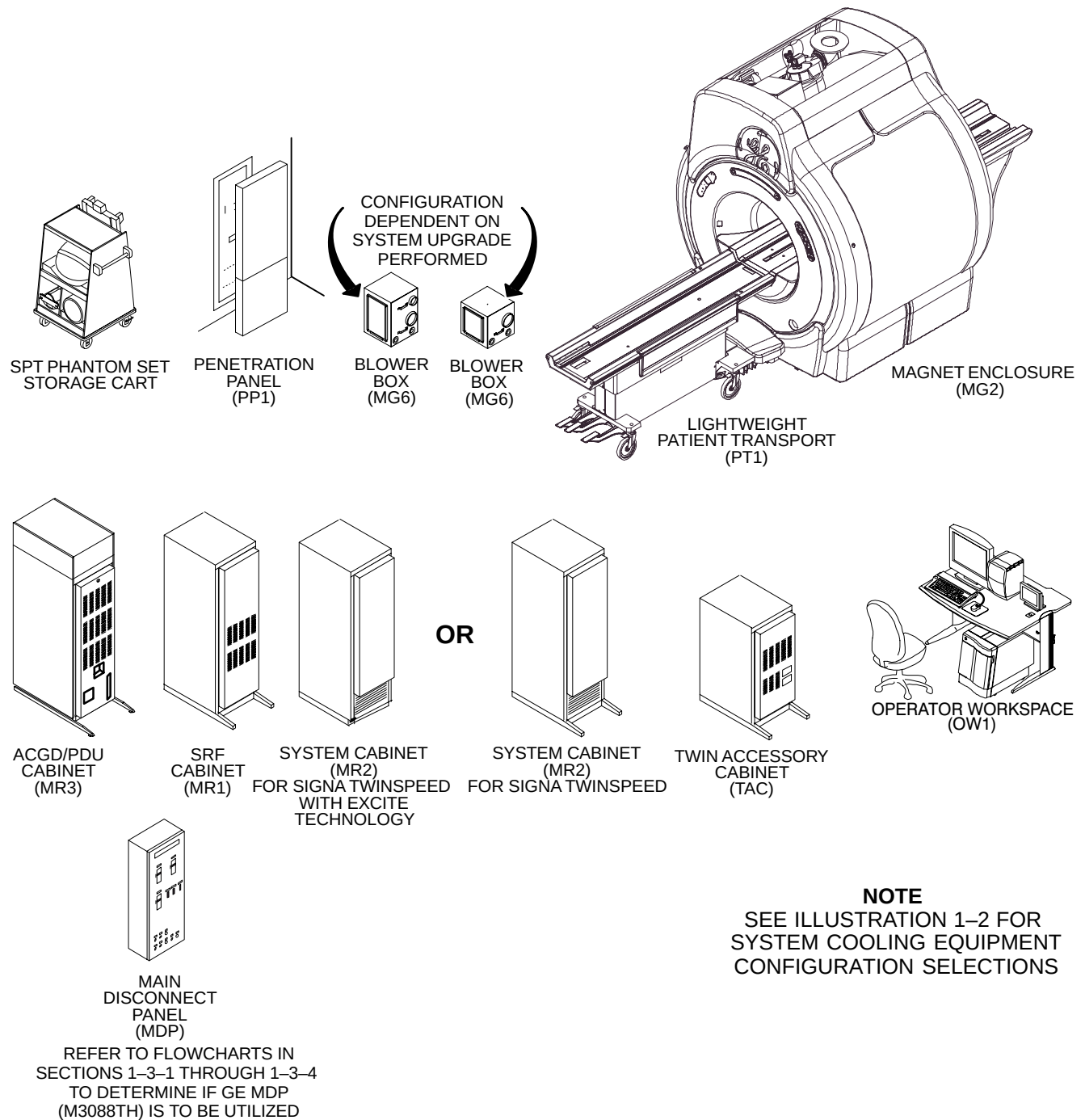
SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

1-1 UPGRADES INTRODUCTION (Continued)

Illustrations 1-1 and 1-2 show the system's major equipment after a Signa 1.5T TwinSpeed upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded. The following sub-sections provide an overview of the various Signa 1.5T TwinSpeed upgrade hardware.



SIGNA 1.5T TWINSPEED SYSTEM MAJOR EQUIPMENT

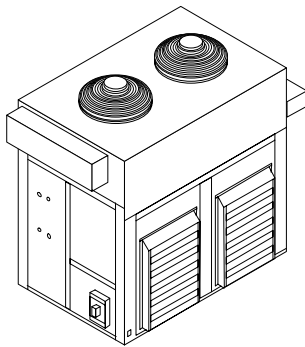
ILLUSTRATION 1-1

1-1 UPGRADES INTRODUCTION (Continued)

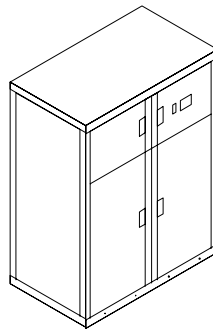
TWIN SYSTEM COOLING CABINET (TSCC)

OUTDOOR TSCC
(M3088TA)

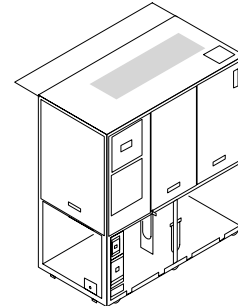
REMOTE
CONTROL
PANEL
(RCP)



INDOOR WATER
COOLED TSCC
(M3088TB)



INDOOR AIR
COOLED TSCC
(M3088TC)



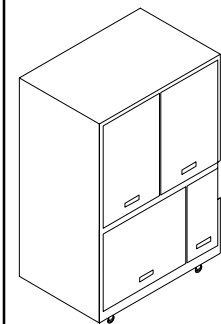
TWIN GRADIENT WATER COOLER
(TGWC)

CAUTION!!

WHEN A TGWC IS SELECTED THEN THE CUSTOMER SITE MUST PROVIDE WATER COOLING FOR THE LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET.

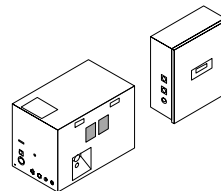
OUTDOOR UNIT
(M3088TF)

REMOTE
CONTROL
PANEL
(RCP)

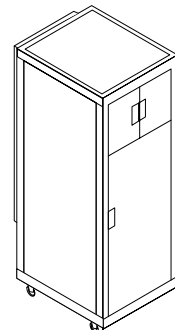


INDOOR WATER
COOLED UNIT
(M3088TE)

REMOTE
CONTROL
PANEL
(RCP)



INDOOR AIR
COOLED UNIT
(M3088TG)



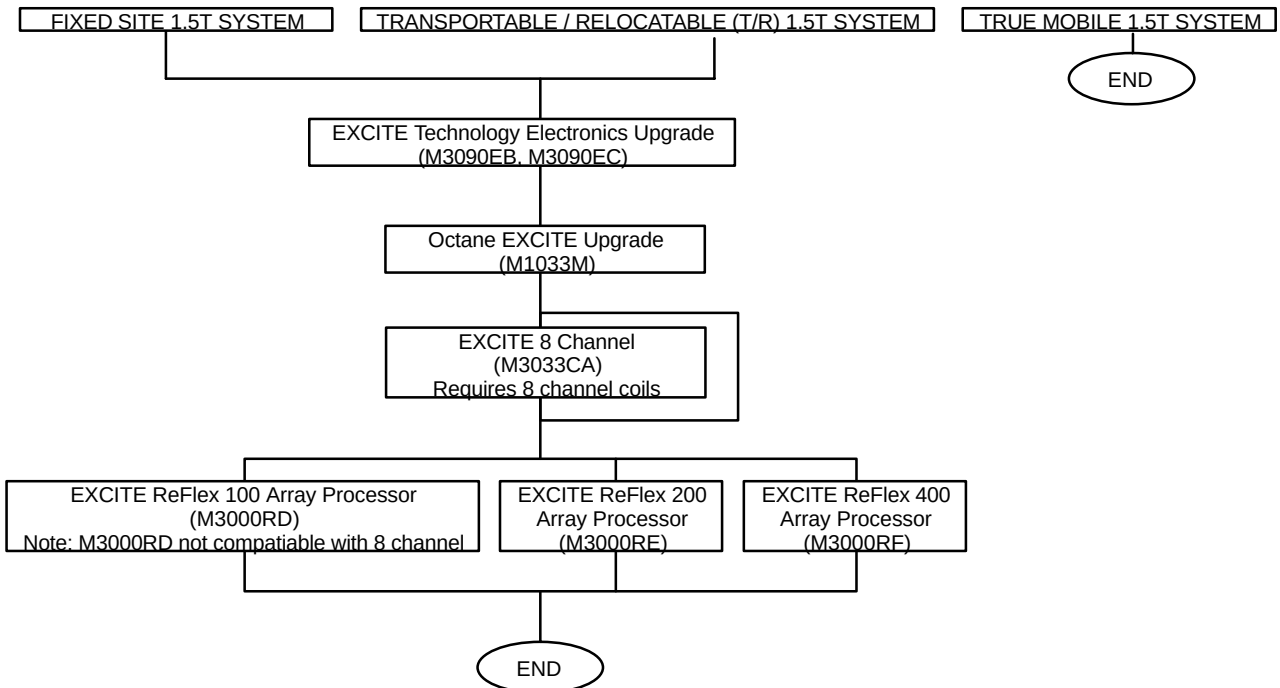
SYSTEM COOLING EQUIPMENT SELECTIONS

ILLUSTRATION 1-2

1-2 UPGRADES TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY

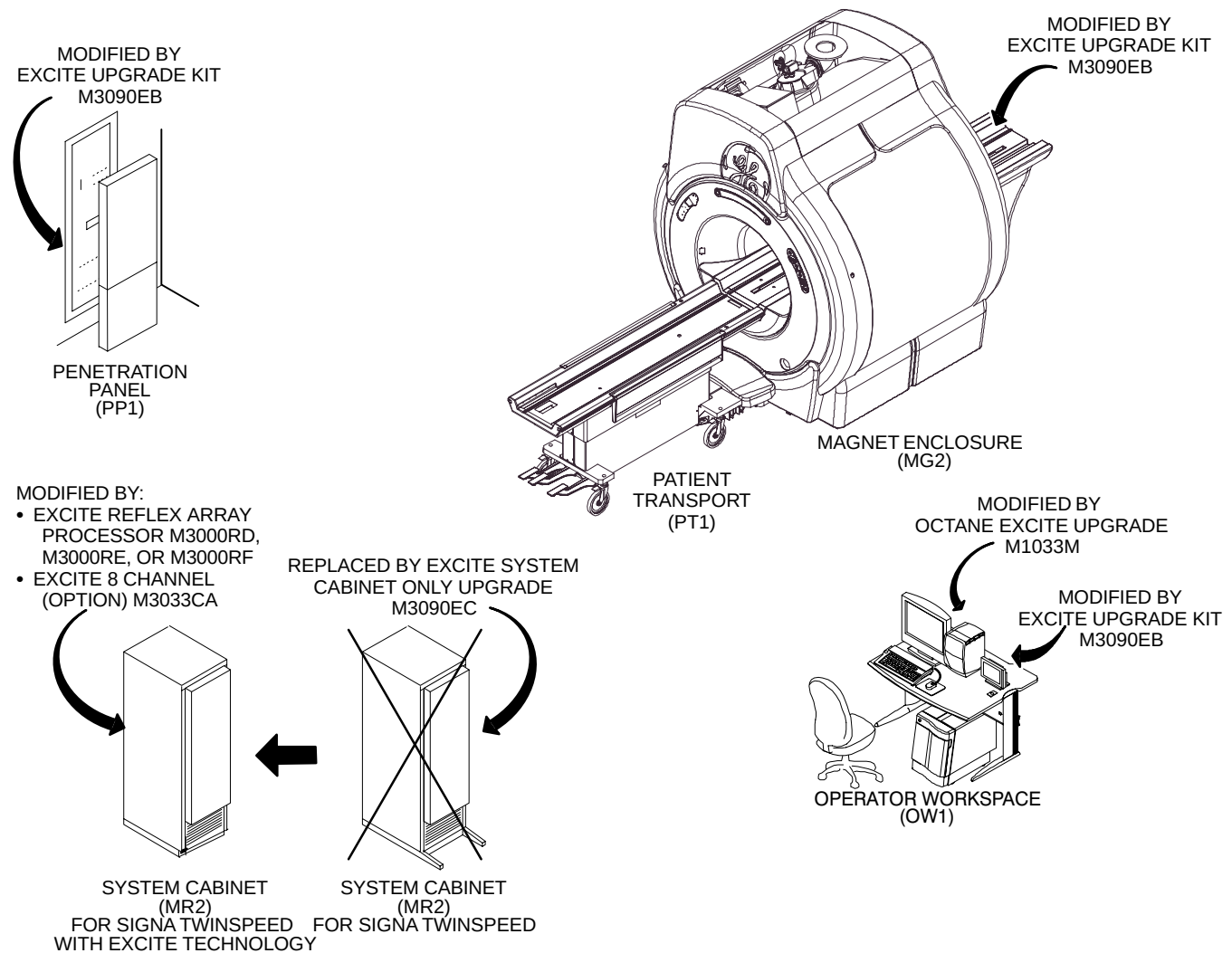
1-2-1 Signa 1.5T TwinSpeed

The flowchart shown in Illustration 1-3 lists the upgrade catalogs for a Signa 1.5T TwinSpeed and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustration 1-4 show the cabinets replaced and modified by the catalogs listed in the flowchart.



SIGNA TWINSPEED UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-3

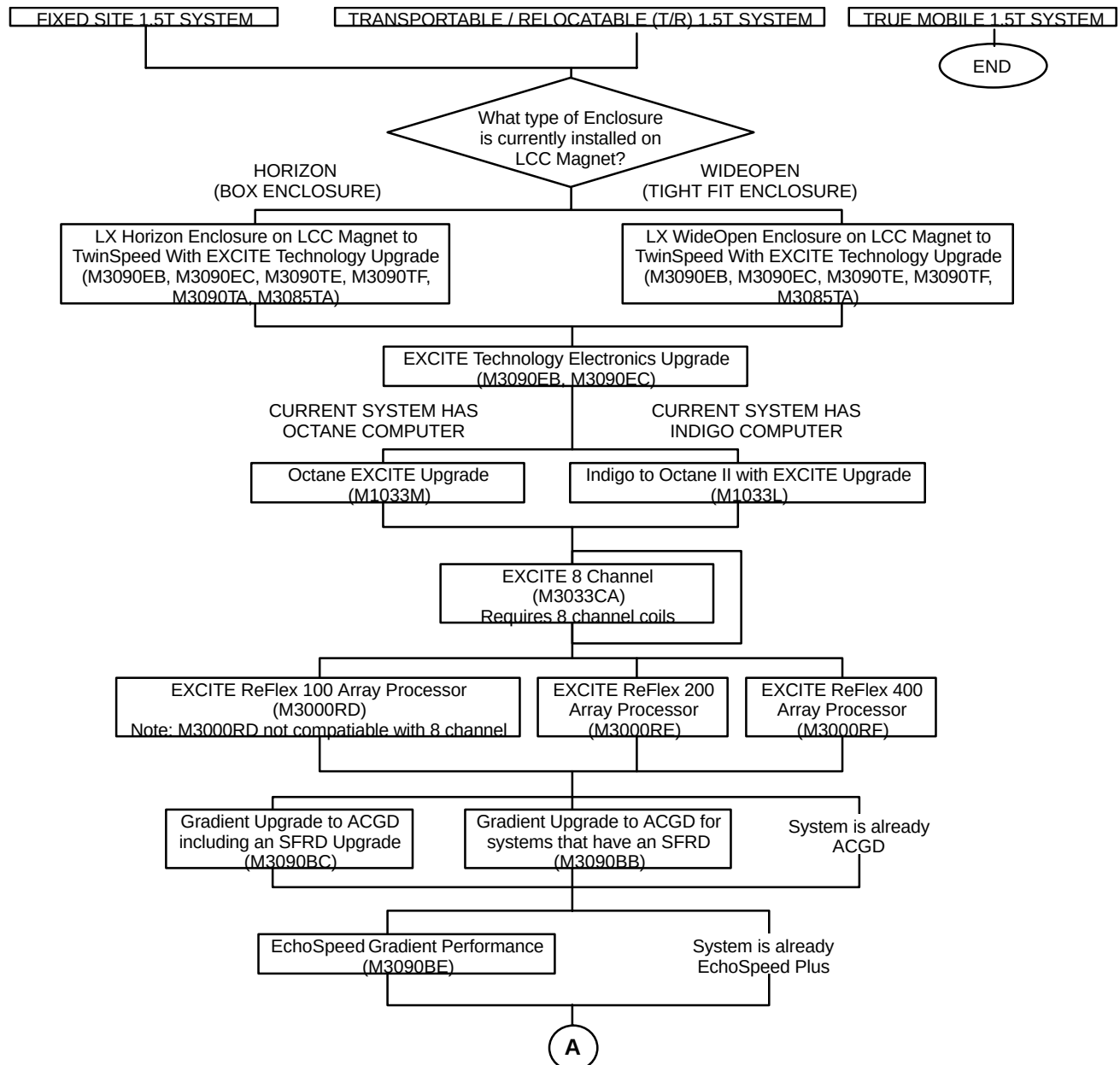
1-2-1 Signa 1.5T TwinSpeed (Continued)



SIGNA TWINSPEED SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-4

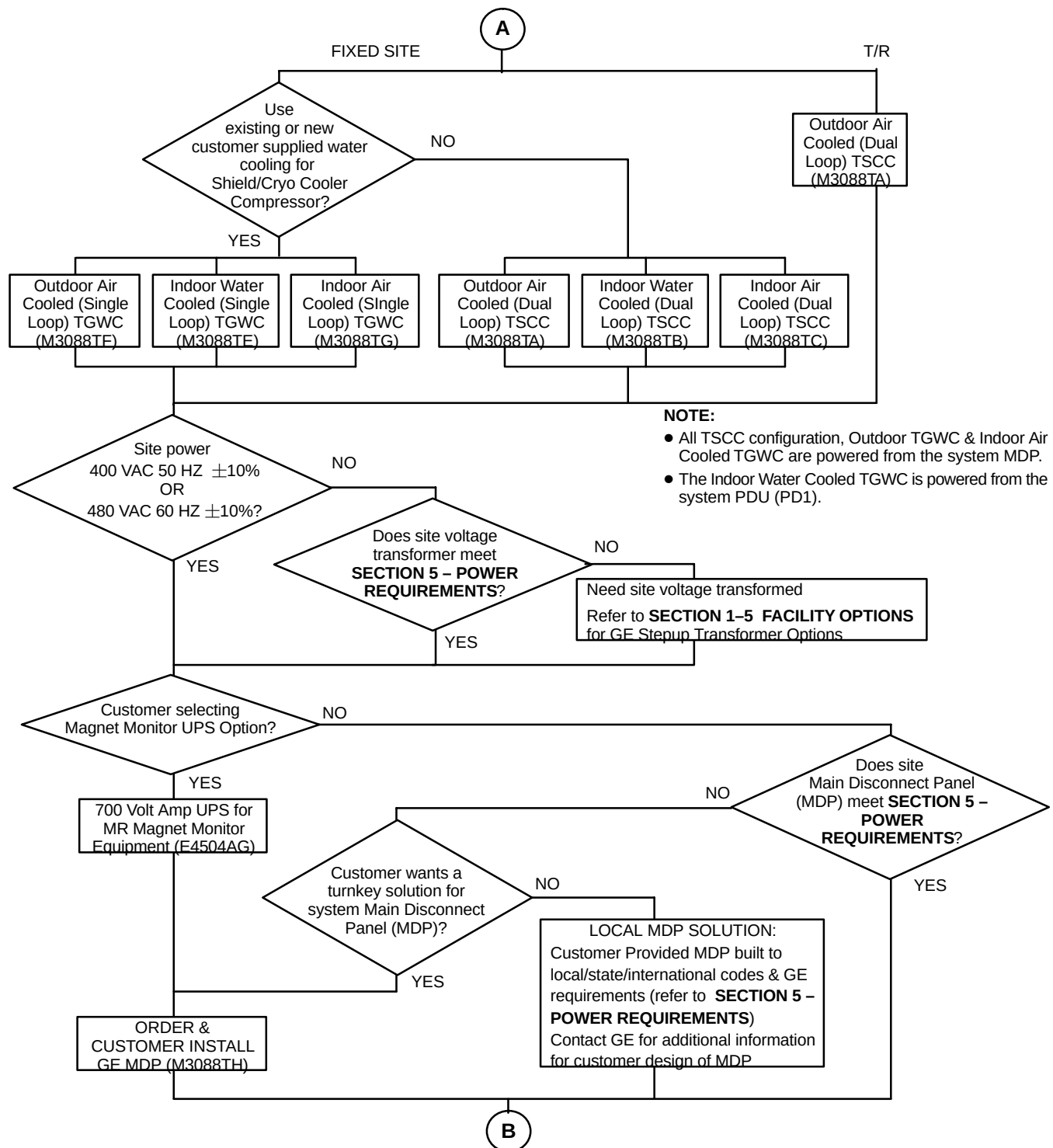
1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet

The flowchart shown in Illustrations 1-5 through 1-7 lists the upgrade catalogs for a Signa MR/i 1.5T or Signa LX 1.5T (with ACGD or SGD) and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustrations 1-8 through 1-9 show the cabinets replaced and modified by the catalogs listed in the flowchart.

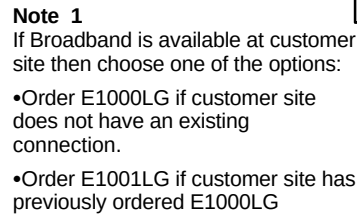


SIGNA MR/I 1.5T & SIGNA LX 1.5T WITH LCC MAGNET UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-5

1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



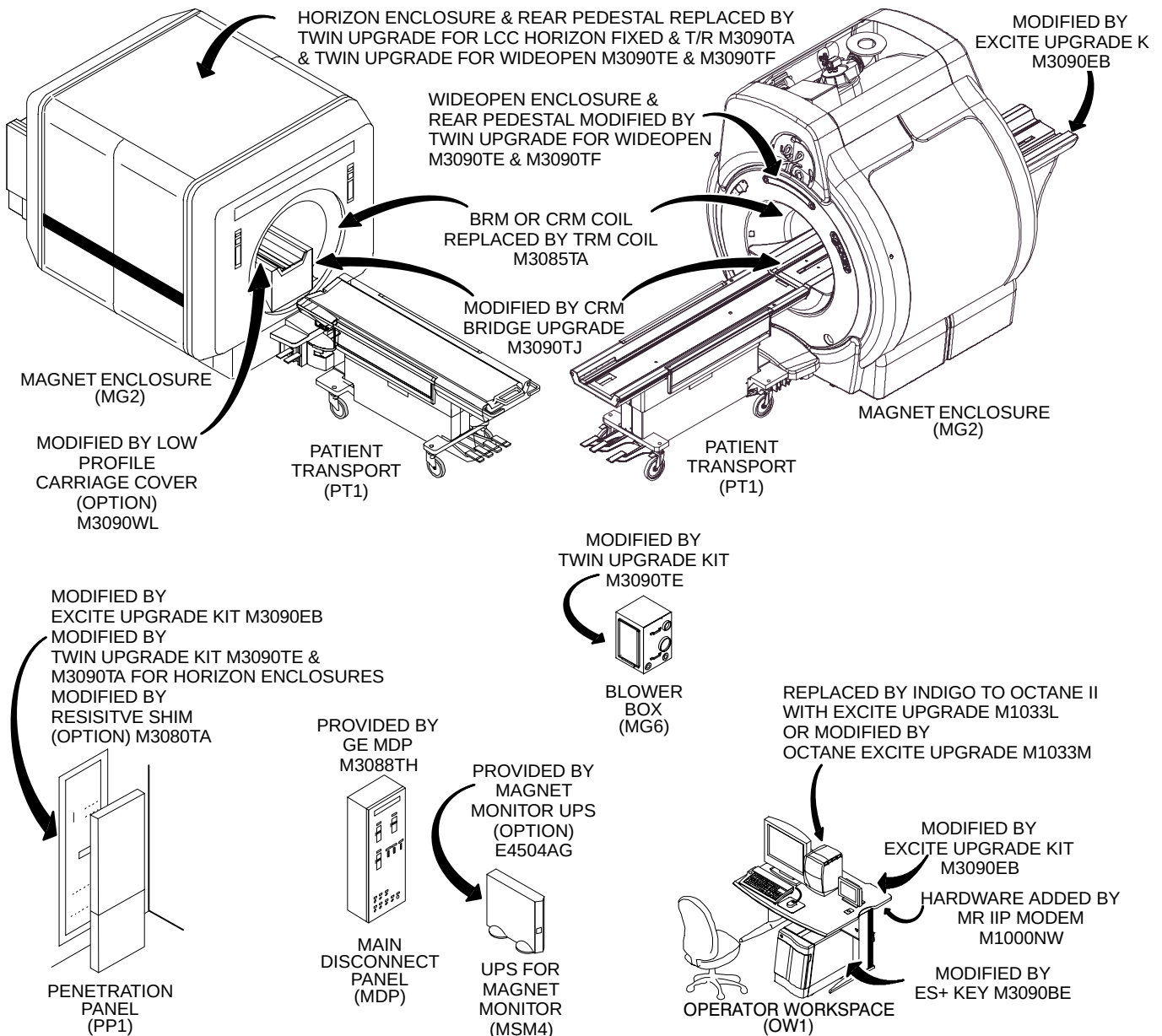
1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



Note 2
If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

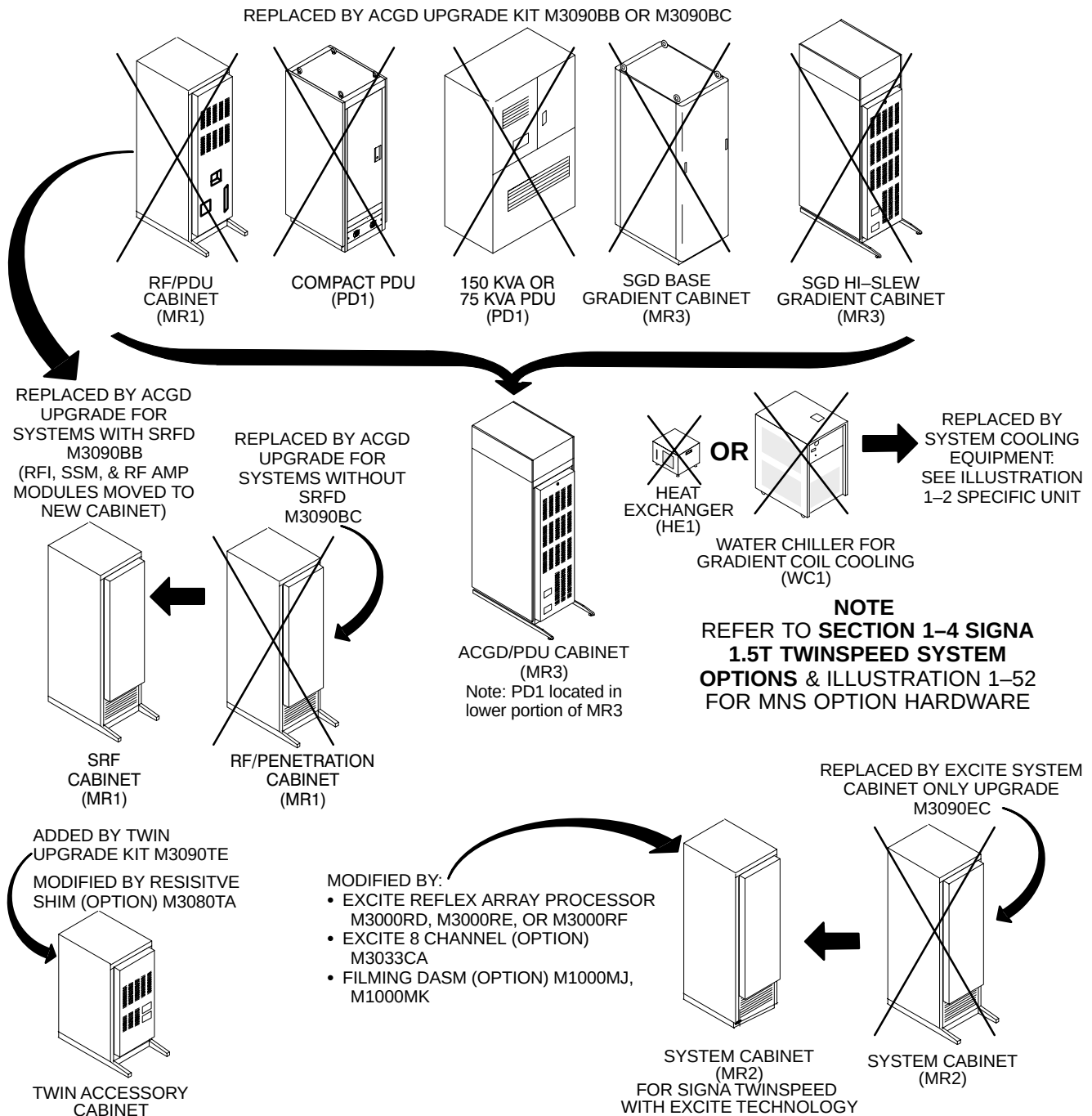
SIGNA MR/I & SIGNA LX WITH 1.5T LCC MAGNET UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-7

1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



SIGNA MR/i & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-8

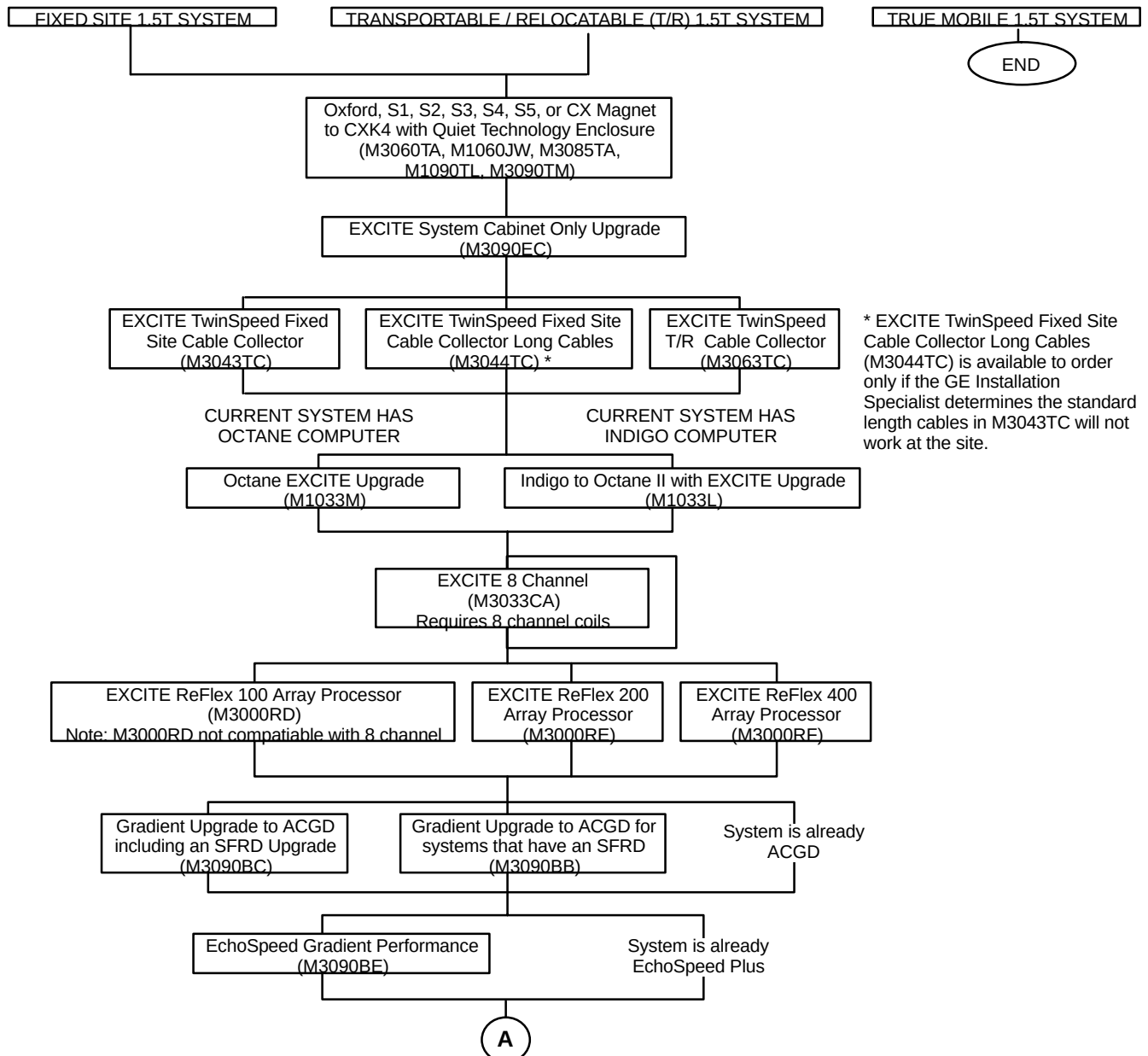
1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



SIGNAL MR/I & SIGNAL LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-9

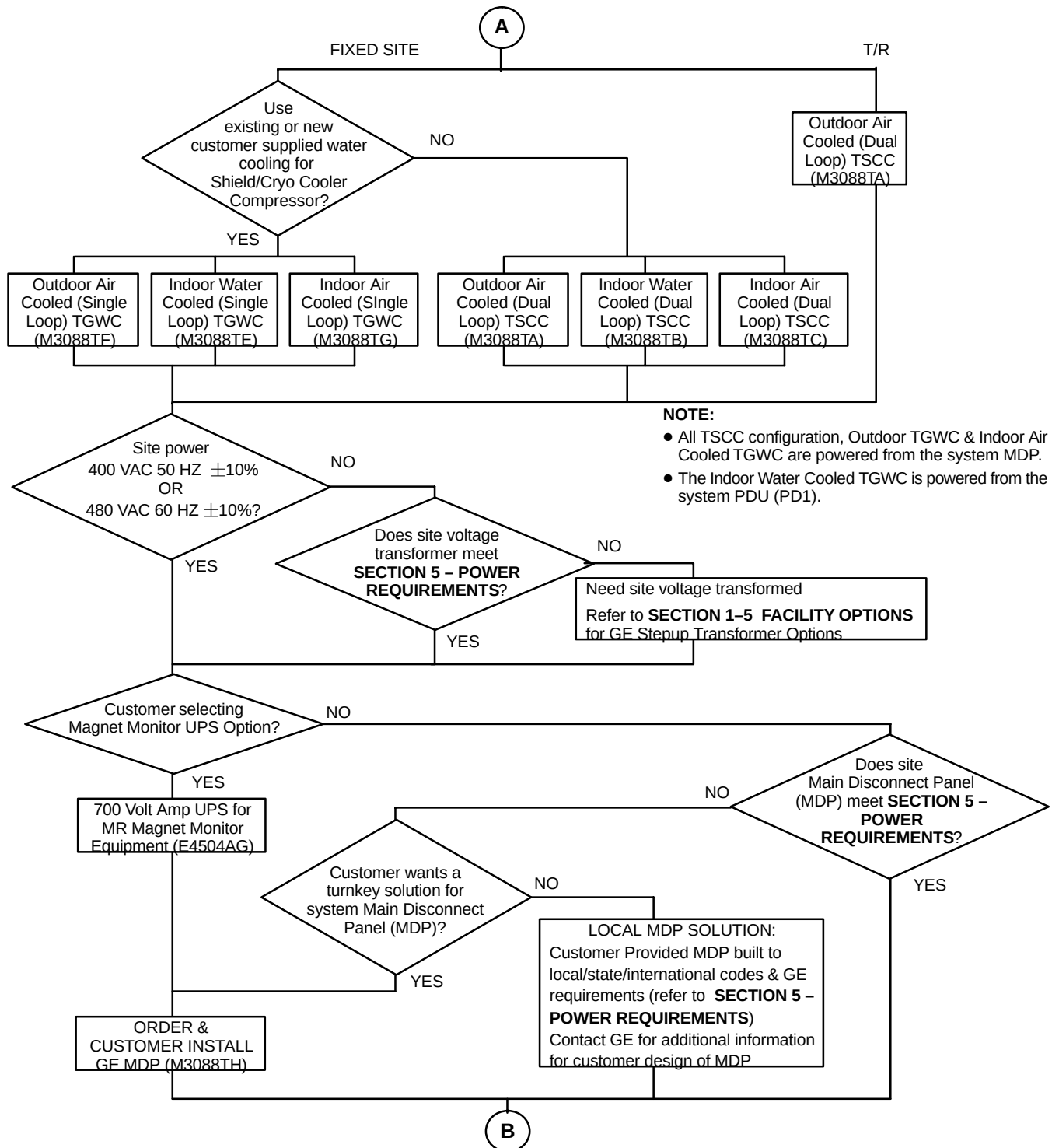
1-2-3 Signa LX 1.5T Without LCC Magnet Installed

The flowchart shown in Illustrations 1-10 through 1-12 lists the upgrade catalogs for a Signa LX or CVMR (with ACGD or SGD) and current system installed magnet is NOT an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustrations 1-13 through 1-14 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.

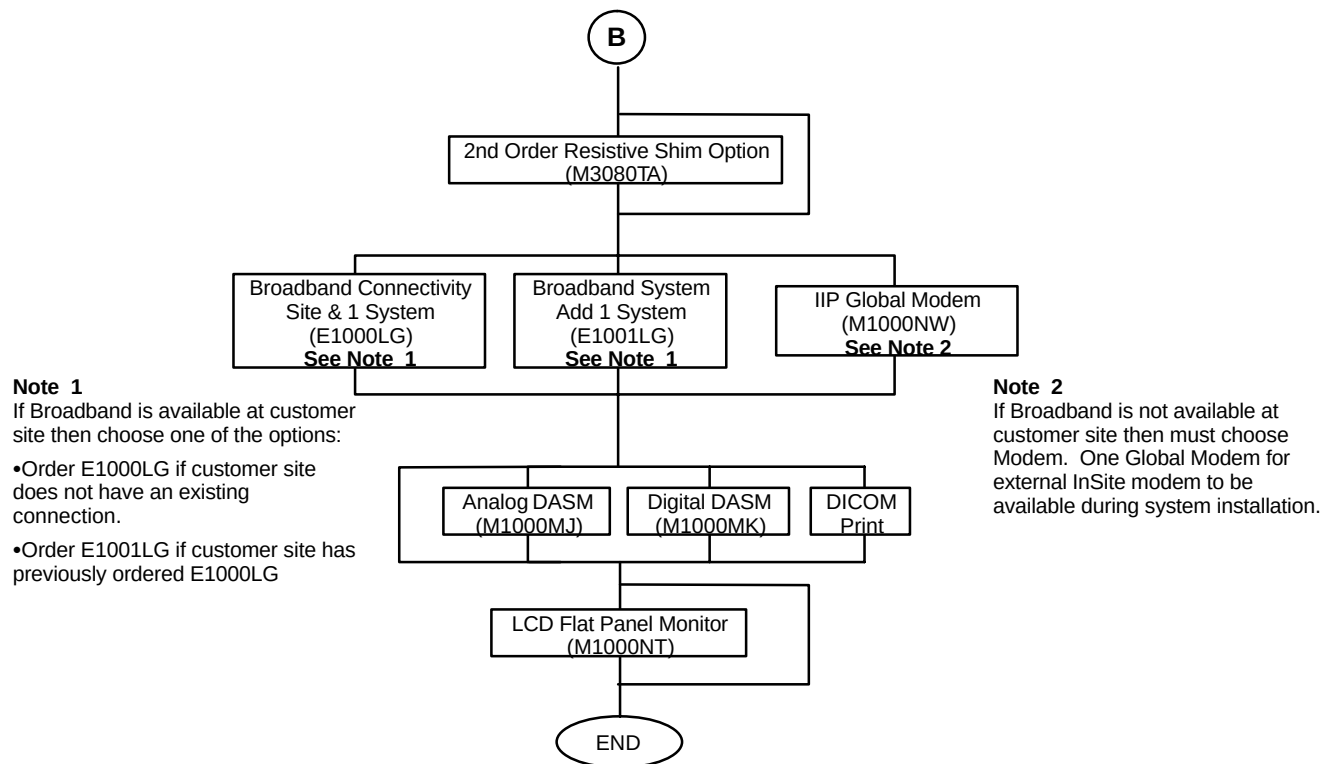


SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWIN SPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-10

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)

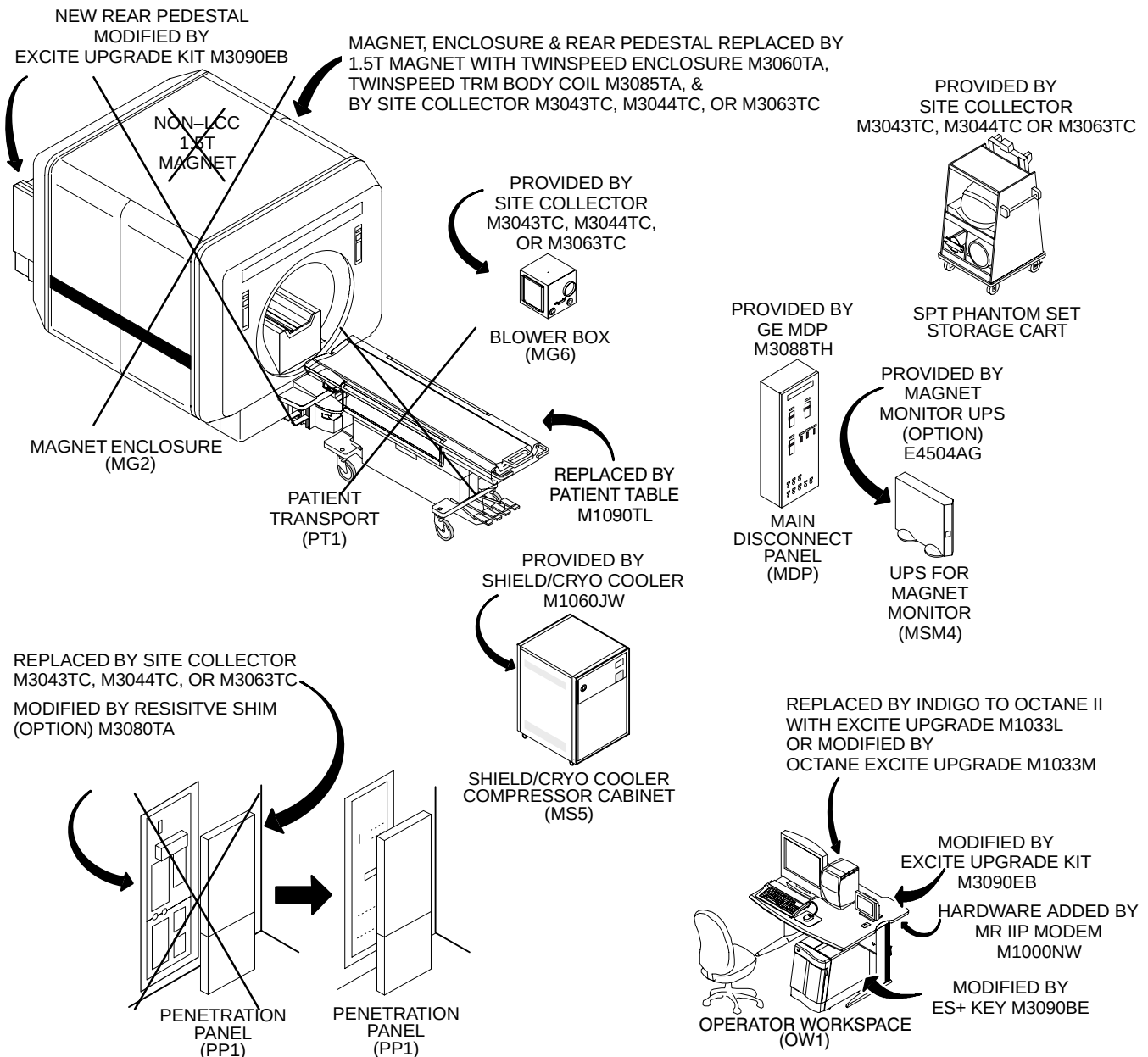


1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



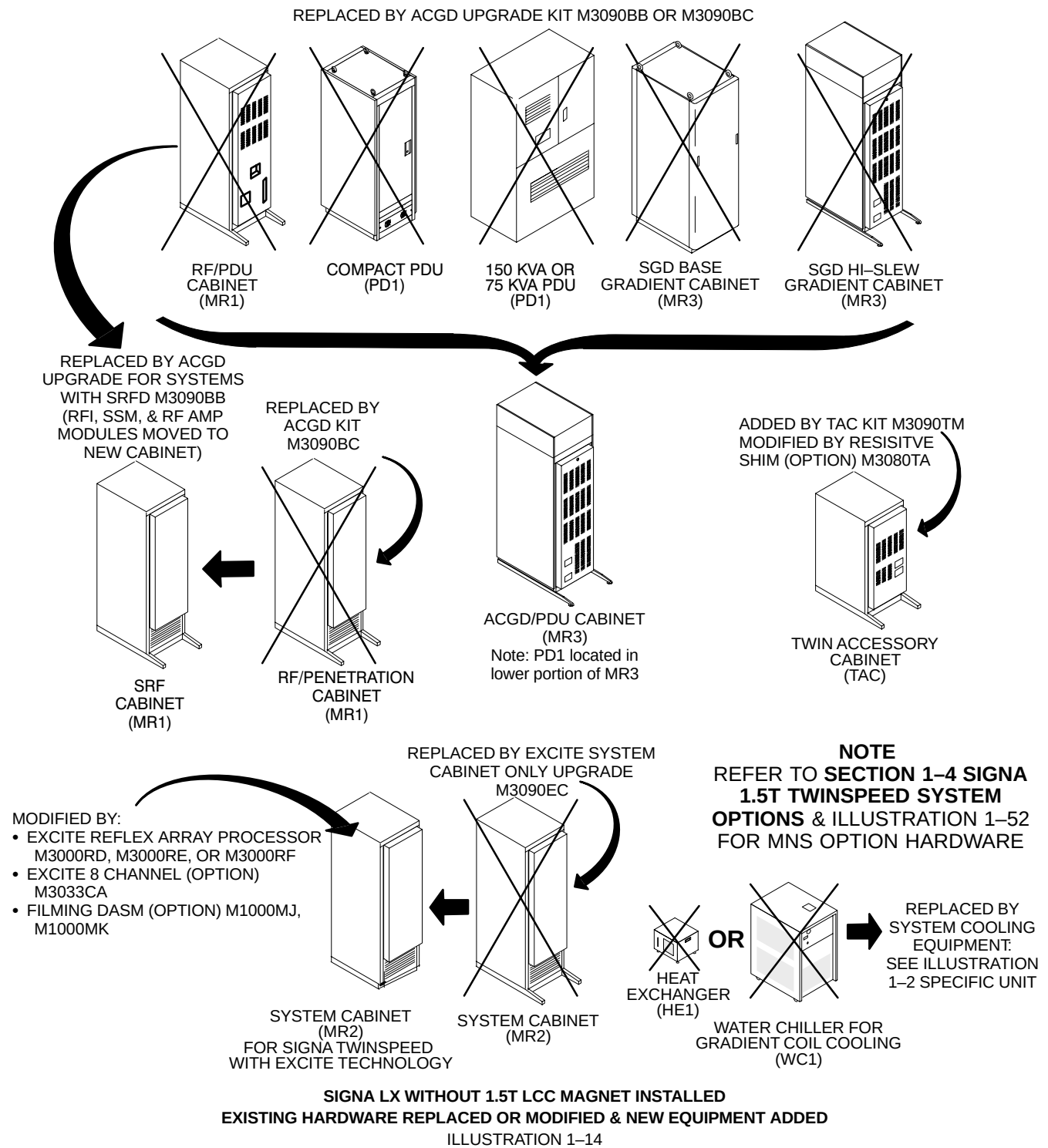
SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-12

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



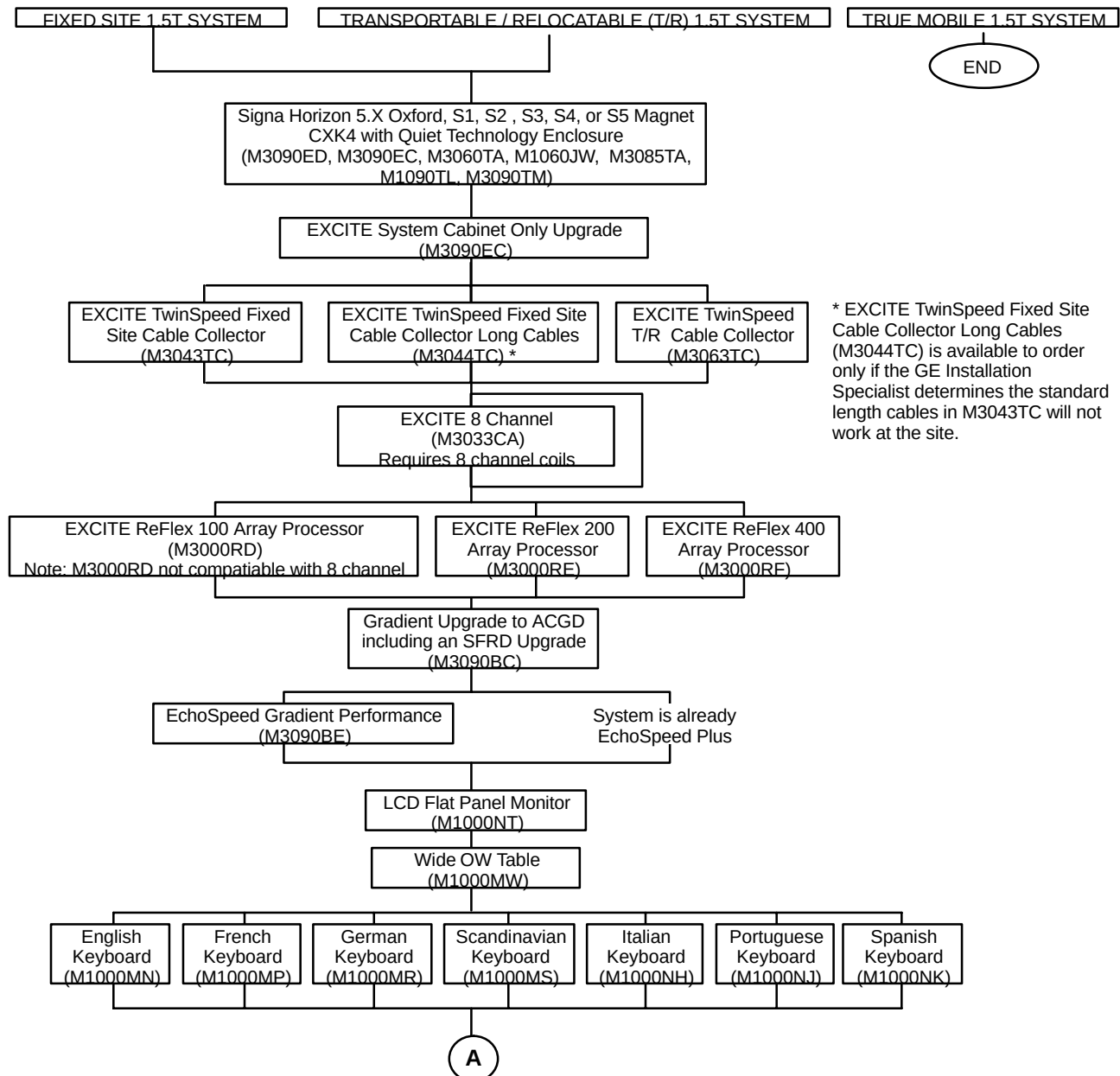
SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-13

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



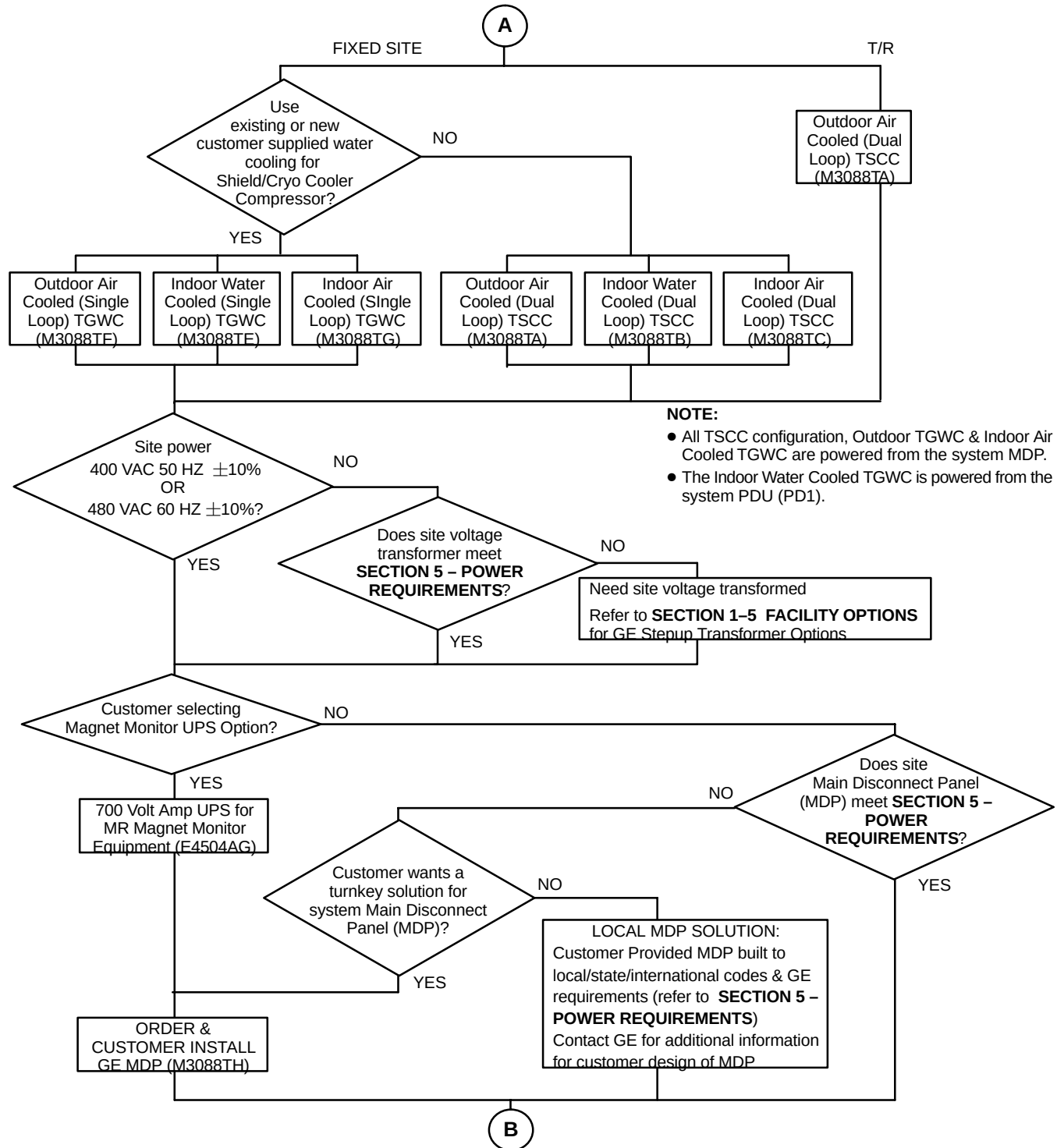
1-2-4 Signa Horizon 1.5T

Upgrades from Signa Horizon 5.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-15 through 1-17 lists the Signa Horizon 5.X upgrade catalogs. Illustrations 1-18 and 1-19 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.

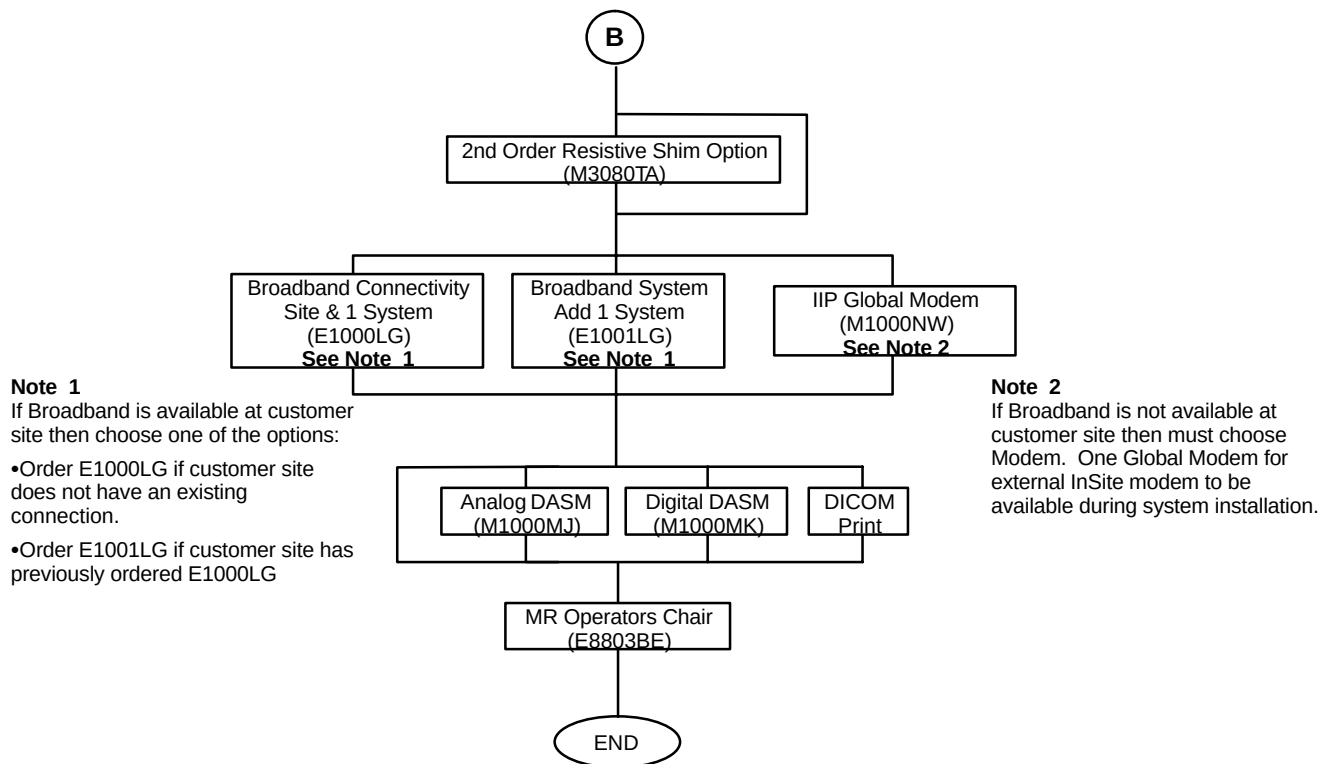


SIGNA HORIZON 1.5T UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-15

1-2-4 Signa Horizon 1.5T (Continued)

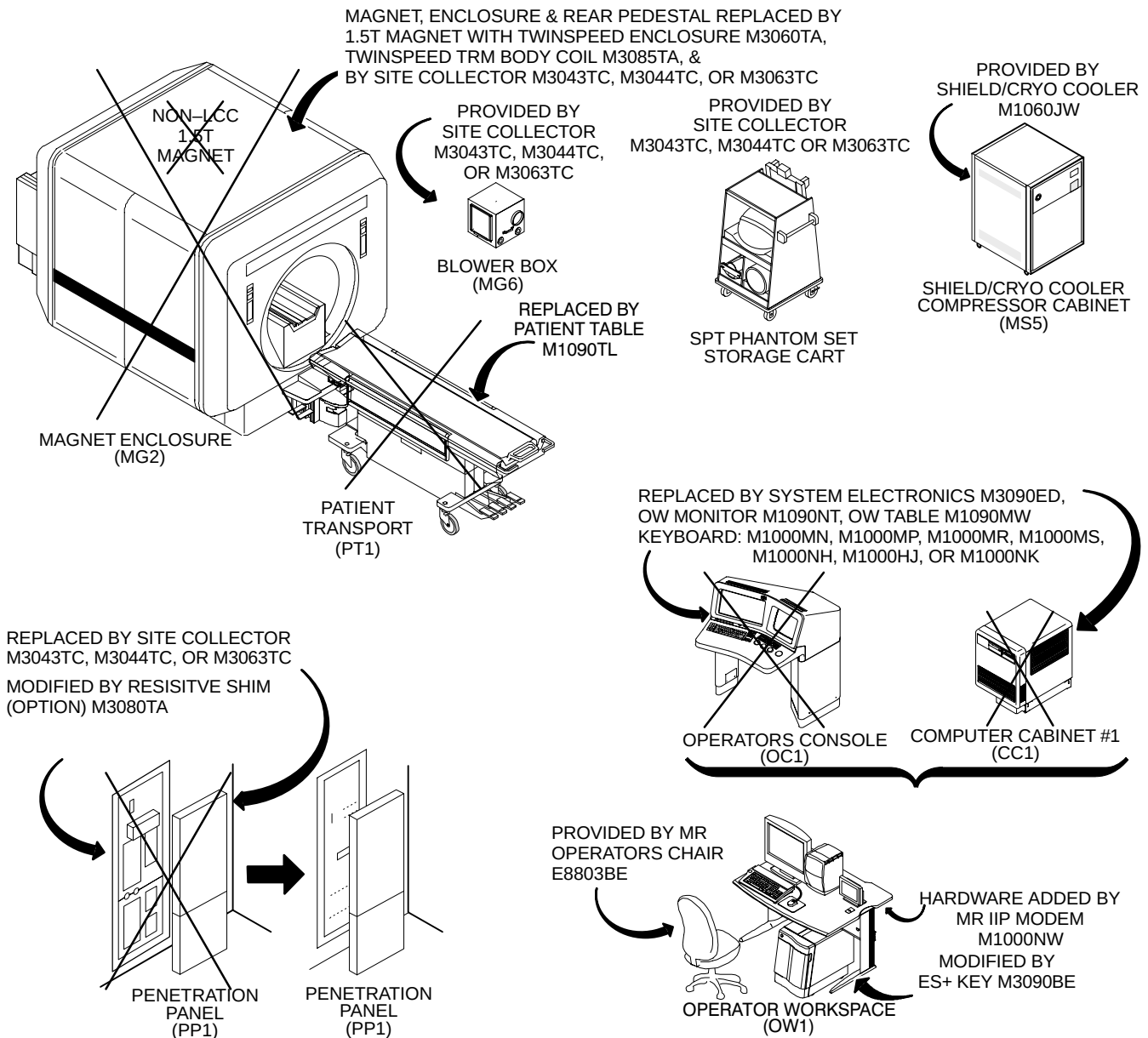


1-2-4 Signa Horizon 1.5T (Continued)



SIGNA HORIZON 1.5T UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1 – 17

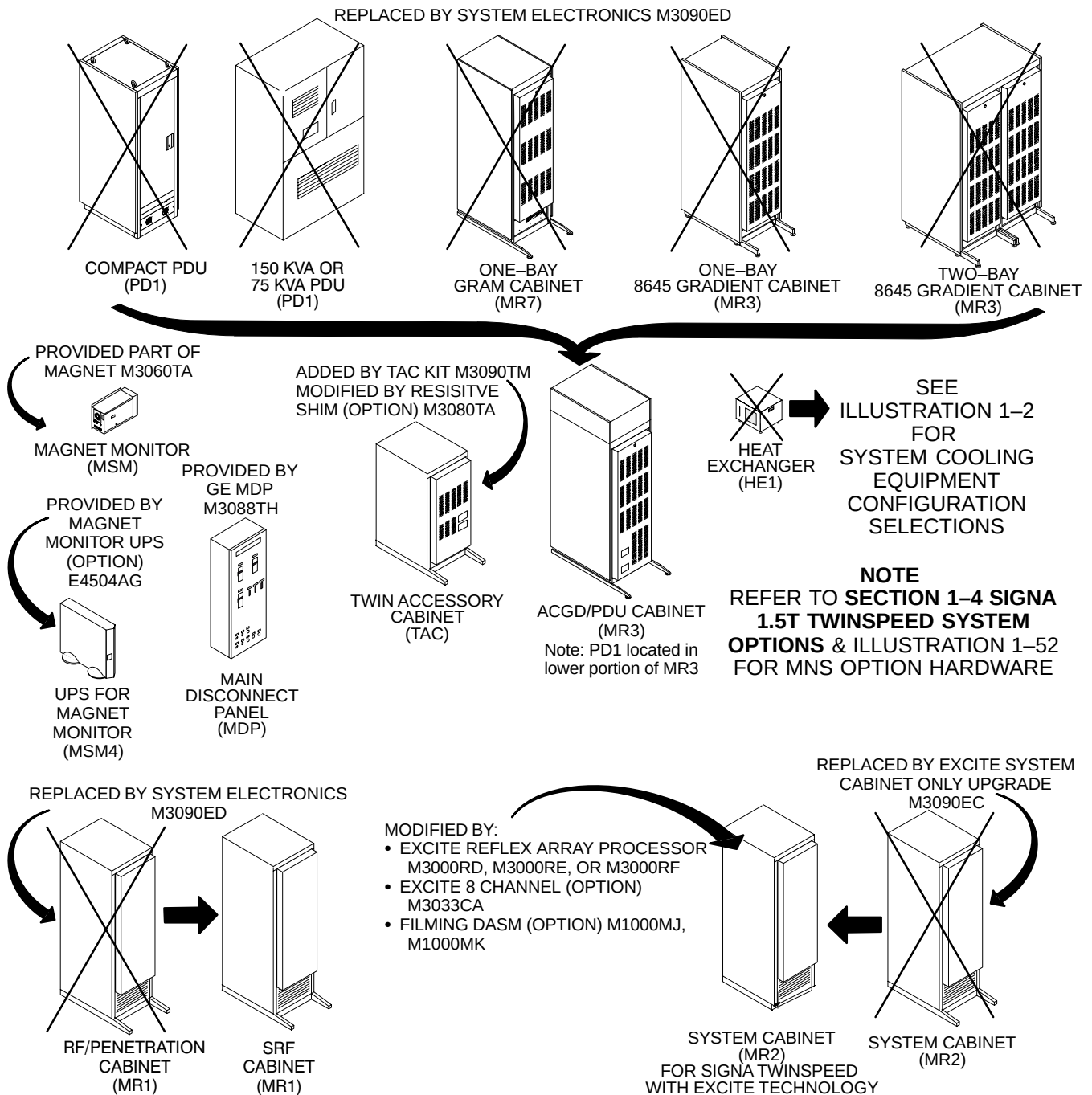
1-2-4 Signa Horizon 1.5T (Continued)



SIGNA HORIZON 1.5T SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-18

1-2-4 Signa Horizon 1.5T (Continued)

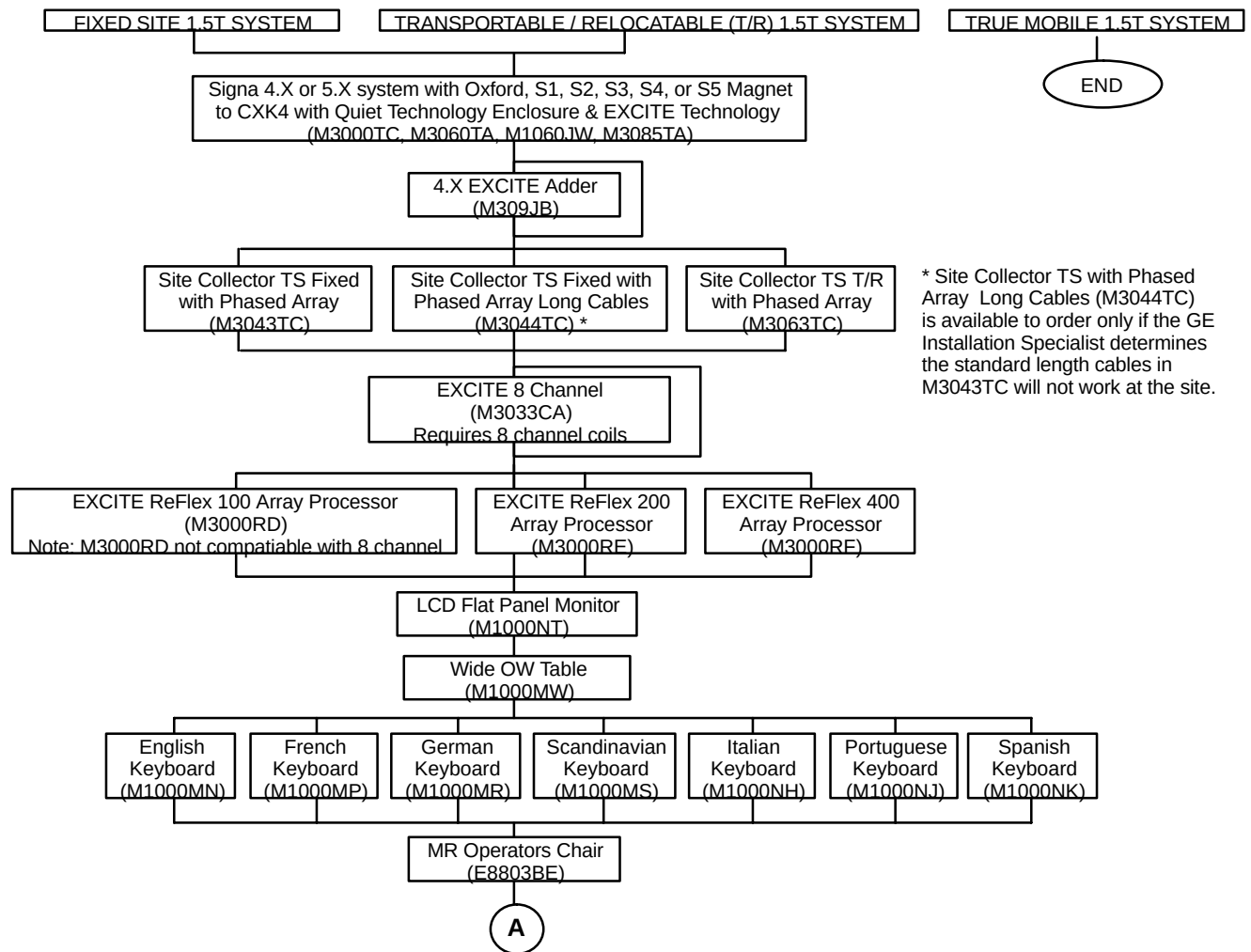


SIGNAL HORIZON 1.5T SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-19

1-2-5 Signa Advantage 1.5T 5.X & 4.X

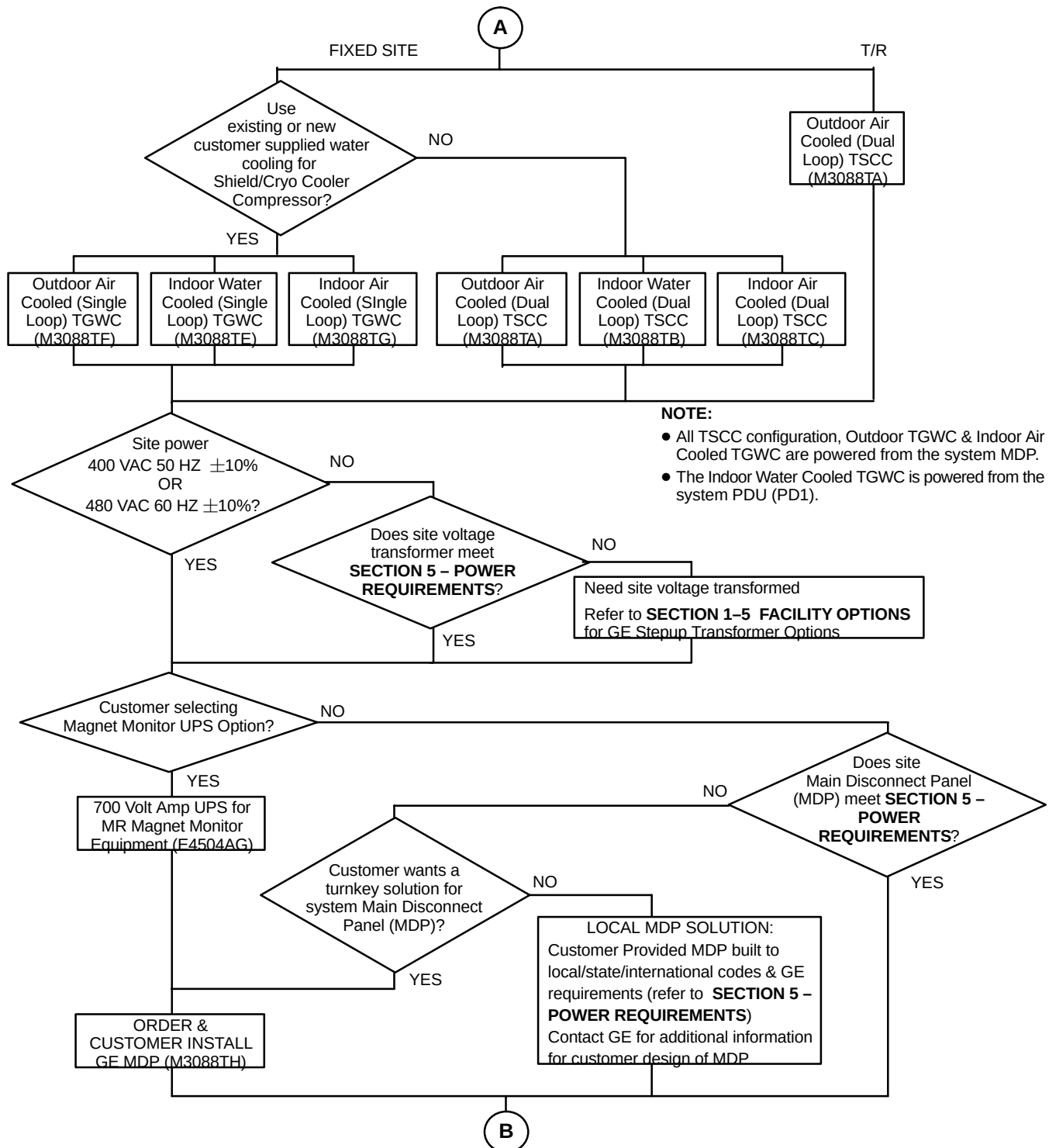
Upgrades from Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed EXCITE Technology require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-20 through 1-22 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 1.5T 5.X Software Only, Advantage 5.X see Illustrations 1-23 and 1-24
- Signa Horizon 1.5T 4.X see Illustrations 1-25 and 1-26



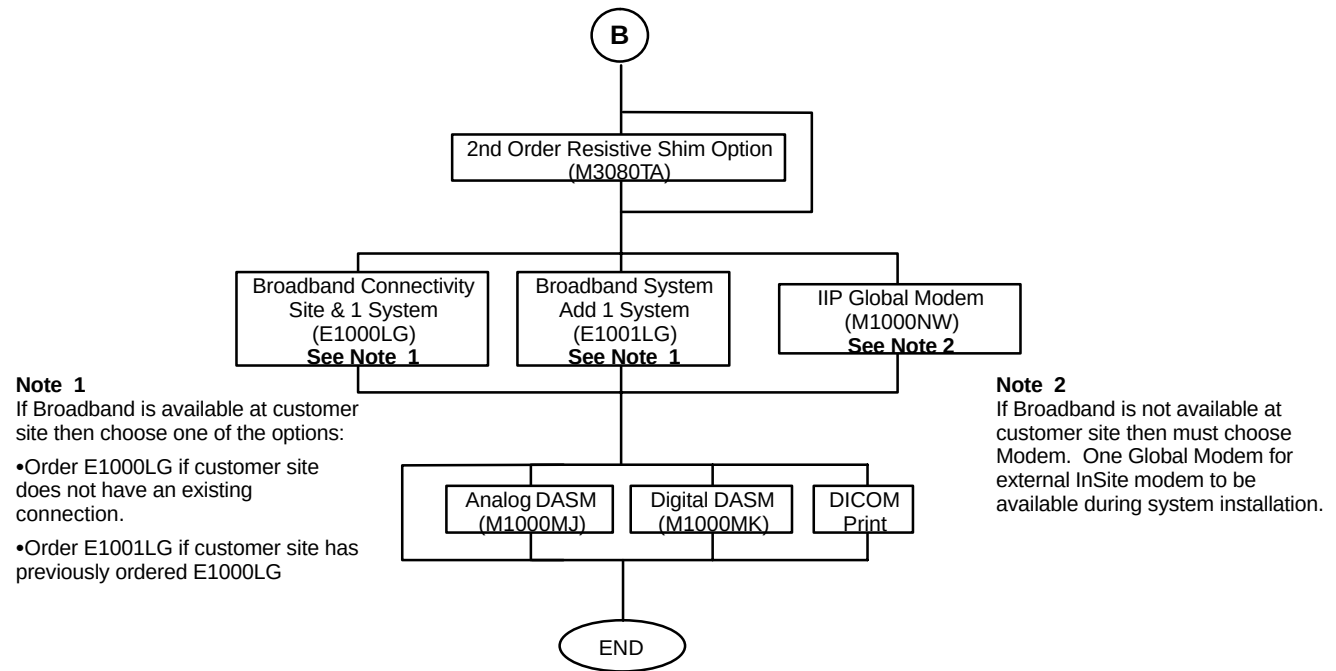
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE
TO SIGNA TWIN SPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-20

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



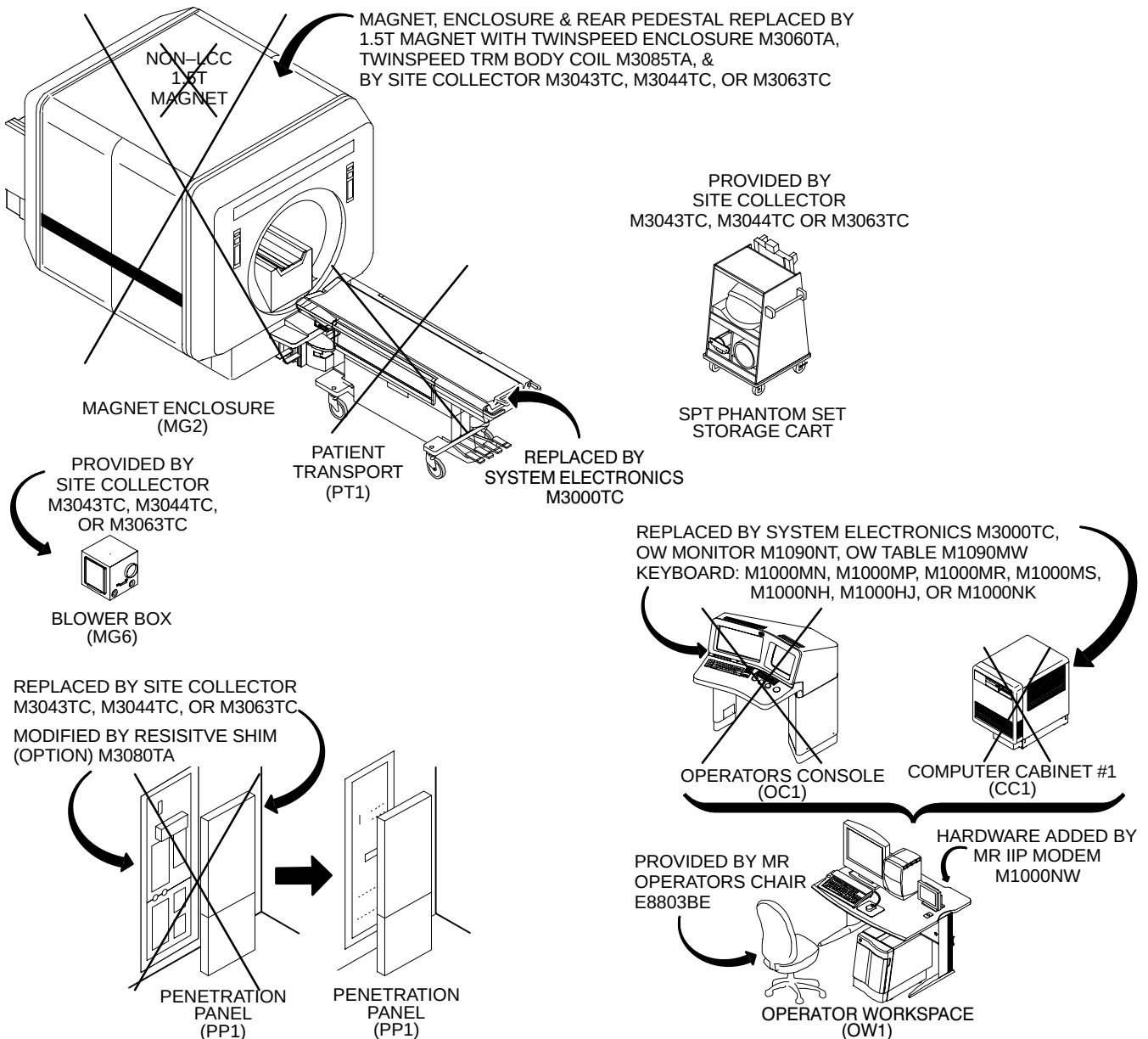
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE
TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-21

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



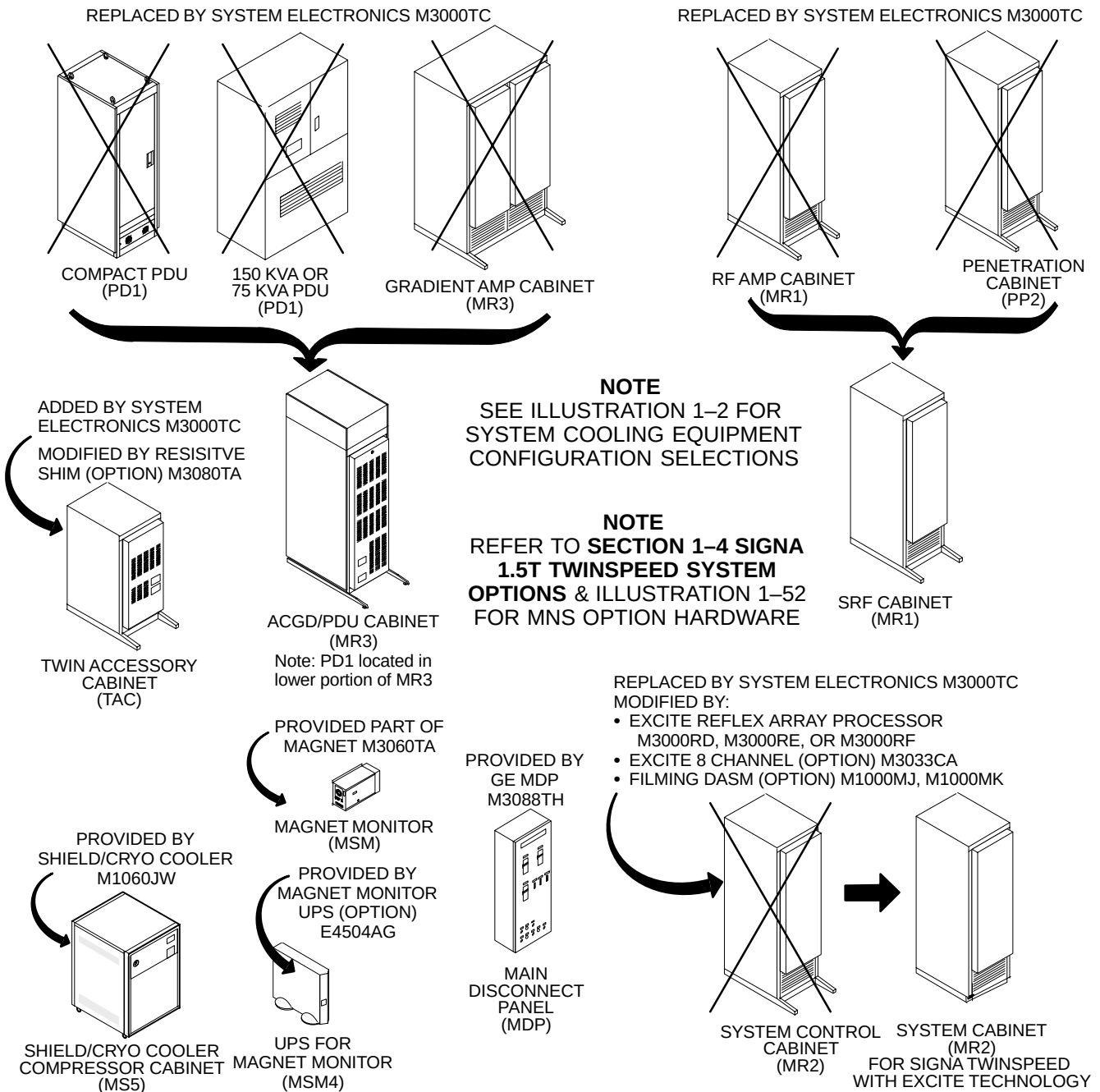
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE
TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-22

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



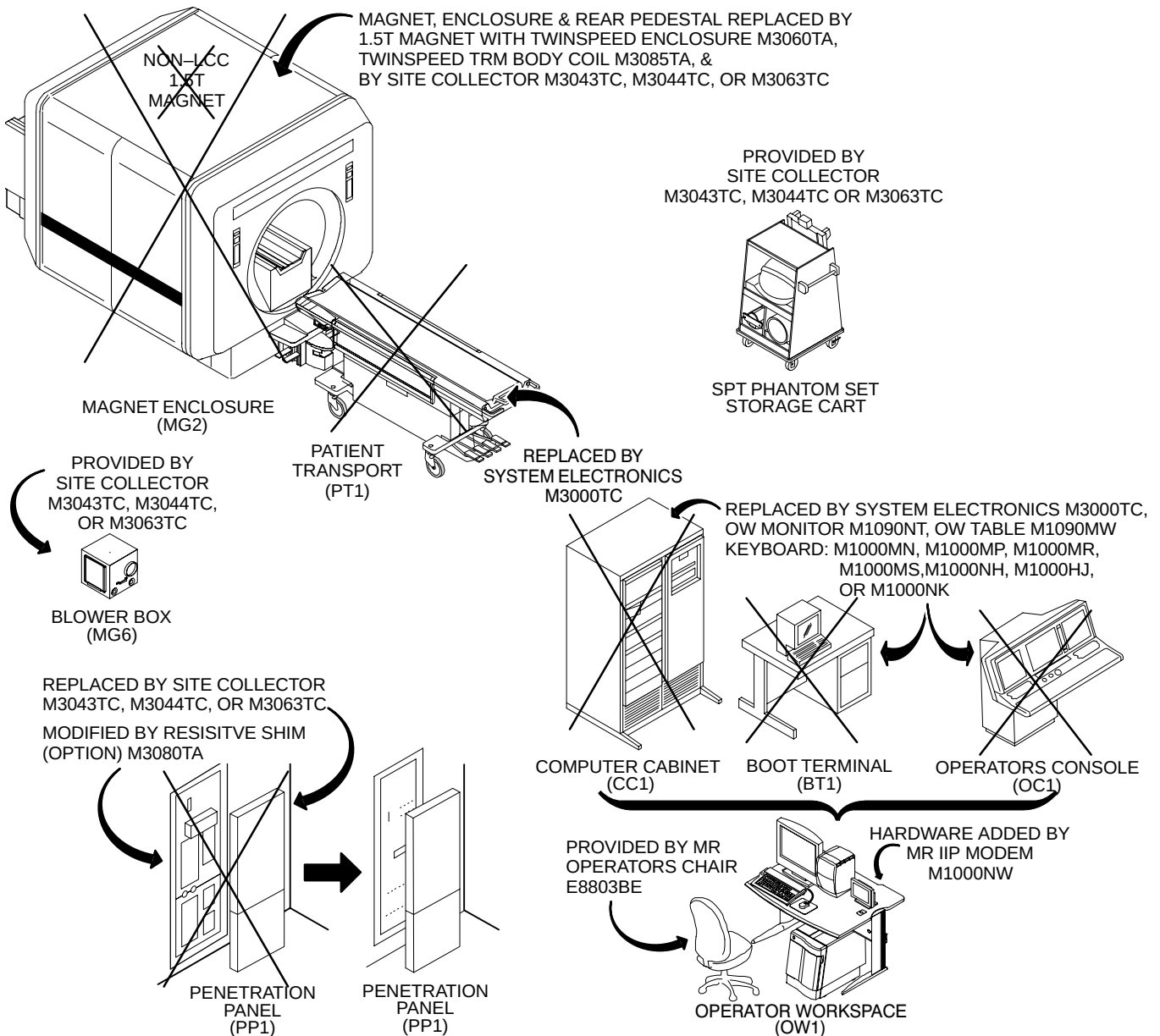
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-23

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



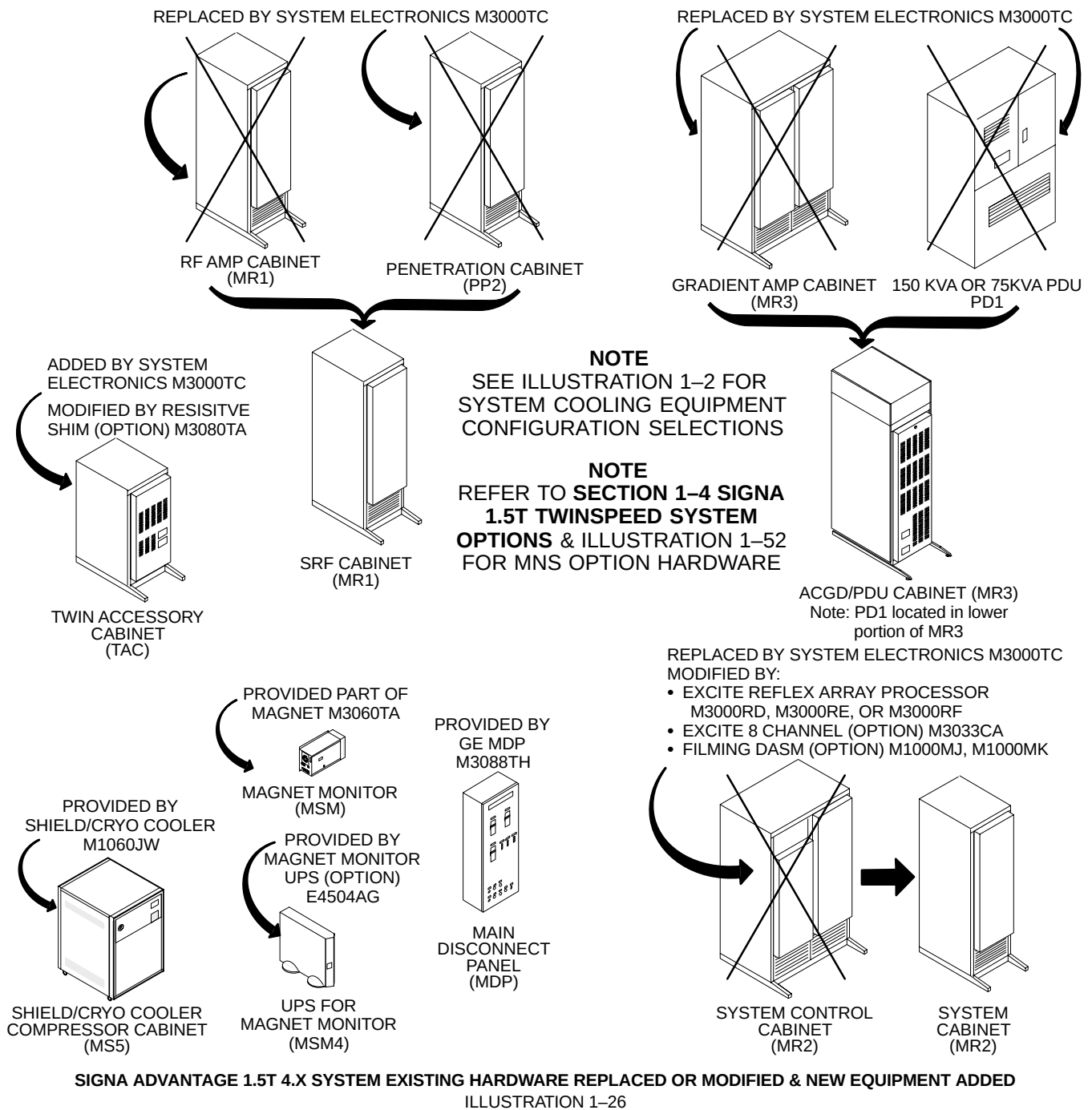
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-24

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 1.5T 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-25

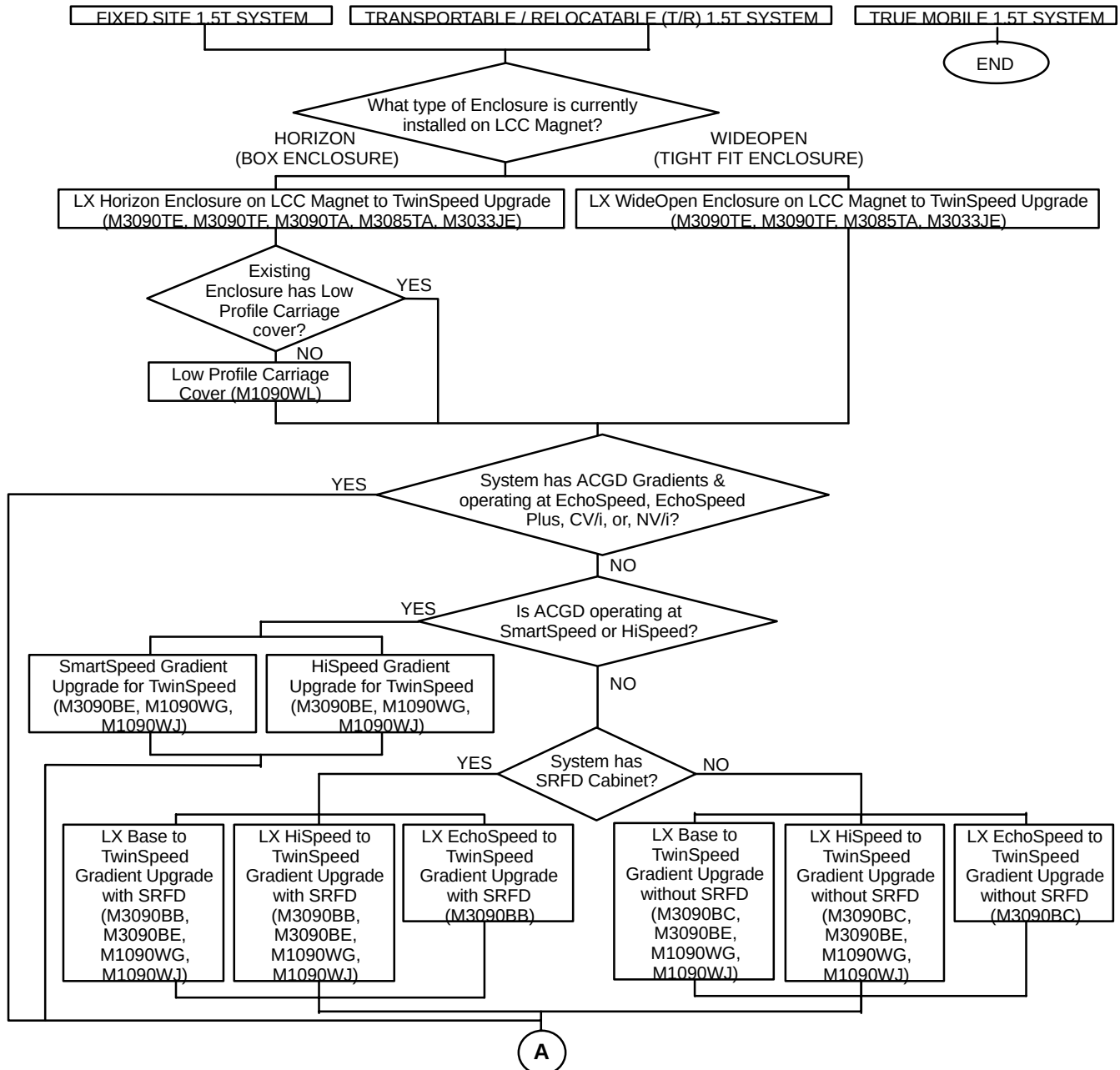
1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



1-3 UPGRADES TO SIGNA TWINSPEED

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet

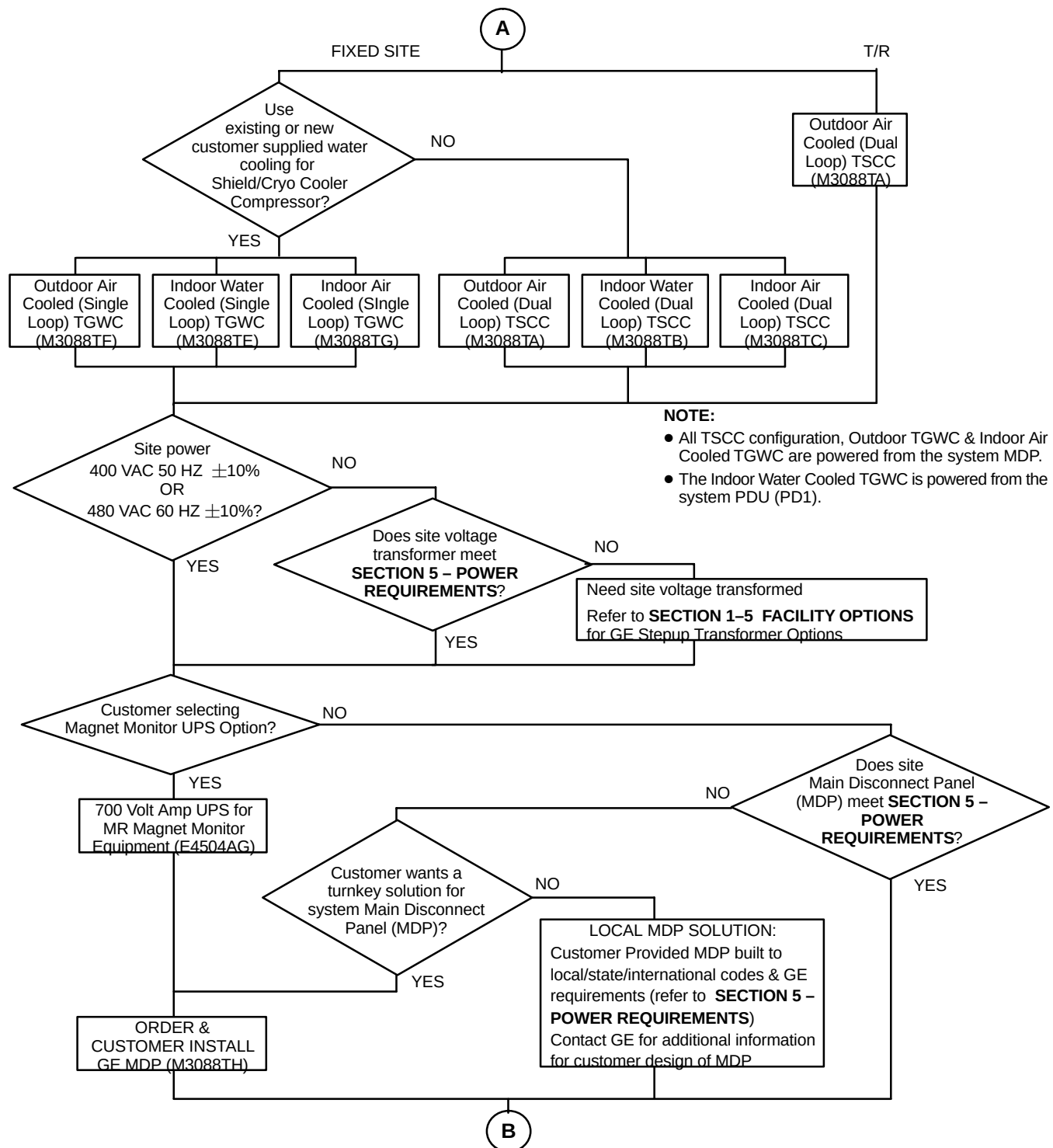
The flowchart shown in Illustrations 1-28 through 1-31 lists the upgrade catalogs for a Signa MR/i, CV/i, & NV/i or Signa LX (with ACGD or SGD) and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed. Illustrations 1-27 through 1-33 show the cabinets replaced and modified by the catalogs listed in the flowchart.



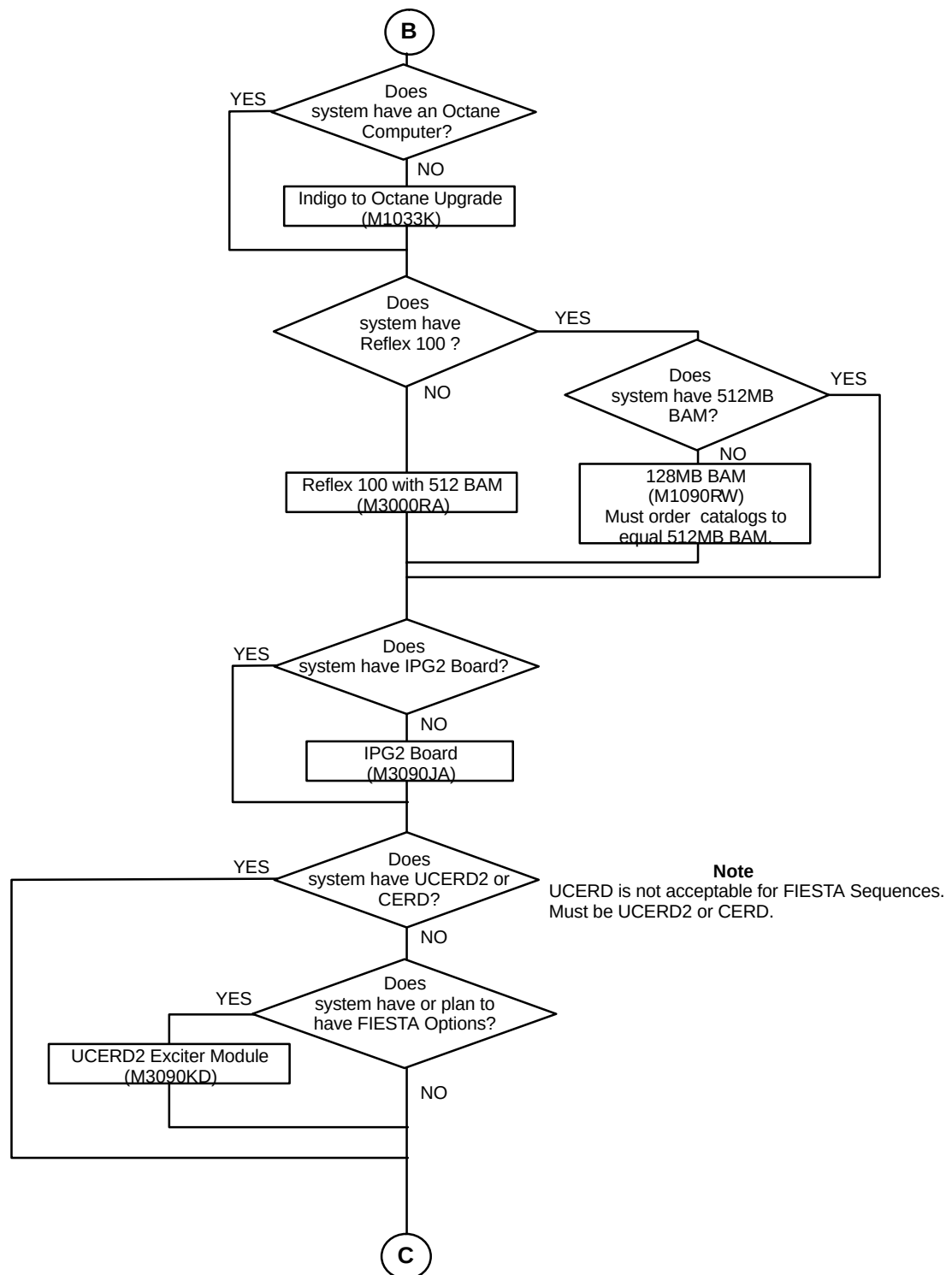
SIGNA MR/i LX UPGRADE TO SIGNA TWINSPEED CATALOGS

ILLUSTRATION 1-28

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



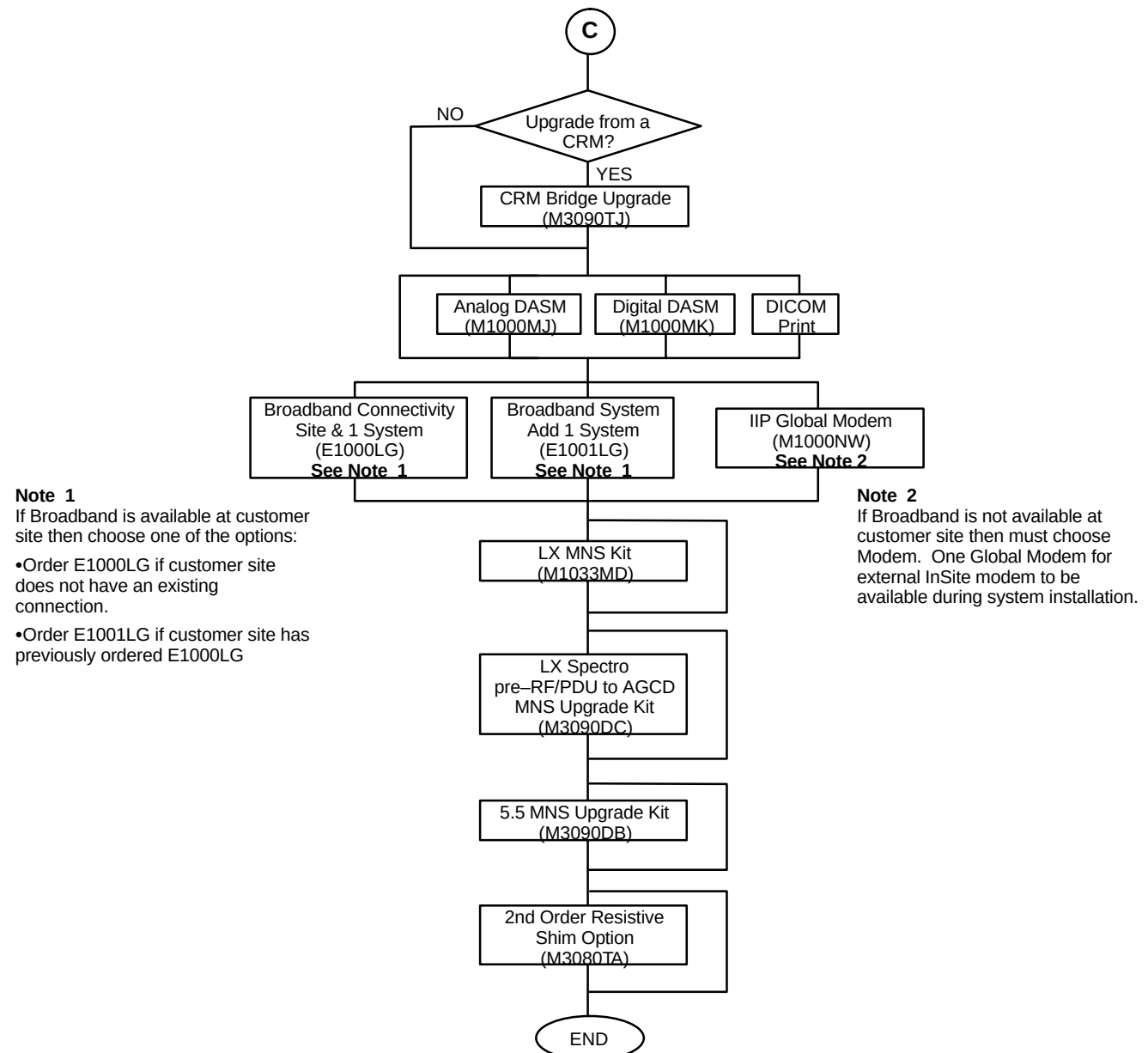
1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



SIGNA LX UPGRADE TO SIGNA TWINSPEED CATALOGS (CONTINUED)

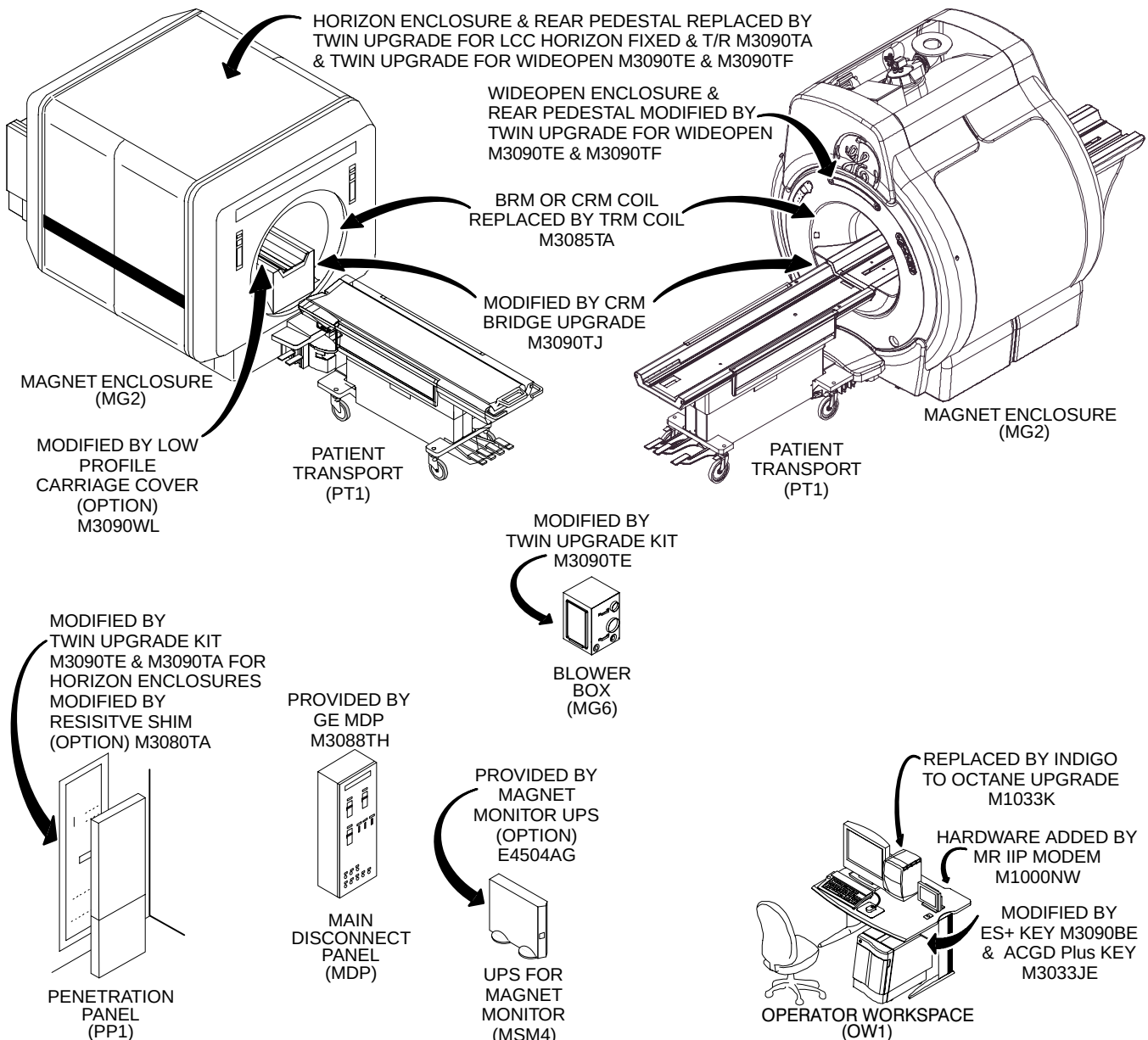
ILLUSTRATION 1-30

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



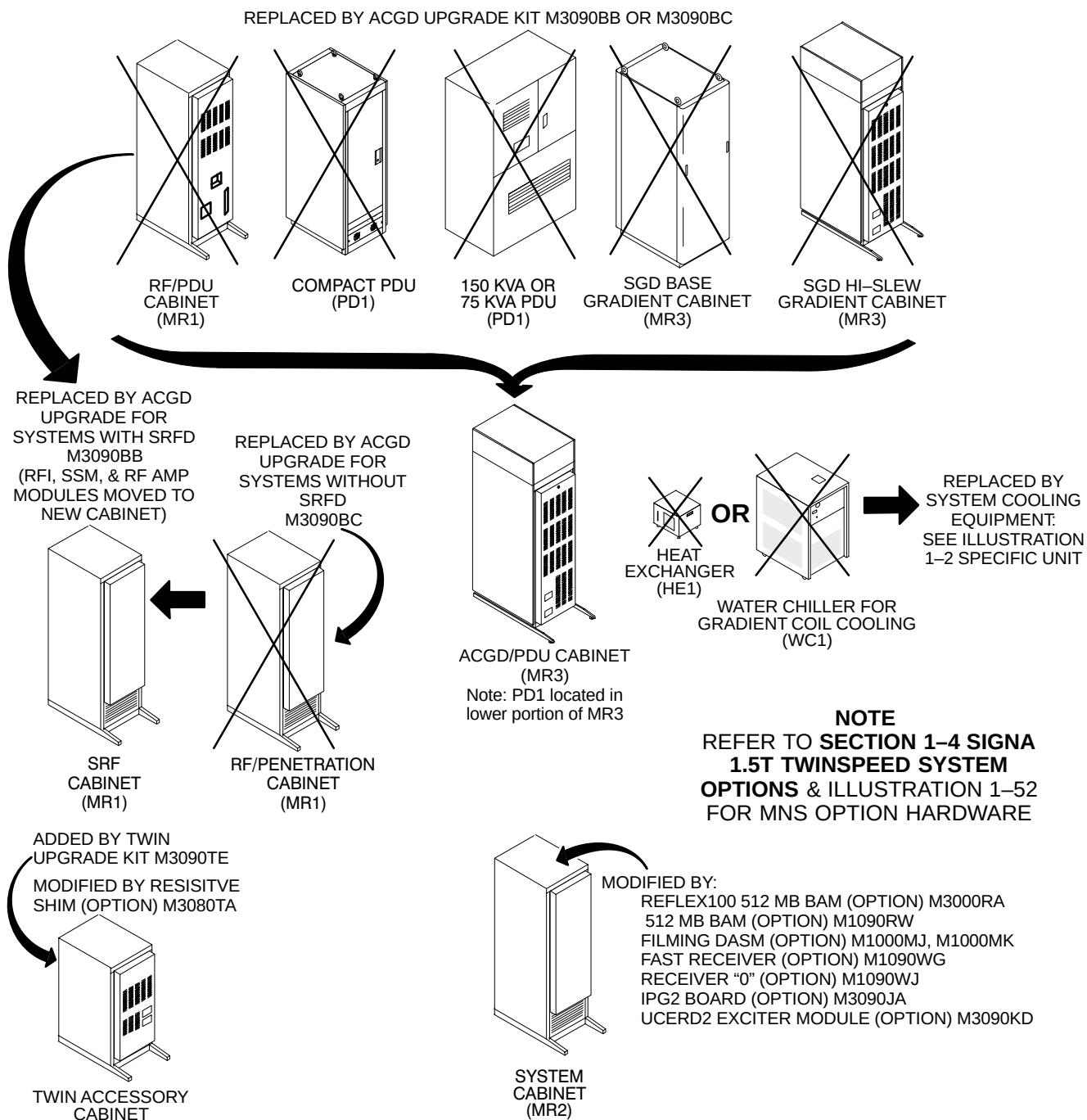
SIGNA HORIZON OR SIGNA LX UPGRADE TO SIGNA TWINSPEED CATALOGS (CONTINUED)
ILLUSTRATION 1-31

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



SIGNA LX & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-32

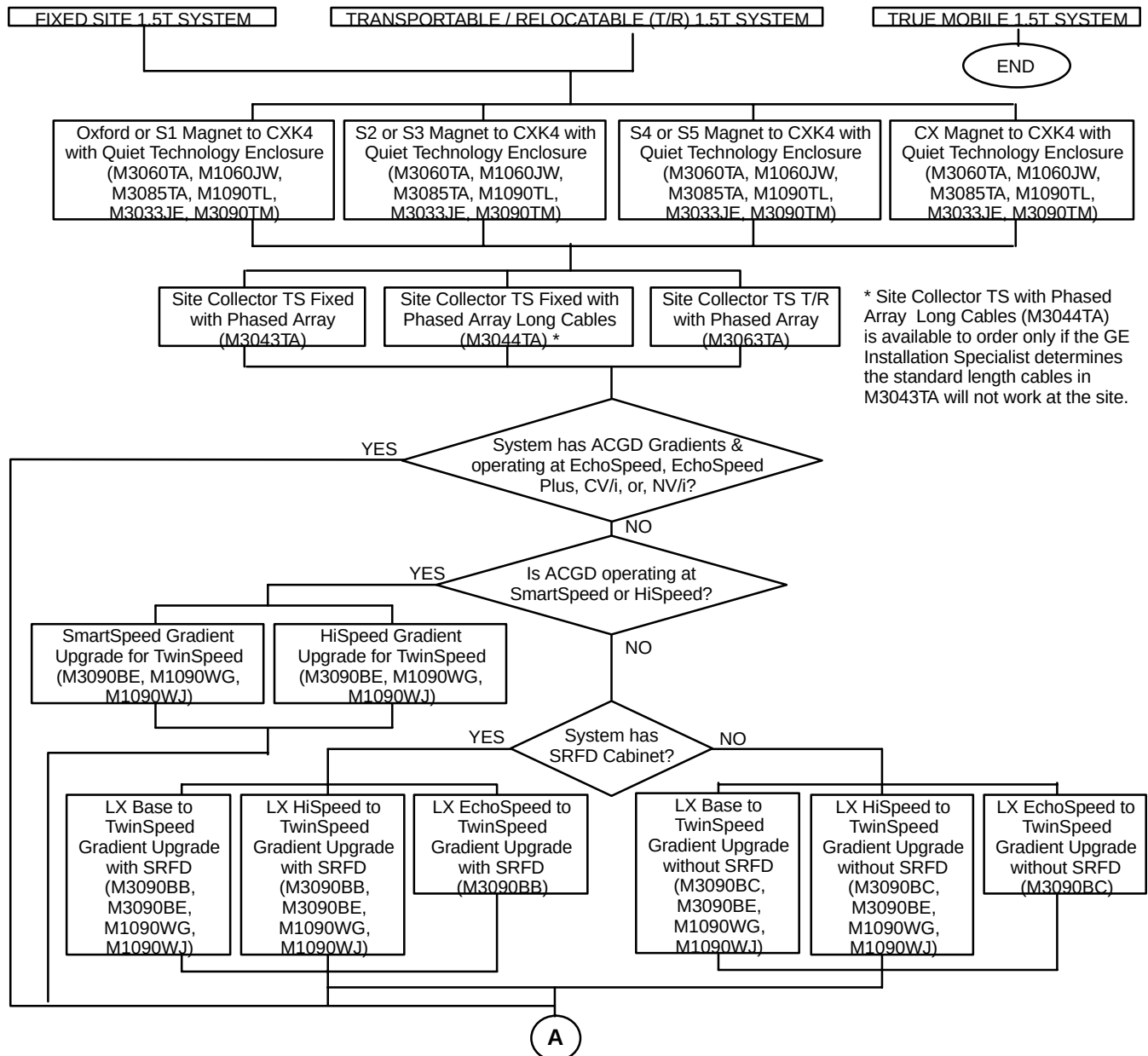
1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



SIGNA LX & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-33

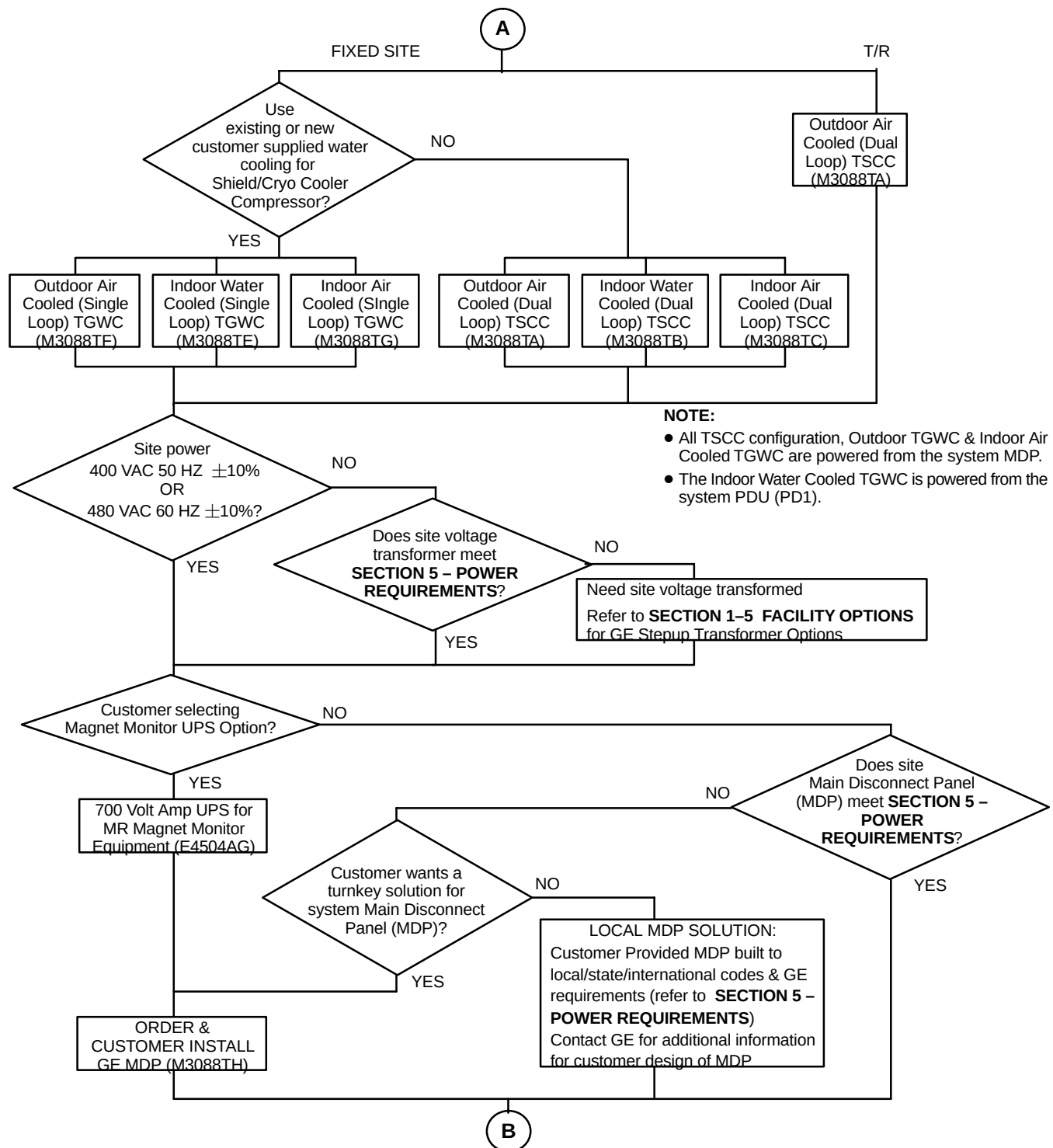
1-3-2 Signa LX & CVMR Without LCC Magnet Installed

The flowchart shown in Illustrations 1-34 through 1-37 lists the upgrade catalogs for a Signa LX or CVMR (with ACGD or SGD) and current system installed magnet is NOT an LCC Magnet upgrading to Signa 1.5T TwinSpeed. Illustrations 1-38 through 1-39 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.



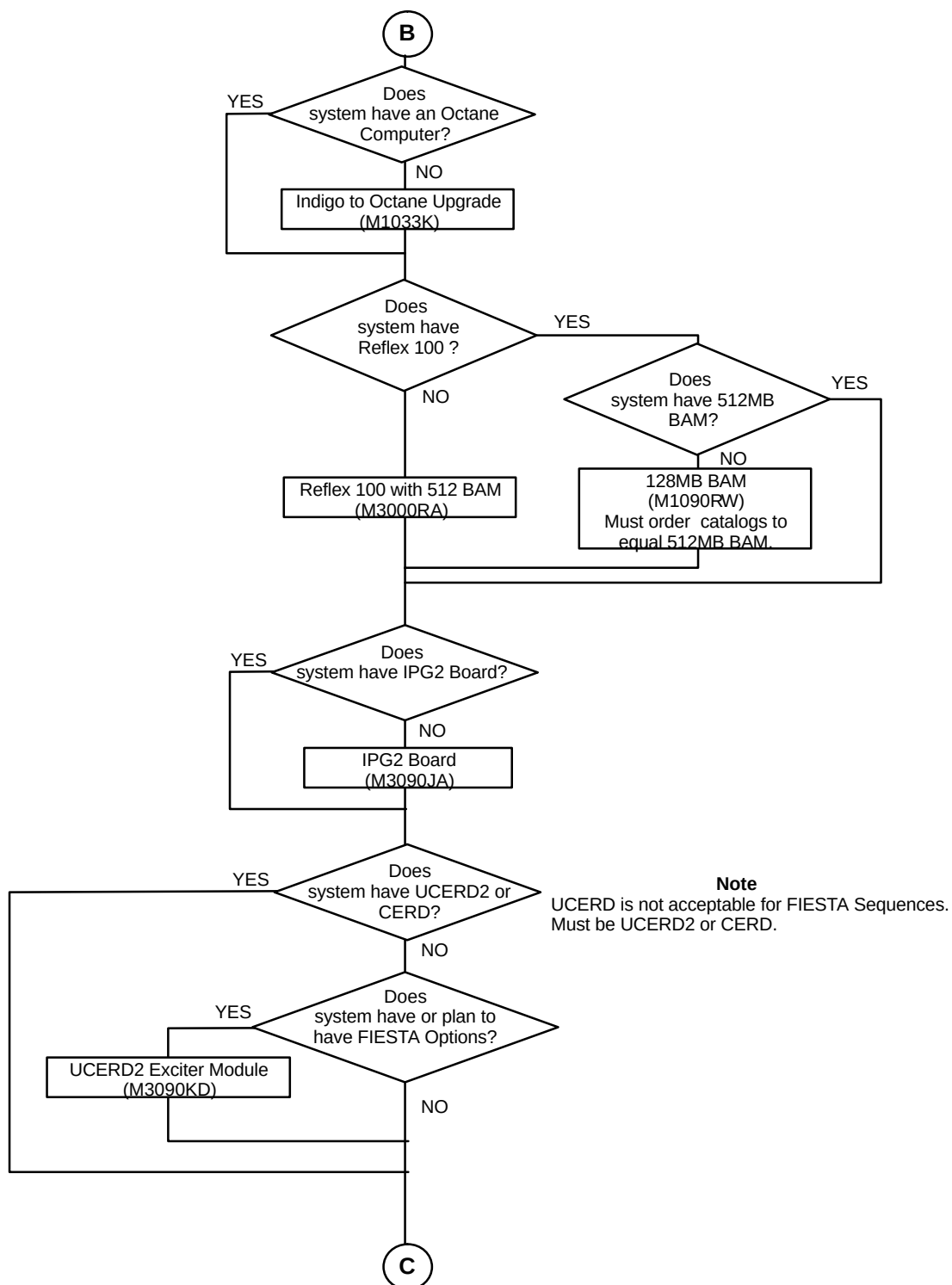
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-34

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



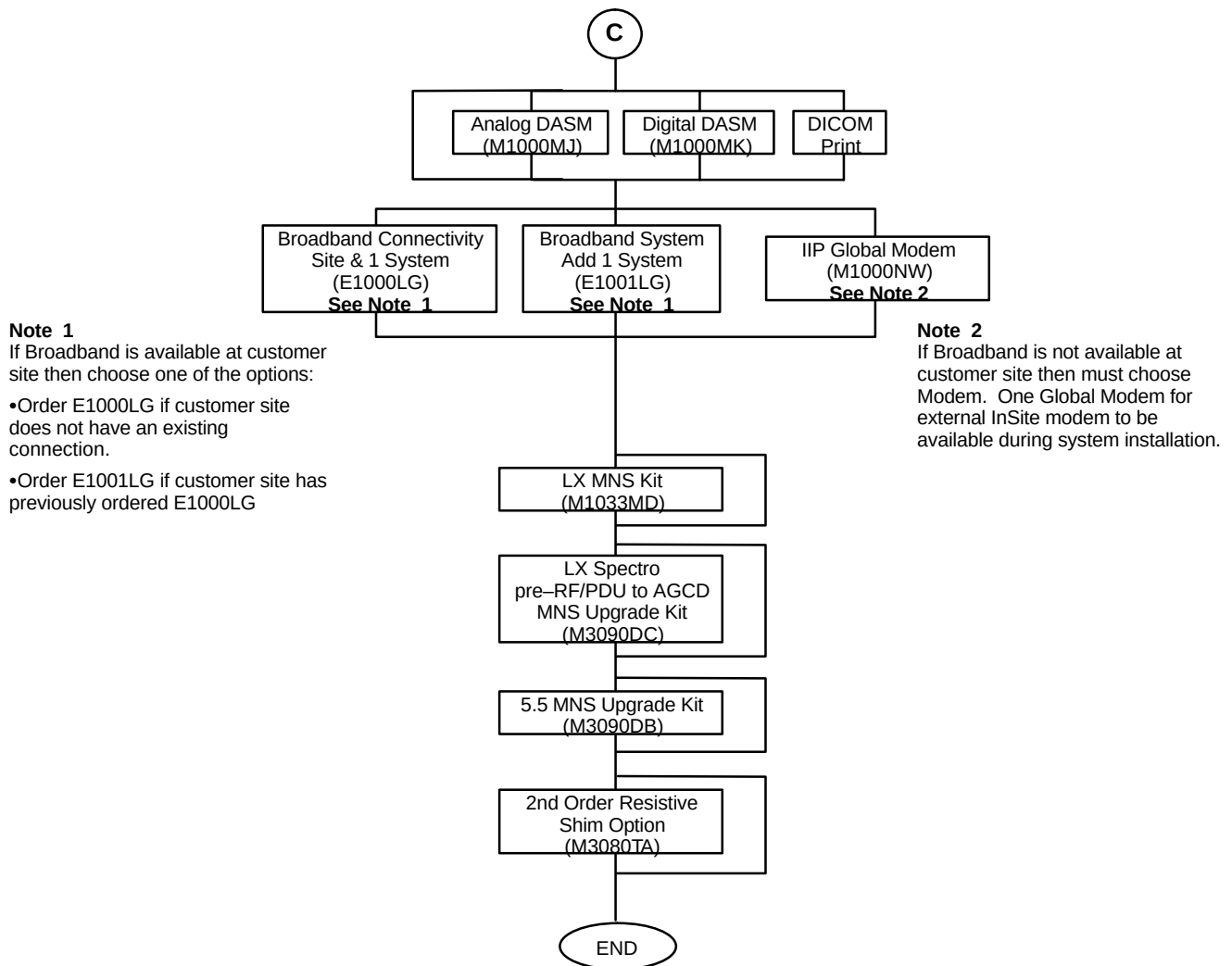
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-35

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



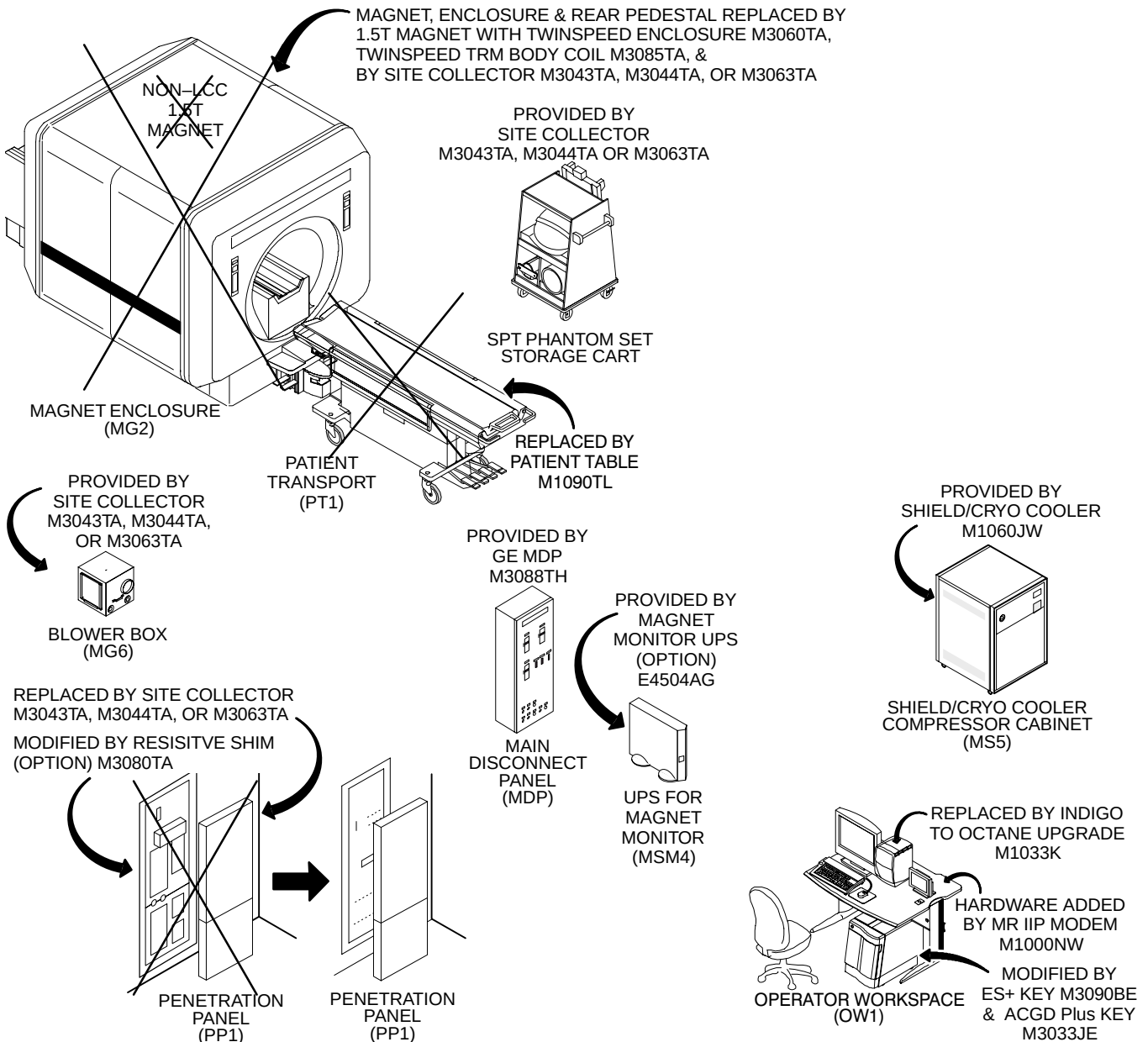
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-36

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



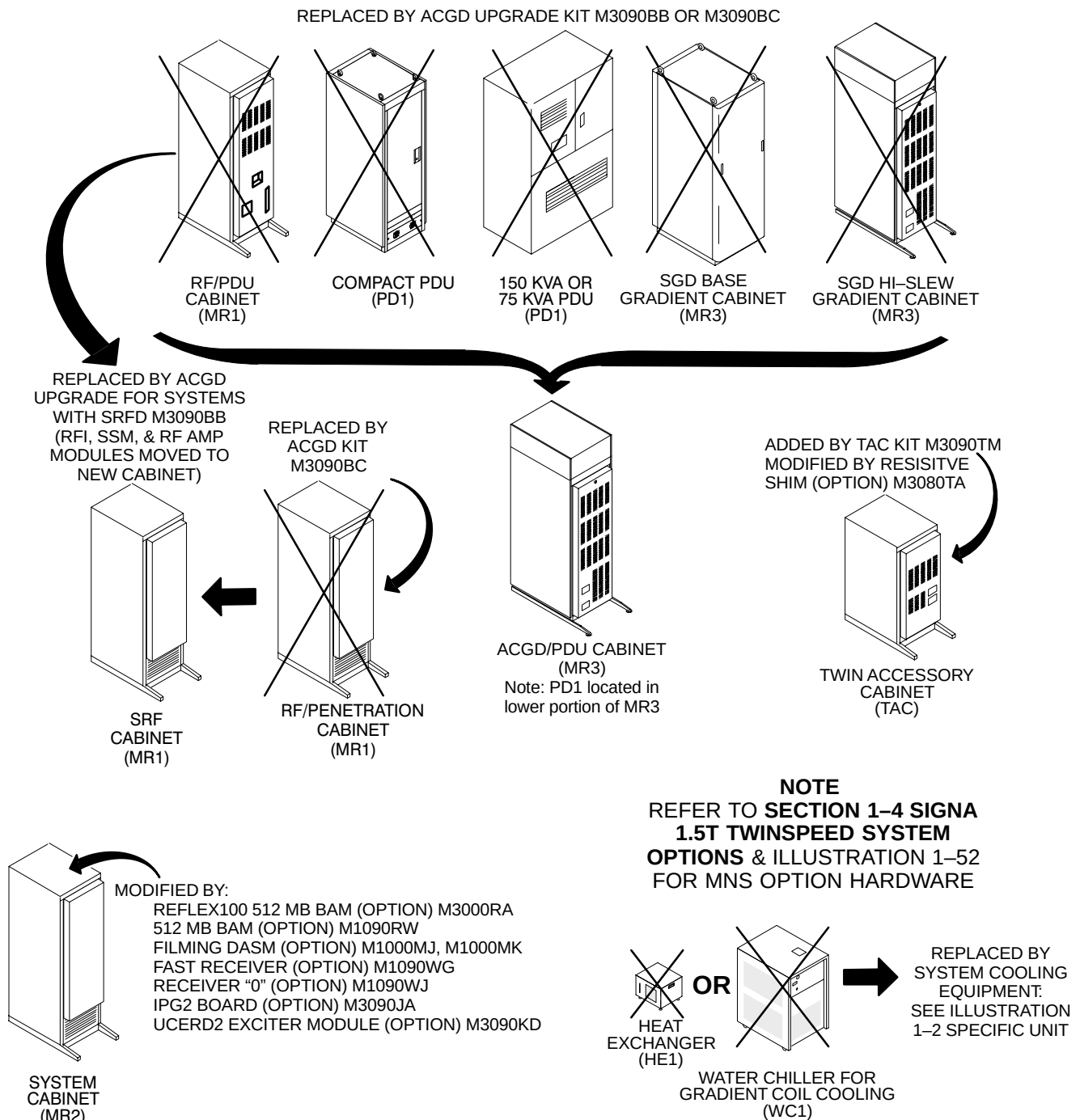
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-37

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-38

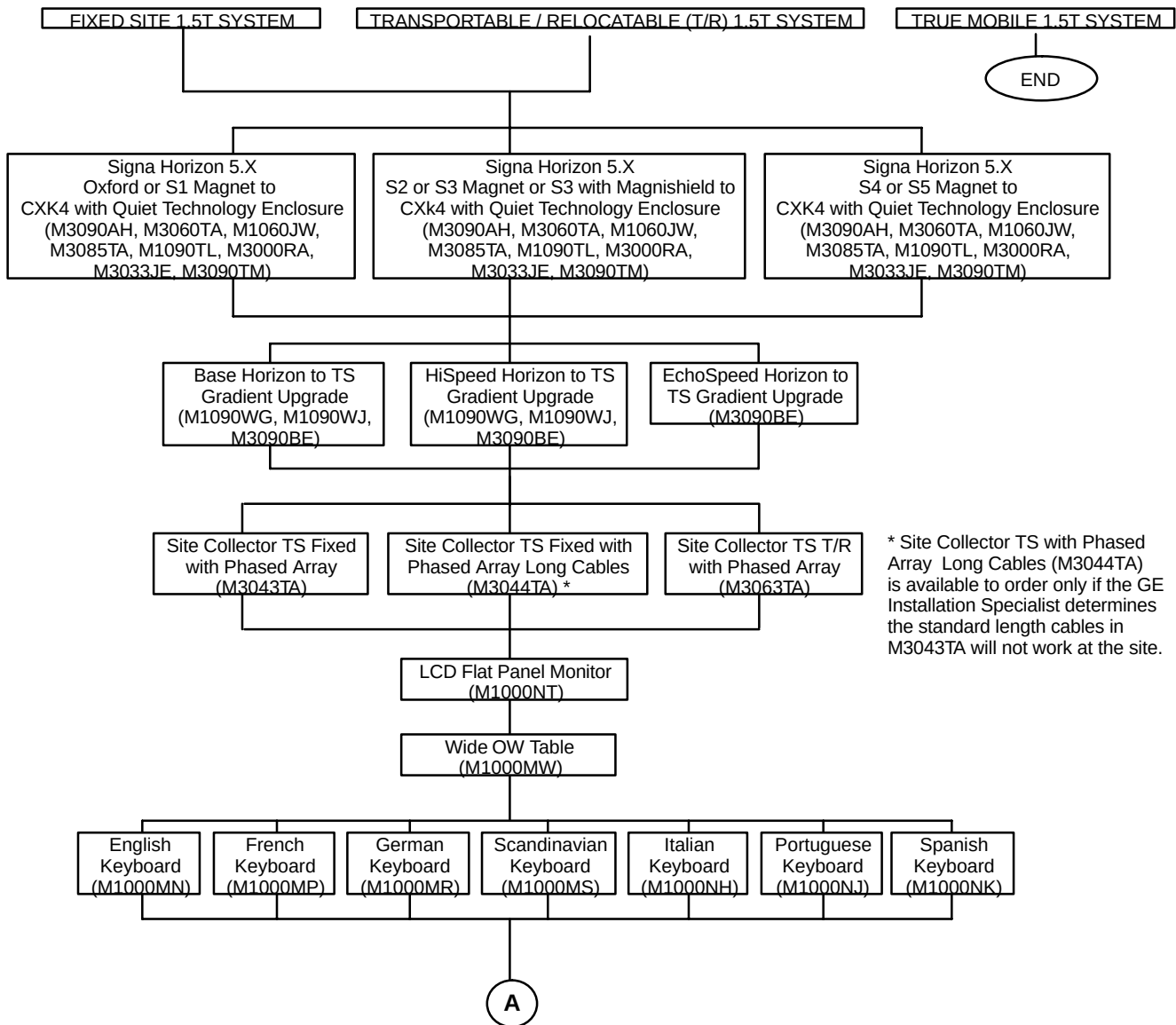
1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-39

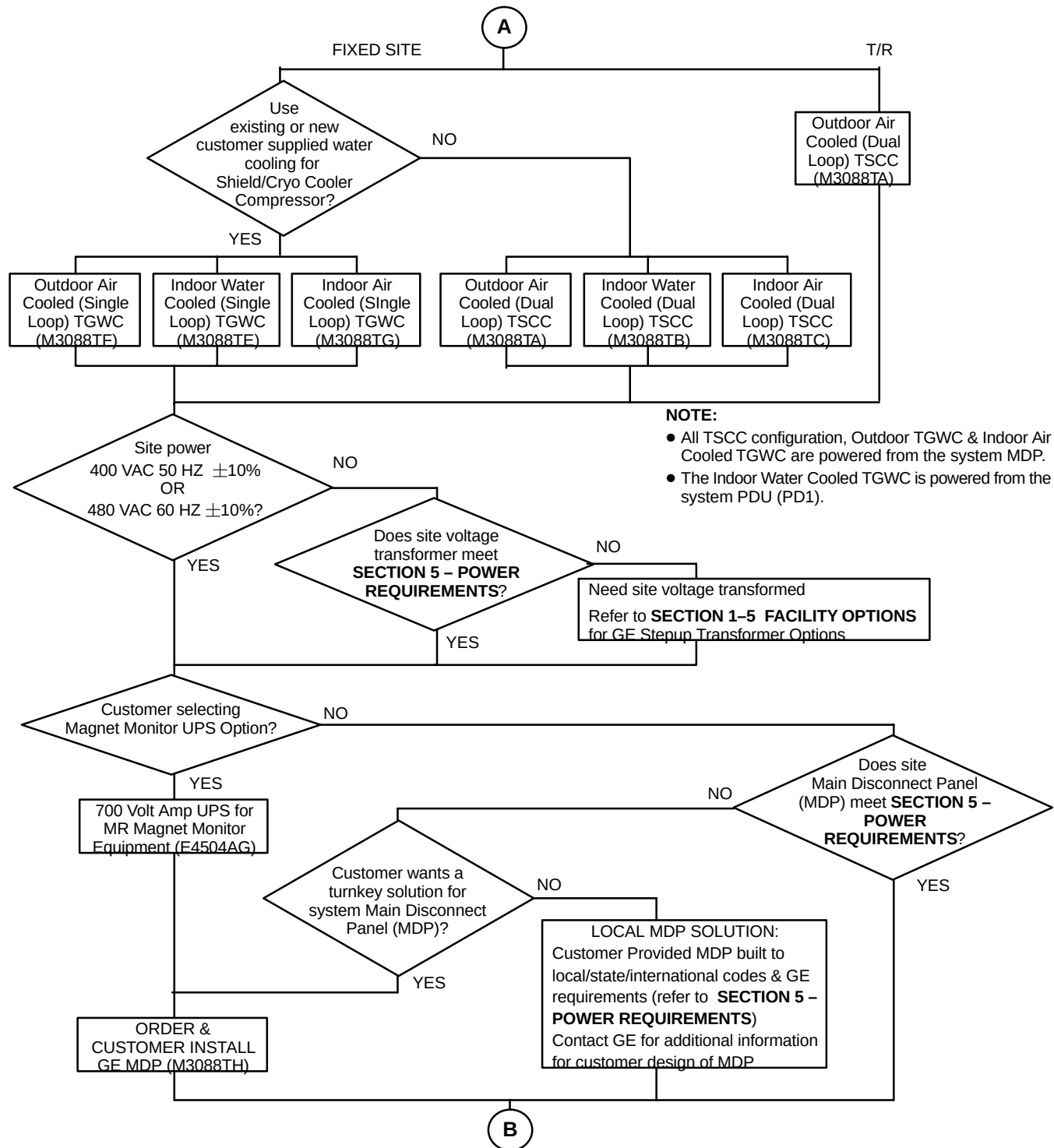
1-3-3 Signa Horizon

Upgrades from Signa Horizon 5.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-40 through 1-42 lists the Signa Horizon 5.X upgrade catalogs. Illustrations 1-43 and 1-44 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.



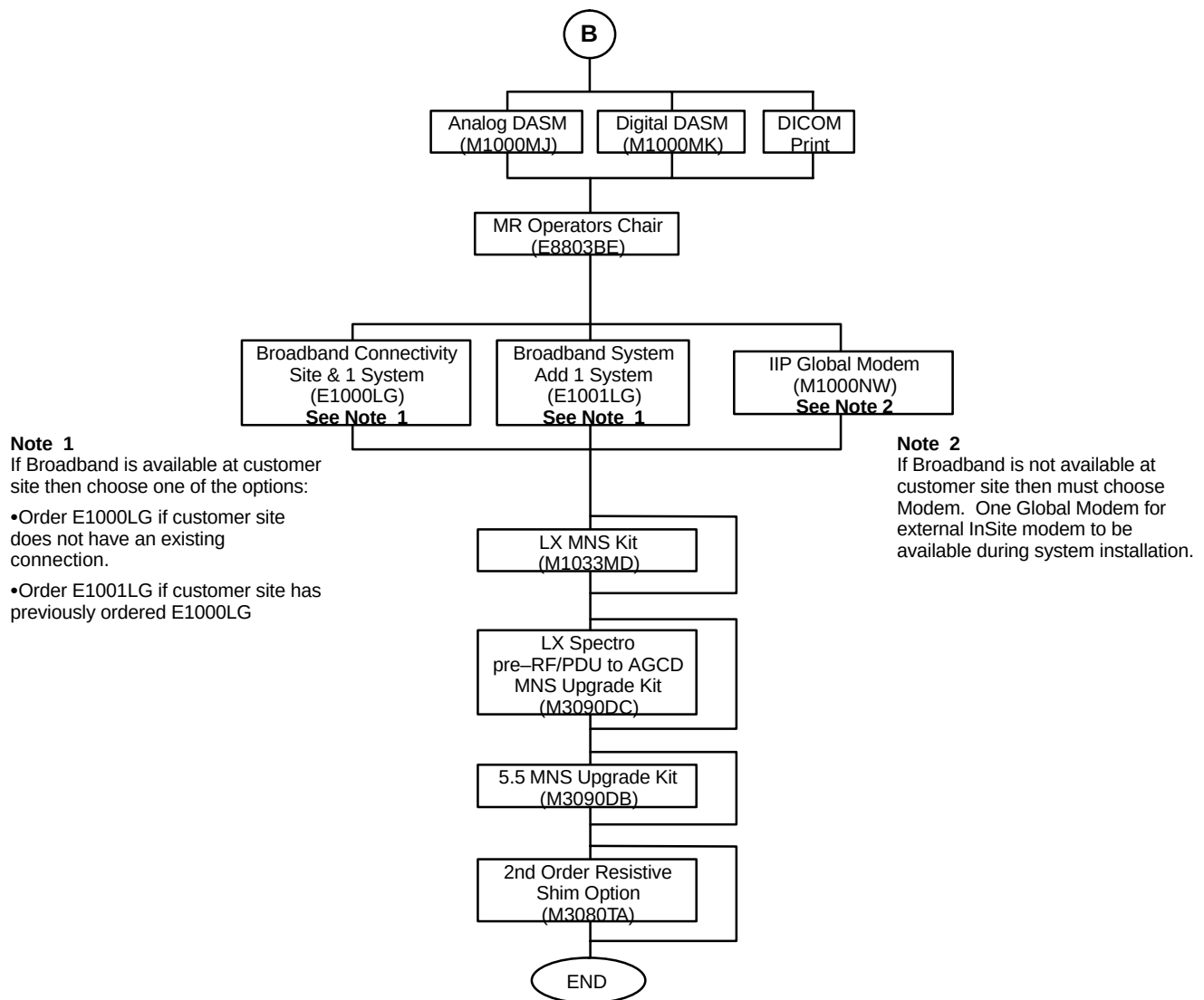
SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-40

1-3-3 Signa Horizon (Continued)



SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-41

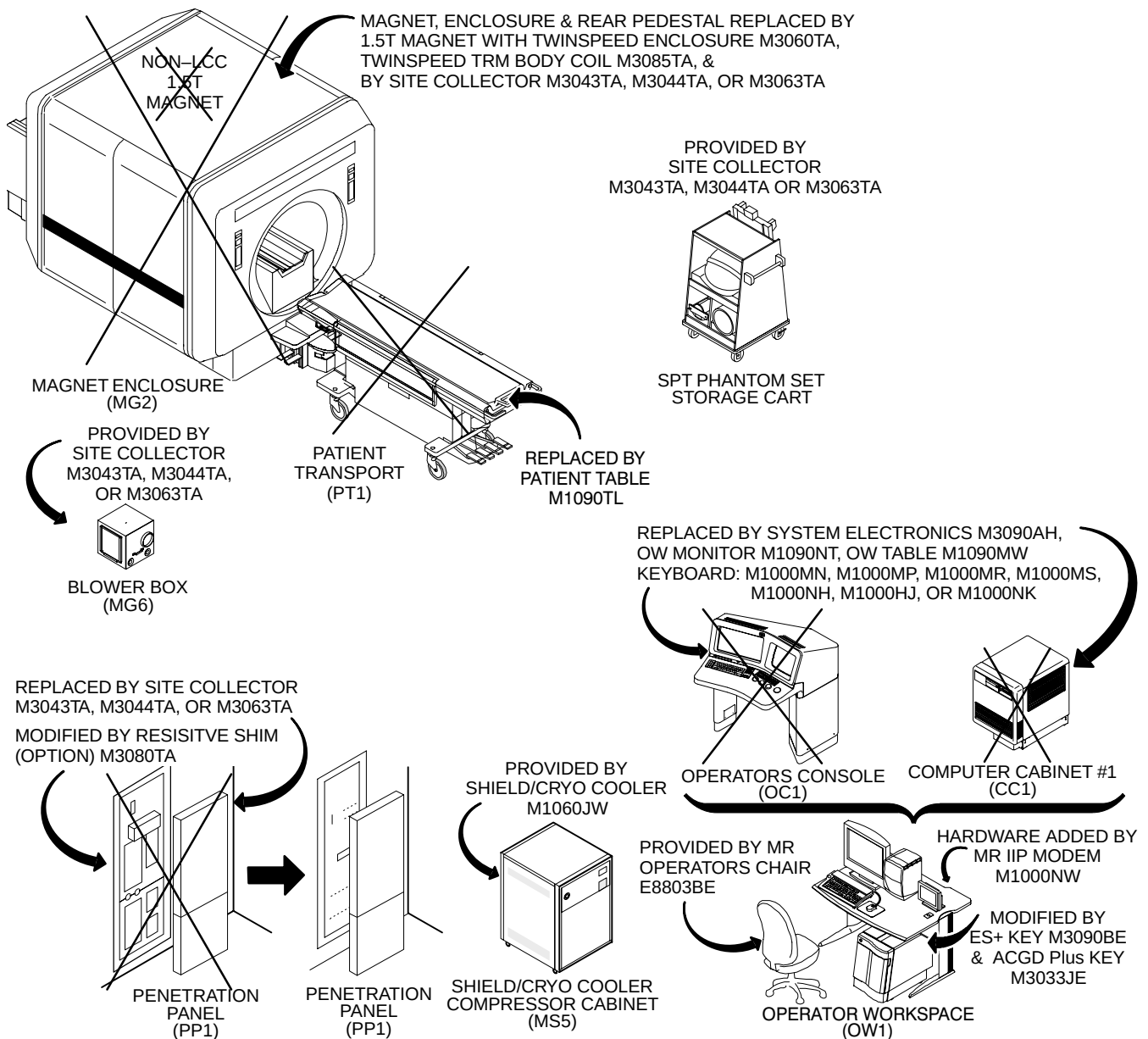
1-3-3 Signa Horizon (Continued)



SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS

ILLUSTRATION 1-42

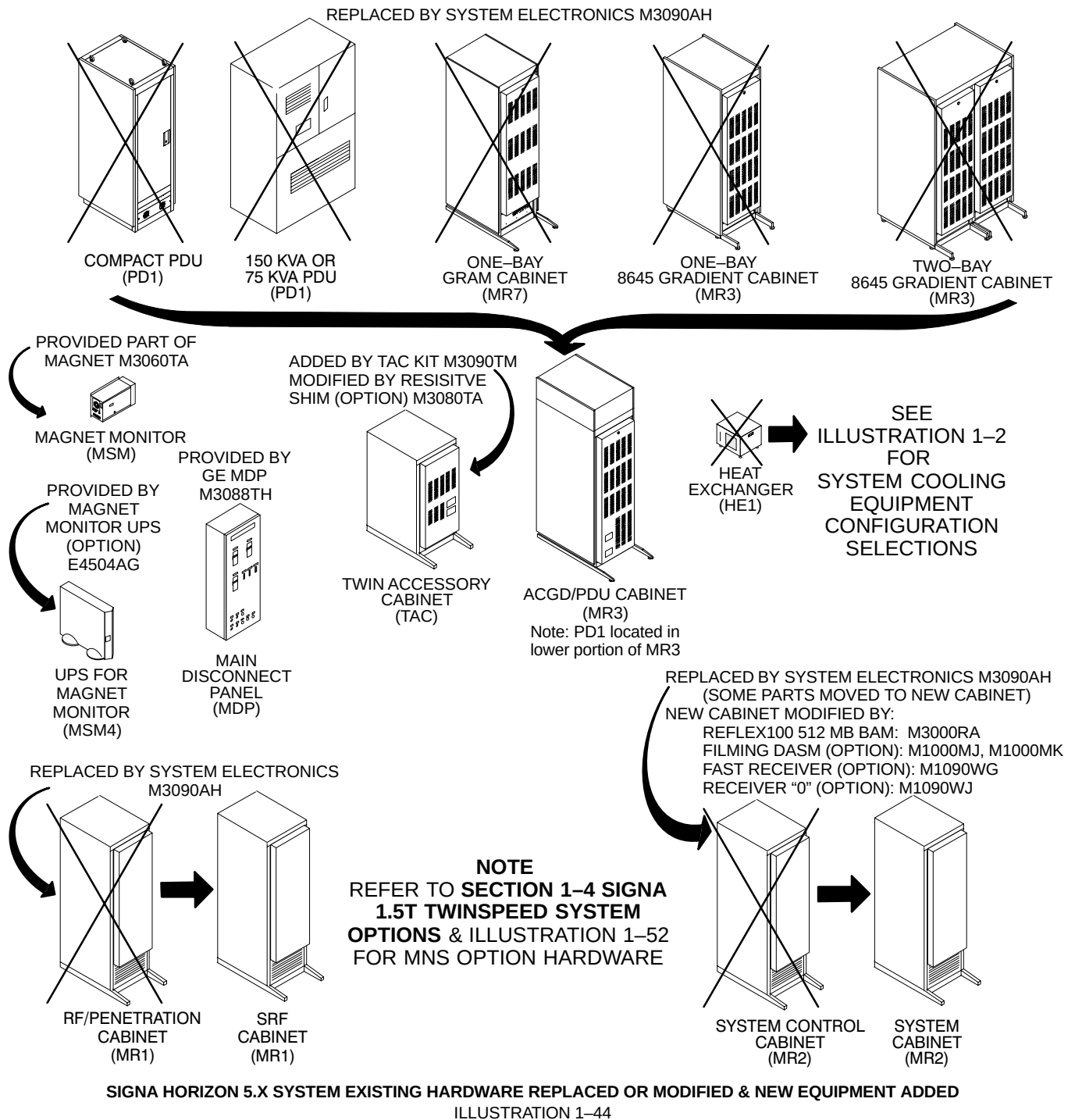
1-3-3 Signa Horizon (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-43

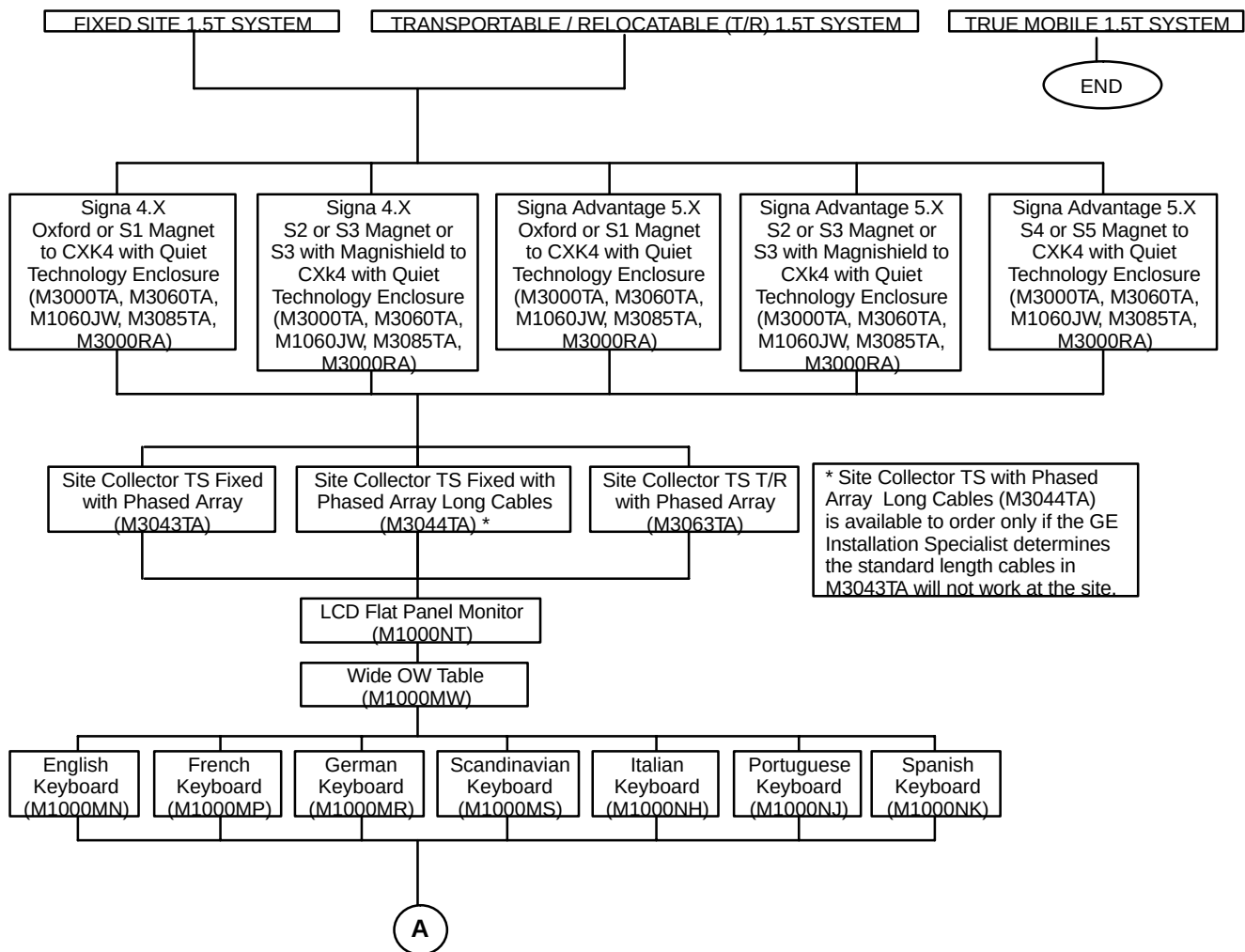
1-3-3 Signa Horizon (Continued)



1-3-4 Signa Advantage 5.X & 4.X

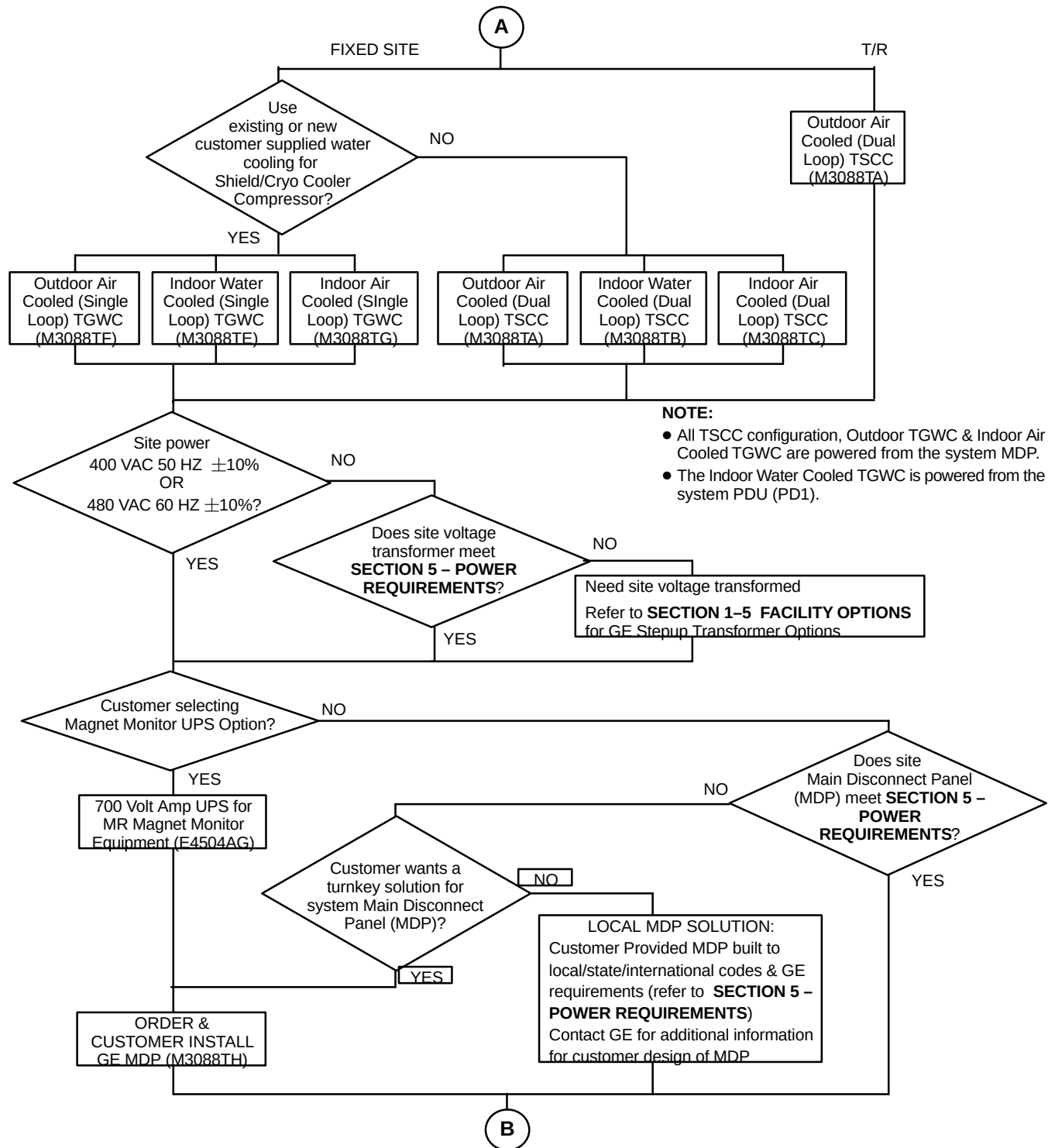
Upgrades from Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-45 through 1-47 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 5.X Software Only & Signa Advantage 5.X see Illustrations 1-48 and 1-49
- Signa Advantage 4.x see Illustrations 1-50 and 1-51



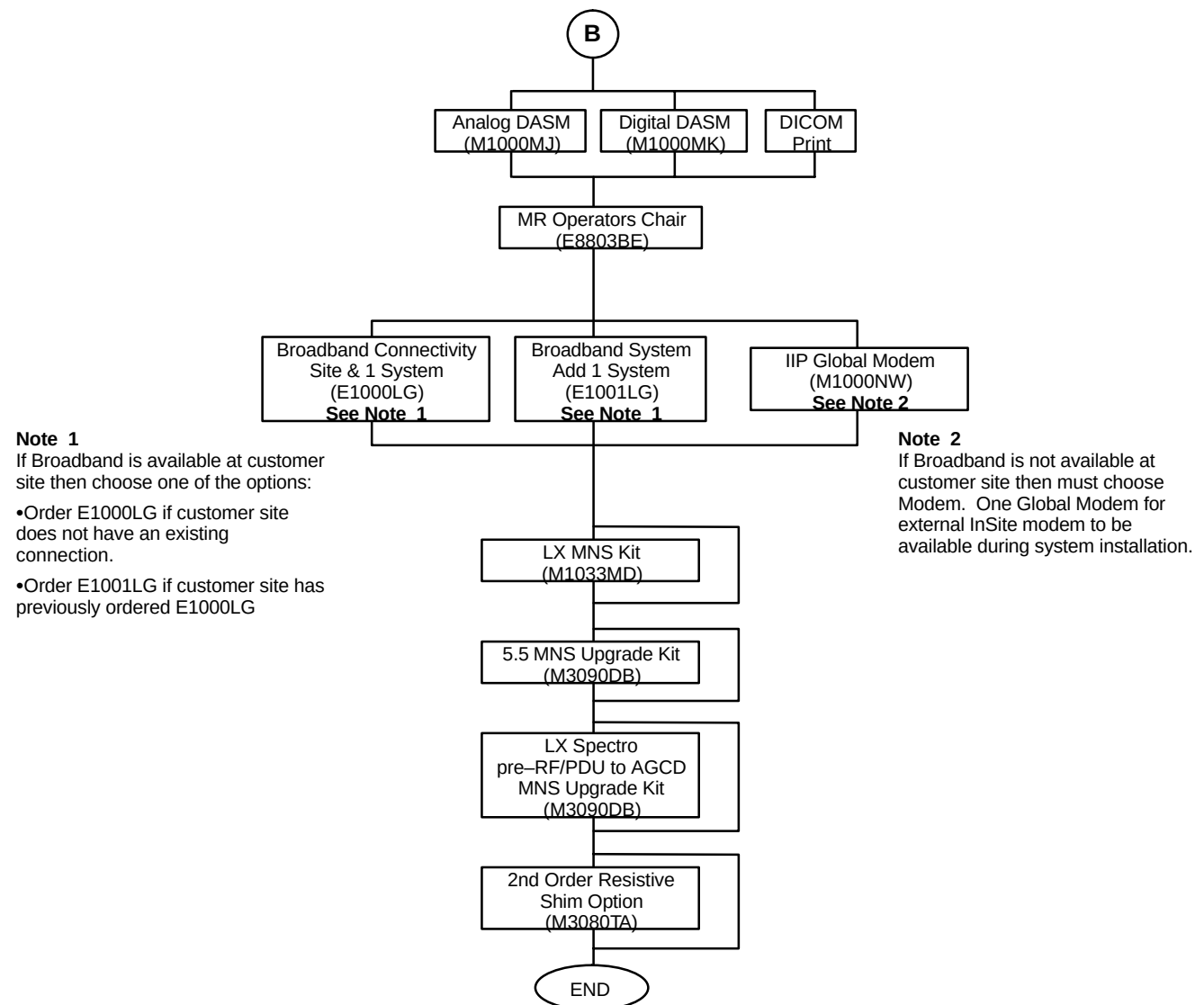
SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-45

1-3-3 Signa Advantage 5.X & 4.X (Continued)



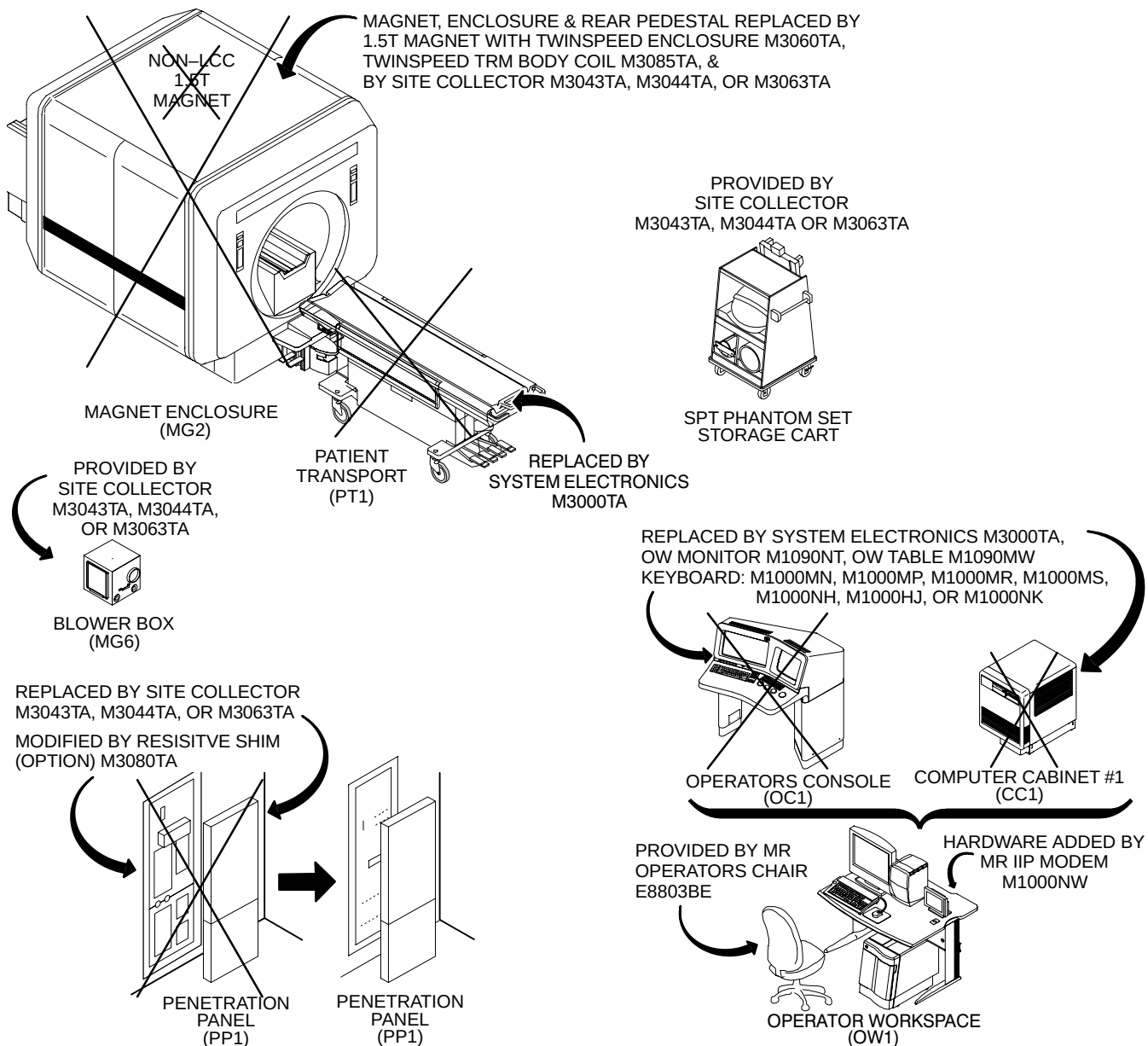
SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-46

1-3-3 Signa Advantage 5.X & 4.X (Continued)

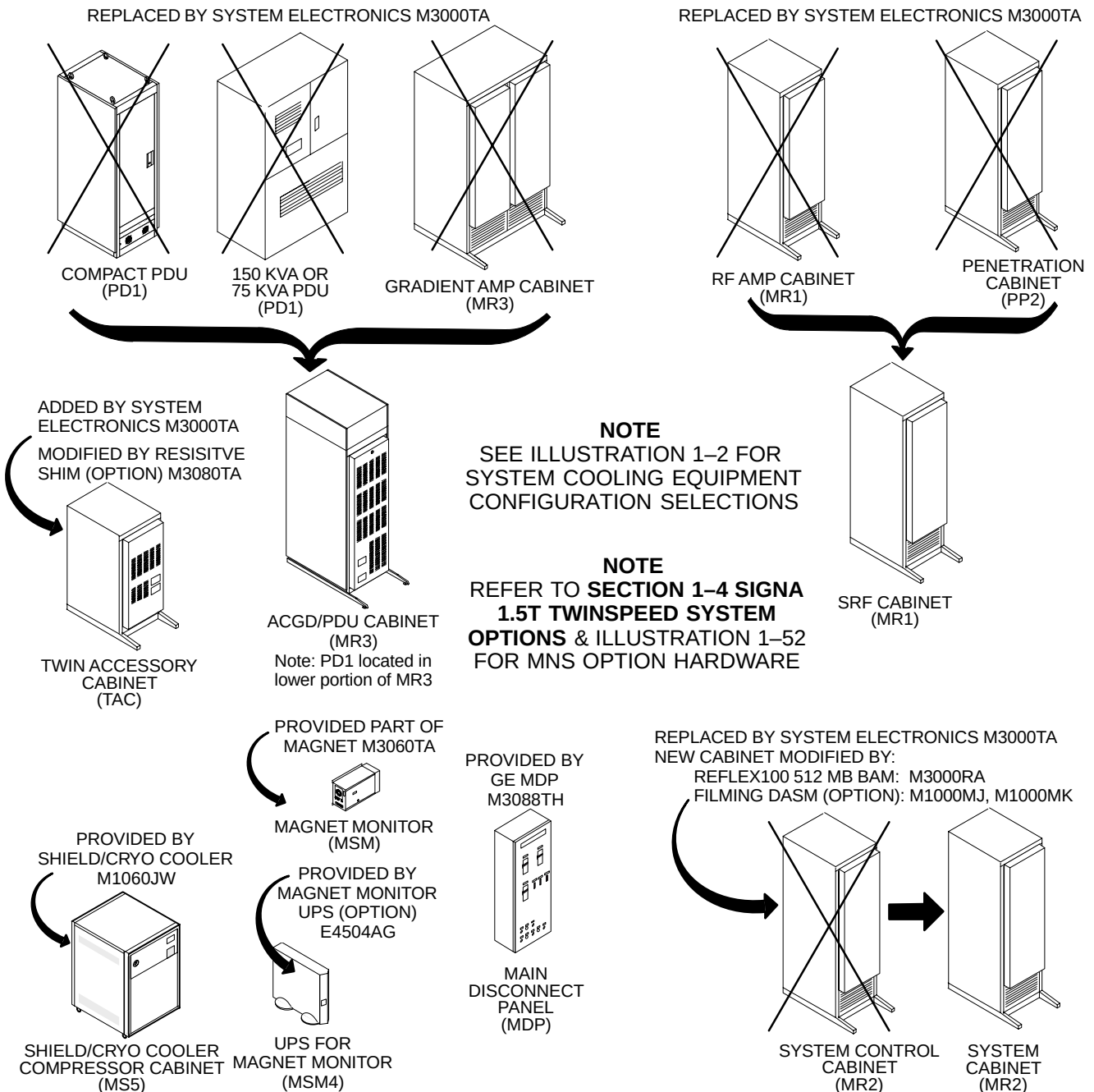


SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
 ILLUSTRATION 1-47

1-3-3 Signa Advantage 5.X & 4.X (Continued)

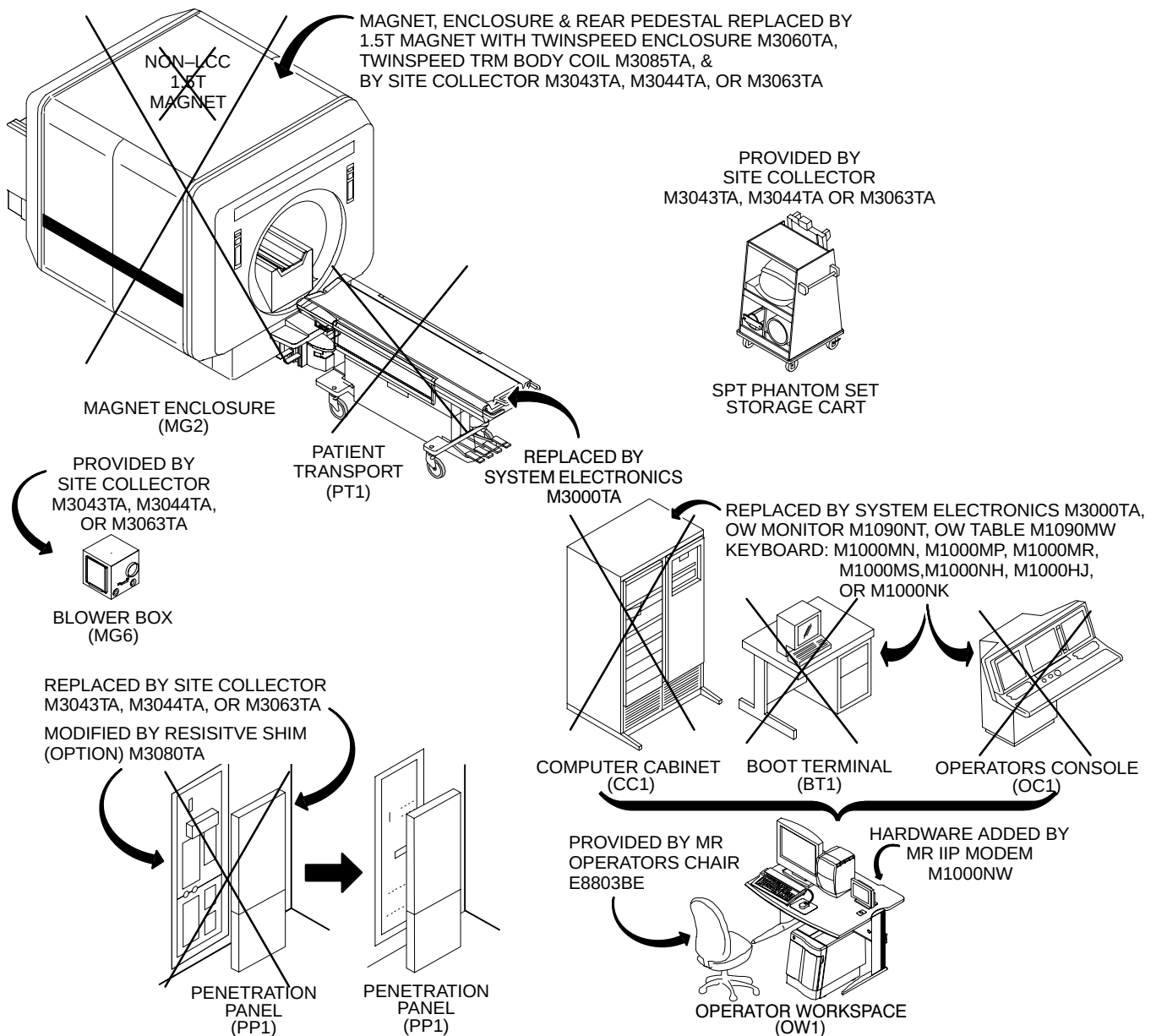


1-3-3 Signa Advantage 5.X & 4.X (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-49

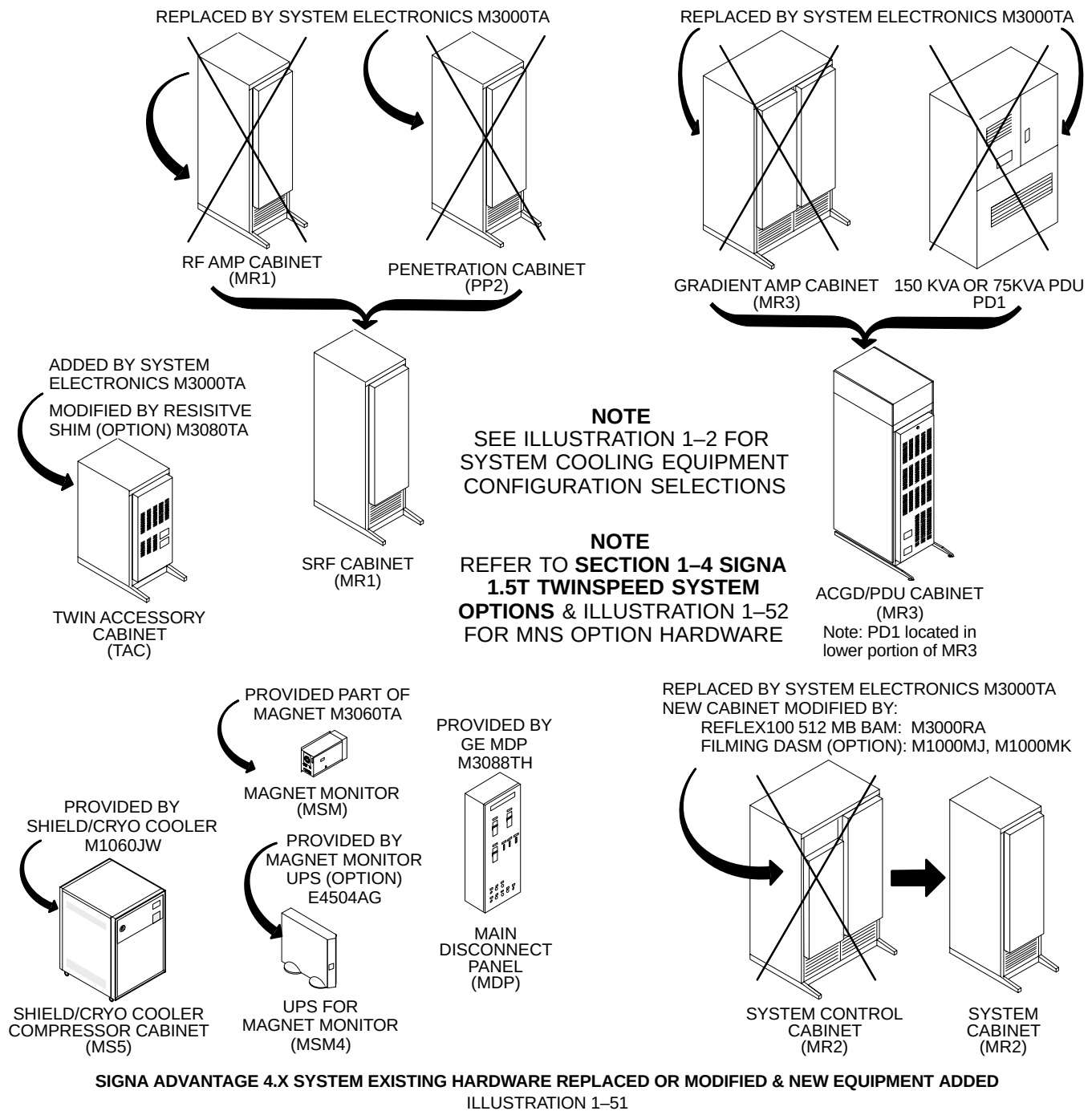
1-3-3 Signa Advantage 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-50

1-3-3 Signa Advantage 5.X & 4.X (Continued)



1-4 SIGNA 1.5T TWINSPEED SYSTEM OPTIONS

Signa 1.5T TwinSpeed system options which have site preparation impact include the following.

- 700 Volt Amp UPS for MR Magnet Monitor Equipment (E4504AG)

Note

The GE MDP (M3088TH) must be ordered and installed when the UPS for MR Magnet Monitor (E4504AG) is installed. The GE MDP provides Emergency Off control of the output of the UPS for system safety.

- 2nd Order Resistive Shim Option for Signa 1.5T TwinSpeed (M3080TA). The 2nd Order Shim Option correct inhomogeneities induced by the patient and restores the original homogeneity of the magnet. The Resistive Shim Module is installed in the Twin Accessory Cabinet.
- Advantage Workstation (AW). Refer to Direction 2261309–100 *Advantage Workstation Ultra60 Pre-Installation*.

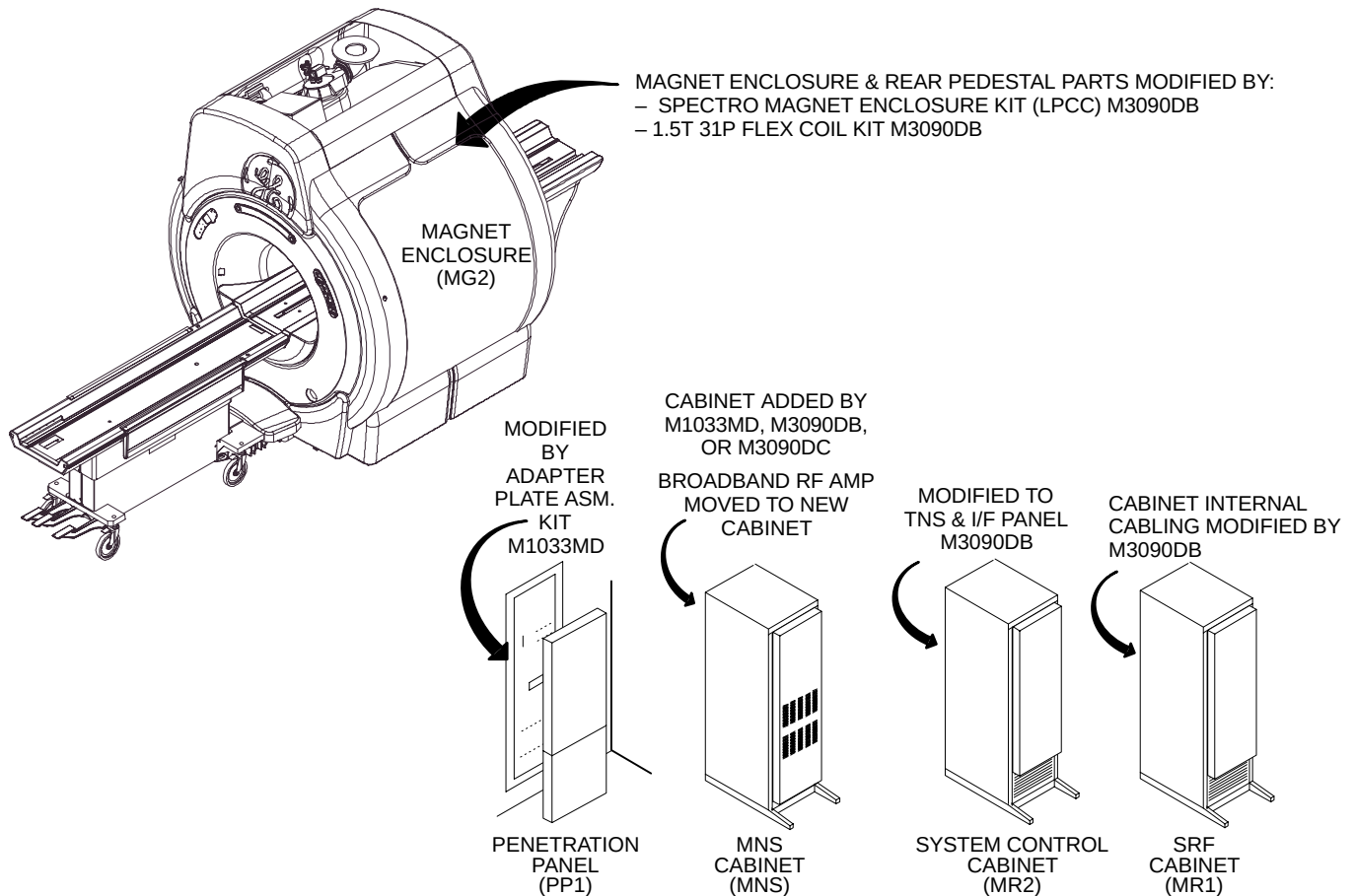
Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- Remote Keyboard Option for use with Operator Workspace PC LCD:
 - English (E6901A)
 - German (E6901AA)
 - Spanish (E6901AB)
 - Italian (E6901AC)
 - French (E6901AD)
 - Scandinavian (E6901AE)
- Additional Patient Transport table (M1090TL).
- Other hard copy devices and patient accessories.

1-4 SIGNA 1.5T TWINSPEED SYSTEM OPTIONS (Continued)

- The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-52):
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems that have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF) and ACGD Cabinet. M1033MD provides MNS software and hardware. The hardware includes MNS Cabinet containing RF Broadband Amplifier, system interconnect cables, modification hardware for the basic 1.5T system, P31 Flexible Coil, and MR Spectroscopy Head Phantom.
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M3090DB).
 - Signa LX with Spectroscopy pre-RF/PDU Cabinet to ACGD Upgrade (M3090DC).



1.5T SYSTEM WITH ACGD/PDU CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-52

1-5 FACILITY OPTIONS

- The GE MDP, all configurations of TSCC, the Outdoor TGWC, and the Indoor Air Cooled TGWC require high voltage power. If system input voltage configuration does not meet specifications in **SECTION 5 – POWER REQUIREMENTS** then customer provided transformer may be needed. Refer to Table 1–1 for low voltage step up transformer options.

TABLE 1–1
LOW VOLTAGE (200 or 208 VOLTS) STEP UP TRANSFORMER OPTIONS

CATALOG See Note 1	DESCRIPTION
R4500AS	•150 KVA 208 to 480Y277 Volt, 60 Hz transformer
R4500BD	•150 KVA 200 to 400Y230 Volt, 50/60 Hz transformer
Note 1 A step-up transformer is required if the site input voltage configuration does not meet specifications in SECTION 5 – POWER REQUIREMENTS . The GE Main Disconnect Panel (MDP) & Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) configurations require high voltage power.	

1-6 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable MR system upgrade catalogs are listed in the previous upgrade sections. Note, the cable kits for Transportable/Relocatable include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the MR system upgrade. The van/system upgrades can be preformed at the Van Manufacturer facility (preferred location) or at customer site. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed at site.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility or customer site. Refer to Section 1–8, VAN MANUFACTURERS CONTACT INFORMATION.
- **Customer is responsible for van upgrade costs with Van Manufacturer.** Contact the Van Manufacturer for Van upgrade cost estimate and Van upgrade timing.

Note

The Van upgrade cost is **NOT** included in the price of the MR System Upgrade.

- Customer responsible for van transportation to and from Van Manufacturer facility for upgrades perform there.
- MR system check and calibration performance:
 - For van/system upgrades performed at Van Manufacturer facility, the installation will be checked by MR Manufacturing. Final calibrations will be performed by GE Field Service.
 - For van/system upgrades performed at customer site, the installation will be check and final calibrations will be performed by GE Field Service.

1-7 TRUE MOBILE UPGRADE

- MR system TwinSpeed upgrade catalogs for True Mobiles are not available at this time.

1-8 VAN MANUFACTURERS CONTACT INFORMATION

- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
USA
Telephone: 513-752-9000
FAX: 513-752-4983

Note

Customers with a MICA Relocatable system need to contact Ellis & Watts.

- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
USA
Telephone: 262-367-7700
FAX: 262-367-7744
- SMIT Mobile Equipment
Buys Ballotstraat 6
3261 La Oud-Beijerland
Holland
Telephone: (31) 186-6-14322
FAX: (31) 186-6-19367
- AK Specialty Vehicles
16745 South Lathrop Avenue
Harvey, IL 60426
USA
Telephone: 708-596-5066
FAX: 708-596-2480

Note

Customers with a Calumet Coach trailers system need to contact AK Specialty Vehicles.

SECTION 2 – ROOM LAYOUTS

TABLE OF CONTENTS

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	2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements	2-6
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2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on equipment. The maximum magnetic field in which the upgrade supplied equipment can be located is listed in Table 2-1.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 2	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 4	Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 4	- Positron Emission Tomography scanner - Linear Accelerator - Cyclotrons - Accurate Measuring scale - Image intensifiers - Color TV	- Video display (color, B/W, monochrome) - CT scanner - Ultrasound - Lithotripter - Electron microscope - Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 4	- Power transformers - Main electrical distribution transformers - Moving steel equipment such as: - Vehicular traffic - Fork lift trucks - Dumb waiters - Electric transport carts	- Loading dock (truck traffic) - Elevators - Escalators - Helicopters (See Note 3)
1 GAUSS (0.1mT) OR LESS	Advantage Workstation with CRT Monitor	
10 GAUSS (1mT) OR LESS	Operator Workspace Host Monitor (Color)	
30 GAUSS (3mT) OR LESS	- System Control Cabinet - Twin Accessory Cabinet	- TSCC equipment - TGWC equipment
50 GAUSS (5mT) OR LESS	- Octane Computer - Operator Workspace Cabinet - LCD Color Monitor for OW (See note 5) - SRF Cabinet - ACGD/PDU Cabinet - Main Disconnect Device	- Magnet Monitor - Telephones - Metal detector for screening - LCD Color Monitor for Advantage Workstation - MNS Cabinet
100 GAUSS (10mT) OR LESS	- Service Tool Magnet Power Supply Cabinet - Service Tool Shim Power Supply Cabinet	- Shield/Cryo Cooler Compressor Cabinet - Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	- Penetration Panel - Blower Box	Magnet Rundown Unit
Note - Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. - Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. - Verify operating limits with provider of helicopter service - The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to <i>Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 4-13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS.</i> - If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.		

2-2 TWO MAGNET SITE LAYOUT

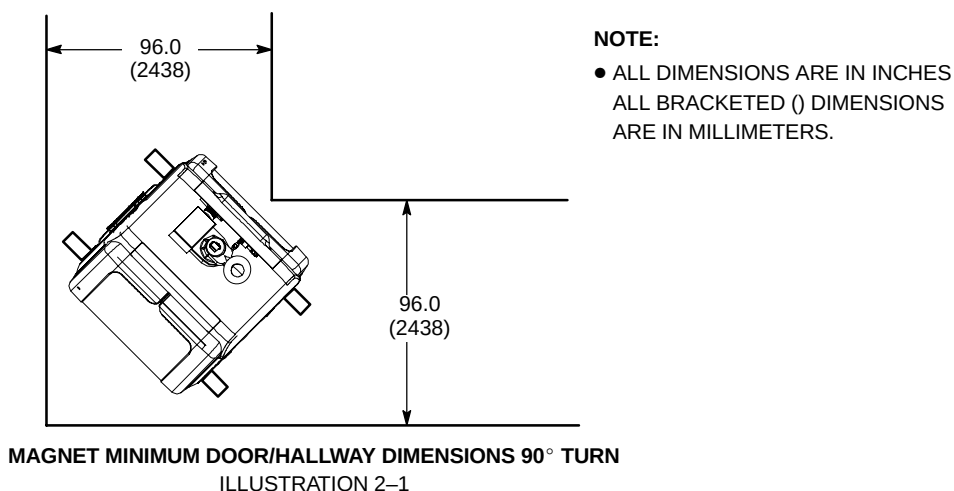
Refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 2-3 TWO MAGNET SITE LAYOUT*** for two magnet site layout information.

2-3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY

Table 2-2 lists minimum actual clearance opening dimensions for doors and hallways required by new equipment installed as part of the various Signa 1.5T TwinSpeed Upgrades. Installation or replacement of components listed in Table 2-3 must be taken into consideration when determining delivery route hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for component shipping dimensions.

TABLE 2-2
MINIMUM HALLWAY/DOOR DIMENSIONS

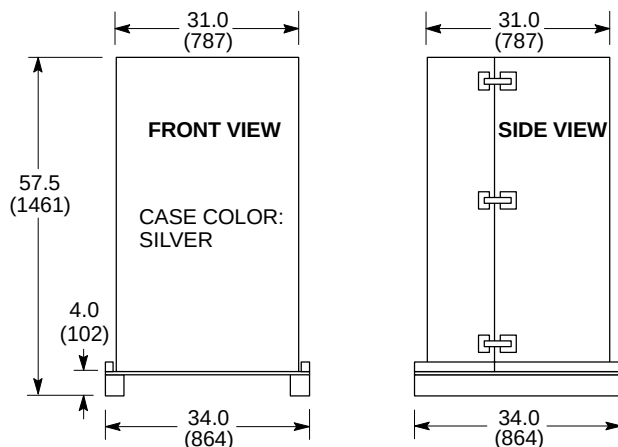
COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Operator Workspace Table	32 (813)	80 (2032)	
Indoor Water Cooled TSCC [Model LC18MT/WC]	48 (1219)	80 (2032)	
Indoor Air Cooled TSCC [Model LC 20MT]	50 (1270)	80 (2032)	
Equipment Cabinets	36 (914)	80 (2032)	Includes Indoor Water Cooled TGWC [Model LTL 10] or Indoor Air Cooled TGWC [Model LC 10M]
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
LCC Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2-3 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension. 1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration 2-1 shows dimensions for 90° turn. 2 Final dimension is dependent on rigger equipment used, refer to Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation SECTION 8, SHIPPING AND DELIVERY DATA.			



2-3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY (Continued)

TABLE 2-3
UPGRADE COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
LCC Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement Gradient (TRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35.5 x 96.09 x 55.88 (902 x 2441 x 1420) See Note 2 & 3	Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2-3-1, Gradient Coil Assembly, Cradle, & Cart Requirements.
Gradient (TRM) Coil Replacement Tool Kit Crate for LCC Magnets	750 (340)	30 x 86 x 28 (762 x 2184 x 711)	See Note 3.
Upgrade or Replacement RF Body Coil	245 (111)	31 x 31 x 57.5 (787 x 787 x 1461)	Replacement coil is shipped in a protective case. Weight & dimensions are for coil & case, see Illustration 2-2.
Enclosure Covers & End Bells	See Comments	See Comments	Refer to Section 2-9 COMPONENT DIMENSIONS, for details of individual assemblies and applicable catalogs.
Rear Pedestal	30 (13.6)	17 x 43 x 33 (432 x 1092 x 838)	
Note 1 The replacement TRM Gradient Coil Assembly weight is approximately 2850 lbs (1293 kg), the shipping cradle is 125 lbs (57 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3830 lbs (1738 kg). The coil assembly outside diameter x length dimensions are 35.0 x 74.6 in. (888 x 1895 mm). 2 Gradient (TRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm). 3 LCC Magnets MUST USE GE Service Tool Gradient Coil Replacement Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet.			



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 245 lbs (111 kg)
WEIGHT AND DIMENSION INCLUDE COIL, CASE, AND SHIPPING PALLET.
- RF COIL CASE HAS WHEELS AND HANDLES FOR EASE MOVING AFTER REMOVAL FROM PALLET.

RF COIL CASE - LCC MAGNET QUIET TECHNOLOGY ENCLOSURE FOR SIGNA 1.5T TWINSPEED UPGRADE
ILLUSTRATION 2-2

2-3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY(Continued)**2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements****Systems With LCC Magnet Currently Installed**

Systems with an LCC Magnet currently installed will require the Body Coil Insertion Tool to upgrade from BRM or CRM to the new TRM coil. The Body Coil Insertion Tool requires the TRM coil to be installed on the aluminum cradle (see Illustration 2-1), and be mated onto the aluminum coil cart (see Illustration 2-2). The new TRM Coil will be shipped on a wooden disposable cradle. The TRM Coil **must** be transferred to an empty aluminum cradle and cart provided GE Service via the Tools Depot. GE Service provided coil lifting bracket will be necessary for transferring the coil and a customer rigger provided forklift or crane/hoist rated for 4,000 lbs (1814 kg). Refer to **Section 10-3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT** for additional rigger/customer supplied equipment. Refer to **Section 10-5 INSTALLATION EQUIPMENT** for GE Service provided equipment.

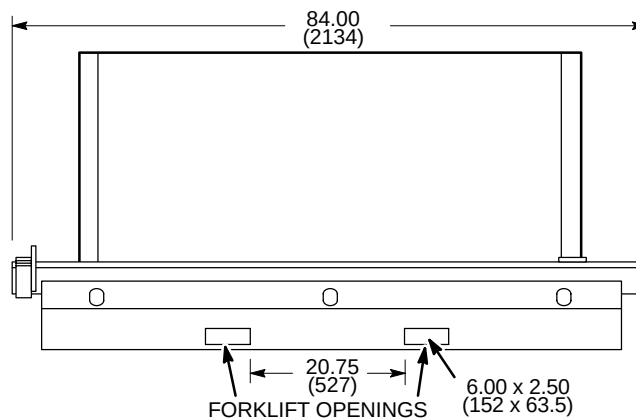
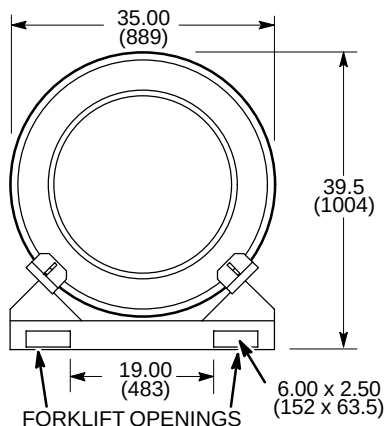
Systems Installing LCC Magnet Upgrade Concurrent With TwinSpeed Upgrade

Systems installing an LCC Magnet Upgrade at the same time as the TwinSpeed Upgrade Systems will receive (M3060TA) LCC Magnet with TwinSpeed Enclosure with the TRM Assembly (M3085TA) installed inside the magnet bore and enclosure.

2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements (Continued)

NOTE:

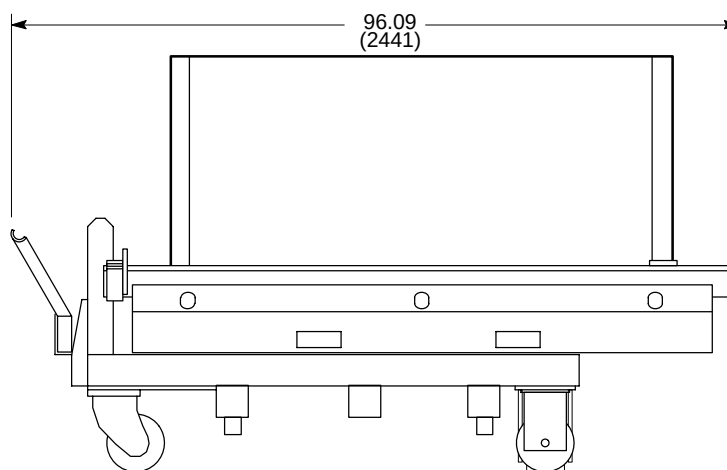
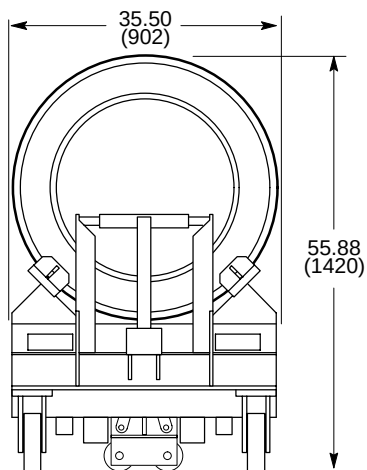
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 2975 lbs (1349 kg)



GRADIENT (TRM) COIL ASSEMBLY & ALUMINUM CRADLE
ILLUSTRATION 2-1

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 3830 lbs (1737 kg)



SHORT ROOM REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE ON CART
ILLUSTRATION 2-2

2-4 CABLING CONSIDERATIONS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-9, COMPONENT DIMENSIONS. Signa 1.5T TwinSpeed upgrade results in several changes to the cabling and hoses routed through the Rear Pedestal cable access areas of the Magnet Enclosure. These changes include the

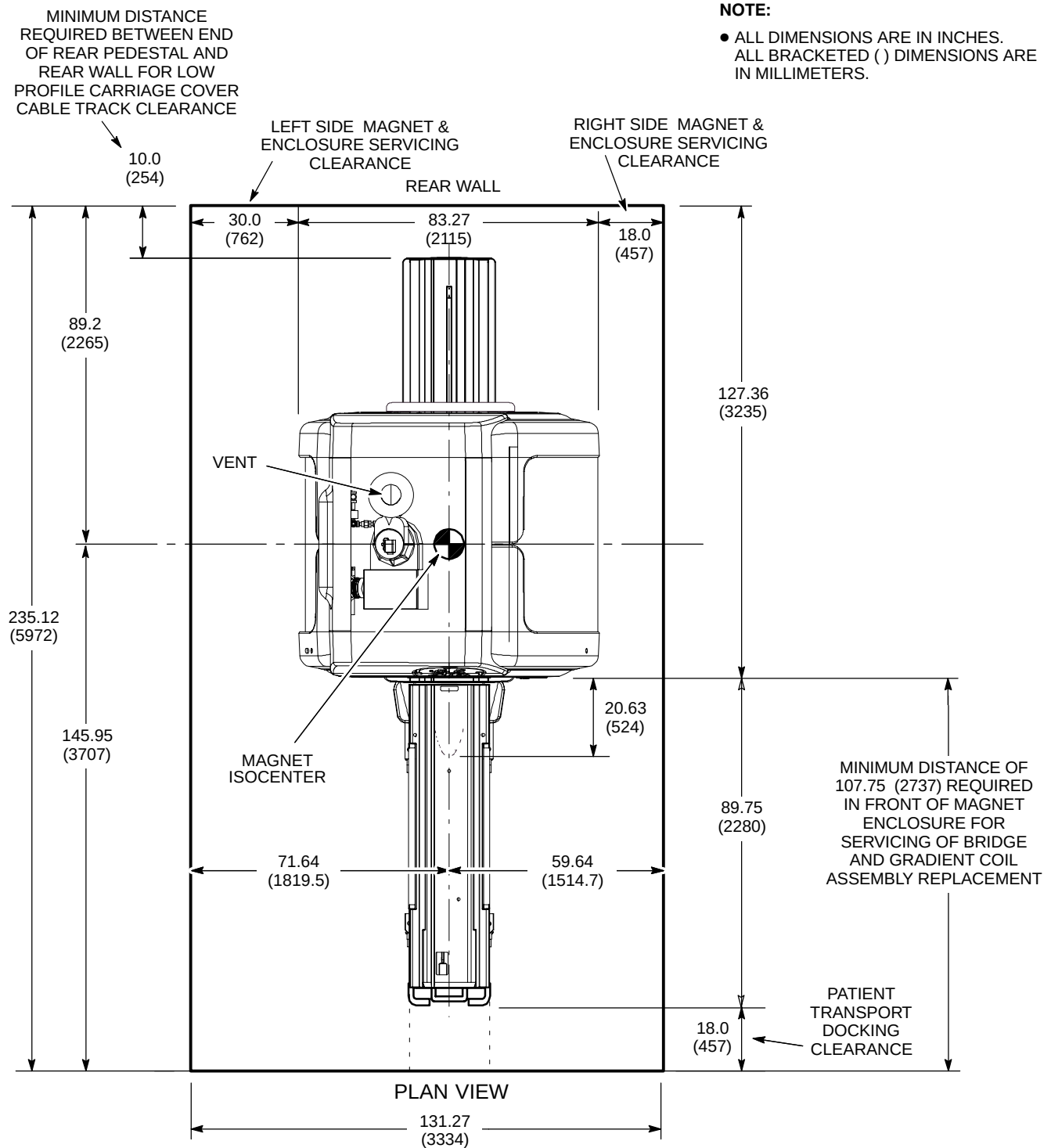
- elimination of 1 Blower Box hose (6.5 inch (165 mm) OD hose)
- installation of 3 additional Gradient cables
- addition of a vacuum hose and a vacuum sensor cable for the Quiet Technology Enclosure
- 2 existing water hoses for Body Coil are replaced with 2 new larger water hoses.

Refer to Section 6 – INTERCONNECT DATA for cabling and hoses details.

The remainder of the Magnet Enclosure system interconnect cables are also still routed through the Rear Pedestal cable access. The existing cable access area utilized to site cable routing should be sufficient for the upgrade system cabling. See Illustration 2-5 for Magnet cable access areas and refer to Section 6 – INTERCONNECT DATA for cables and their dimensions information.

If an LCC Magnet upgrade is being performed with the system TwinSpeed upgrade there will be additional system interconnects changes for the new magnet. Refer to Section 6 – INTERCONNECT DATA for additional information and details.

2-4 CABLING CONSIDERATIONS (Continued)



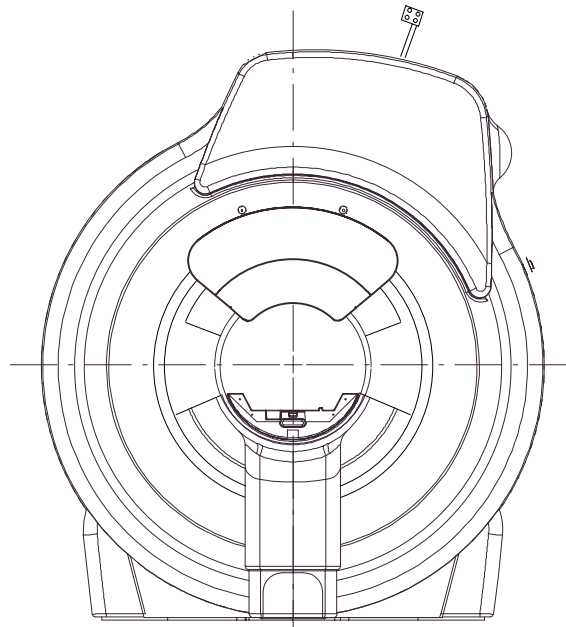
SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)

ILLUSTRATION 2-3

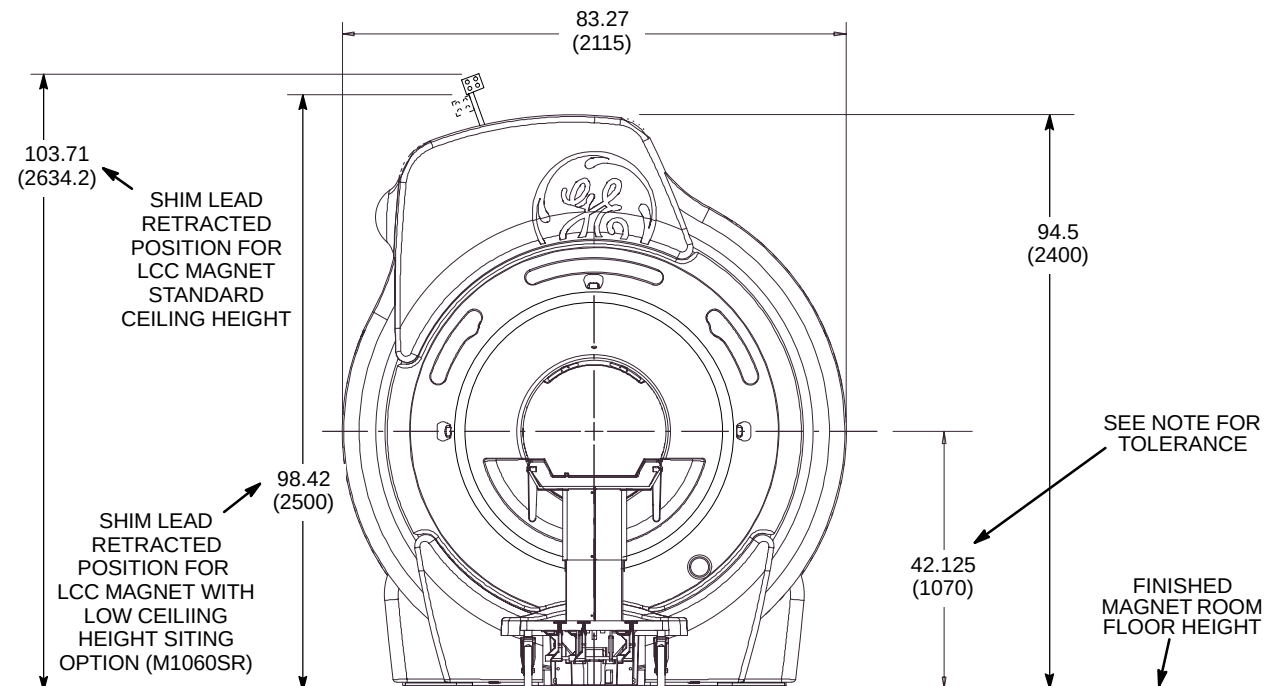
2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- FINISHED FLOOR TO MAGNET CENTER LINE HEIGHT DIMENSION MUST BE 42.125 ± 0.25 inches (1070 ± 6.35 mm) TO ALLOW PATIENT TABLE TO PROPERLY DOCK TO THE ENCLOSURE.



REAR VIEW



FRONT VIEW

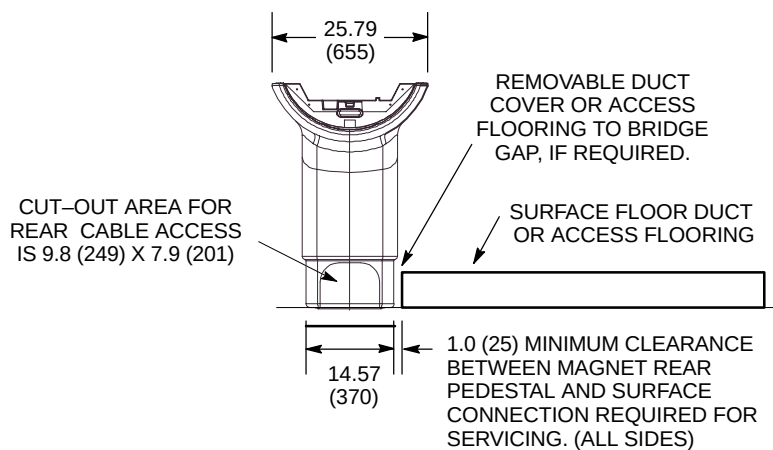
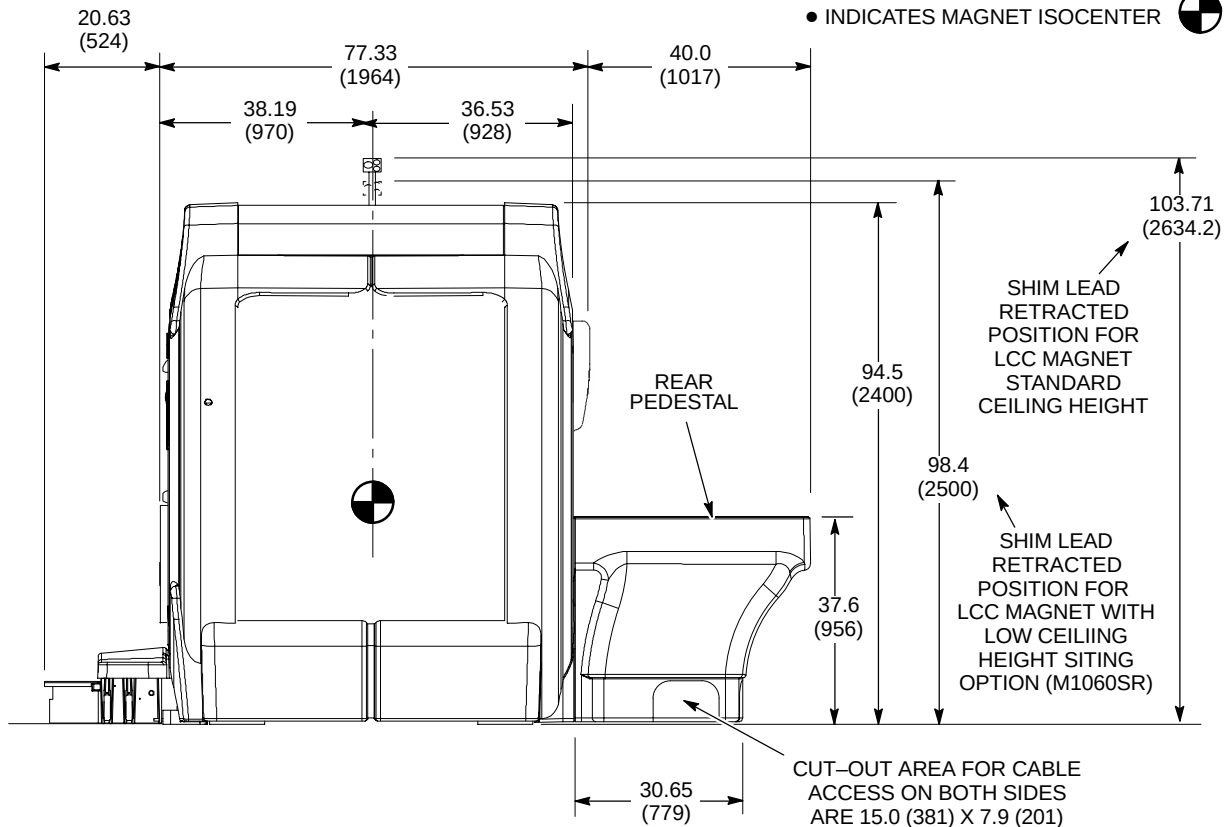
SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)

ILLUSTRATION 2-4

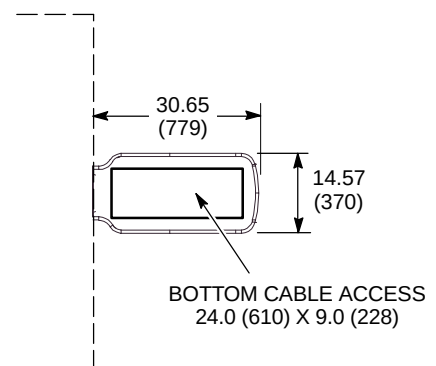
2-4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



REAR PEDESTAL
(REAR VIEW)



REAR PEDESTAL
(FLOOR CONTACT VIEW)

SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS

ILLUSTRATION 2-5

2-5 System Cooling Equipment

The Signa TwinSpeed system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The Signa TwinSpeed system requires a different capacity water cooling unit for the TRM Gradient Coil located in the Magnet bore from the units utilized in prior system configurations. If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then the Shield/Cryo Cooler Compressor will be required. Refer to **Section 1 – SYSTEM CONFIGURATION** flowcharts for listing of catalogs dependent on system site configuration.

- Twin System Cooling Cabinet (TSCC) which provides water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. Refer to Section 2-5-1 TSCC Siting Considerations for details.
- Twin Gradient Water Cooler (TGWC) which provides water cooling for ONLY the Gradient Coil located in the Magnet bore. When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Refer to Section 2-5-2 TGWC Siting Considerations for details.

Note

Transportable/Relocatable Site Signa 1.5T TwinSpeed upgrade will replace the existing van manufacturer provided water cooling unit with the Outdoor Air Cooled Twin System Cooling Cabinet (TSCC) unit.

2-5-1 TSCC Configurations Siting Considerations



Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. Therefore the Twin System Cooling Cabinet (TSCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TSCC is available in three fixed site configuration options:

- Outdoor Air Cooled TSCC [Model LC 20MT/O] has the TSCC unit located outside the building and a Remote Control Panel (RCP) located in the MR system Equipment Room; see Illustration 2-10 for Outdoor Air Cooled TSCC and Illustration 2-12 for the Remote Control Panel.
- Indoor Water Cooled TSCC [MODEL LC 18MT/WC]; see Illustration 2-13 and refer to Section 4-4-3 Water Cooled TSCC Requirements for water cooling requirements.
- Indoor Air Cooled TSCC [MODEL LC 20MT]; see Illustration 2-15.

Note

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet. Refer to Section 4-4-2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation.

2-5-1 TSCC Configurations Siting Considerations (Continued)

TSCC & System Interconnects/Separation Limitations

Location of the Outdoor TSCC must meet the following limitations for the water lines (customer provided copper line and E&W/GE provided flexible tubing combined).

- Outdoor TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- Outdoor TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 160 ft (48.8 m) water line length.
- The Outdoor TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 100 ft (30.5 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.
- The Outdoor TSCC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total length.

Location of the Indoor Water Cooled TSCC or Indoor Air Cooled TSCC must meet the following limitations for the water lines (E&W/GE provided flexible tubing).

- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 60 ft (18.3 m) water line length.
- The Indoor Water Cooled TSCC or Indoor Air Cooled TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 15 ft (4.6 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.

The TSCC is powered from the system Main Disconnect Panel (MDP) via customer supplied wiring. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2-5-1 TSCC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TSCC Equipment

The TSCC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2-4 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2-4
**TSCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED**

APPLICABLE CONFIG			INSTALLATION TASKS FOR TSCC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC20MT/O)	INDOOR WATER COOLED (LC18MT/W)	INDOOR AIR COOLED (LC20MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN TSCC IS AT SITE & PRIOR TO MAGNET DELIVERY						
X	X	X	Unload TSCC from truck.	X		
X			Move TSCC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TSCC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TSCC to Equipment Room and mount. (Use seismic tie down brackets & vibration isolators where required by law)	X		
X	X		Install customer water supply and return lines to TSCC	X		
X	X	X	Connect customer supplied power wiring from MDP to TSCC.	X		
X			Install power and control cable between TSCC & RCP	X		
WHEN MAGNET IS DELIVERED/INSTALLED						
X			Install water cooling lines from customer supplied/installed ball valves & hose barbs on copper lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
	X	X	Install water cooling lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X	X	X	Fill TSCC with Propylene Glycol		X	
X	X	X	See APPENDIX B Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed and review for completeness. Send email request for initial performance verification to service@elliswatts.com . Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X	X	X	Verify installation and operation of TSCC per E&W Chiller Startup Checklist			X
Note 1 Refer to Ellis & Watts <i>Technical Publication 470 Pre installation/Installation/Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment</i> for additional Outdoor TSCC information and details. 2 Refer to Ellis & Watts <i>Technical Publication 471 Pre installation/Installation/Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment</i> for additional Indoor Water Cooled TSCC information and details. 3 Refer to Ellis & Watts <i>Technical Publication 472 Pre installation/Installation/Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment</i> for additional Indoor Air Cooled TSCC information and details.						

2-5-2 TGWC Configurations Siting Considerations



When a TGWC is selected for the system then the customer site must still provide cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet; refer to Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TGWC is available in three fixed site configuration options:

- Outdoor Air Cooled TGWC [MODEL LC 10M/O] has the TGWC unit located outside the building and a Remote Control Panel (RCP) is located in the MR system Equipment Room; see Illustration 2-17 for Outdoor Air Cooled TGWC and Illustration 2-19 for the RCP.
- Indoor Water Cooled TGWC [MODEL LTL 10] has the TGWC unit and a Remote Control Panel both located in the MR system Equipment Room; see Illustration 2-20 and refer to **Section 4-4-5 Water Cooled TGWC Requirements** for water cooling requirements.
- Indoor Air Cooled TGWC [MODEL LC 10M]; see Illustration 2-22.

TGWC & System Interconnects/Separation Limitations

Location of the Outdoor TGWC must meet the following limitations for the water lines (customer provided copper line and GE provided flexible tubing combined).

- Outdoor TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- The Outdoor TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 100 ft (30.5 m) above or below the Gradient Coil.
- The Outdoor TGWC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total length.

Location of the Indoor Water Cooled TGWC or Indoor Air Cooled TGWC must meet the following limitations for the water lines (GE provided flexible tubing).

- Indoor Water Cooled TGWC or Indoor Air Cooled TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- The Indoor Water Cooled TGWC or Indoor Air Cooled TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

The Outdoor TGWC and Indoor Air Cooled TGWC are powered from the Main Disconnect Panel (MDP) via customer supplied wiring. The Indoor Water Cooled TGWC is powered from the MR system PDU. Refer to Section 6, INTERCONNECT DATA, for additional interconnects information and requirements.

2-5-2 TGWC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TGWC Equipment

The TGWC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2-5 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2-5
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG			INSTALLATION TASKS FOR TGWC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN TGWC IS AT SITE & PRIOR TO MAGNET DELIVERY						
X	X	X	Unload TGWC from truck.	X		
X			Move TGWC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TGWC and Remote Control Panel to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TGWC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
X	X		Install customer water supply and return lines to TGWC	X		
X		X	Connect customer supplied power wiring from MDP to TGWC.	X		
X	X		Install power and control cable between TGWC & RCP	X		
WHEN MAGNET IS DELIVERED/INSTALLED						
X			Install water cooling lines from customer supplied/installed ball valves & hose barbs on copper lines from TGWC to Gradient Coil. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
	X	X	Install water cooling lines from TGWC to Gradient Coil. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X		X	Fill TGWC with Propylene Glycol		X	
X		X	Verify installation and operation per E&W Chiller Startup Check List			X
(Continued)						

2-5-2 TGWC Configurations Siting Considerations (Continued)

TABLE 2-5 (Continued)
**TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
 PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED**

APPLICABLE CONFIG			INSTALLATION TASKS FOR TGWC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN MR PDU IS DELIVERED/INSTALLED						
	X		Connect E&W supplied power cable from PDU to TGWC.	X		
	X		Fill TGWC with Propylene Glycol		X	
X	X	X	See APPENDIX B Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed and review for completeness. Send email request for initial performance verification to service@elliswatts.com . Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
Note 1 Refer to Ellis & Watts Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Outdoor TGWC information and details. 2 Refer to Ellis & Watts Technical Publication 469 Pre installation/Installation/Operating LTL 10 Chilled Water System for GE Twin Magnet MRI Equipment for additional Indoor Water Cooled TGWC information and details. 3 Refer to Ellis & Watts Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Indoor Air Cooled TGWC information and details.						

2-6 SPECIAL SITING CONSIDERATIONS

The following system components have special siting concerns which need to be considered.

2-6-1 Power Distribution Unit (PDU)

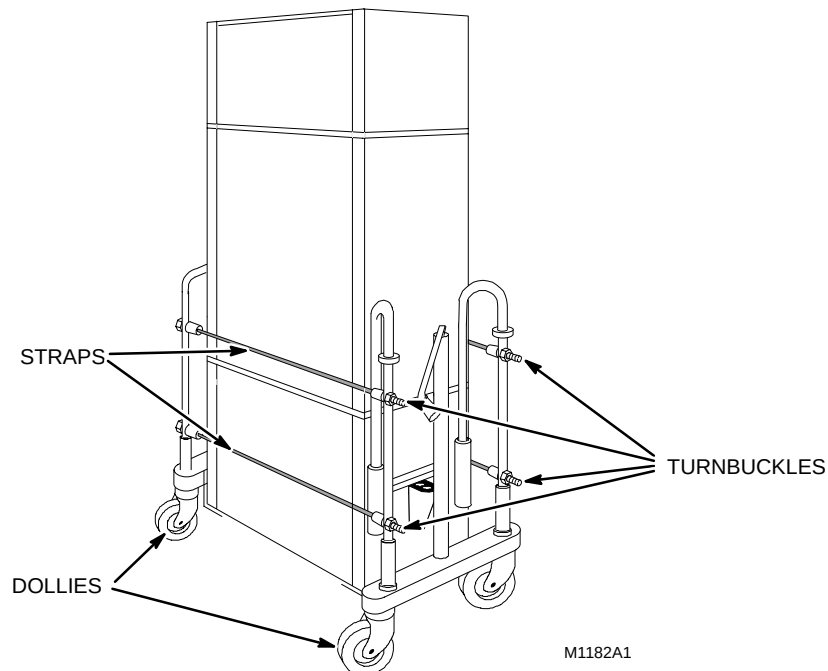
Sites that are upgrading to ACGD require the Full-size PDU, Compact PDU, or RF/PDU Cabinet be de-installed. The Compact PDU and RF/PDU Cabinet have built-in casters for ease of movement. The Full-size PDU does not have casters and weighs approximately 1000 lbs (455 kg). Sites with a Full-size (double bay) PDU must plan for removal of the PDU.

- **Sites located outside the USA:**

The Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel/equipment.

- **Sites located in the USA:**

Can contact Techtrans at telephone 800-851-9929 to order a "MR PDU Double Bay Cabinet Dolly Set". The PDU can be moved with two shipping dollies placed one on each side of the PDU and clamped in place using straps with turnbuckles as shown in Illustration 2-6.



FULL-SIZE PDU SHIPPING DOLLIES

ILLUSTRATION 2-6

2-6-2 Blower Box (MG6)

Systems with the LCC Magnet with a WideOpen Enclosure installed: The Signa 1.5T TwinSpeed upgrade will modify the existing system Blower Box (MG6) mounted within the RF Shielded Room. The 6.5 inch (165.1 mm) OD flexible vinyl air ducting to the body coil is eliminated since the TRM Body Coil is only water cooled. The Blower Box will still be connected to the the Patient Comfort Module of the Quiet Technology Magnet Enclosure with the existing 4.5 inch (114.3 mm) OD flexible vinyl air ducting.

Systems without a WideOpen Enclosure currently installed: The Signa 1.5T TwinSpeed upgrade will add a new Blower Box (MG6) to be mounted within the RF Shielded Room. The new Blower Box will be connected to the the Patient Comfort Module of the Quiet Technology Magnet Enclosure with a 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** Refer to Illustrations 2-33 for the new Blower Box and mounting dimensions.

2-6-3 Advantage Workstation Additional Hardware Features (Optional)

The VP Workstation offering is replaced by utilizing an Advantage Workstation with cardiac and neurovascular software offerings. Refer to Direction 2261309-100 *Advantage Workstation Ultra60 Pre-Installation* for more information.

The customer must determine the desired location for the AW relative to the imaging system. The following factors should be considered in determining AW location.

- If the customer is purchasing RTIP (neurovascular) option, then the customer may want to observe the experiment real time while operating the Signa system versus reviewing the data at a later time. If the former, then consideration should be made to co-locate the AW near the system Operator Workspace (OW).

Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- If the customer is going to perform cardiac studies only, then co-location is not necessary at this time. However, co-location of the AW may be recommended for future applications.

2-6-4 Spectroscopy (Optional)

Signa 1.5T TwinSpeed requires the ACGD/PDU Cabinet. Therefore the spectroscopy option hardware configuration must be compatible with the ACGD system cabinet. Multi-Nuclear Spectroscopy (MNS) option For Systems With ACGD/PDU Cabinet provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet which has mounting space for Broadband RF Amplifier and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling. Refer to Section 6-7, SPECTROSCOPY INTERCONNECTIONS – 1.5T SYSTEM OPTION, cable length information and other cable data.

2-6-5 Magnet Monitor

Refer to **Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 2-8-7 Magnet Monitor** for Signa 1.5T TwinSpeed LCC Magnet Magnet Monitor and Magnet Monitor UPS option information.

2-6-6 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2-6 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2-6
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Network connection with a static IP address	Access located near the Operator Workspace (OW) in the Control Room.
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring.
Multiple telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Note 1 & 2
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Note 1 & 2
Note 1 A dedicated direct-distance-dialing <u>voice-grade</u> telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. 2 If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

2-7 ARCHITECTURAL REMINDERS

- The operator seated at the Operator Workspace must have an unobstructed view of the patient on the transport table when table is docked to the magnet.
- It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a "honeycomb-type pattern") for better visibility of the patient from the Operator Workspace.
- Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2-6-6 System Monitoring & Support Connectivity Requirements**.
- Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, TRM Body Coil Assembly, TSCC or TGWC).

2-8 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for Signa 1.5T TwinSpeed upgrade components. Listed in Table 2-7 are the weights, floor loading, and normal mounting methods for components.

TABLE 2-7
SIGNA 1.5T TwinSpeed FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
LCC Magnet, Quiet Technology Enclosure, and Coils	See Notes 1 & 3 & Refer to Section 2-8-1	83.27 x 117.67 x 101.96 (2115 x 2989 x 2587.21)	See Illustration 2-7	Magnet & Table Dock Asm. Resting on base, bolted to floor with RF Screen Room vendor supplied/ installed bolts & anchors.
Patient Table (See Note 4)	630 (286)	27 x 89 x 38 (686 x 2261 x 965)	Rectangular base 23 x 65 (584 x 1651). Four casters 8 (203) dia. x 1.5 (38) wide.	Mobile.
Existing Blower Box modified	47 (21)	16 x 16 x 22 (406 x 406 x 559)	Square base 15 x 15 (381 x 381)	Anchor to floor, shelf, or wall.
New Blower Box added	20 (9)	14.5 x 14 x 13.7 (368 x 356 x 348)	Square base 14.5 x 14 (368 x 356)	Anchor to floor or shelf.
SRF Cabinet	525 (239)	24 x 40 x 70.7 (610 x 1016 x 1793)	Rectangular base 24 x 30 (610 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 TwinSpeed with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 TwinSpeed System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	1844 (838)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 5	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Twin Accessory Cabinet	600 (273)	23.32 x 38.75 x 50.13 (592 x 984 x 1273)	Rectangular base 22 x 30 (584 x 762). Four casters 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor and rest on casters.
Shield/Cryo Cooler Compressor Cabinet	275 (125)	17.72 x 19.7 x 26.93 (450 x 500 x 684)	Rectangular base 17.72 x 19.7 (450 x 500). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
Twin System Cooling Cabinet (TSCC) See Note 6				
Outdoor Air Cooled	1200 (544)	74.83 x 45.5 x 66.88 (1901 x 1156 x 1699)	See Illustration 2-10.	Bolted to mounting pad external to building.
Indoor Water Cooled	1000 (454)	55.4 x 33.59 x 73 (1407 x 358 x 1854)	See Illustration 2-13.	Set on floor and rest on casters.
Indoor Air Cooled	1410 (640)	65 x 32.5 x 71.26 (1651 x 825.5 x 1810)	See Illustration 2-15.	Set on floor and rest on casters.

(Continued)

TABLE 2-7 (Continued)
SIGNA 1.5T TwinSpeed FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Twin Gradient Water Cooler (TGWC) See Note 6				
Outdoor Air Cooled	780 (354)	49.5 x 34 x 76.2 (1257 x 864 x 1936)	See Illustration 2-17.	Bolted to mounting pad external to building.
Indoor Water Cooled	175 (79)	33.94 x 19.66 x 23.67 (86.21 x 49.94 x 60.13)	See Illustration 2-20.	Set on floor and rest on 4 feet.
Indoor Air Cooled	780 (354)	28.4 x 33 x 73 (721 x 838 x 1854)	See Illustration 2-22.	Set on floor and rest on casters & front pads on anti-tip legs
*Advantage Workstation	Refer to Direction 2261309-100 <i>Advantage Workstation Ultra60 Pre-Installation</i> . Also refer to Section 2-6-3 Advantage Workstation Additional Hardware Features to determine location of workstation.			
MNS Cabinet Option	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559).	Casters for location. Set on floor on four leveling pads.
Note 1 Weight of 1.5T LCC (Active Shield) magnet with Quiet Technology Enclosure, RF/Gradient (TRM) Body Coil, and cryogenics is 12,826 lbs (5818 kg). 2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading. 3 Refer to Section 2-3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY for Gradient Coil Assembly replacement weight and dimension requirements. 4 630 lbs (286 kg) includes 350 lbs (159 kg) patient. 5 ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm). 6 System Cooling Cabinet is available in the following configuration options (must choose one): TSCC (Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil). - Outdoor Air Cooled TSCC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room - Indoor Water Cooled TSCC located in or near the Signa system Equipment Room - Indoor Air Cooled TSCC located in or near the Signa system Equipment Room. - True Mobile system equipment, contact Van Manufacturer for details. TGWC (Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor, refer to Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC. - Outdoor Air Cooled TGWC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room. - Indoor Water Cooled TGWC and Remote Control Panel (RCP) wall mounted both located in the Signa system Equipment Room - Indoor Air Cooled TGWC located in or near the Signa system Equipment Room. 7 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2-40. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table. * Optional Equipment.				

2-8-1 Magnet Loading Considerations

Refer to **Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 2-10-1 Magnet Loading Considerations** for Signa 1.5T TwinSpeed LCC Magnet information.

2-8-2 Operator Workspace Mounting Requirements

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2-40. Note, the Workspace Table must be bolted to the floor for safe use of the table and equipment.

2-9 COMPONENT DIMENSIONS

To assist in completing your room layout, refer to Table 2-8 for list of component Illustrations and the upgrade catalog numbers which provide the new equipment. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included, catalogs may list the same new equipment but differ due to other modification hardware.

TABLE 2-8
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1000MN M1000MP M1000MR M1000MS M1000MW M1000NH M1000NJ M1000NK M1000NT	OPERATOR WORKSPACE (OW1)	2-40
M1033K M1033L	OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2-43
M1033MD	MNS CABINET (MNS) – 1.5T OPTION	2-45
M1060JW	LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2-9
M1090TL	PATIENT TRANSPORT TABLE (PT1)	2-32
M3000TA	PATIENT TRANSPORT TABLE (PT1)	2-32
	TWIN ACCESSORY CABINET (TAC)	2-35
	ACGD/PDU CABINET (MR3)	2-36
	SRF CABINET (MR1)	2-37
	SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED	2-39
	OPERATOR WORKSPACE (OW1)	2-40
M3000TC	PATIENT TRANSPORT TABLE (PT1)	2-32
	TWIN ACCESSORY CABINET (TAC)	2-35
	ACGD/PDU CABINET (MR3)	2-36
	SRF CABINET (MR1)	2-37
	SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY	2-38
	OPERATOR WORKSPACE (OW1)	2-40
M3043TA	UPGRADE NEW BLOWER BOX (MG6)	2-33
M3044TA	PENETRATION PANEL (PP1)	2-46
M3063TA	PENETRATION PANEL COVER	2-47
M3043TC	SPT PHANTOM SET SHIPPING/STORAGE CABINET	2-48
M3044TC		
M3063TC		
M3060TA	SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)	2-4
	SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS	2-5
	1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-7
	MAGNET MONITOR (MSM1)	2-8
(Continued)		

2-9 COMPONENT DIMENSIONS (Continued)

TABLE 2-8 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3085TA	UPGRADE TRM GRADIENT COIL & CRADLE ON CART Note: Systems installing M1060TA 1.5T Magnet With TwinSpeed Enclosure will receive the magnet with the TRM installed in the magnet bore.	2-2
M3088TA	OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)	2-10
	OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING	2-11
	REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]	2-12
M3088TB	INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2-13
	INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING	2-14
M3088TC	INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2-15
	INDOOR AIR COOLED TSCC MOUNTING	2-16
M3088TE	INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC)	2-20
	REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]	2-21
M3088TF	OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M/O] (TGWC)	2-17
	OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O] MOUNTING	2-18
	REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC-10M/O]	2-19
M3088TG	INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)	2-22
M3088TH	SIGNA TWINSPEED MAIN DISCONNECT PANEL (MDP) M3088TH	2-49
M3090AH	ACGD/PDU CABINET (MR3)	2-36
	SRF CABINET (MR1)	2-37
	SYSTEM CABINET (MR2)	2-39
	OPERATOR WORKSPACE (OW1)	2-40
M3090BB	ACGD/PDU CABINET (MR3)	2-36
	SRF CABINET (MR1)	2-37
M3090BC	ACGD/PDU CABINET (MR3)	2-36
	SRF CABINET (MR1)	2-37
M3090DB M3090DC	MNS CABINET (MNS) – 1.5T OPTION	2-45
M3090EC	SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY	2-38
M3090ED	SRF CABINET (MR1)	2-37
	SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY	2-38
	OPERATOR WORKSPACE (OW1)	2-40
M3090TA	TWIN ACCESSORY CABINET (TAC)	2-35
	REAR PEDESTAL	See Note 1
M3090TE	UPGRADE MODIFIED BLOWER BOX (MG6) See Note 2	2-34
	TWIN ACCESSORY CABINET (TAC)	2-35
(Continued)		

2-9 COMPONENT DIMENSIONS (Continued)

TABLE 2-8 (Continued)

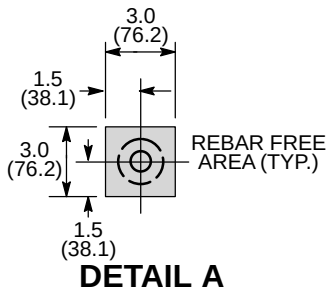
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3090TF	FRONT RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-23
	FRONT LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-24
	REAR RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-25
	REAR LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-26
	FRONT ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-27
	FRONT COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-28
	REAR ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-29
	REAR COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2-30
M3090TJ	FRONT SPLIT BRIDGE	2-31
M3090TM	TWIN ACCESSORY CABINET (TAC)	2-35
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-3, MINIMUM DOOR/HALLWAY SIZES, for dimension information. 2 Catalog provides a kit to modify existing system equipment to the configuration shown in the illustration.		

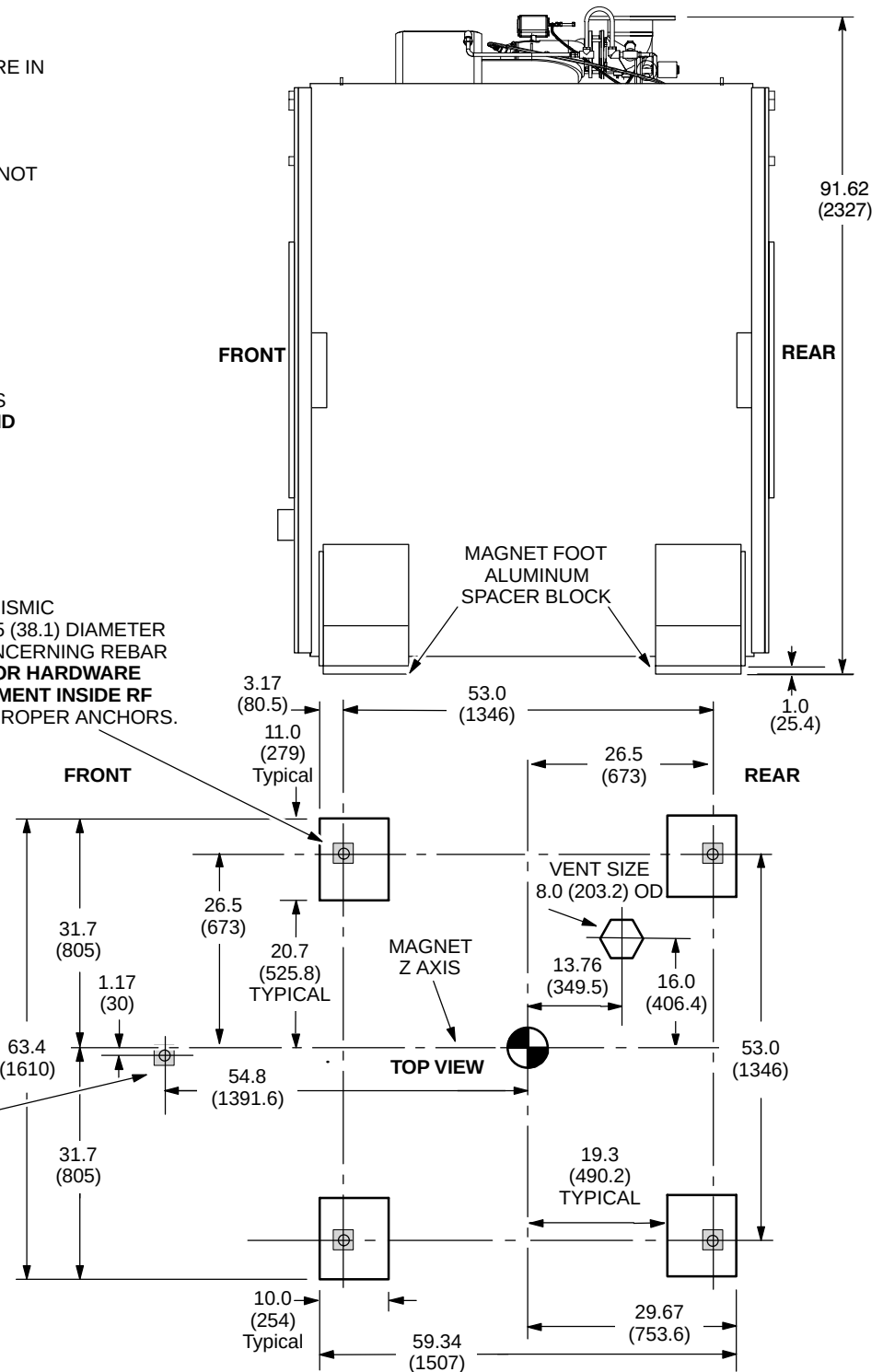
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8-1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**

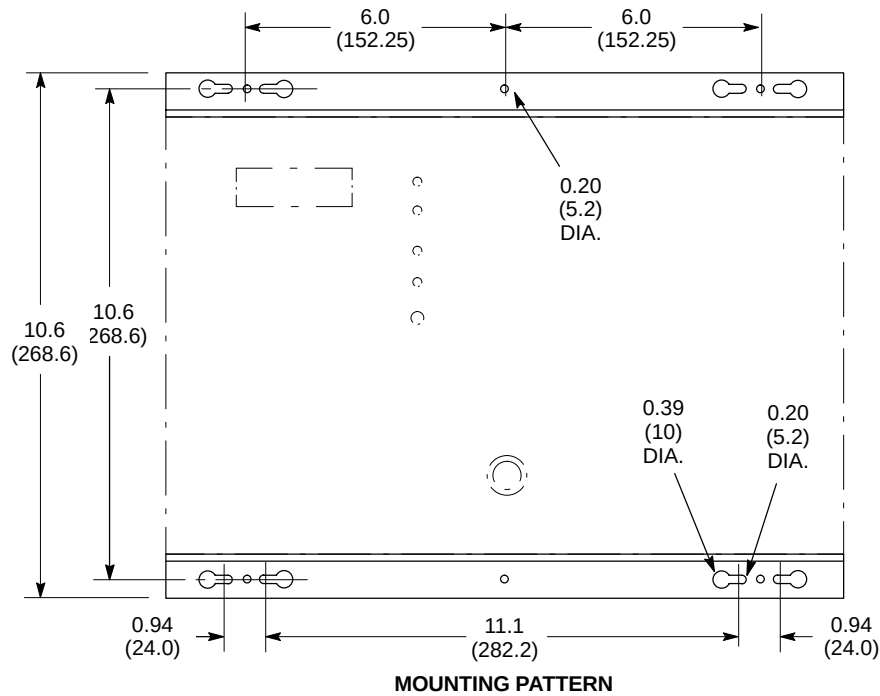
SEISMIC AND NON-SEISMIC
MAGNET MOUNTING HOLES (4) 1.5 (38.1) DIAMETER
SEE NOTE ABOVE & DETAIL A CONCERNING REBAR
REFER TO **SECTION 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** TO DETERMINE PROPER ANCHORS.



M10 ANCHOR FOR
TABLE DOCK MOUNTING,
LOCATED OFFSET FROM
MAGNET CENTER LINE.
FOR ADDITIONAL
MOUNTING INFORMATION
REFER TO SECTION
NO TAG
**ANCHOR HARDWARE
REQUIREMENTS FOR MR
EQUIPMENT INSIDE RF
SHIELDED ROOM.**

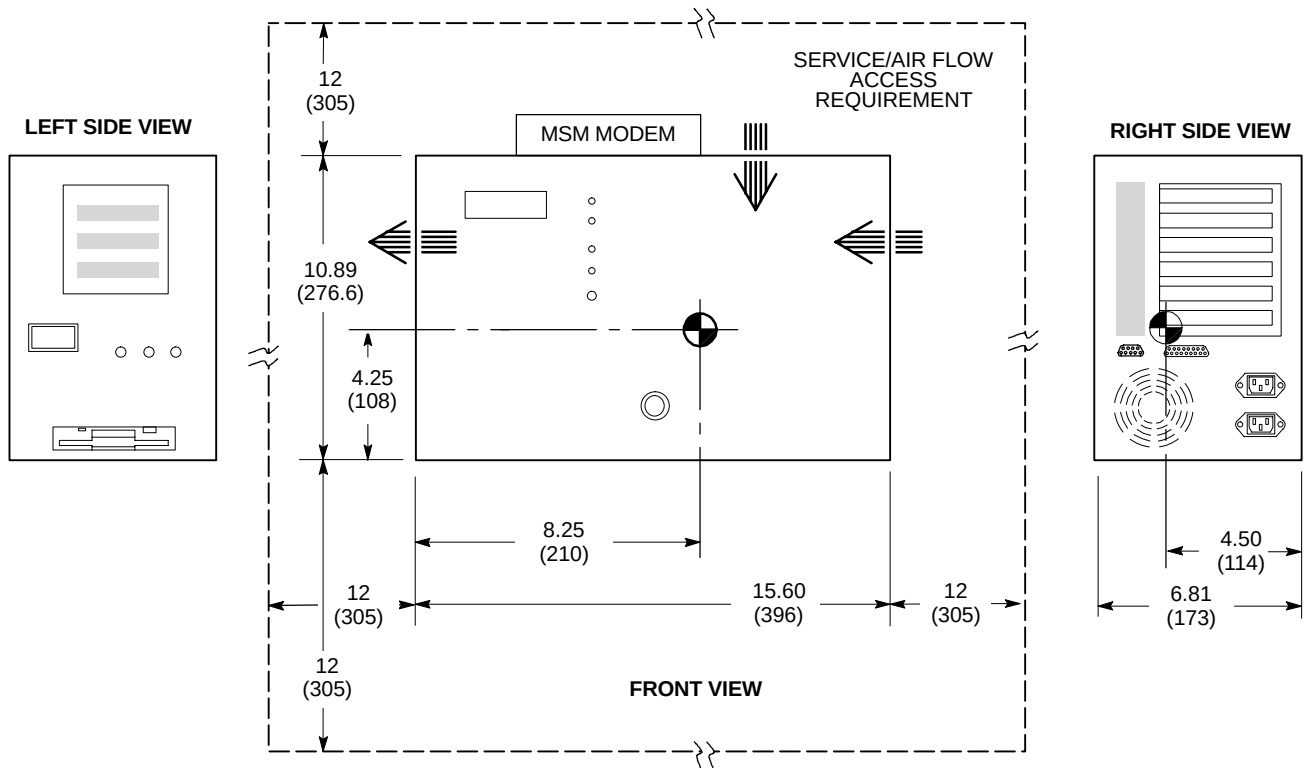


1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN
ILLUSTRATION 2-7



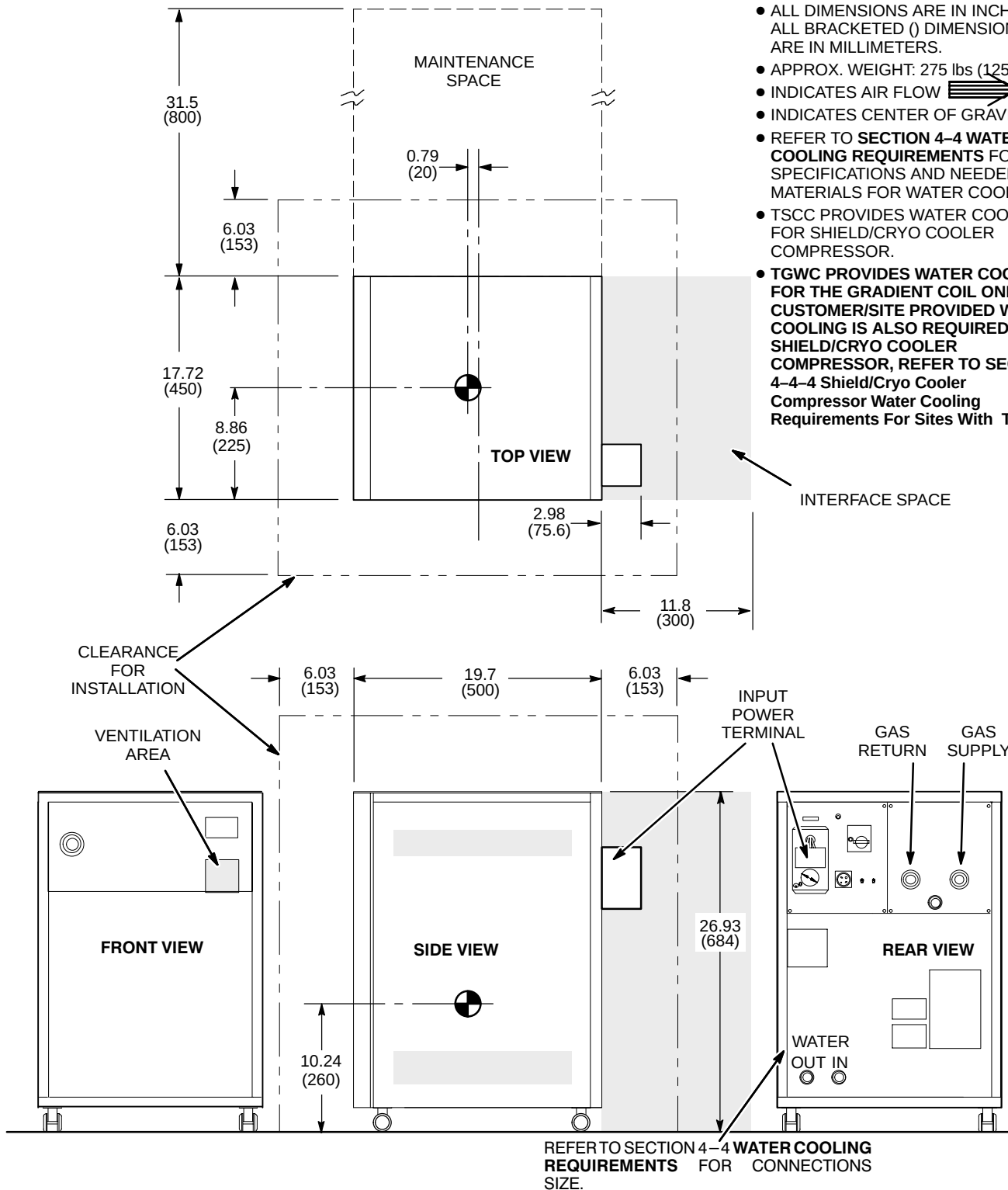
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 22 lbs (10 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 275 lbs (125 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- REFER TO **SECTION 4-4 WATER COOLING REQUIREMENTS** FOR SPECIFICATIONS AND NEEDED MATERIALS FOR WATER COOLING.
- TSCC PROVIDES WATER COOLING FOR SHIELD/CRYO COOLER COMPRESSOR.
- TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.

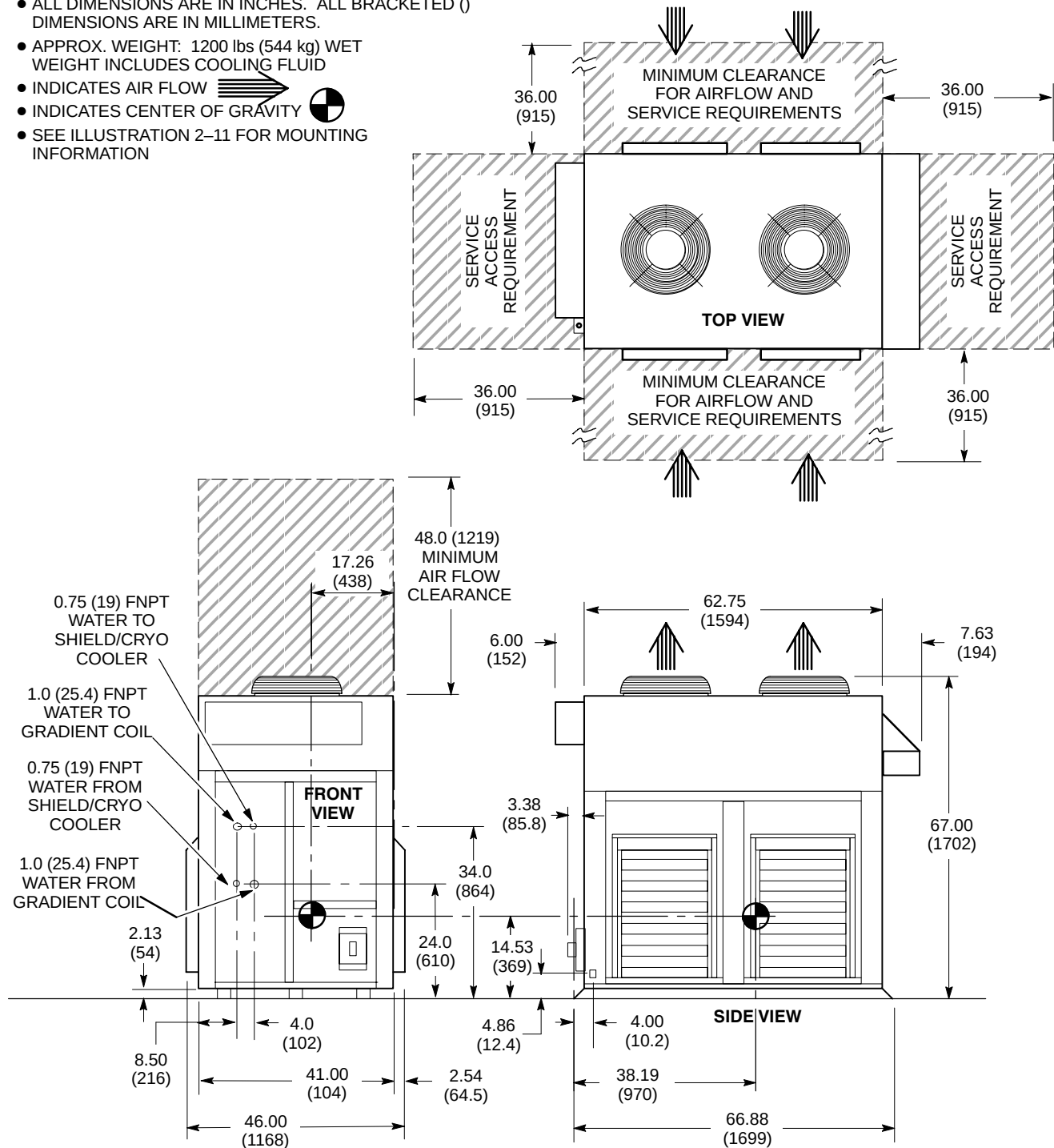


LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)

ILLUSTRATION 2-9

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1200 lbs (544 kg) WET WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-11 FOR MOUNTING INFORMATION

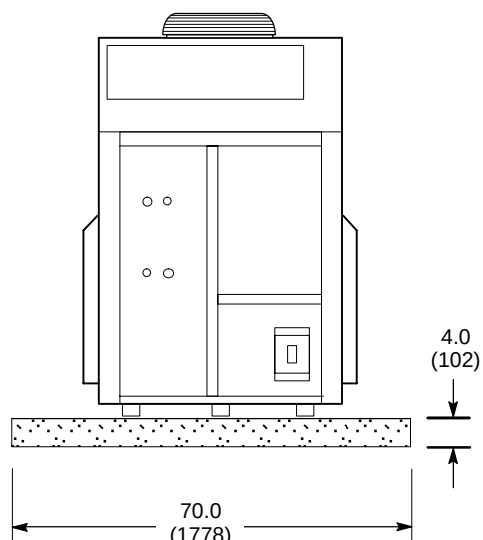
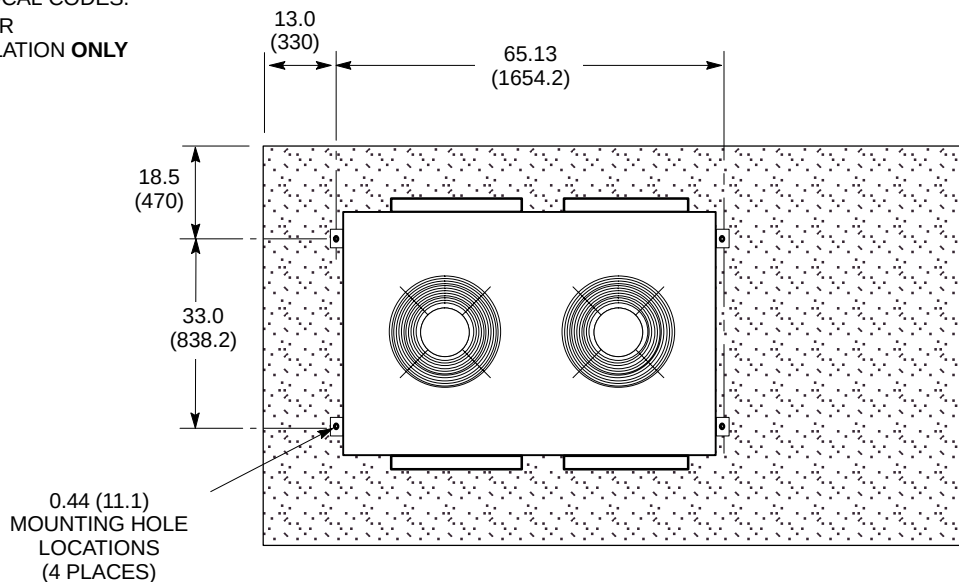


OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)

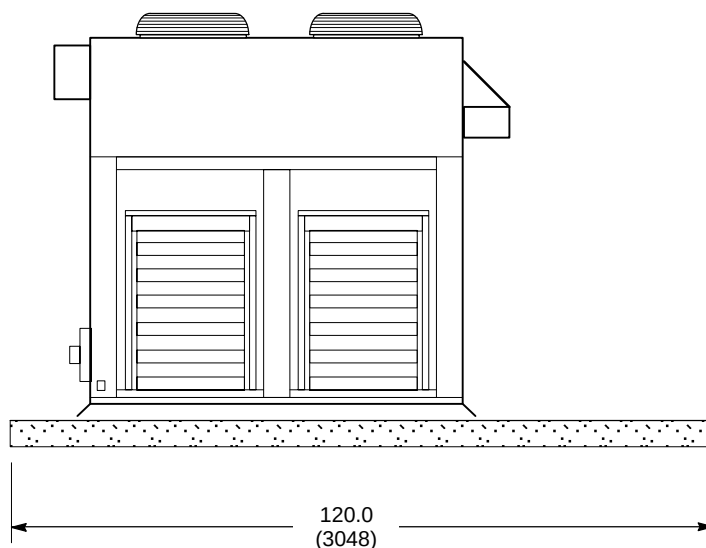
ILLUSTRATION 2-10

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON-GRADE (GROUND) INSTALLATION **ONLY**



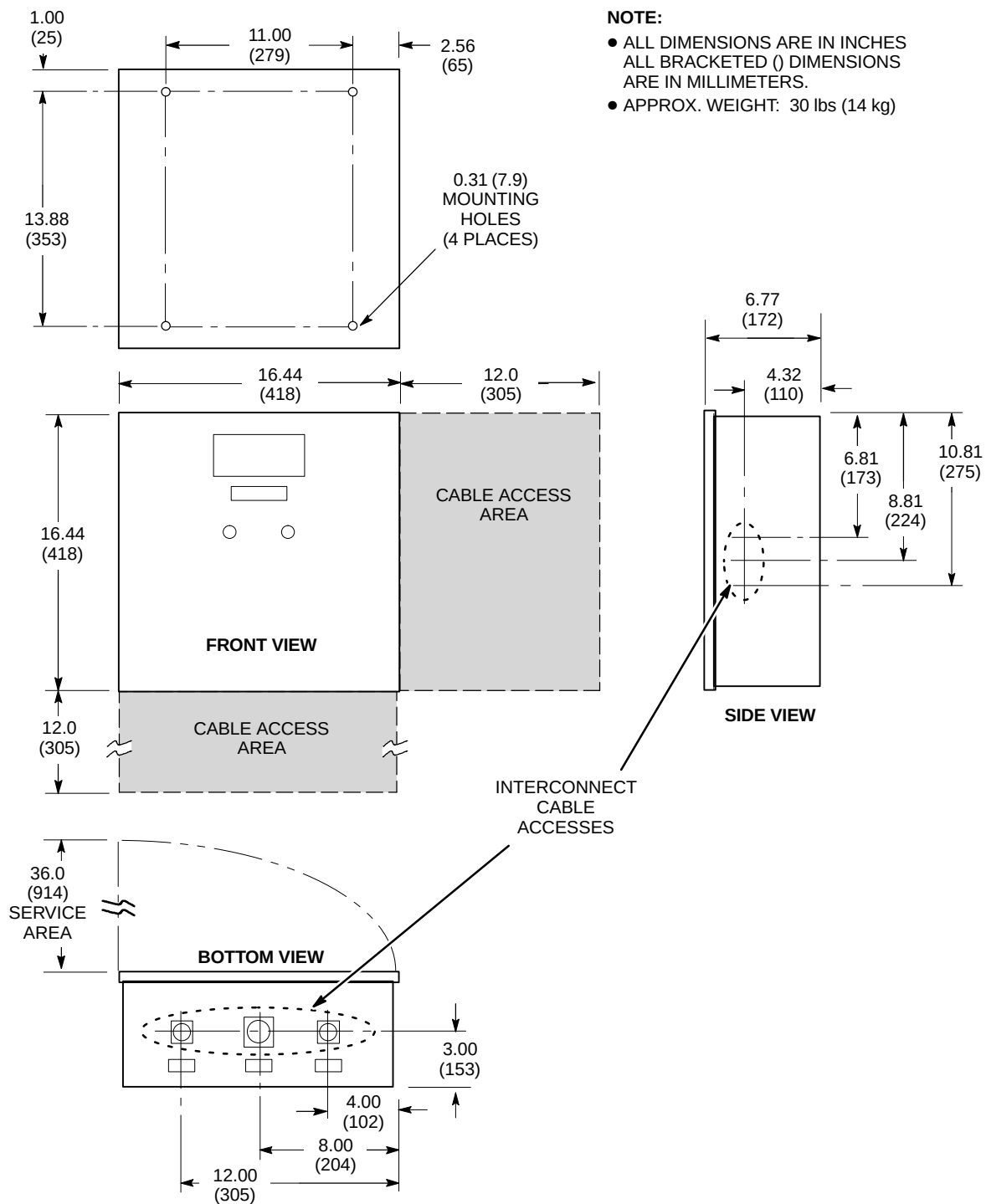
FRONT VIEW



LEFT SIDE VIEW

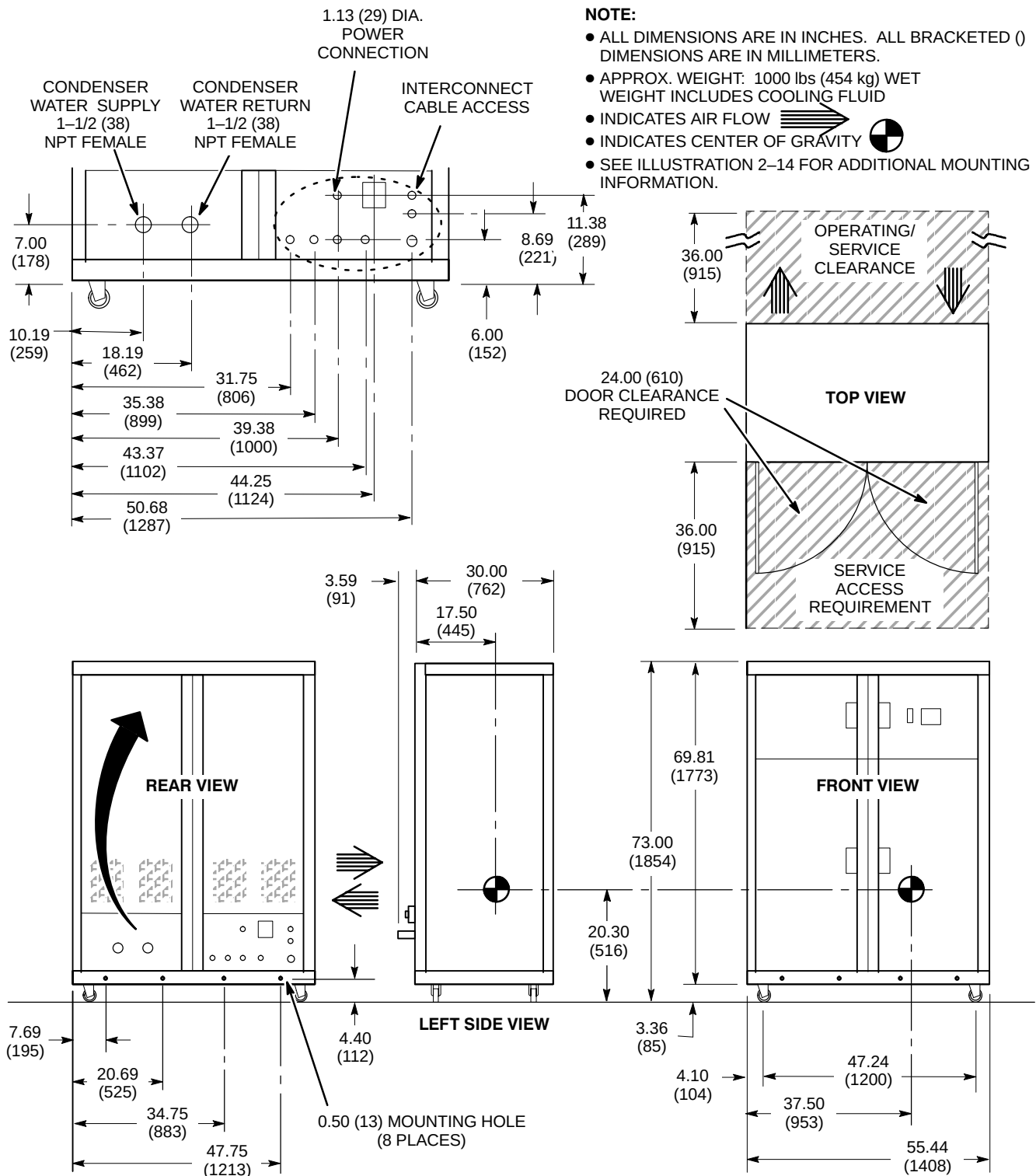
OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING

ILLUSTRATION 2-11



REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]

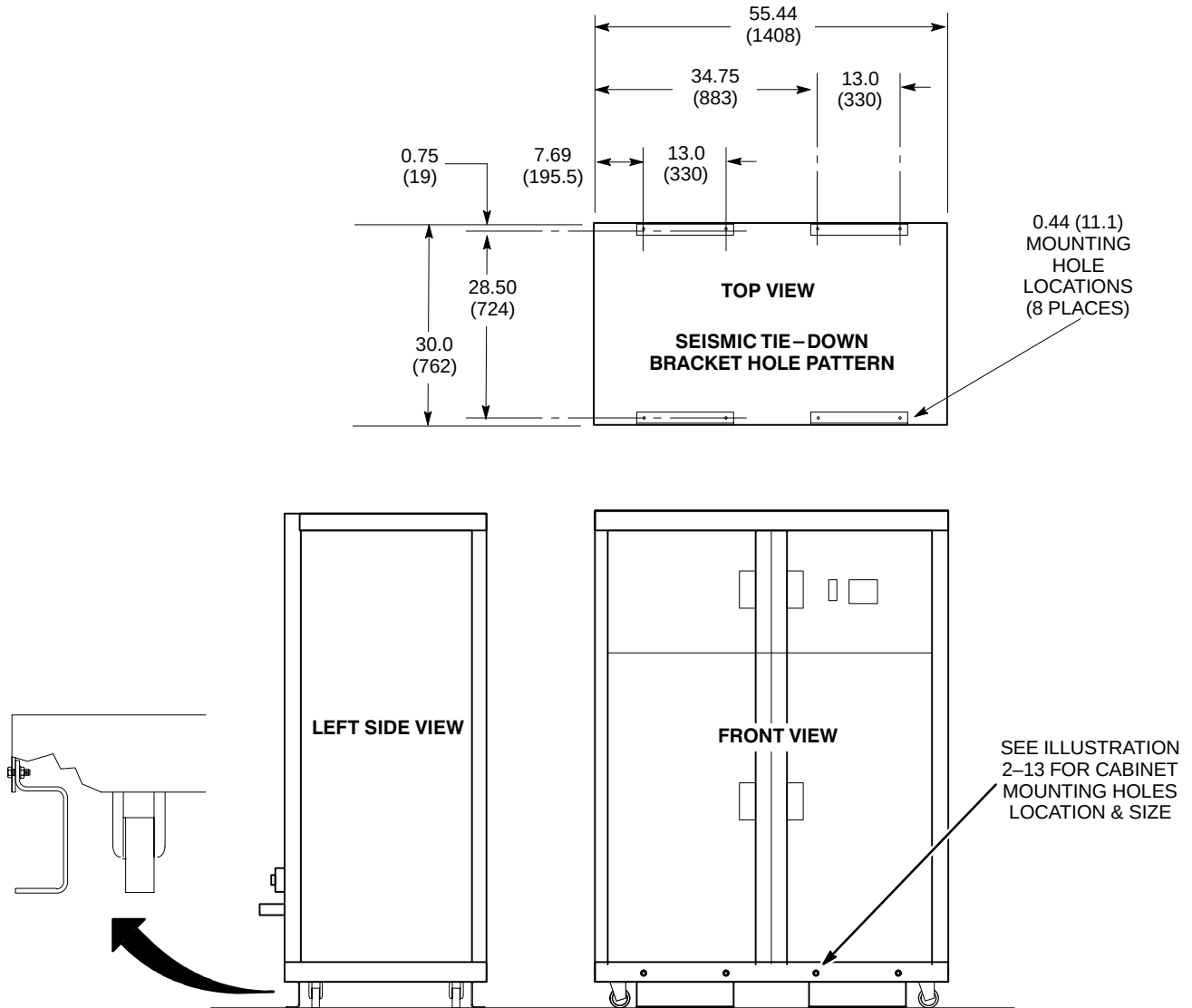
ILLUSTRATION 2-12



INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2-13

NOTE:

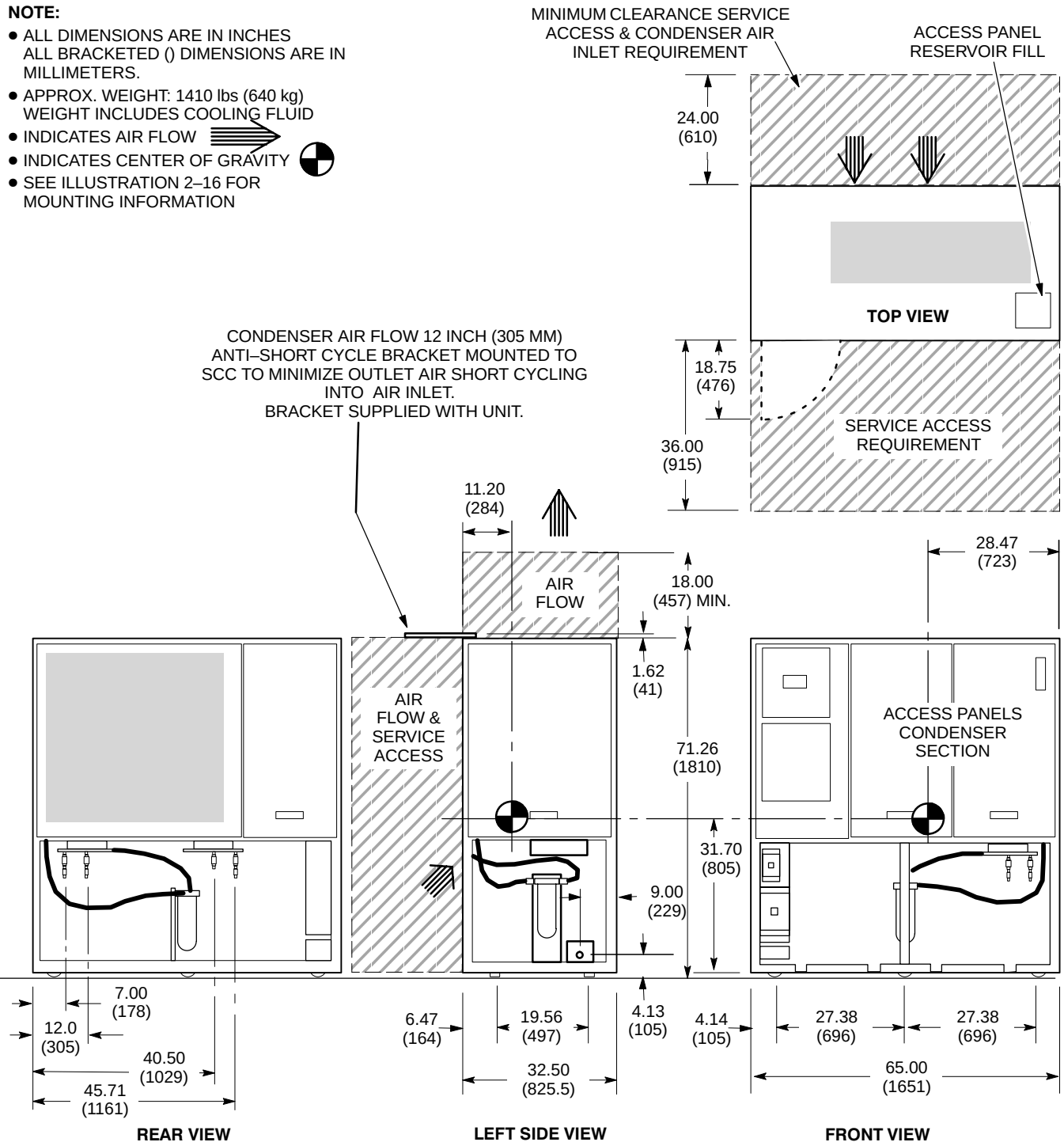
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING
ILLUSTRATION 2-14

NOTE:

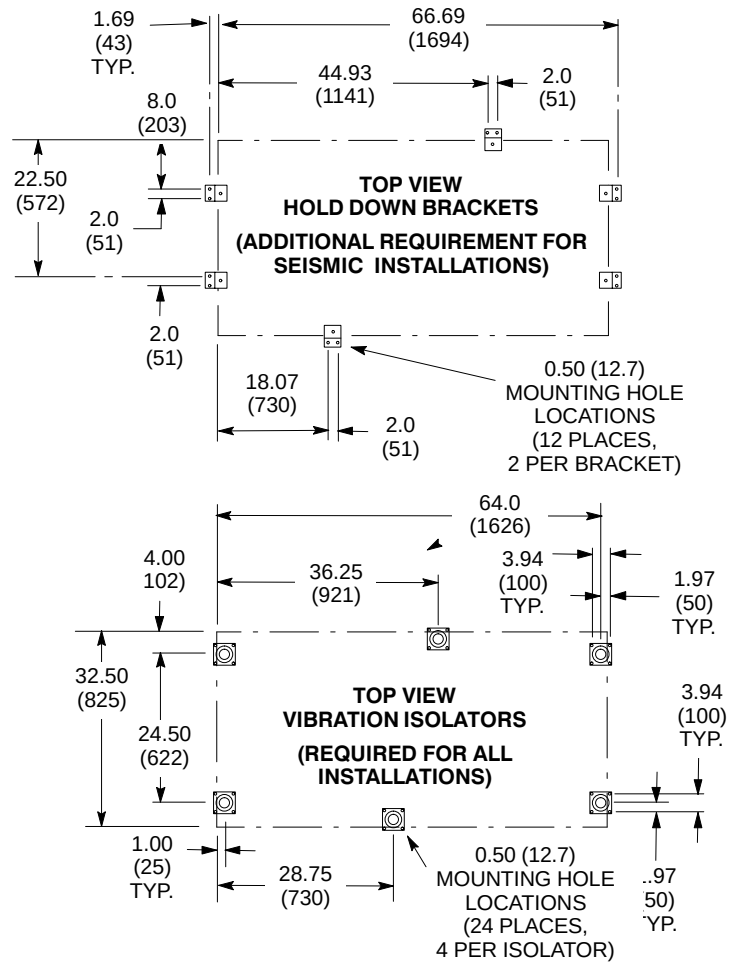
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1410 lbs (640 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-16 FOR MOUNTING INFORMATION



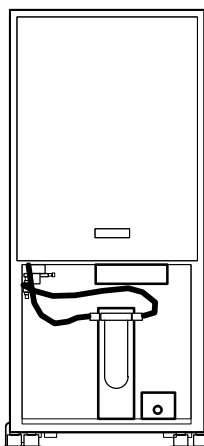
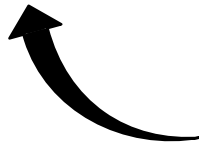
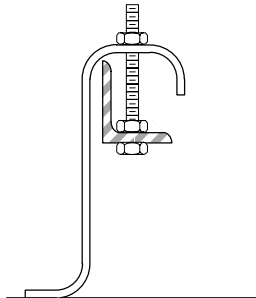
INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2-15

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

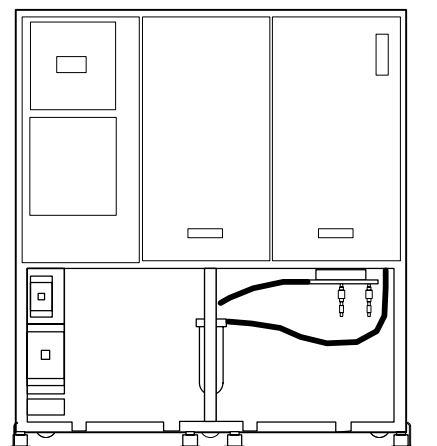


**HOLD DOWN BRACKET
DETAIL**



LEFT SIDE VIEW

3.8
(96.5)



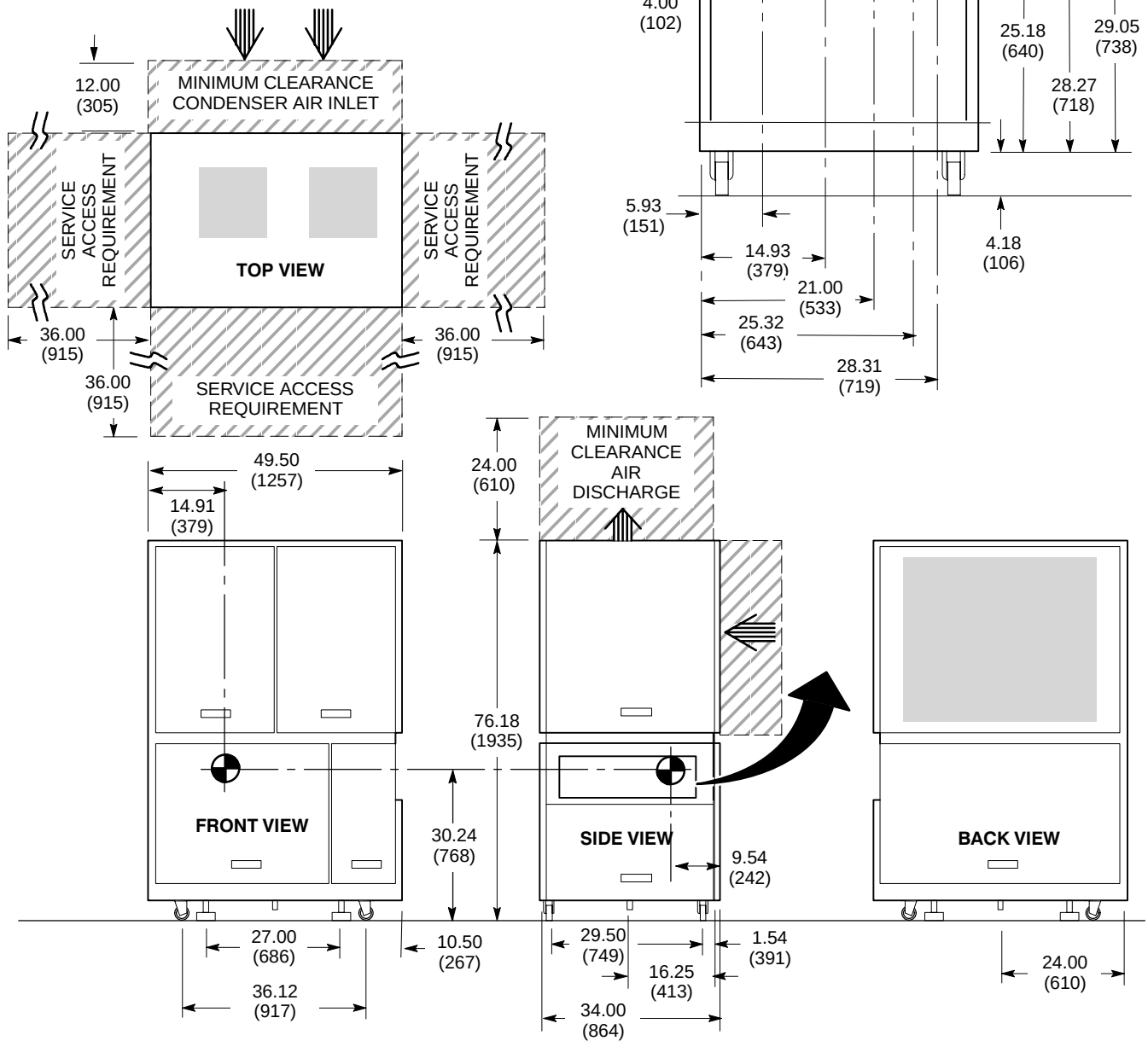
FRONT VIEW

INDOOR AIR COOLED TSCC MOUNTING

ILLUSTRATION 2-16

NOTE:

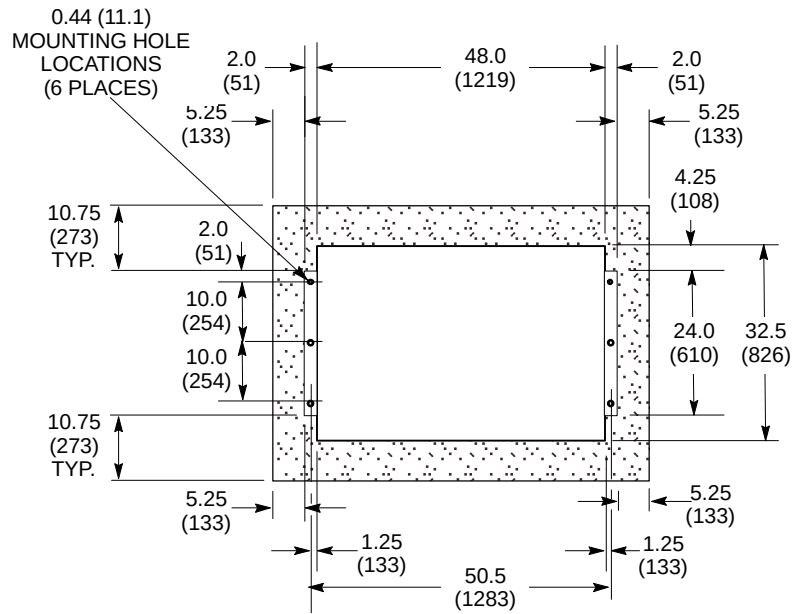
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 780 lbs (345 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-18 FOR MOUNTING INFORMATION
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



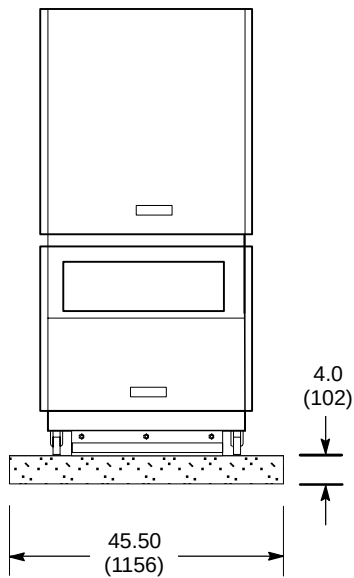
OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MO] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2-17

NOTE:

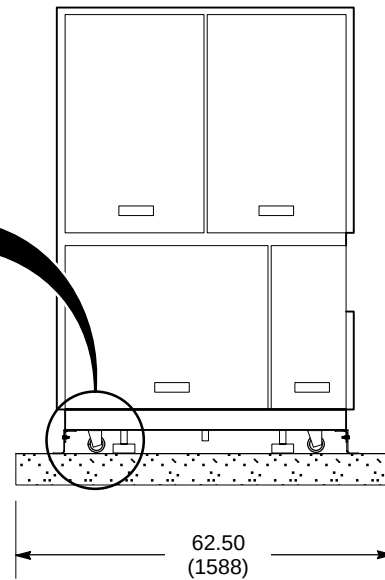
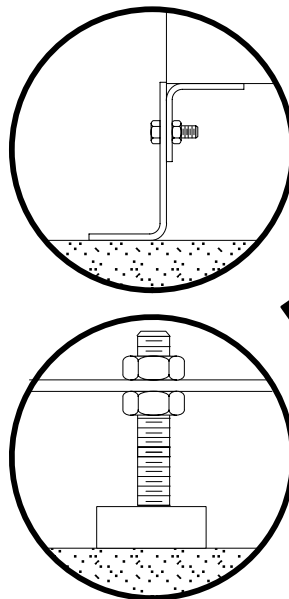
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON-GRADE (GROUND) INSTALLATION **ONLY**



SEISMIC TIE-DOWN BRACKET HOLE PATTERN

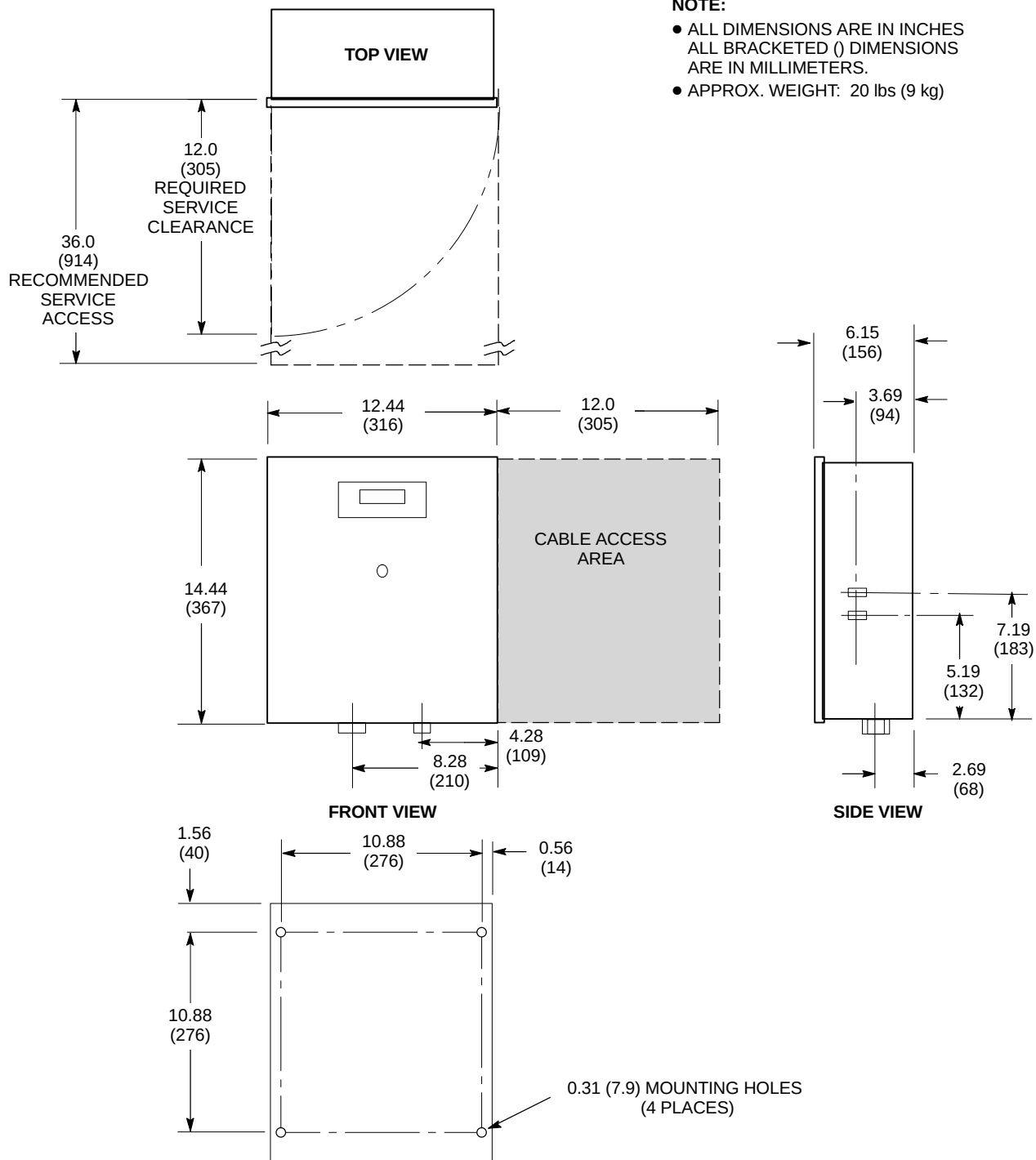


RIGHT SIDE VIEW



FRONT VIEW

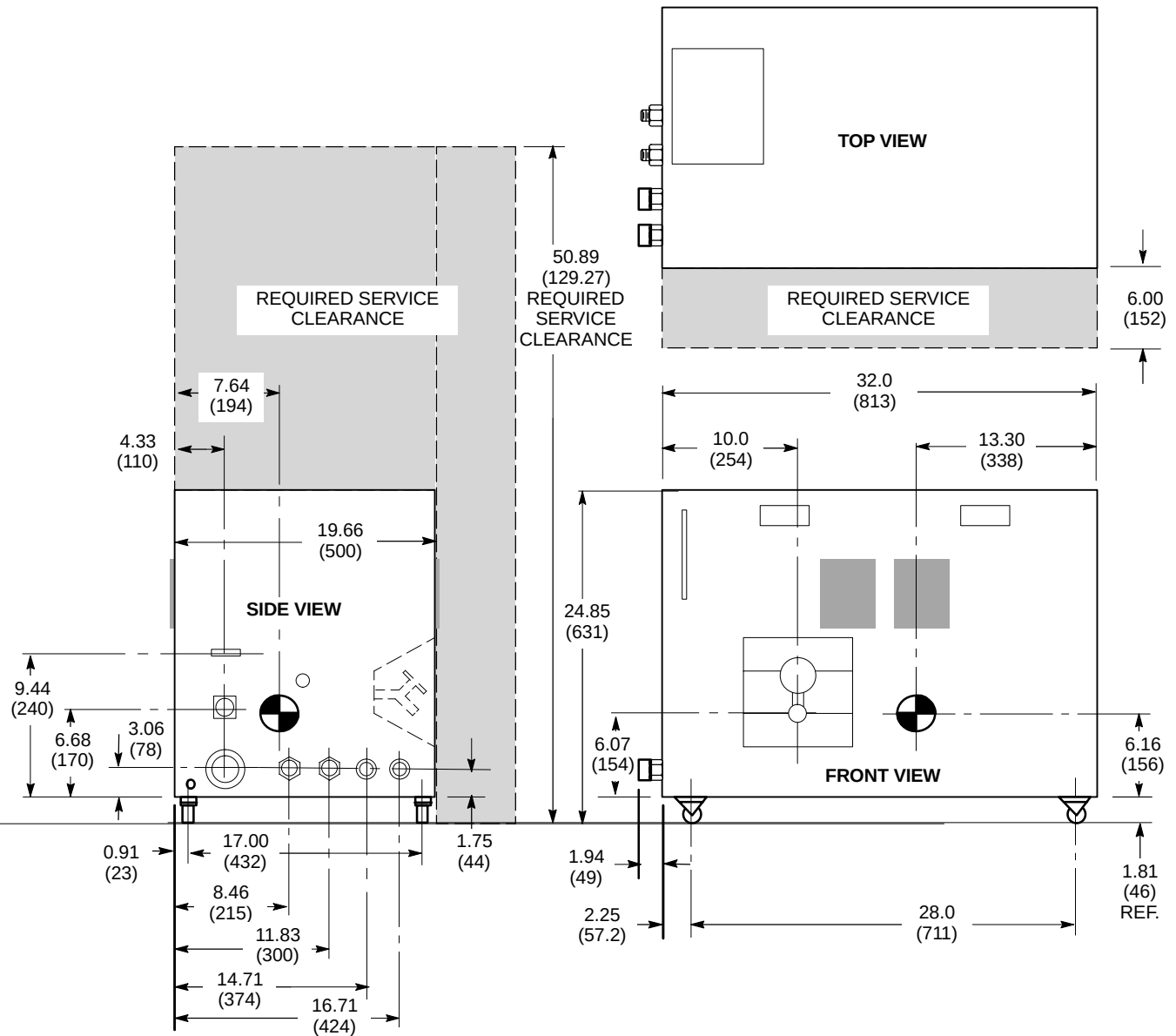
OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O] MOUNTING
ILLUSTRATION 2-18



REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O]
ILLUSTRATION 2-19

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: TGWC 175 lbs (79 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.

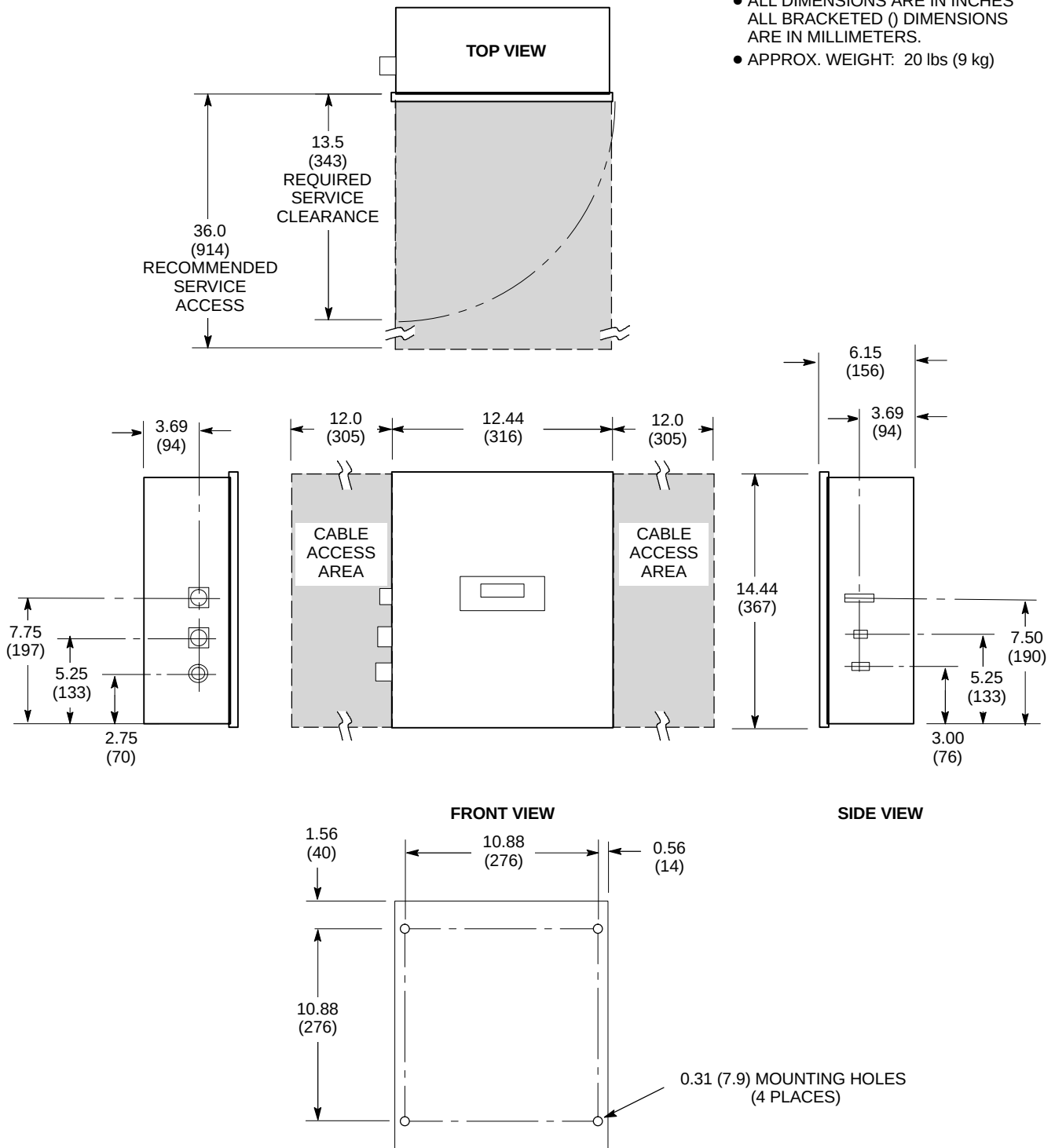


INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL-10] (TGWC)

ILLUSTRATION 2-20

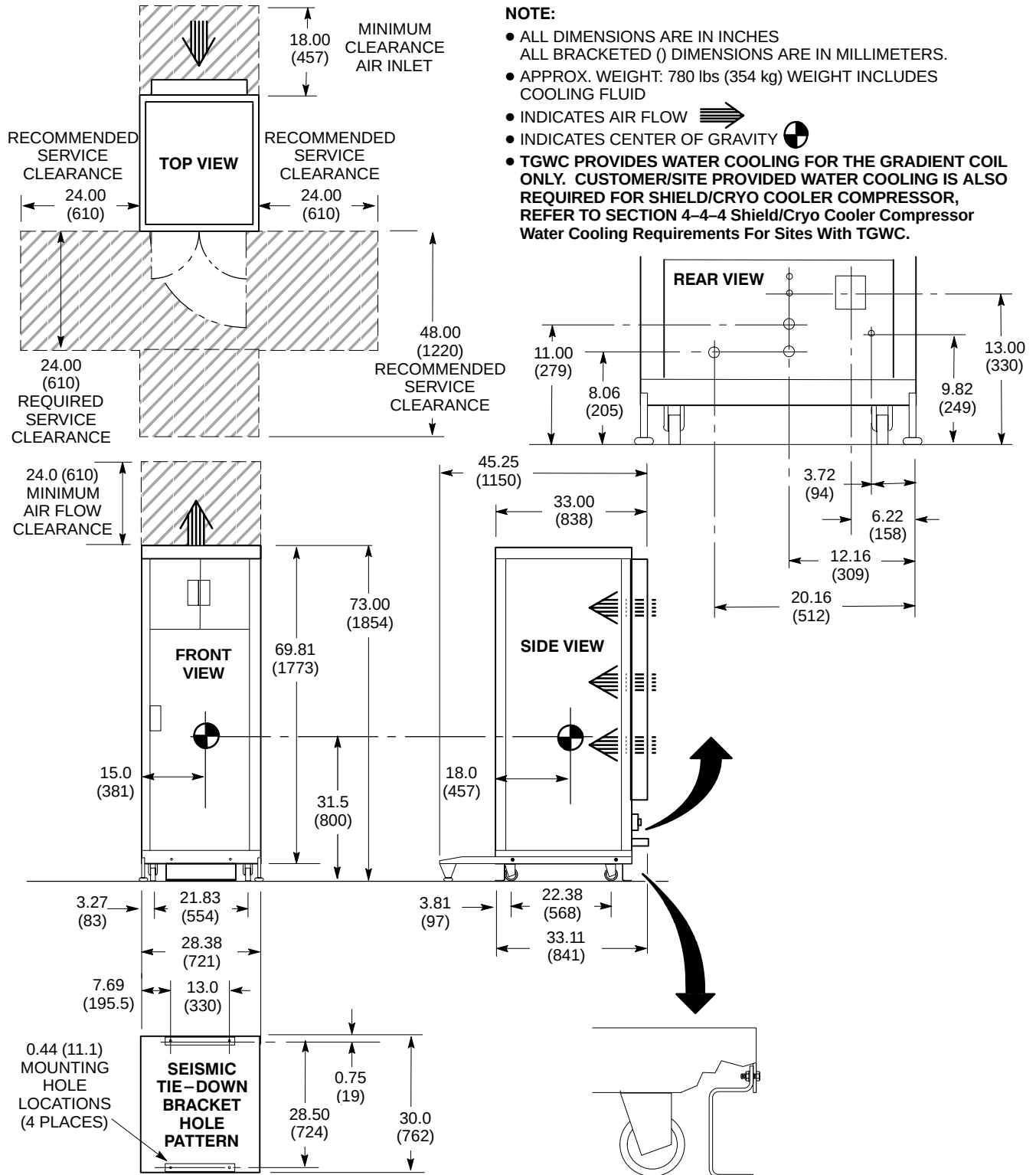
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL-10]

ILLUSTRATION 2-21

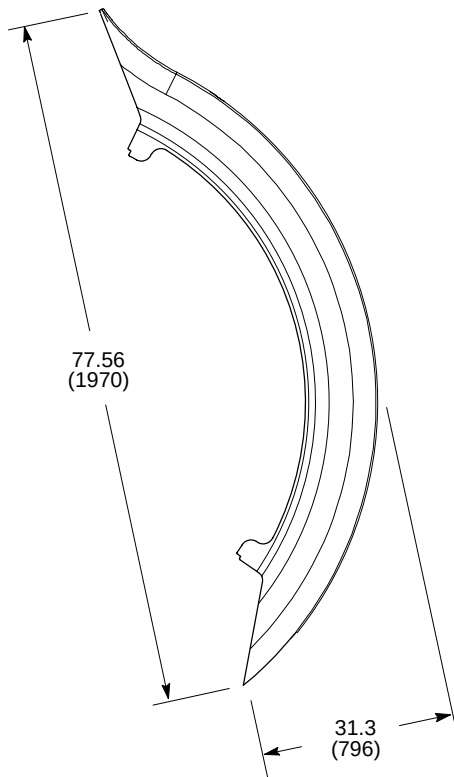


INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)

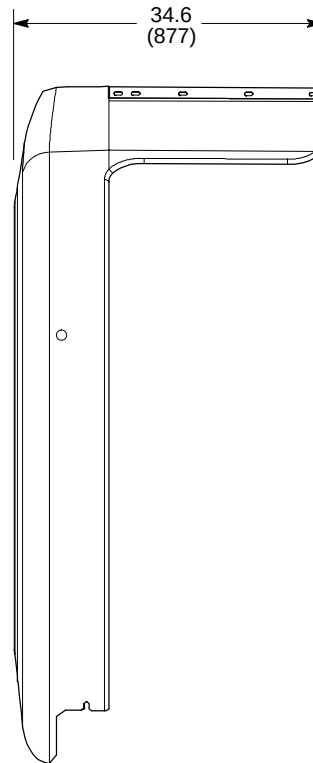
ILLUSTRATION 2-22

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT VIEW

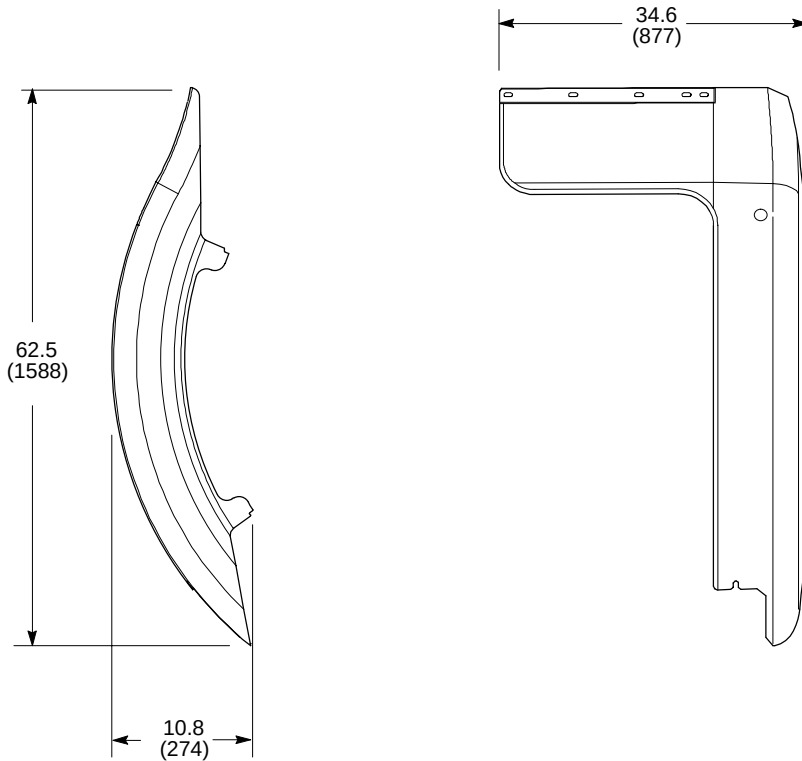


SIDE VIEW

FRONT RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-23

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



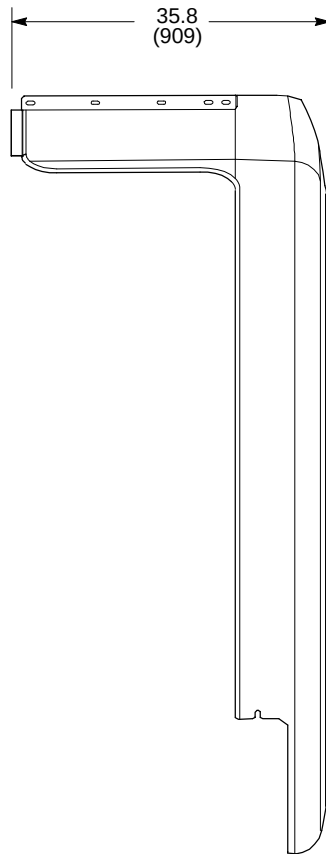
FRONT VIEW

SIDE VIEW

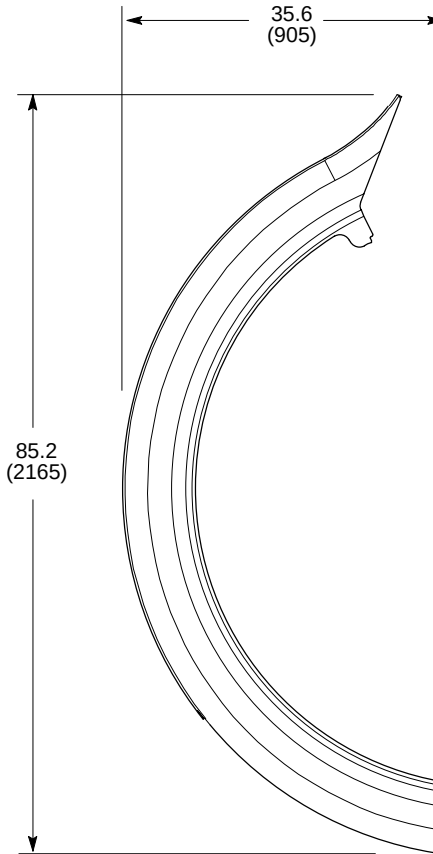
FRONT LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-24

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



SIDE VIEW

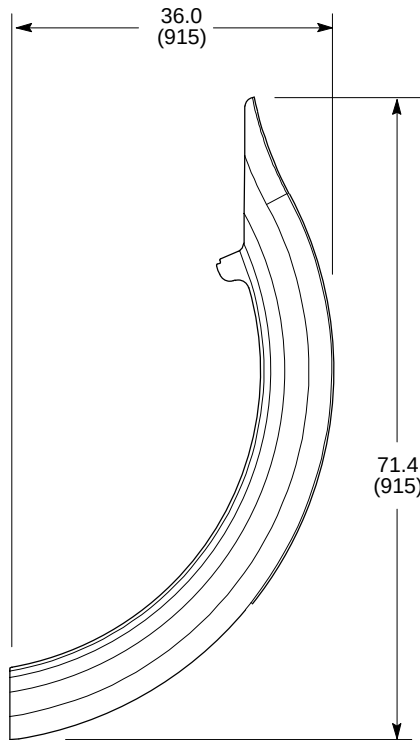


FRONT VIEW

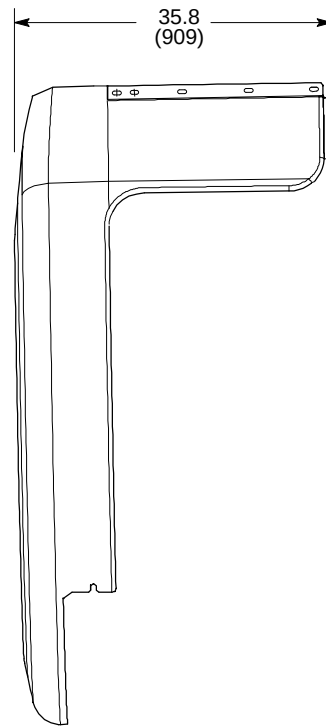
REAR RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-25

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT VIEW

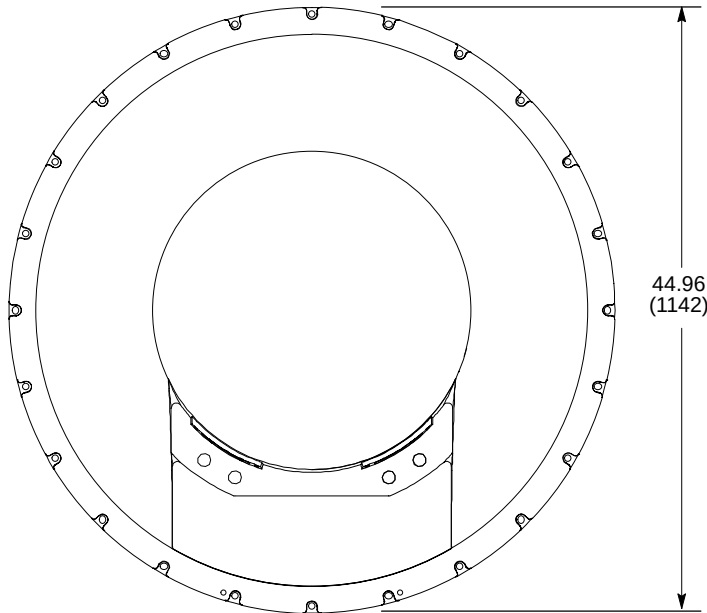


SIDE VIEW

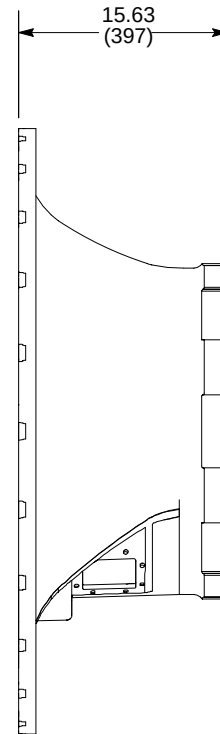
REAR LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-26

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 70 lbs (32 kg)



FRONT VIEW

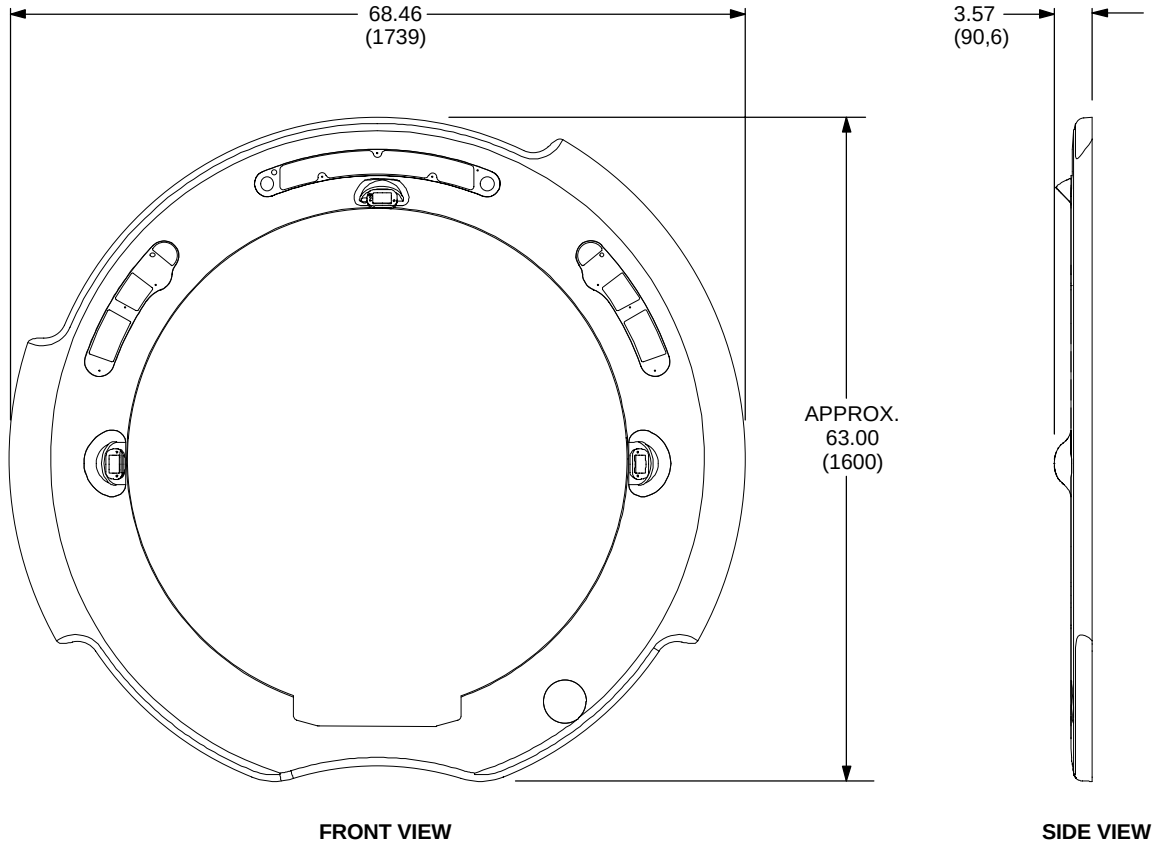


SIDE VIEW

FRONT ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-27

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 35 lbs (16 kg)



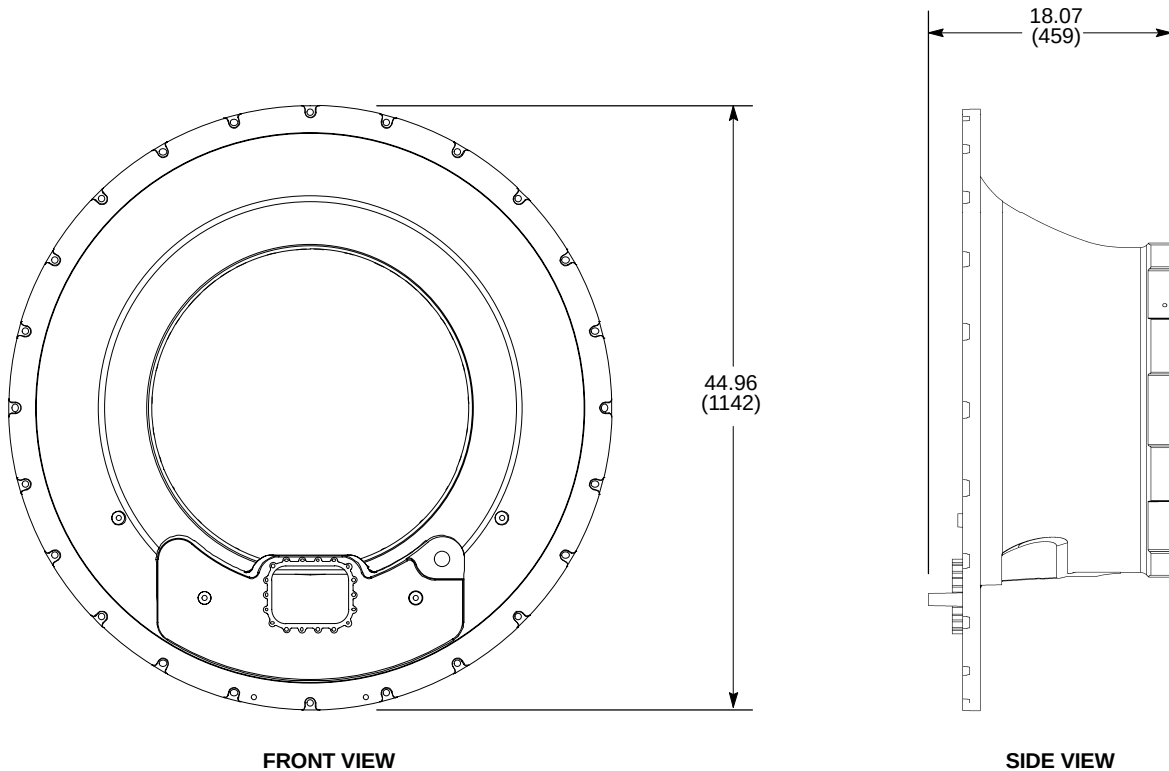
FRONT VIEW

SIDE VIEW

FRONT COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-28

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



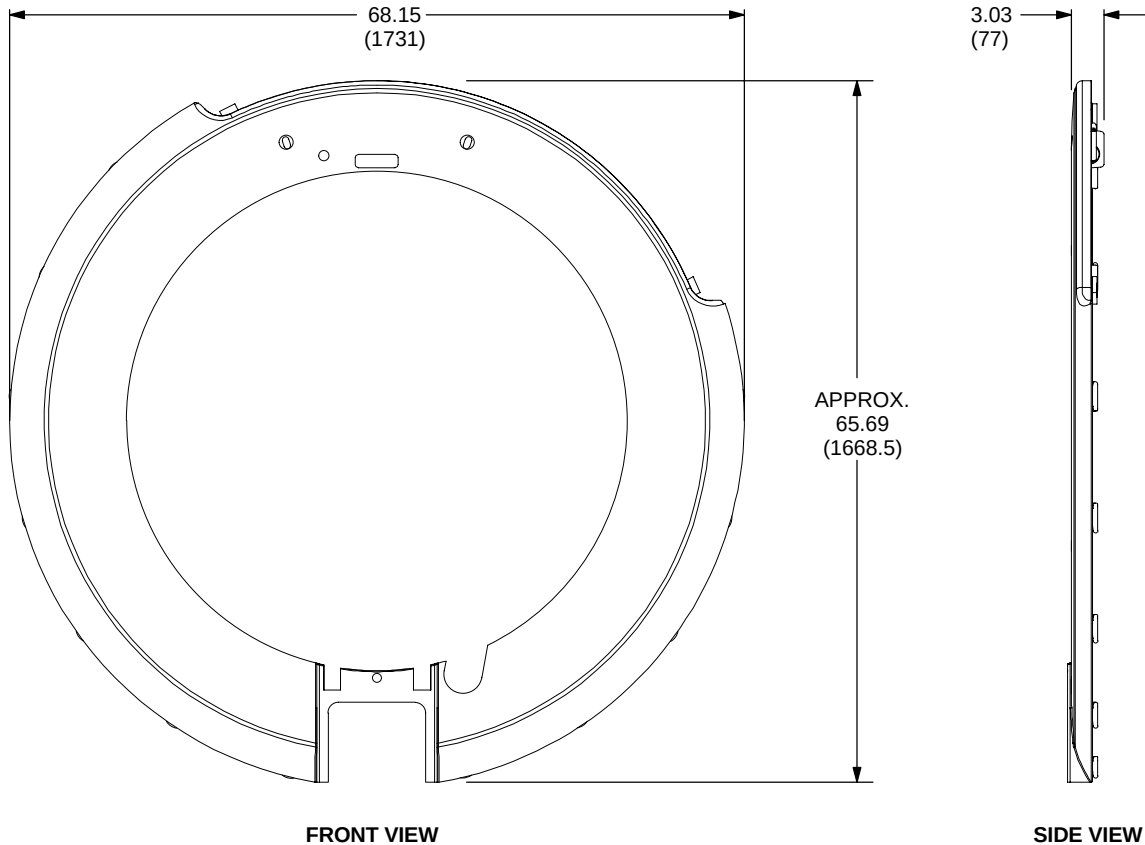
FRONT VIEW

SIDE VIEW

REAR ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-29

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 35 lbs (16 kg)



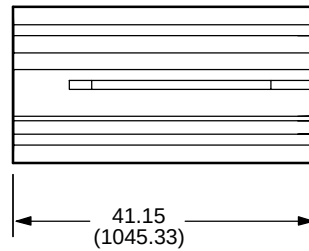
FRONT VIEW

SIDE VIEW

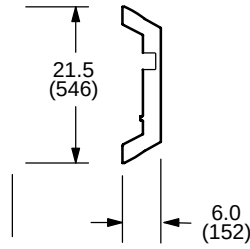
REAR COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2-30

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)

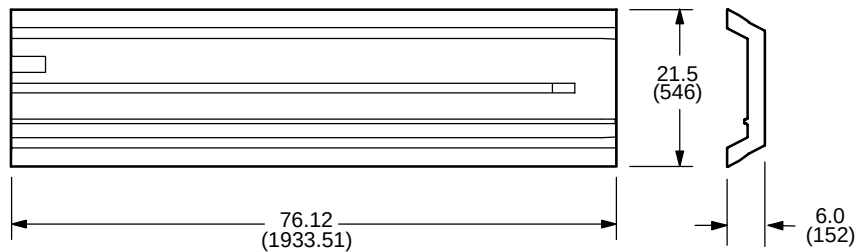


TOP VIEW



END VIEW

REAR PORTION SPLIT BRIDGE



TOP VIEW

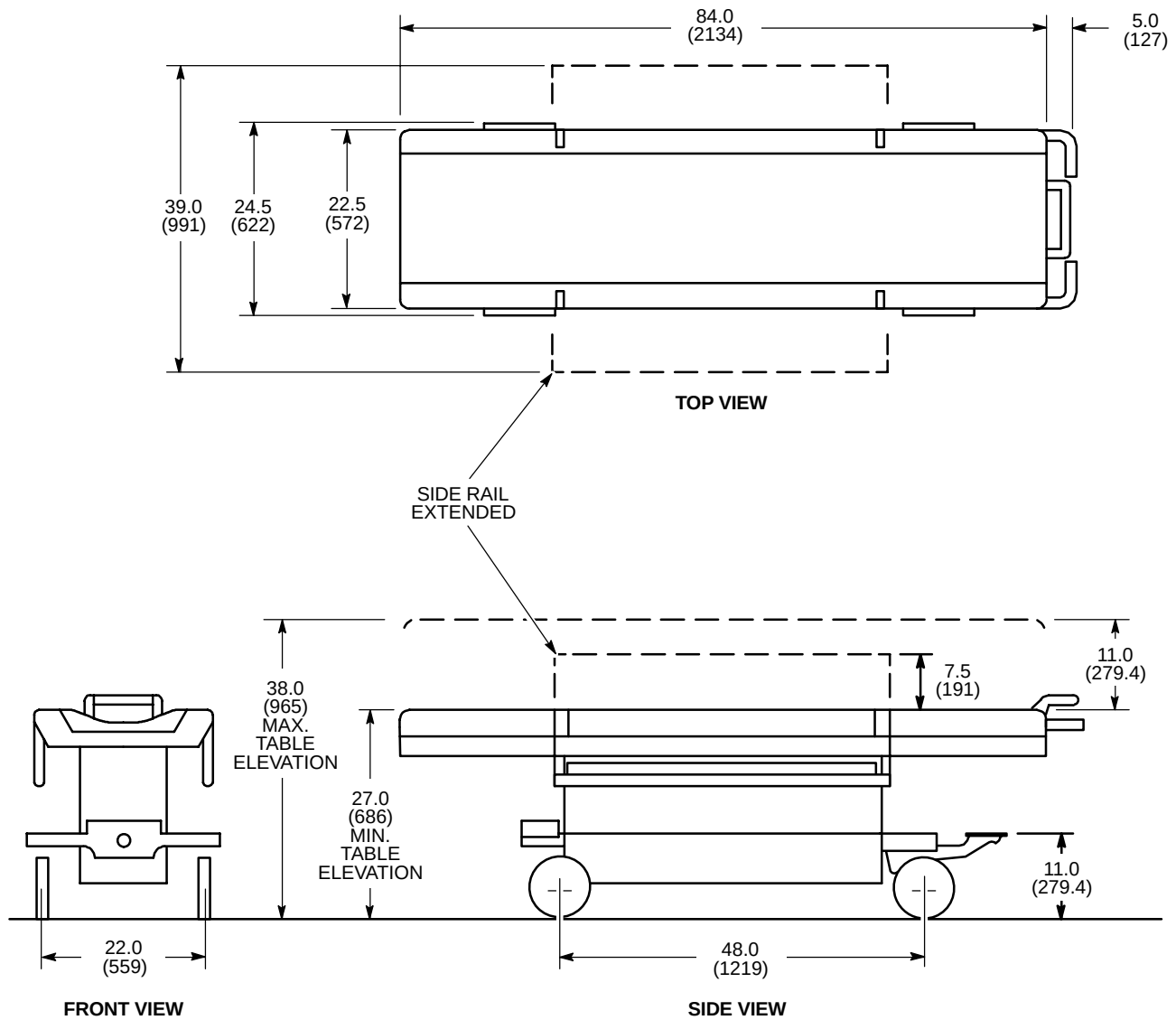
END VIEW

FRONT PORTION SPLIT BRIDGE

SPLIT BRIDGE
ILLUSTRATION 2-31

NOTE:

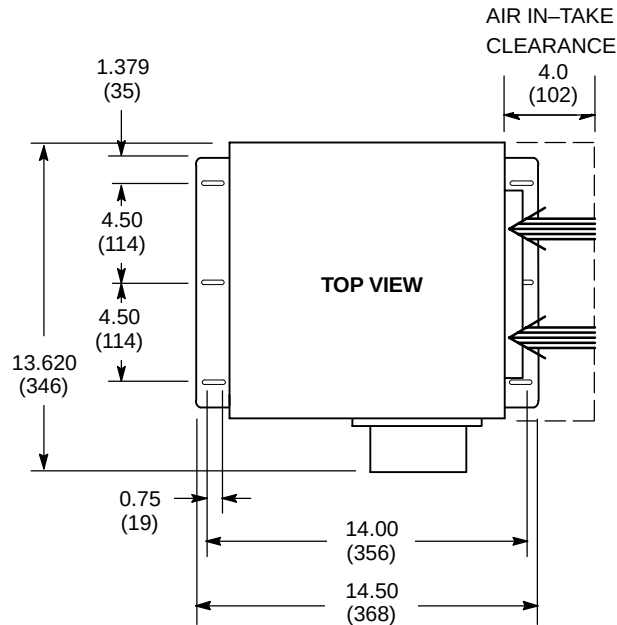
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)



M1081A2

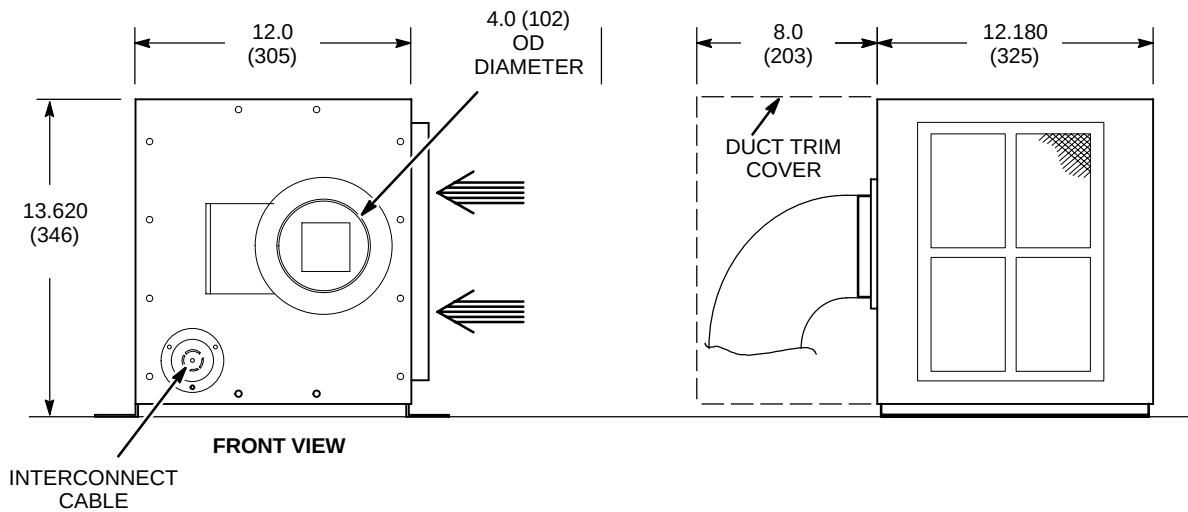
PATIENT TRANSPORT TABLE (PT1)

ILLUSTRATION 2-32



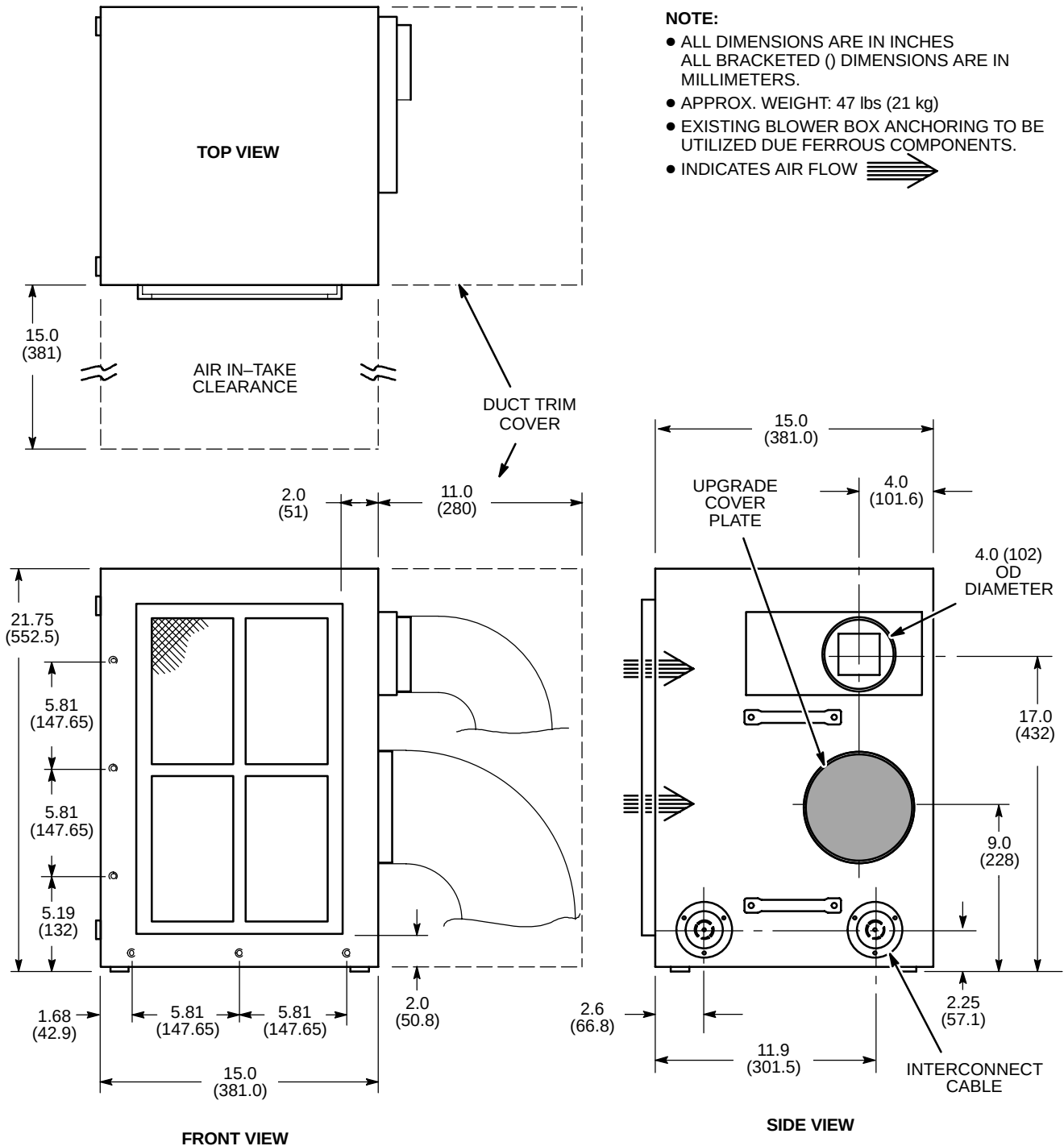
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW ➡



UPGRADE NEW BLOWER BOX (MG6)

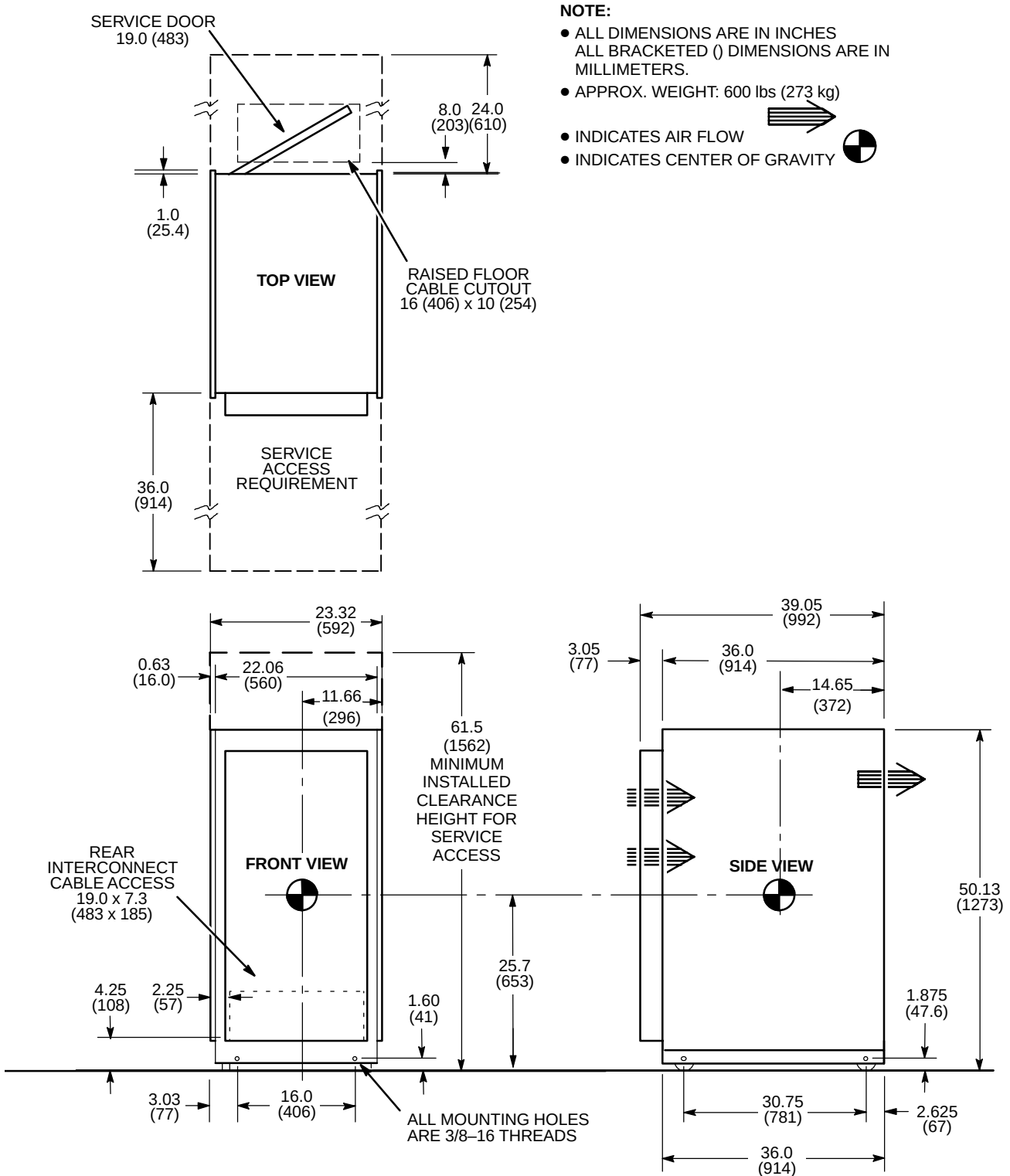
ILLUSTRATION 2-33



M4447AM

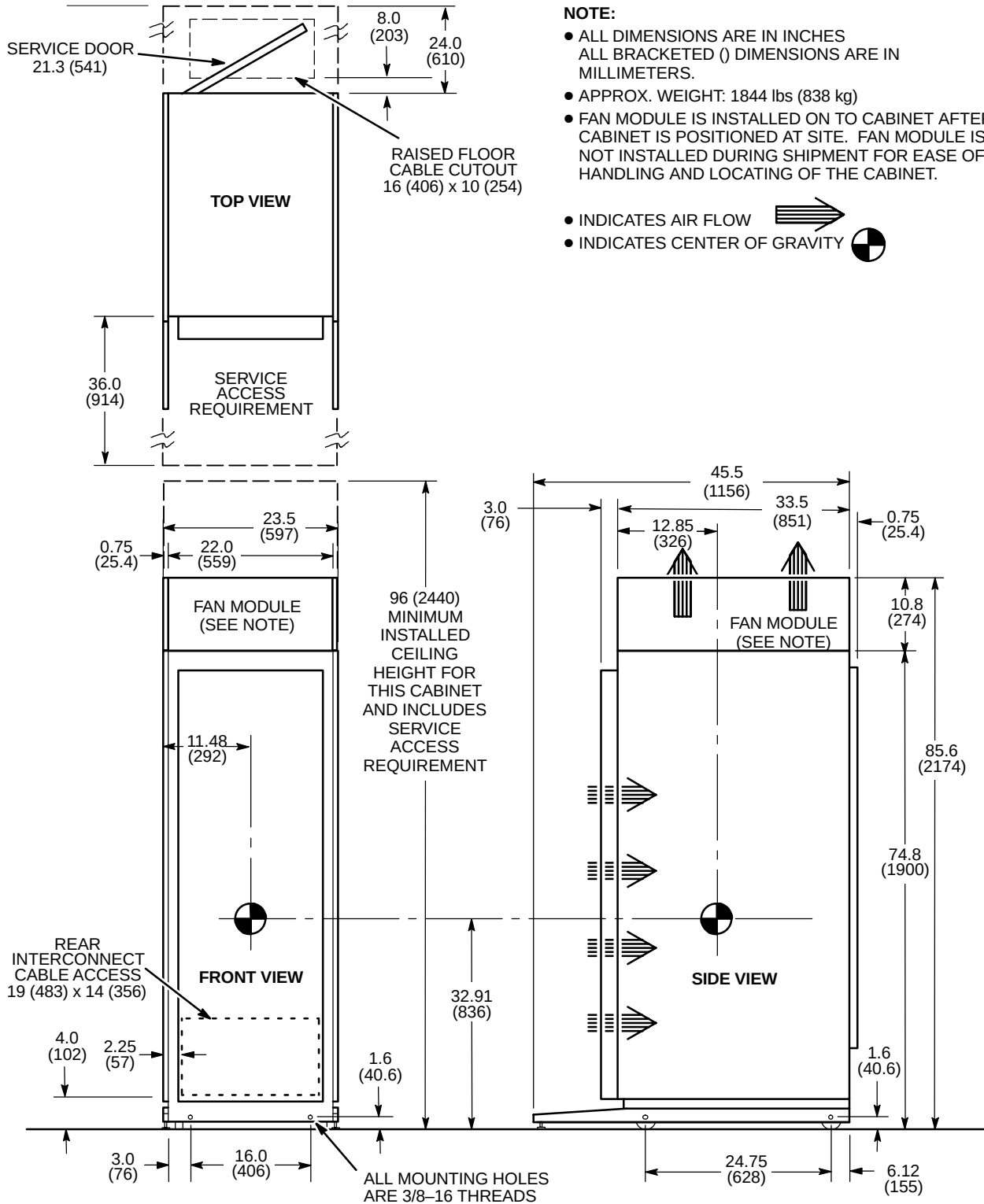
UPGRADE MODIFIED BLOWER BOX (MG6)

ILLUSTRATION 2-34



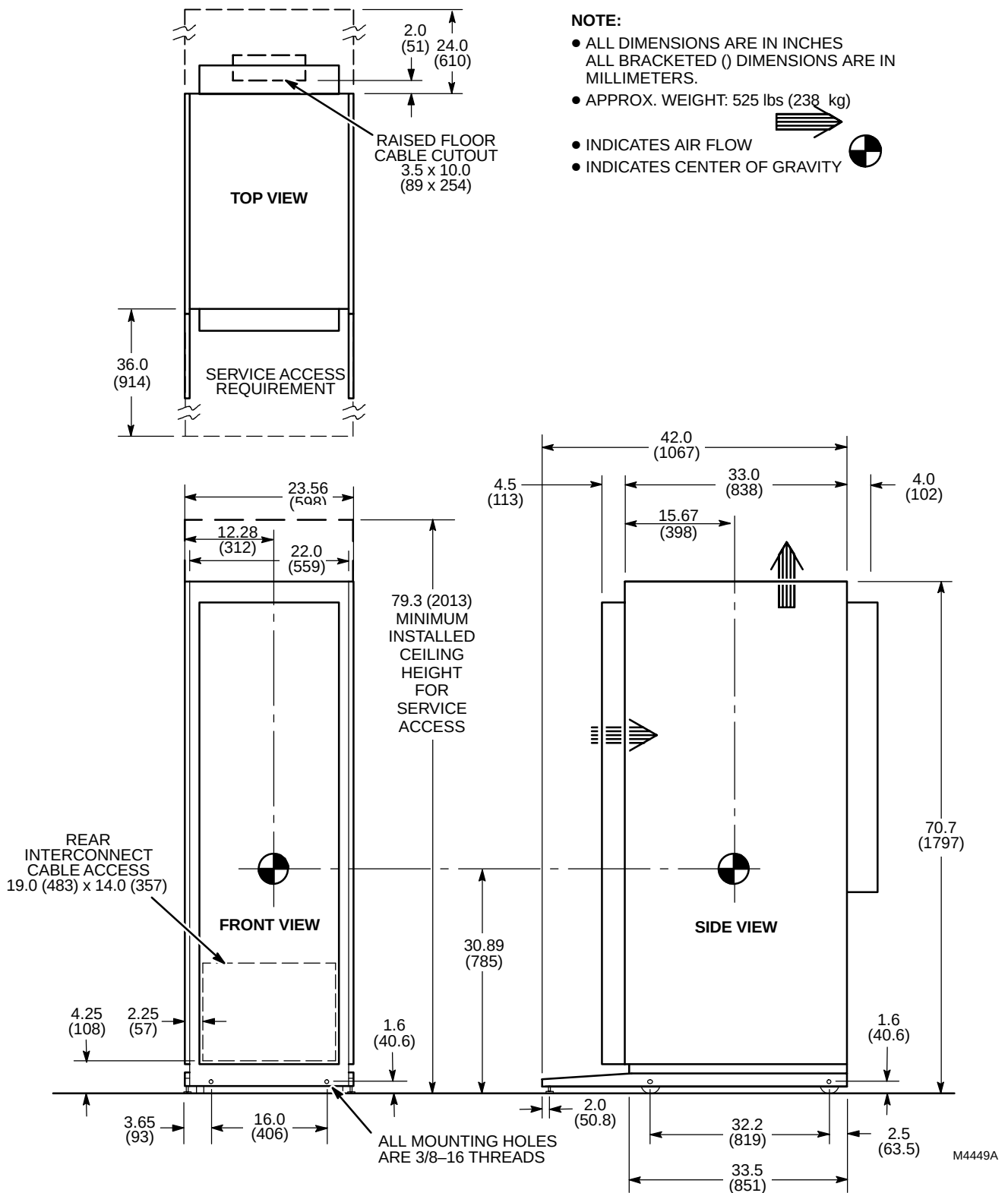
TWIN ACCESSORY CABINET (TAC)

ILLUSTRATION 2-35

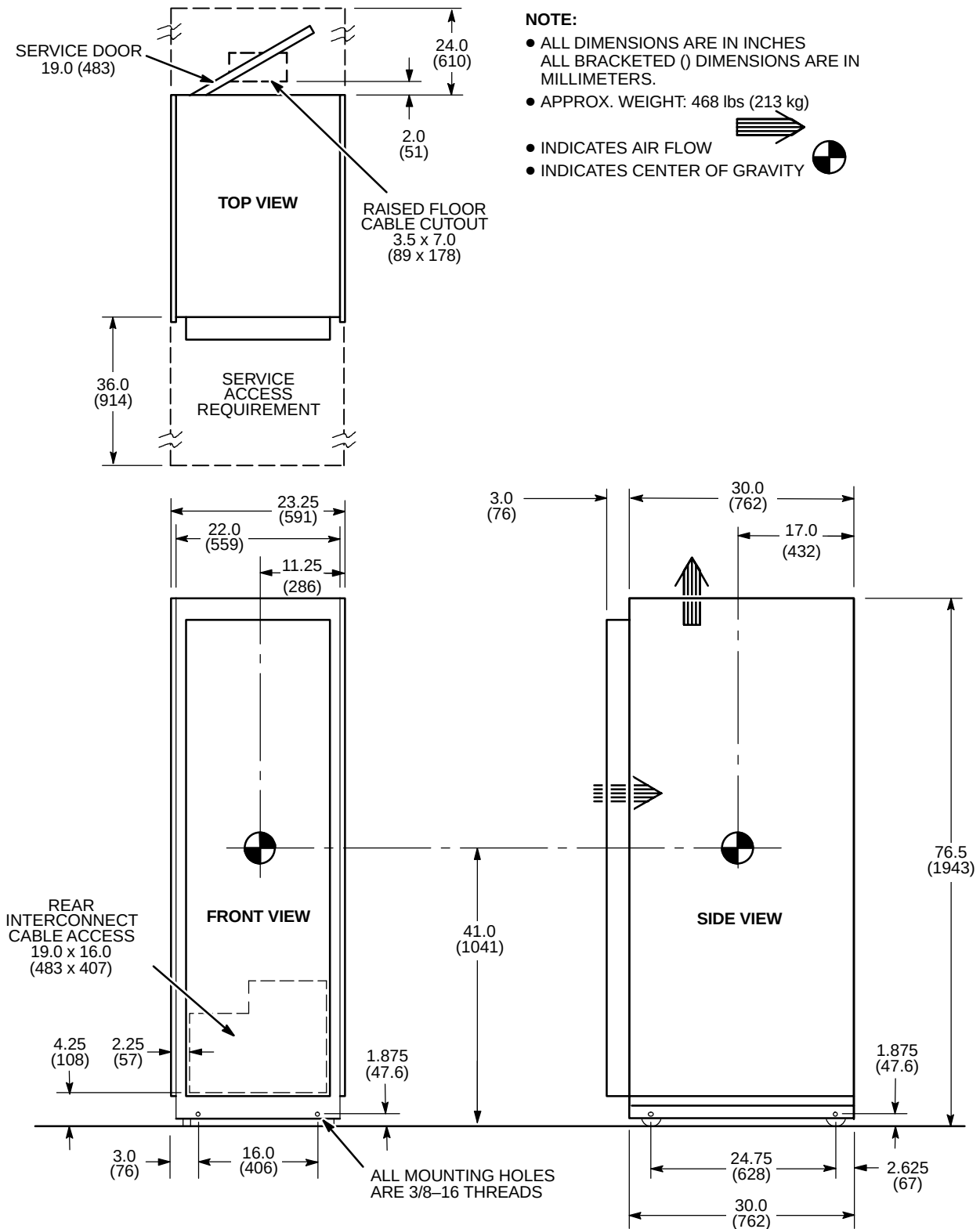


ACGD/PDU CABINET (MR3)

ILLUSTRATION 2-36

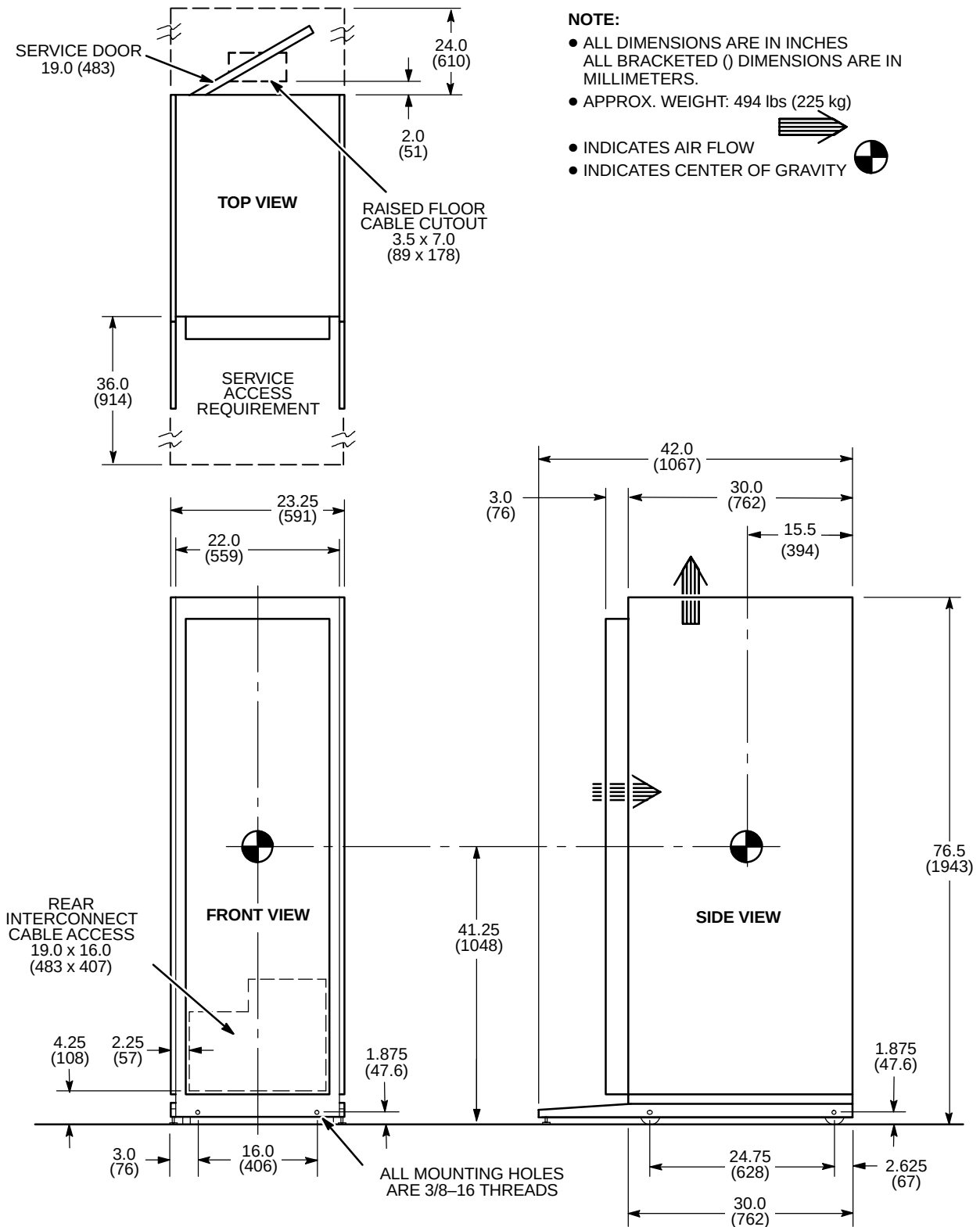


SRF CABINET (MR1)
ILLUSTRATION 2-37



SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY

ILLUSTRATION 2-38

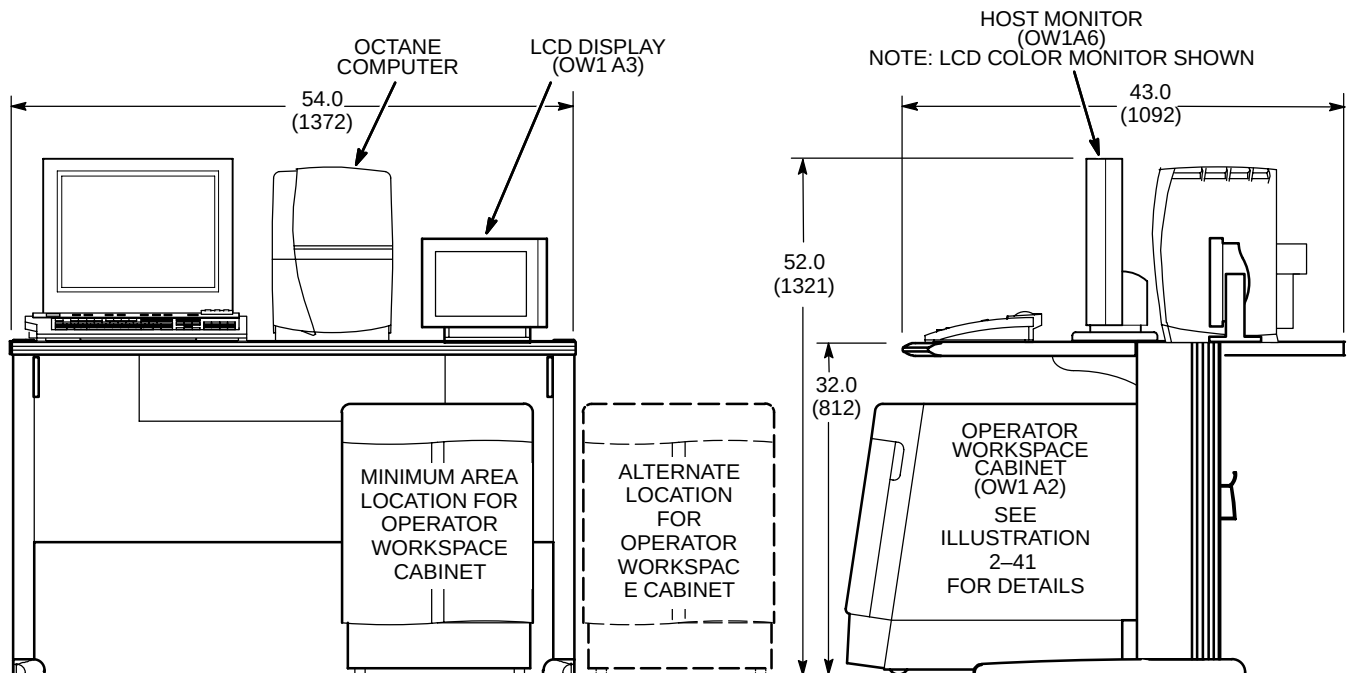
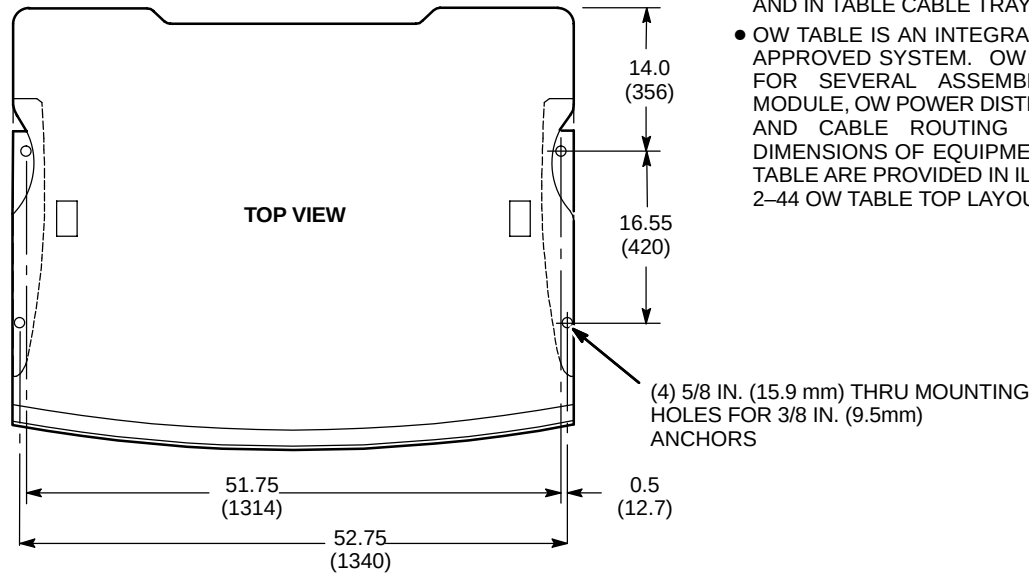


SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED

ILLUSTRATION 2-39

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS 2-42 THROUGH 2-44 OW TABLE TOP LAYOUT PLANNING PURPOSES.

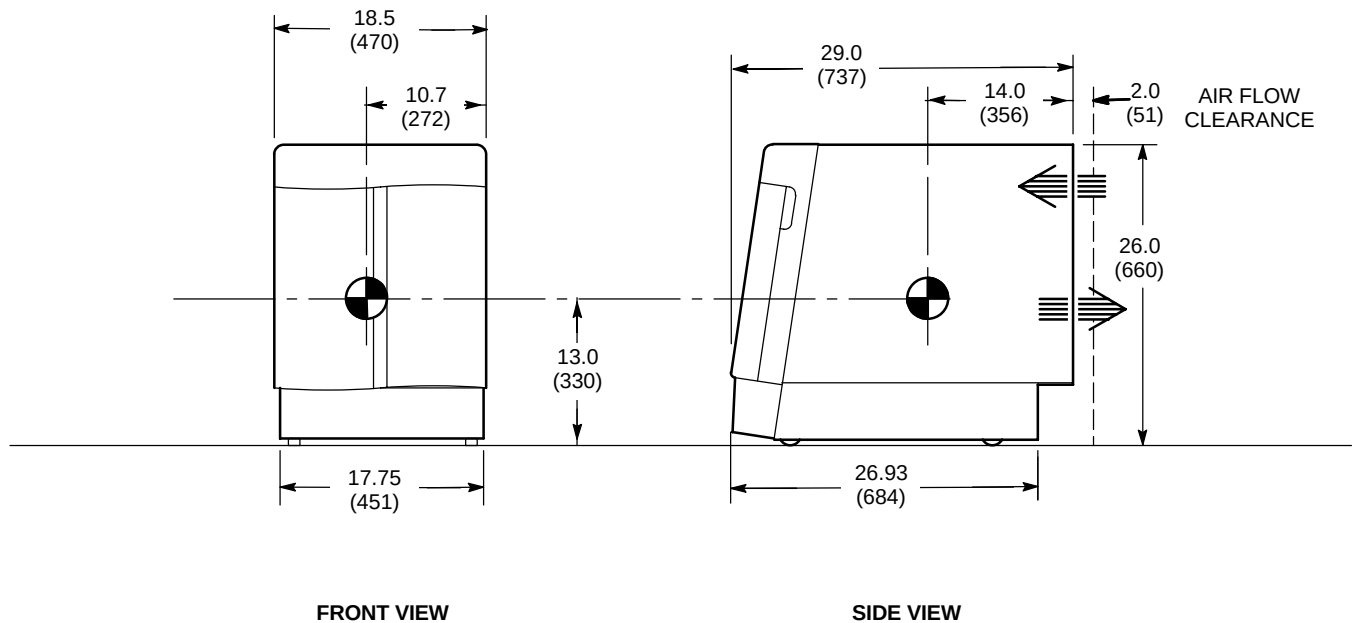


OPERATOR WORKSPACE (OW1)

ILLUSTRATION 2-40

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 

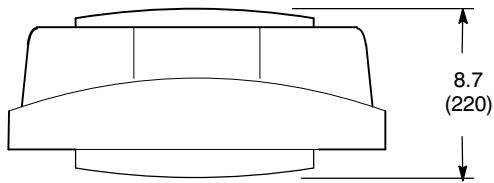


OPERATOR WORKSPACE CABINET (OW1 A2)

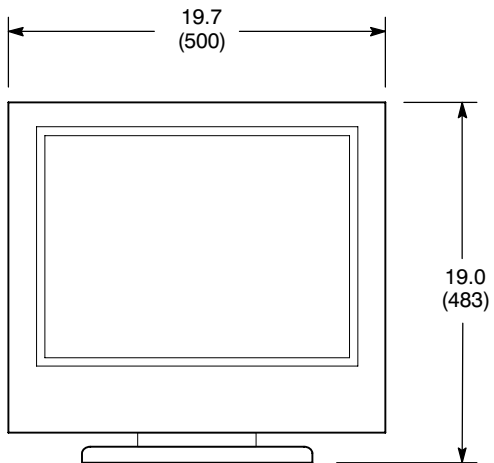
ILLUSTRATION 2-41

NOTE:

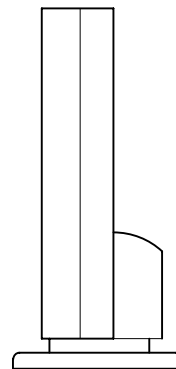
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



TOP VIEW



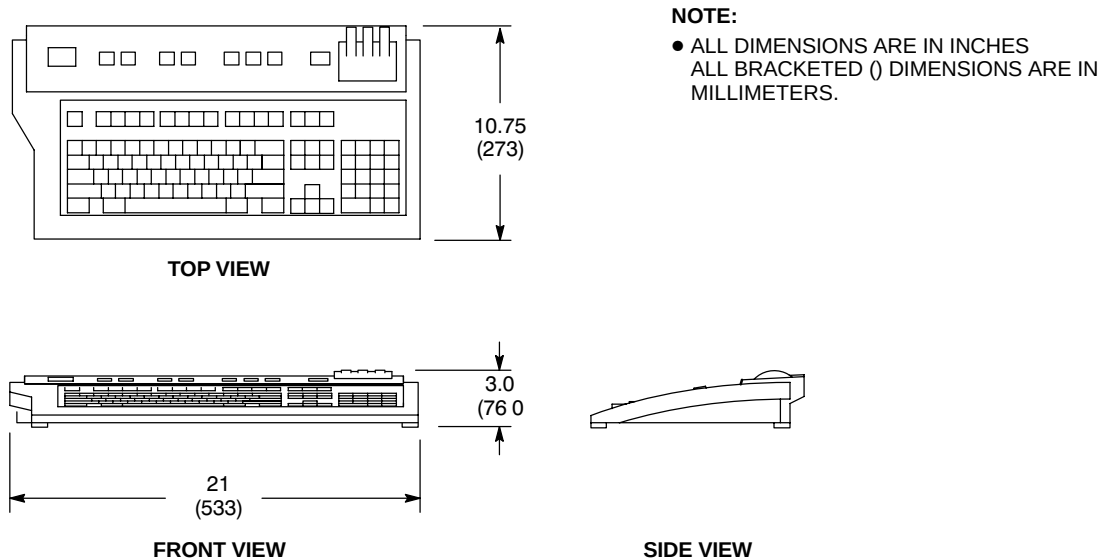
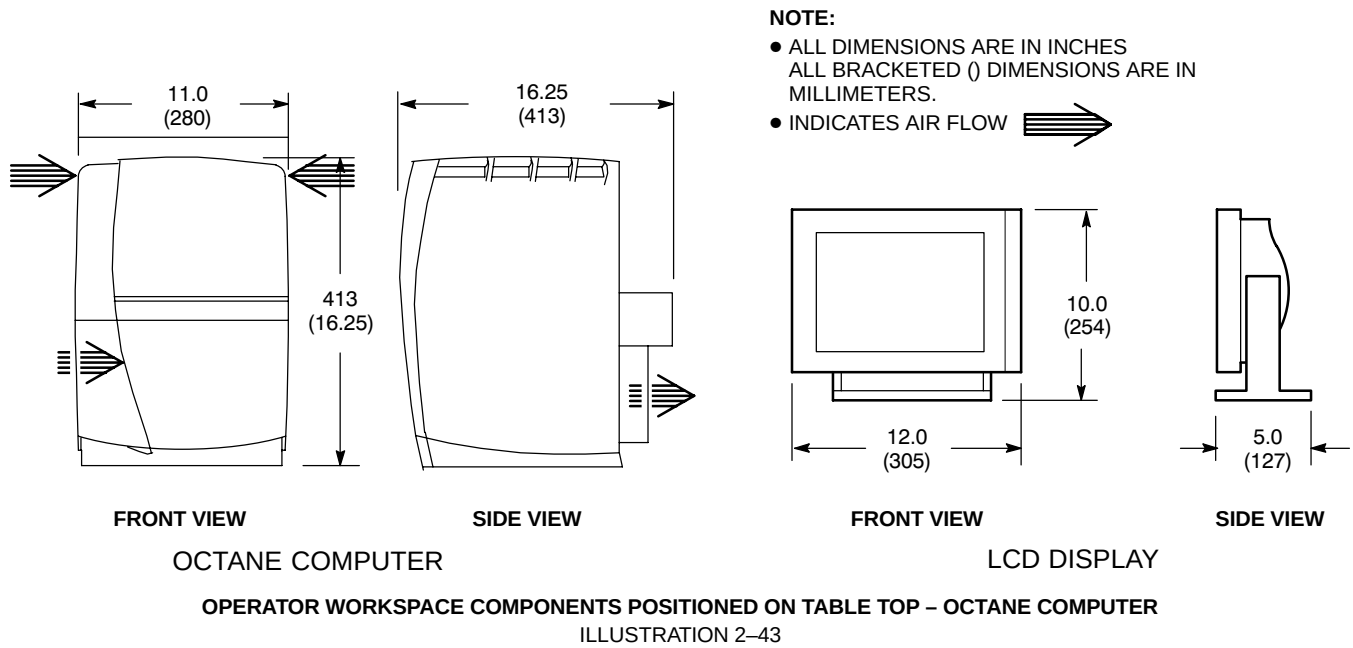
FRONT VIEW

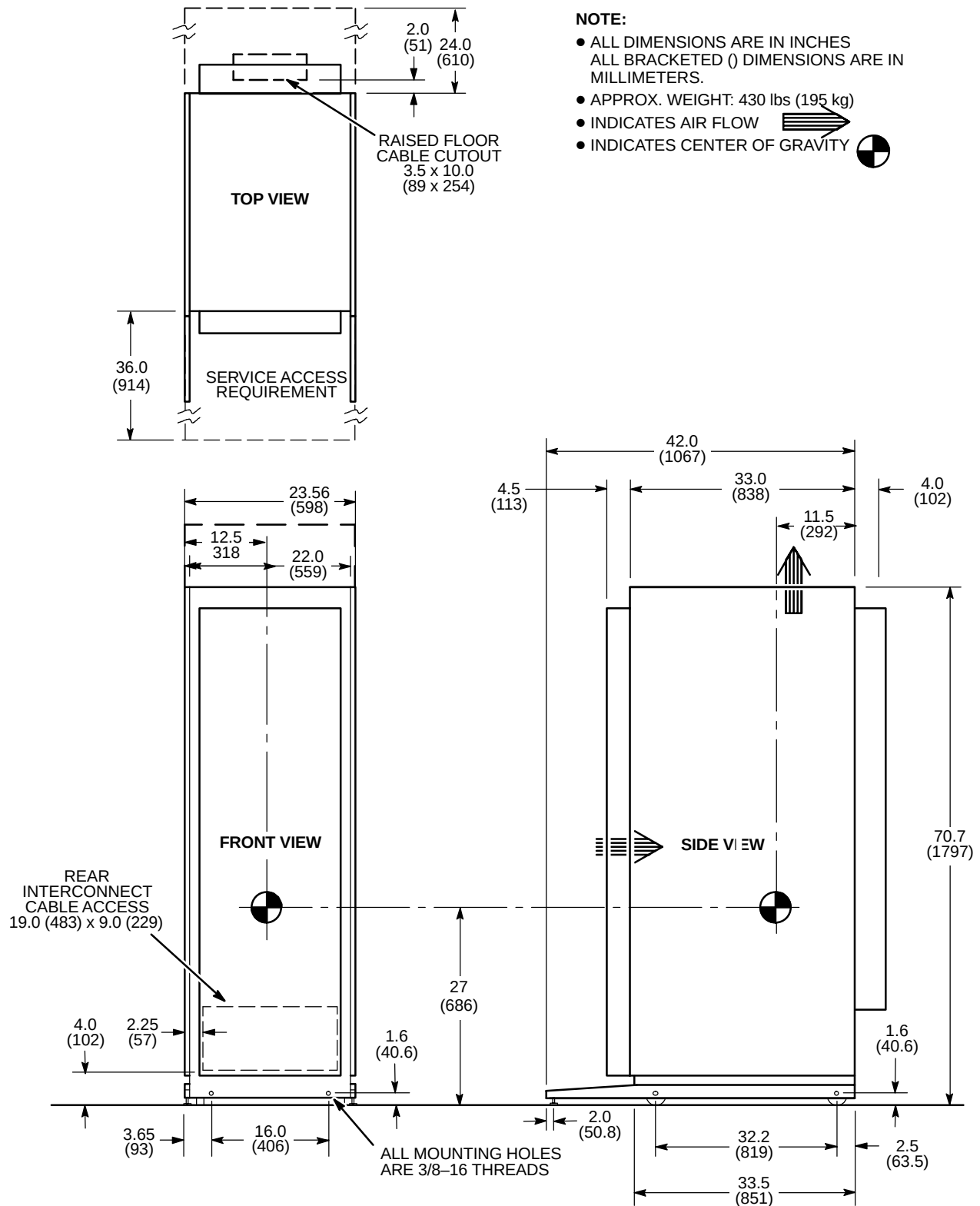


SIDE VIEW

OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION

ILLUSTRATION 2-42



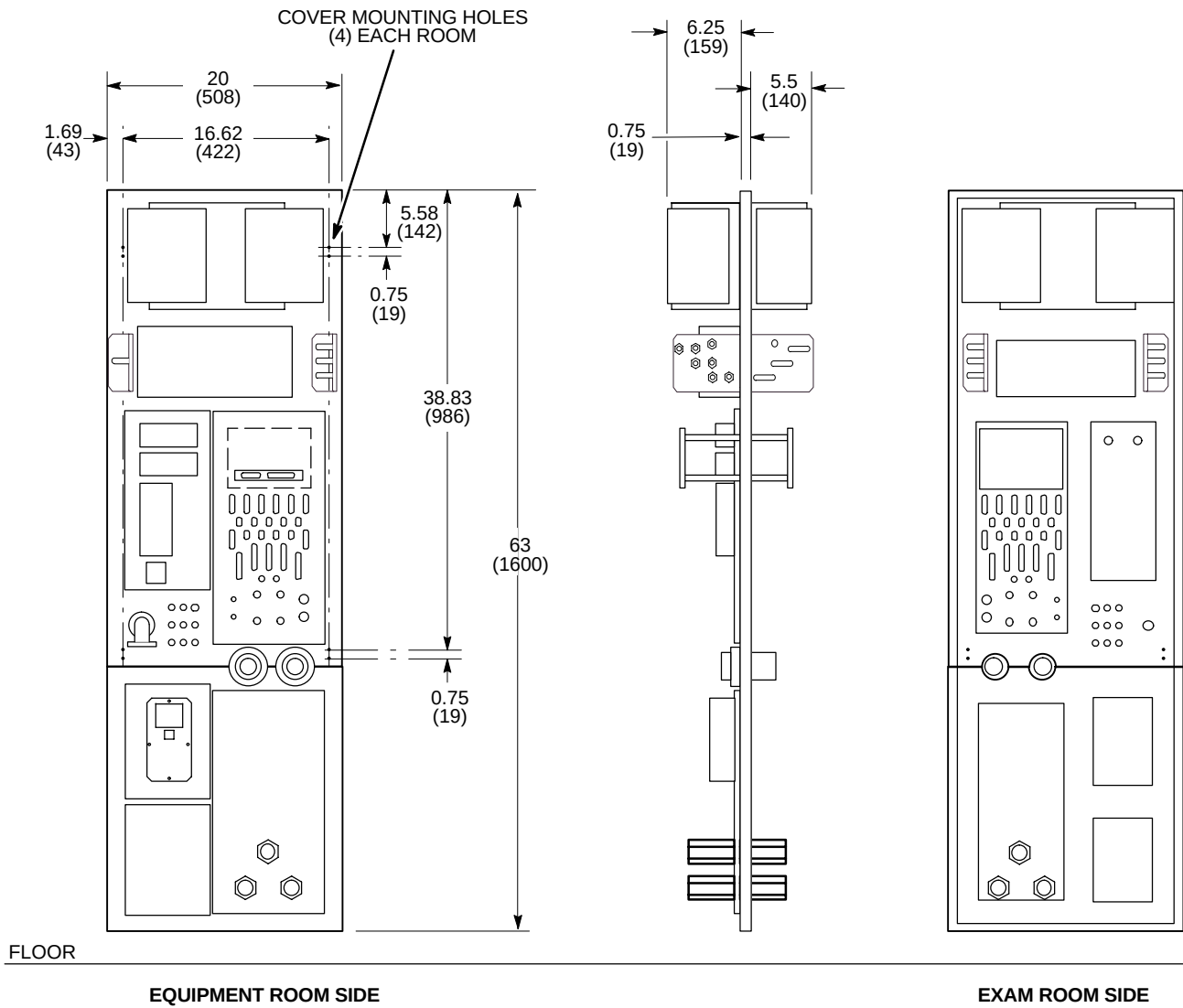


MNS CABINET (MNS) - SPECTROSCOPY OPTION

ILLUSTRATION 2-45

NOTE:

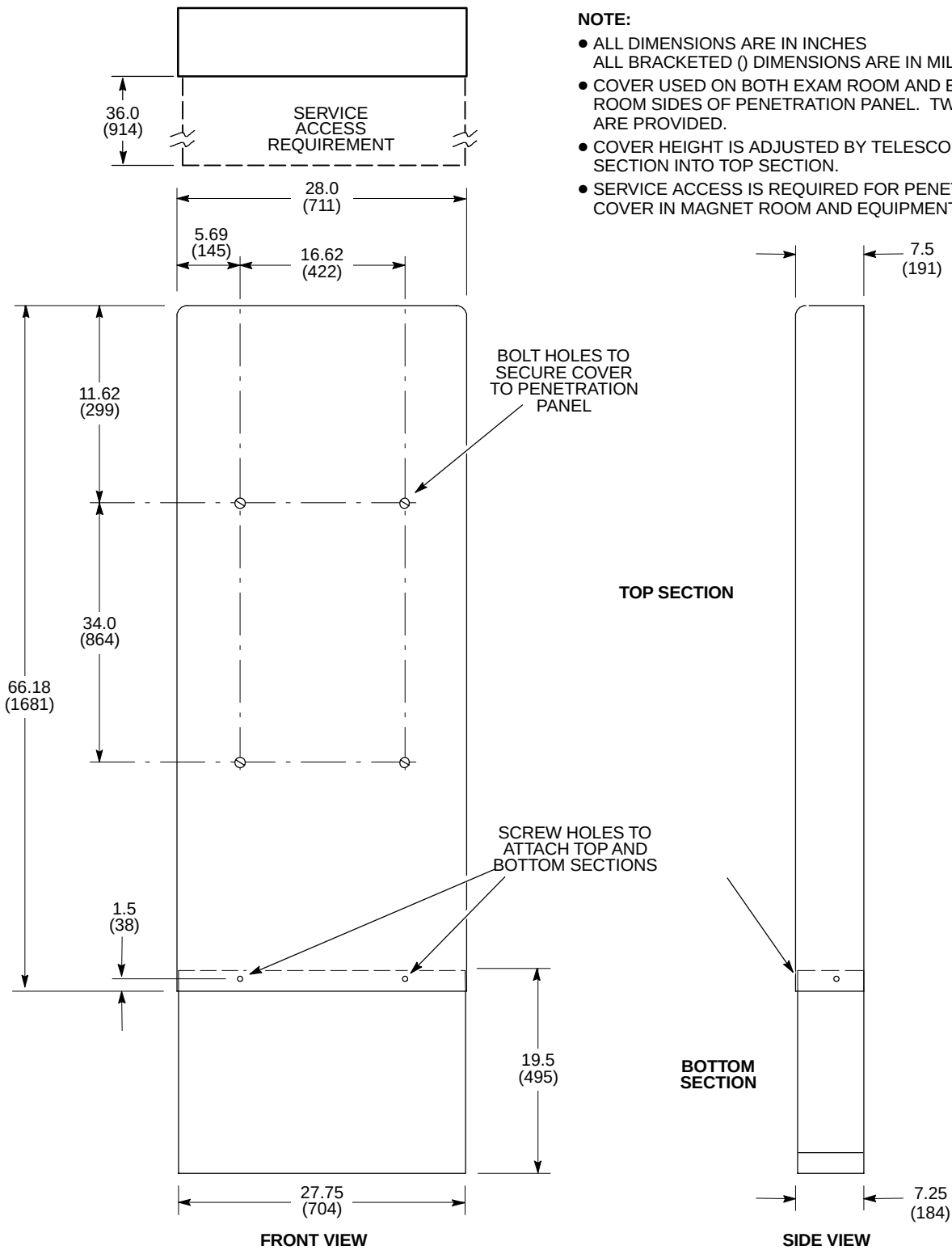
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)



M4494A

PENETRATION PANEL (PP1)

ILLUSTRATION 2-46

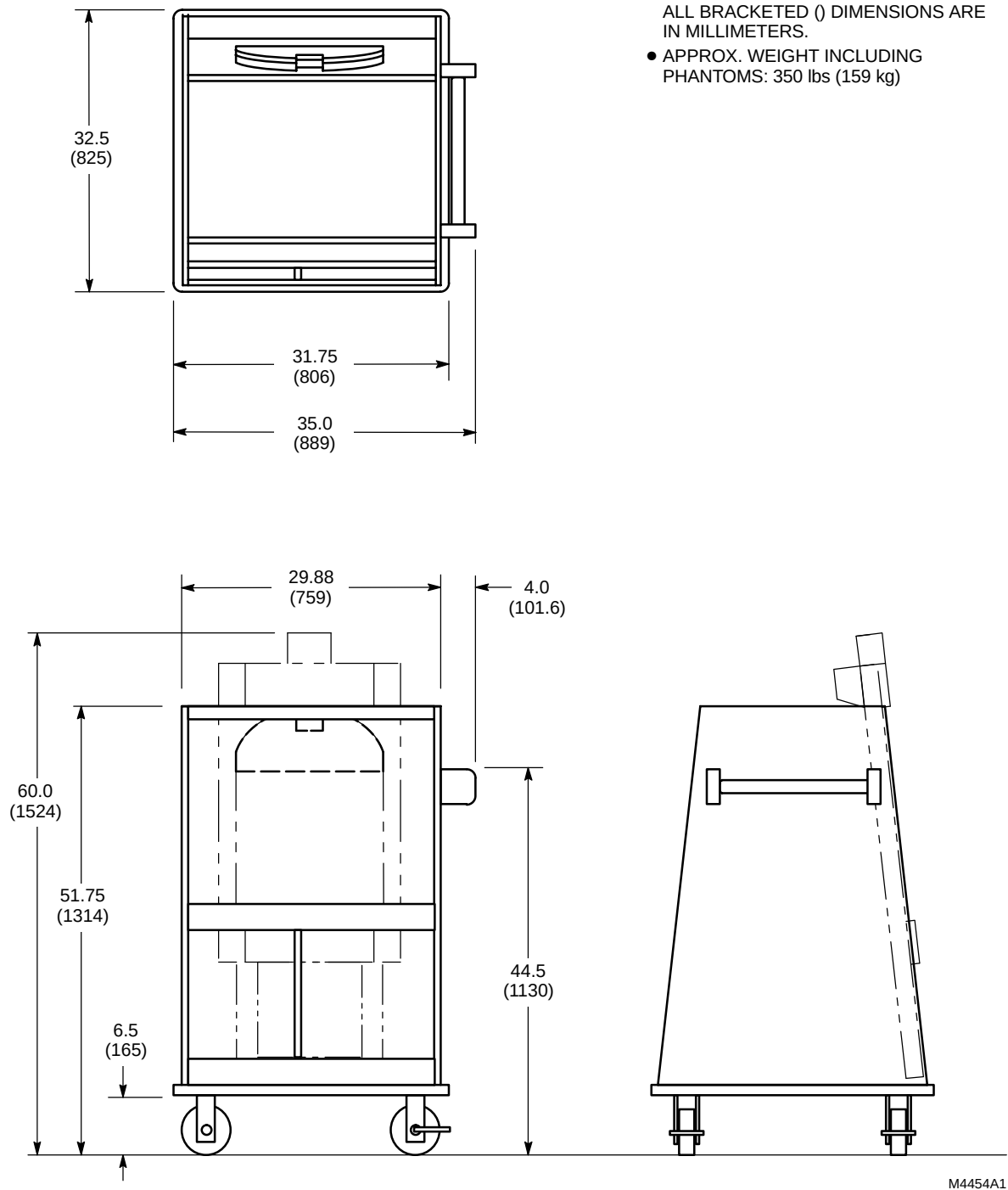


M4009A1M

PENETRATION PANEL COVER
ILLUSTRATION 2-47

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING
PHANTOMS: 350 lbs (159 kg)

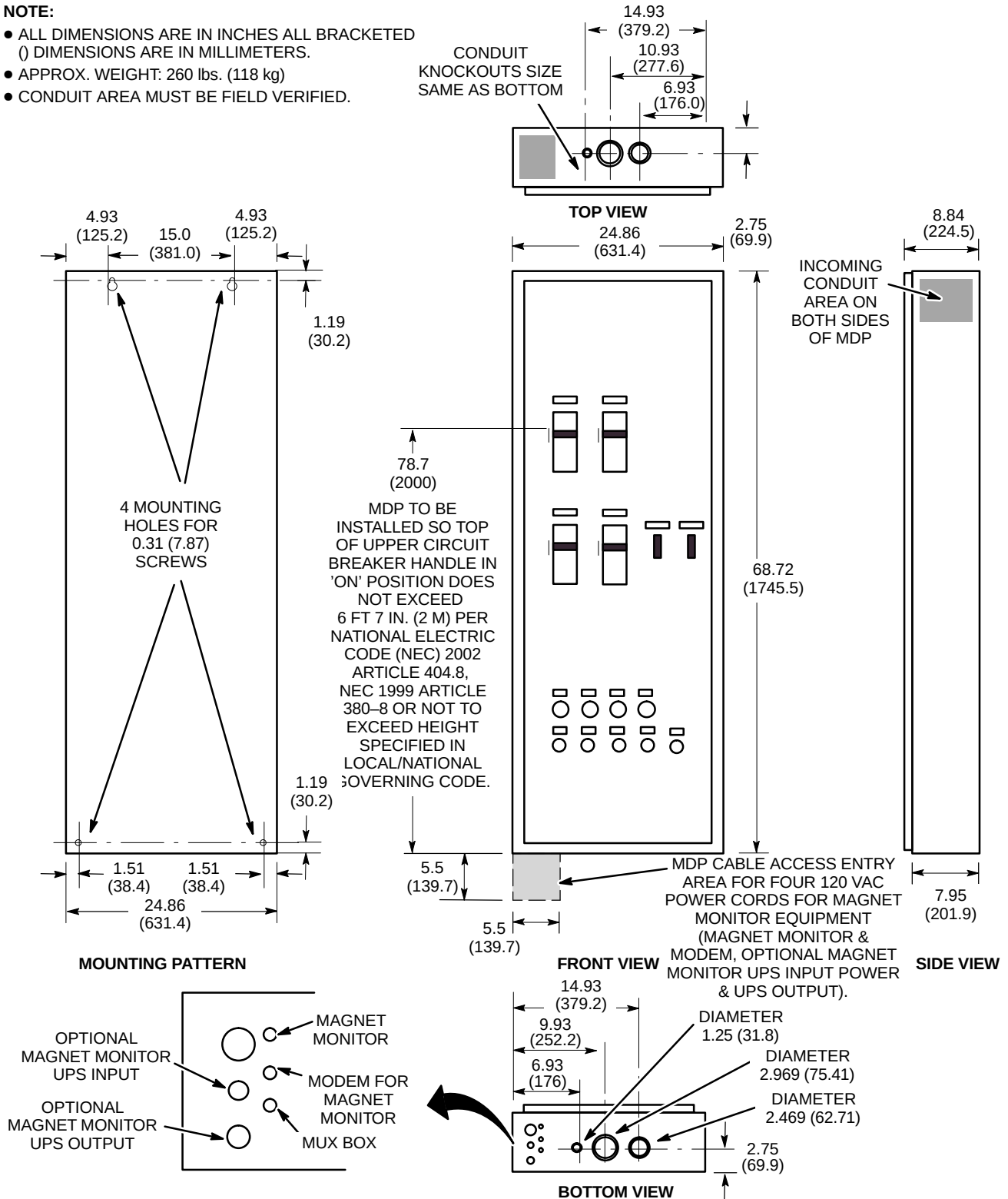


M4454A1

SPT PHANTOM SET SHIPPING/STORAGE CABINET
ILLUSTRATION 2-48

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 260 lbs. (118 kg)
- CONDUIT AREA MUST BE FIELD VERIFIED.



SIGNA TWINSPEED MAIN DISCONNECT PANEL (MDP)

ILLUSTRATION 2-49

SECTION 3 – MAGNETIC FIELD CONSIDERATIONS

Note

There may be changes to magnetic field considerations if an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade. Refer to *Direction 2294003 Signa TwinSpeed Pre-Installation* for 1.5T Magnet magnetic field information.

SECTION 4 – SITE ENVIRONMENT

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4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS

4-2-1 System Suite

Note

If specifications in this section are not strictly adhered to, failures of the ACGD modules in the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmentally controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8°F (21°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS FOR SYSTEM WITH ACGD/PDU CABINET

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4-2-2 Outdoor TSCC & TGWC Operating Environment

The Outdoor TSCC and Outdoor TGWC are designed to be located external to the building and operate in environments meeting the following specifications.

- Operating Ambient Temperature: –22°F (–30°C) to 122°F (50°C)
- Operating Humidity: 5–100%

4-3 AIR COOLING REQUIREMENTS

Table 4-2 contains the heat output of the upgrade equipment listed in the typical site location. These values do not include people, lights, magnet equipment cabinets, and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Physical placement of the air conditioning equipment (compressor, etc.) is an important factor due to the homogeneous field requirements of the magnet. Therefore, it is important this equipment be located outside the 10 gauss line. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for plot of gauss lines.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal, location of coil/bore cooling fans intake.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch access flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM (SEE NOTE 2)		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	8189	2400				
Patient Blower Box	1366	400				
Main Disconnect			900	264		
SRF Cabinet			8530	2500		
ACGD/PDU Cabinet			34,120	10,000		
System Cabinet			6140	1800		
Twin Accessory Cabinet			2354	690		
TSCC Configurations for Magnet & Body Coil water cooling:						
• Outdoor Air Cooled Remote Control Panel (See Note 3 & 5)			34	10		
• Water Cooled (See Note 3 & 6)			12,000	3517		
• Indoor Air Cooled (See Note 3)			120,000	35,140		
TGWC Configurations for only Body Coil water cooling (requires site cooling for Shield Cooler Compressor):						
• Outdoor Air Cooled Remote Control Panel (See Note 4 & 5)			34	10		
• Water Cooled TGWC (See Note 4 & 6) RCP for water cooled TGWC			10,240 34	3000 10		
• Indoor Air Cooled (See Note 4)			61,000	17,880		
Shield/Cryo Cooler Compressor			Heat dissipation to air negligible			
Magnet Monitor			205	60		
Operator Workspace (See Note 7) with LCD Color Monitor					4095	1200
Magnet Monitor UPS* & Modem			450	132		
*MNS Cabinet			2730	800		
*Advantage Workstation Option					See Note 8	

Note * Optional equipment.

† Maximum heat output is defined for temperature and humidity as defined in Table 4-1.

1 Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan.

2 FOR EQUIPMENT ROOM ONLY: The air cooling load averaged over a working day (~12 hours) is typically 1/2 of the maximum value.

3 The TSCC provides water cooling for the Shield/Cryo Cooler Compressor Cabinet (MS5) and the Gradient Coil.

4 The TGWC provides water cooling for the Gradient Coil only. **When a TGWC is selected then the customer/site must still provide water cooling for the Shield/Cryo Cooler Compressor Cabinet.**

5 The Outdoor Air Cooled TSCC and Outdoor Air Cooled TGWC are designed to be installed external to building and exhaust heat to outside air. A Remote Control Box is located in the Equipment Room.

6 The water cooled TSCC or TGWC require customer provided chilled water cooling in addition to the air cooling, refer to Section 4-4 Water Cooling Requirements for details.

7 Operator Workspace equipment includes the following: LCD Color Monitor, Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table

8 Refer to Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation.

4-4 WATER COOLING REQUIREMENTS**4-4-1 Gradient Coil Temporary Backup Water Cooling Recommendation**

There are no options available to support temporary backup water cooling for the TRM Gradient Coil. Water cooling must be provided by the cooling cabinet (TSCC or TGWC) supplied with the system to prevent contamination/damage to the coil and for proper image quality.

4-4-2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet (MS5). The backup cooling design can utilize open loop city water only as temporary backup during loss of the closed loop water cooling from the TSCC or customer provided water cooling to the Shield/Cryo Cooler Compressor. Long term open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching that can take place which will eventually contribute to failure. Water system capacity must be selected to make sure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Cooler Compressor Cabinet, refer to Tables 4-3 and 4-4 and Illustration 4-1 for water cooling specifications.



Switching the Shield/Cryo Cooler Compressor inlet/outlet cooling from the TSCC to a temporary water backup supply will result in approximately 1.5 gallons (5.5 liters) of 50% mixture of Dowfrost HD and de-ionized water being discharged. This discharge may have site impacts due to local regulatory codes. Make sure to understand and follow local regulatory requirements when designing and implementing a temporary backup water system.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

Note

Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

TABLE 4-3
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.

4-4-2 Shield/Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation (Continued)

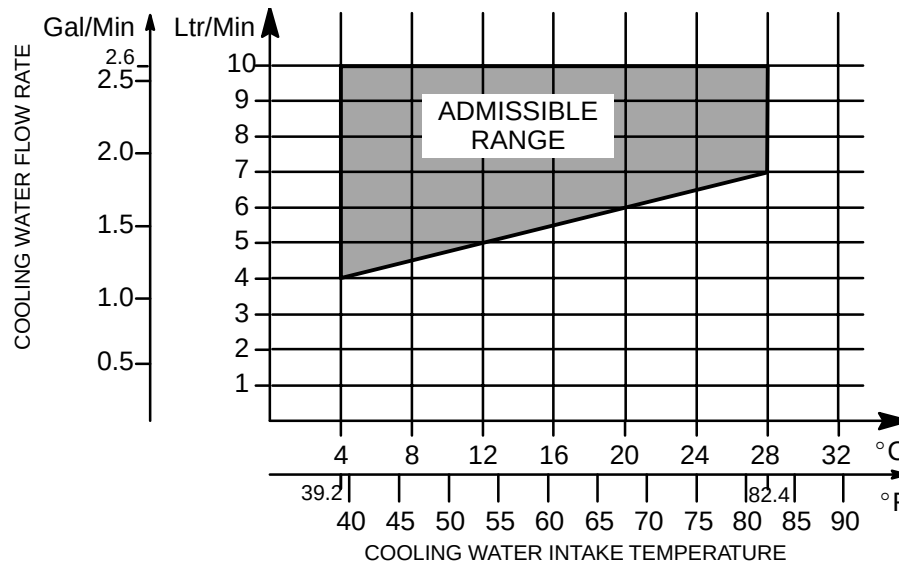
TABLE 4-4
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Note 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28)	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2 & 3	7.5 (0.5) [52] at minimum flow rate See Note 5	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2 & 3	47 (3.2) [324] at maximum flow rate See Note 5	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The backup cooling system design must have allowances for pressure/temperature changes due to distance located from the Shield/Cryo Cooler Compressor.

- 1 Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- 2 Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- 3 Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- 4 Water flow rate and temperature rise value are based on water.
- 5 Pressure drop values based on new system, may rise due to calcification.
- 6 Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.



SHIELD/CRYO COOLER COMPRESSOR REQUIREMENTS FOR TEMPORARY BACKUP WATER COOLING
ILLUSTRATION 4-1

4-4-3 Water Cooled TSCC Requirements

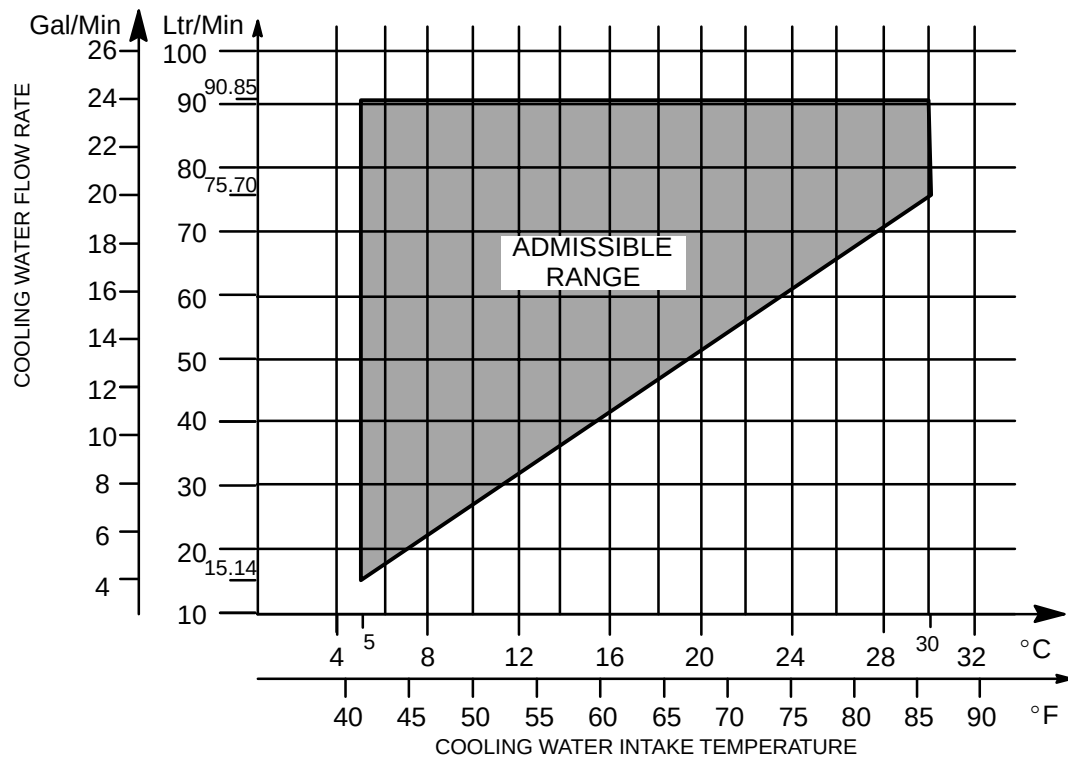
The water cooled configuration of the Twin System Cooling Cabinet (TSCC) contains a dedicated, closed loop, liquid-to-liquid water chiller system providing water cooling for the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. The water cooled TSCC configuration requires customer provided water cooling, refer to Table 4-5 and Illustration 4-2 for cooling water requirements. The water cooled TSCC has 1-1/2 inch (38 mm) NPT female connections at the lower back of the TSCC unit for customer water cooling connection. Refer to Illustration 2-13 for water supply and return connection location details. The water supply and return lines must be cleared of debris (solder, dirt, etc.) before connecting to the TSCC, refer to **APPENDIX C Prerequisites for Scheduled Start-up of Chillers For Signa TwinSpeed**.

Note

Continuous water cooling is critical for the TSCC and Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

TABLE 4-5
WATER COOLED TSCC COOLING WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TSCC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TSCC filter at cabinet inlet.
Anti-freeze	Propylene Glycol (50% maximum concentration in water)	
Inlet Pressure	150 psi maximum	
Pressure Drop	1.6 psi (11 kPa) at minimum flowrate of 4 gpm (15.14 L/min) 5.6 psi (38.5 kPa) at maximum flowrate of 20 gpm (75.70 L/min)	
Maximum Heat Output	100,000 BTU/Hr 29,307 Watts to water 12,000 BTU/Hr 3517 Watts to air	Refer also to Section 4-3 AIR COOLING REQUIREMENTS



TSCC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT

ILLUSTRATION 4-2

4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC**Note**

Shield/Cryo Cooler Compressor Water Cooling information is provided here for the convenience of customers performing an LCC Magnet upgrade with the Signa 1.5T TwinSpeed upgrade.

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

The Shield/Cryo Cooler Compressor customer provided water cooling system must be closed loop design. The water cooling design can utilize open loop city water, with required filtering, only as temporary backup during loss of closed loop water cooling system. Open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching to take place which will eventually contribute to failure. Water system capacity must be selected to insure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Compressor.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

The Shield/Cryo Cooler Compressor closed loop system may be shared with other equipment in the MR suite. The number of sharing systems should be kept to as low as possible in order to minimize contamination and reliability problems. Flow gauges and valves are recommended at all branch lines to control distribution and allow servicing of equipment.

Refer to Tables 4-3 and 4-4 for water cooling specifications.

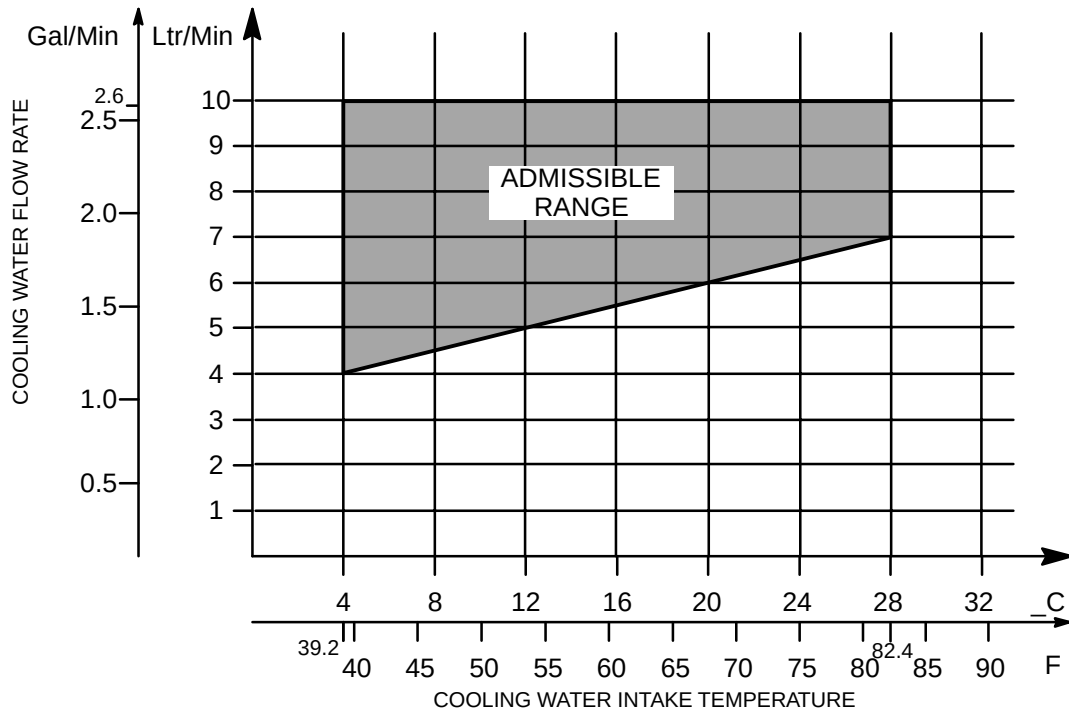
Note

The **Shield/Cryo Cooler Compressor** requires customer supplied flexible hose for mating with water cooling. Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4-6
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.
Anti-freeze	Preferred minimum 25% by Volume Maximum 50% by volume	Use stabilized product to reduce corrosion and organic growth. Preferred minimum value to minimize organic growth.



SHIELD/CRYO COOLER COMPRESSOR COOLING WATER REQUIREMENT
ILLUSTRATION 4-3

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4-7
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Notes 3, 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28) See Note 9	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2, 3 & 9	7.5 (0.5) [52] at minimum flow rate See Notes 2, 5, & 9	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2, 3 & 9	47 (3.2) [324] at maximum flow rate See Notes 2, 5, & 9	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The cooling system design must have allowances for pressure/temperature changes due to distance from the Shield/Cryo Cooler Compressor.

- Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- Water Flowmeter Kit (46-294052G1) is available to check/monitor flow rate for Shield/Cryo Cooler Compressor. Add 2 psi to total system pressure drop if flowmeter is permanently installed in system.
- Recommend a flowmeter be permanently installed in system, include flowmeter drop in total system pressure drop.
- Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- Water flow rate and temperature rise value are based on water. Laboratory grade Ethylene Glycol or Propylene Glycol anti-freeze may be used (do not mix Ethylene Glycol with Propylene Glycol). Preferred concentration is 75% water and 25% Glycol to minimize organic growth. Concentration of 50/50 is acceptable with a derate of 0.8 in specific heat calculations and a 20% increase in flow.
- Pressure drop values based on new system, may rise due to calcification.
- Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.
- Water Cooling Circuit typical values:
 – water inlet flow 1.8 to 2.1 gal/minute (7 to 8 liter/minute)
 – water inlet temperature 53.6 to 59°F (12 to 15°C)
 There is a risk of damaging the Shield/Cryo Cooler Compressor with water inlet low temperature and low flow range.

4-4-5 Water Cooled TGWC Requirements

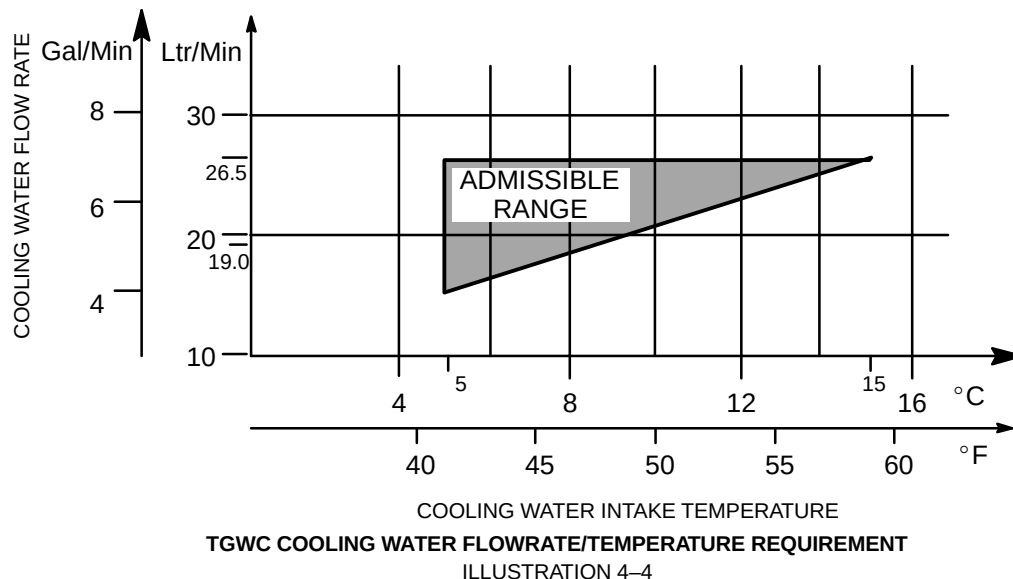
The Water Cooled configuration of the Twin Gradient Water Cooler (TGWC) contains a dedicated, closed loop, liquid-to-liquid heat exchanger system providing water cooling for the Gradient Coil ONLY. The Water Cooled TGWC configuration requires customer provided water cooling, refer to Table 4-8 for cooling water requirements. The water cooled TGWC has 0.5 inch (12.7 mm) female thread connections at the lower left side of the unit for customer water cooling connections. Refer to Illustration 2-20 for water supply and return connection location details.

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor. Refer to **Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC** for customer provided water cooling requirements.

TABLE 4-8
WATER COOLED TGWC WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
Composition	Water	Maximum of 50% glycol in water mixture
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TGWC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TGWC filter at cabinet inlet.
Maximum Heat Output	37,000 BTU/Hr 10,843 Watts	Refer also to Section 4-3 AIR COOLING REQUIREMENTS
Pressure Drop	5.9 psi at minimum flowrate of 2.8 gpm 20.3 psi at maximum flowrate of 5.2 gpm	
Inlet Pressure	150 psi maximum	
Cooling Water Flowrate & Temperature	5 gpm at 41 – 59 °F (18.9 liter/min at 5 – 15 °C)	



4-5 ALTITUDE

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-5 ALTITUDE*** for additional information and specifications.

4-6 LIGHTING

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-6 LIGHTING*** for additional information and specifications.

4-7 ACOUSTICS

Note

Design improvements in the MR system (RF/Gradient Body Coil, Magnet, RF and Gradient subsystems) balances the impact of the higher performance scan protocols such that there may be slight increases in the acoustics output. The following information is supplied for design of new installations but can also be used to assess the expected performance of the upgrade MR system.

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the system operator manual.

4-7-1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustics issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 5% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustics issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

4-7-2 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, GE requires that patients wear appropriate hearing protection for all scans because levels could exceed $L_{eq} \geq 99\text{dBA}$. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR system operator manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA
- Outdoor TSCC (Twin System Cooling Cabinet) 75 dBA
- Outdoor TGWC (Twin Gradient Water Cooler) 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet iso-center) as follows:

Average SPL 115 dBA SPL= Sound Pressure Level
Peak 120 dB
Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso-center) somewhat lower as follows:

Average SPL 85 to 95 dBA
Peak 105 to 112 dB
Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4-7-3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4-7-2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter-Spacial Areas

Acoustic Noise Control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

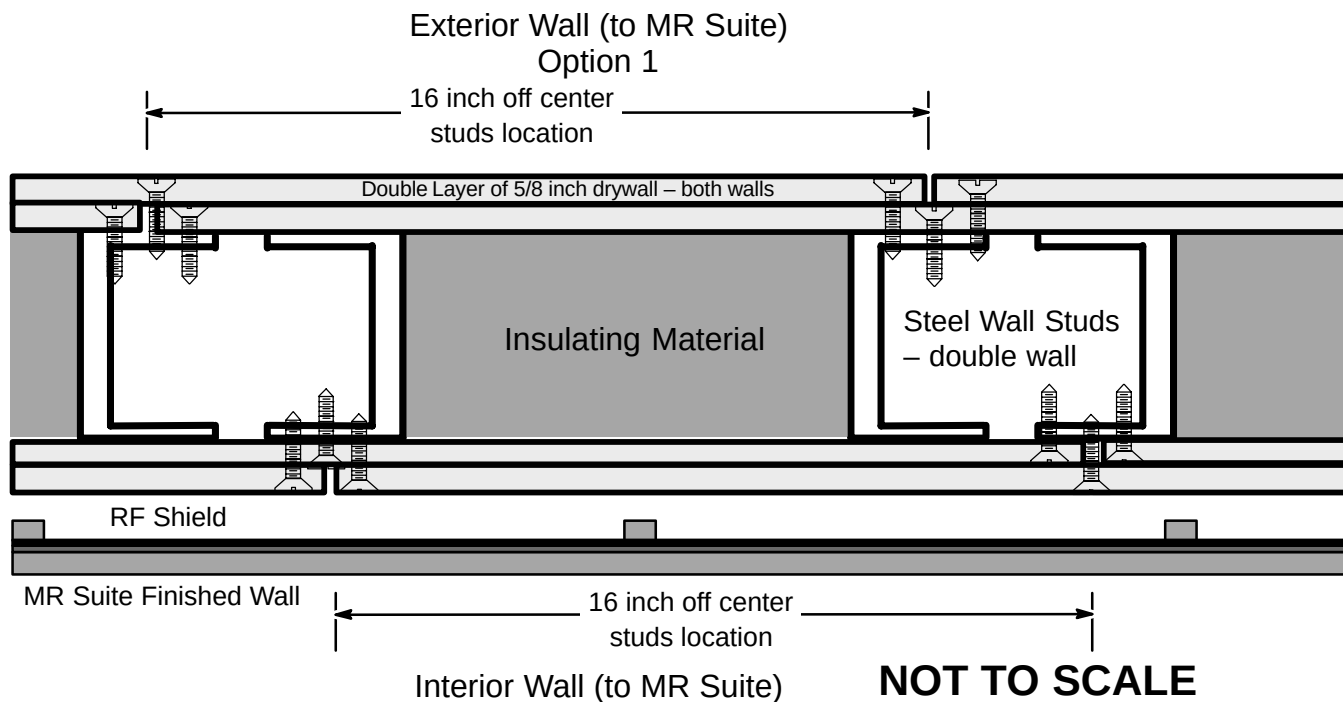
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

Wall Construction

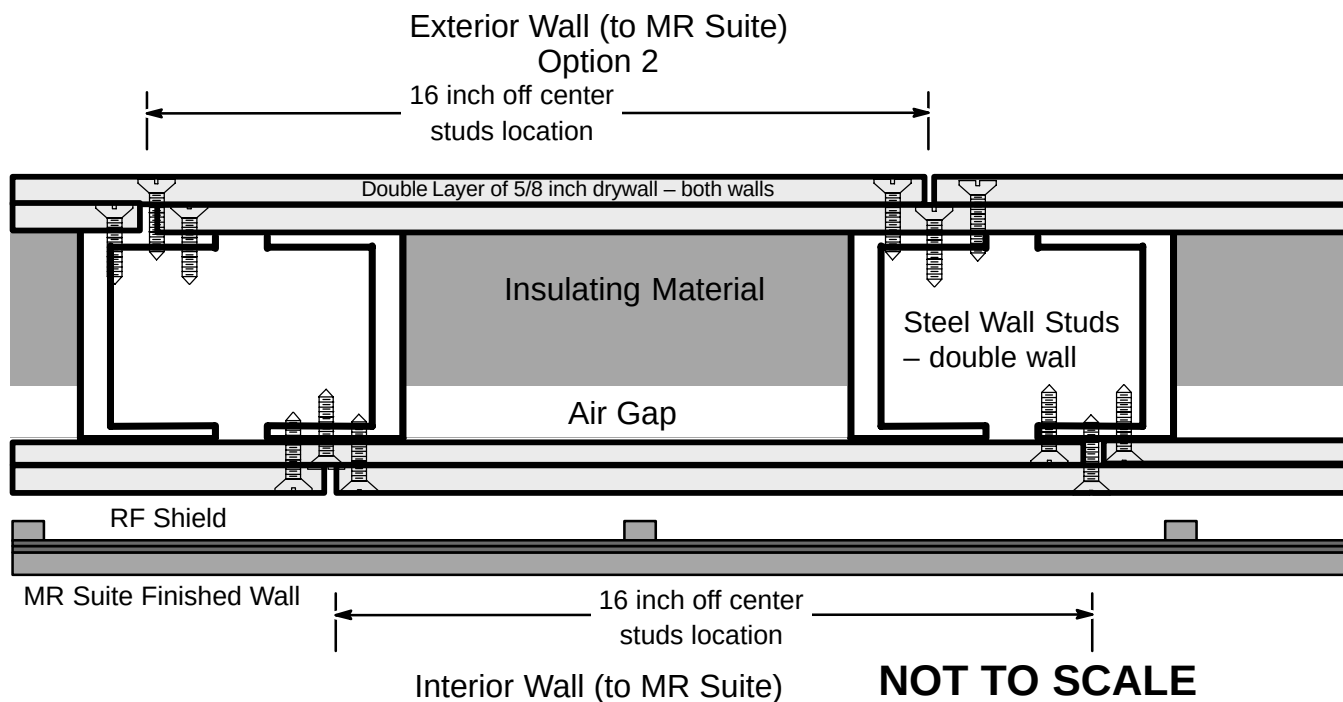
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4-5 and 4-6.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

4-7-3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 1
ILLUSTRATION 4-5



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 2
ILLUSTRATION 4-6

4-7-3 Design Guidelines (Continued)

High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 55 or 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide and STC 45 or 50 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

4-8 ROOM VENTILATION

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-8 ROOM VENTILATION*** for requirements and recommendations.

4-9 CRYOGENIC VENTING



WARNING!

IF AN UPGRADE TO AN LCC MAGNET IS BEING PERFORMED WITH THE SIGNA TWINSPEED SYSTEM UPGRADE THEN REFER TO ***DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION SECTION 4-9 CRYOGENIC VENTING FOR REQUIREMENTS AND RECOMMENDATIONS FOR LCC MAGNET CRYOGENIC VENTING.***

4-10 ALARM DEVICES, WATER SENSORS AND THERMOSTATS

4-10-1 System Cabinet

For Signa TwinSpeed with EXCITE Technology the System Cabinet has 104°F (40°C) and 122°F (50°C) temperature sensors. When the temperature in the System Cabinet reaches 104°F (40°C) a Warning Message will be displayed on the Operator Workspace. When the temperature in the System Cabinet reaches 122°F (50°C) or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the System Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

For Signa TwinSpeed the System Cabinet has one temperature sensor and one control module which sounds an alarm located in the System Cabinet when temperature reaches 94°F (34.4°C) and a Warning Message will be displayed on the Operator Workspace. After 3 minutes of alarm condition PDU will revert to full off condition. Any external alarm device other than mentioned above must be supplied by the customer.

4-11 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS.***

4-12 CONSTRUCTION MATERIALS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-14 CONSTRUCTION MATERIALS*** for requirements and recommendations.

4-13 VIBRATION

4-13-1 Definition

Improvements in MR imaging technology has increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning for Signa Horizon upgrades.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The stability tests (HSS) currently used by local GE Field Service to evaluate the existing system performance and image quality can be used to assess the vibration environment. Sites which currently pass their system stability criteria (HSS) may proceed with the upgrade. Sites which have marginal stability performance require vibration measurements to be performed to identify vibration source(s).** Then it is the customer's responsibility to contract a vibration consultant to perform vibration measurements as defined in Appendix A, MR SITE VIBRATION TEST GUIDELINES. With the vibration consultant, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design a site solution.

**WARNING!**

REFER TO *DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION* SECTION 4-15 VIBRATION SPECIFICATIONS FOR SIGNA 1.5T TWINSPEED SYSTEMS.

SECTION 5 – POWER REQUIREMENTS

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5-1 INTRODUCTION

Signa 1.5T TwinSpeed requires the Power Distribution module (PD1) located in the lower portion of the ACGD/PDU Cabinet. Prior PDU configurations are not compatible with the Signa 1.5T TwinSpeed.

For systems with ACGD/PDU Cabinet installed the existing PDU will be used. For systems without ACGD/PDU Cabinet the ACGD Upgrade will replace the existing system PDU. The ACGD upgrade includes the new Power Distribution module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3) which distributes power to most MR system components and provide the proper voltage configuration for the ACGD modules. Refer to Table 5-1 for the required customer power specifics. Refer to Section 5-2 CRITICAL POWER REQUIREMENTS for specifications of required facility input for the Signa 1.5T TwinSpeed system.



WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Customers should carefully consider the advantages and disadvantages of raised flooring, access flooring, conduits, floor ducts, and surface raceways for running cables in accordance with local and national codes. If used, conduits should be large enough to pass any cable and its connector through with all other cables in the conduit.

To reduce voltage regulation problems and wiring costs, minimize the cable length between the primary power source and the Power Distribution Unit. When routing cables, keep all phase conductors and ground for a circuit in the same trough. Whenever possible, keep power cables away from signal and data cables. Use separate trough or dividers in duct.

5-1 INTRODUCTION (Continued)

TABLE 5-1
REQUIRED CUSTOMER POWER

MR COMPONENT	VOLTAGE (VAC)	FREQUENCY	PHASE	MAX. AMPS	COMMENTS
GE Main Disconnect Panel (MDP) See Note1, 2 & 3 See Note 4 if customer MDP	480Y/277 VAC ±10% or 400Y/230 VAC ±10%	60 Hz 50 Hz	(3+GND) See Com- ments	See Note 5	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the Power Cabinet. (See Note 6) Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Magnet Rundown Unit	100–120 or 200–240	50/60 Hz	1	1.0	Hard wired in unit.
Pneumatic Patient Alert Control Box	115 or 230	50/60 Hz	1	0.2	An outlet on the Operator Workspace Table equipment can be used.
Service Receptacle in Magnet Room	110–120 See Comments	50/60 Hz	1	2.0	Receptacle required for small power tools. Local voltage and portable transformers for voltages values.
*O ₂ Monitor	110–120 or 200–240	50/60 Hz	1	3.0	Hard wired in monitor
*MRI Music System	120	60 Hz	1	1.2	Receptacle required in Control Room
*Advantage Windows	Refer to Direction 2261309–100 Advantage Workstation Ultra60 Pre–Installation.				
Note * Optional equipment. 1 Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable. or cord per National Electric Code (NEC) 2002 Articles 250.134, 300.3, 517.13 or 1999 Articles 250–134, 300–3, 517–13. 2 Signa TwinSpeed MDP controls power to the following system equipment: – Power Distribution Unit – System Cooling Cabinet – Shield/Cryo Cooler Compressor – Magnet Monitor equipment including the Magnet Monitor, Modem, Uninterruptible Power Supply (UPS*) (optional) for Magnet Monitor, Multiplexer Box (optional). 3 MDP power circuits for Twin System Cooling Cabinet and Shield/Cryo Cooler Compressor Cabinet, and cooling for these units, are required immediately upon magnet arrival. If permanent site power is not ready, temporary power drop line and cooling must be made available. If site voltage is not any of the voltages listed above, customer must provide transformer and secondary circuit breaker to provide correct voltage and/or configuration. Refer to Section 1–5 FACILITY OPTIONS for listing of step up transformers options. 4 If customer provided MDP has been selected based on meeting SECTION 1 – SYSTEM CONFIGURATION upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements. 5 Maximum amps dependent on voltage selected. Refer to Section 5–2, CRITICAL POWER REQUIREMENTS , for configuration. 6 PDU Module is located in the lower portion of the ACGD/PDU Cabinet (MR3).					

5-2 CRITICAL POWER REQUIREMENTS

The system Main Disconnect Panel is a Low Voltage Low Energy local control unit, with multi-point remote control capability, connected to the feeder lines that supply input power to the following:

- Power Distribution Unit (PD1)
- Shield/Cryo Cooler Compressor Cabinet
- System cooling equipment: Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC)
- Magnet Monitor equipment.

The Main Disconnect Panel is provided as part of the system per **SECTION 1 – SYSTEM CONFIGURATION** flowcharts for upgrades system cooling equipment configurations options and relationship to other system equipment/options selections.



WARNING!

IF CUSTOMER PROVIDED MDP HAS BEEN SELECTED BASED ON MEETING THE REQUISITES IN SECTION 1 – SYSTEM CONFIGURATION UPGRADE FLOWCHARTS THEN CUSTOMER PROVIDED MDP MUST MEET ALL MDP REQUIREMENTS.

All work is to be done in accordance with national and local electrical codes. Refer to Section 5-3, POWER DISTRIBUTION SYSTEM, for Main Disconnect Panel capability and set up.



WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

5-2 CRITICAL POWER REQUIREMENTS (Continued)**Configuration:**

- Recommend input configuration 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet.
- Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.

Frequency: 50 ± 3 Hz or 60 ± 3 Hz

Regulation: 4% maximum at system maximum 5 seconds power demand from source to PDU (i.e. includes all feeders and transformers to utility).

Phase Balance: Difference between the highest phase line-to-line voltage and the lowest phase line-to-line voltage must not exceed 2%.

Daily Voltage Variation: $\pm 10\%$ from nominal under worst case line and load regulation.

PDU Voltage: 200, 400, 480 VAC $\pm 10\%$

Shield/Cryo Cooler Compressor Voltage: 380/400/415 VAC 50 Hz or 460/480 VAC 60Hz

TSCC or TGWC (excludes Water Cooled TGWC powered from PDU) Voltage: 400 VAC $\pm 10\%$ 50 Hz or 480 VAC $\pm 10\%$ 60Hz

Magnet Monitor equipment Voltage: 100/120 or 200/240 VAC 50/60 Hz

Voltage Transients: Phase-to-phase voltages must be within 2% of the lowest phase-to-phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.

Facility Zero Voltage Reference Ground:

- Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger.
- Main facility ground wire to be insulated.
- Ground impedance to earth at power source to be 2 ohms or less.
- Main facility ground wire to be bonded at every distribution box in an approved grounding block.

5-2 CRITICAL POWER REQUIREMENTS (Continued)

Maximum Momentary Demand: The power demands for the Signa 1.5T TwinSpeed system with ACGD are specified as a function of the duration of the power demand. Table 5-2 lists points on the curve.

The power system feeding the Signa system with ACGD must be designed to meet the specifications of less than 4% regulation when loaded at 92.4 KVA (corresponds to 5.0 second allowable consumption). For short intervals the Signa system with ACGD power demands can exceed 56.2 KVA and the line voltage delivered to the system will sag below the 4% regulation. The Signa system with ACGD is designed to tolerate these **short** voltage sags.

TABLE 5-2
SYSTEM PEAK POWER DEMAND

COOLING EQUIPMENT SYSTEM EQUIPMENT	TSCC <i>See Note 1</i>			TGWC <i>See Note 2</i>		
	OUTDOOR TSCC	INDOOR WATER COOLED TSCC	INDOOR AIR COOLED TSCC	OUTDOOR TGWC	INDOOR WATER COOLED TGWC	INDOOR AIR COOLED TGWC
PDU draw for 5.0 sec	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA
PDU draw for 1.0 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.1 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.04 sec <i>See Note 3</i>	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
Magnet Monitor <i>See Note 4</i>	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	23 KVA	20 KVA	22.7 KVA	16 KVA	0 KVA <i>See Note 5</i>	11.6 KVA
TOTAL for 5.0 sec	~92.7 KVA	~89.7 KVA	~92.4 KVA	~85.7 KVA	~69.7 KVA	~81.3 KVA
TOTAL for 1.0 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.1 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.04 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
Note 1 TSCC provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil. 2 TGWC provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values. 3 The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA. 4 The Magnet Monitor equipment power is 1.5 KVA 1 phase on an unbalanced leg of 3 phase input (4.5 KVA 3 phase equivalent). 5 The Indoor Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value.						

5-2 CRITICAL POWER REQUIREMENTS (Continued)**Average (while scanning) Power Demand:** Refer to Table 5-3

TABLE 5-3
SYSTEM AVERAGE (CONTINUOUS) SCANNING POWER DEMAND

COOLING EQUIPMENT SYSTEM EQUIPMENT	TSCC See Note 1			TGWC See Note 2		
	OUTDOOR TSCC	INDOOR WATER COOLED TSCC	INDOOR AIR COOLED TSCC	OUTDOOR TGWC	INDOOR WATER COOLED TGWC	INDOOR AIR COOLED TGWC
PDU draw See Note 3	43.1 KVA	43.1 KVA	43.1 KVA	43.1 KVA	44.5 KVA	43.1 KVA
Magnet Monitor	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	18.3 KVA	14 KVA	13.0 KVA	15 KVA	0 KVA See Note 4	8.8 KVA
TOTAL See Note 5	74.9 KVA	70.6 KVA	69.6 KVA	71.6 KVA	58.0 KVA	65.4 KVA
Note 1 TSCC Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil. 2 TGWC Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values. 3 The PDU is rated for 45 KVA continuous power. 4 The Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value. 5 GE Main Disconnect Panel (MDP) is rated continuous power draw is 77 KVA but MDP continuous draw does not exceed list demands.						

Standby (no scan) Power Demand: 26.9 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU, 9 KVA for system cooling equipment, 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet, and 1.5 KVA 1 phase for Magnet Monitor equipment (4.5 KVA 3 phase equivalent).

5-3 POWER DISTRIBUTION SYSTEM

5-3-1 Main Disconnect Panel (MDP)

WARNING!

If customer provided MDP has been selected based on meeting **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements.

The Signa TwinSpeed system includes a Main Disconnect Panel (based on **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts) with multi-point remote control capability which is shown in Illustration 5-1. The Main Disconnect Panel consists of the following:

- A three-pole circuit breaker rated for the current of the PDU circuit. The short-circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three-pole circuit breaker rated for the current of the System Cooling Cabinet circuit. The short-circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three-pole circuit breaker rated for the current of the Shield/Cryo Cooler Compressor Cabinet circuit. The short-circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A circuit to provide 120VAC single phase power to the Magnet Monitor, Modem, UPS for Magnet Monitor (optional), and Multiplexer Box (optional). The short-circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate available fault current. The MDP includes a single phase step down transformer for 120VAC loads such as Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box).
- The MDP Panel has receptacles inside the panel enclosure for connections of the UPS for Magnet Monitor input and output, Multiplexer Box, Magnet Monitor, and modem. The enclosure has provision for these cables to enter through the access panels in the bottom left side of the enclosure. Mounting of the panel must allow for 5-6 inch (127-152 mm) of free space to allow for cable bending and installation. Strain relief bushings are provided with the individual equipment for each of these cables, not provided with the MDP.

Main Disconnect Panel (MDP) is to be located so the top of the upper circuit breaker handle does not exceed 79 inches (2000 mm) from the floor when in the ON position and is visible to the customer and service personnel from the Power Distribution Unit (PD1), System Cooling Cabinet (TSCC or TGWC), Shield/Cryo Cooler Compressor Cabinet. The height of the MDP location is per National Electric Code (NEC) 2002 Article 404.8 or NEC 1999 Article 380-8 or the height specified in local/national governing code. The optional UPS for the Magnet Monitor may be located below the MDP if sufficient space is available or adjacent if sufficient space is not available, refer to Section 2-6-5 Magnet Monitor.

5-3-1 Main Disconnect Panel (MDP) (Continued)**Note**

The MDP circuits for the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and the single phase transformer for Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box) have auto restart feature where normal power is returned after a time delay of 3 to 30 seconds (field adjustable) to reduce operating costs and minimize cryogen consumption of the system thus decreasing system downtime. The MDP Emergency Off circuit turns off power to all branch circuits including the Magnet Monitor UPS option output and turns off the auto restart function.

The PDU circuit has low voltage release feature which disconnects power from the PDU upon the first loss of power. Power to the PDU is not restored automatically after a power interruption. Emergency Off operation disconnects power from all circuits including the PDU.

For the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and Magnet Monitor equipment the auto restart function is disabled by the Emergency Off pushbuttons. Extended power outages of 2 days or more will result in the auto restart function loss of battery backup and will require pressing of the Main Power ON push button manual restart after restoration of normal power.

The circuit breakers or fuses ahead of the MDP must be capable of handling the magnetizing inrush currents of the System Cooling Cabinet, Shield/Cryo Cooler Compressor, Magnet Monitor equipment, and transformer of the PDU module (PD1) in the ACGD/PDU Cabinet (MR3). If fuses are used time delayed fuses are recommended.

Check local and national codes to determine if an interlock to the air-conditioning unit in the Computer/Equipment Room is required in the protective disconnect set-up.

The MDP provides two Emergency Off buttons to be connected to the MDP to disable the power to all system equipment in emergency situations. The Emergency Off buttons need to be mounted near each exit 60 in. (1524 mm) from the floor, near the exit in the Magnet Room and Equipment Room. The Emergency Off buttons are clearly labeled "Emergency Off" and visible to personnel. It is important that the buttons are labeled "OFF" and not "STOP" since there exists an "Emergency Stop" button in the Signa system which powers down only a portion of system equipment for patient safety.

Note

The emergency off circuit disconnects power to the PDU, Twin System Cooling Cabinet, Shield/Cryo Cooler Compressor Cabinet, and from the single phase transformer for Magnet Monitor equipment. Power can be restored to the MDP outputs by pressing the MAIN POWER ON pushbutton on the MDP for the TSCC, Shield/Cryo Cooler Compressor Cabinet, Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box). Power to the PDU is restored by pressing the PDU POWER ON pushbutton and also requires pressing the EMO Reset button on the PDU.

5-3-1 Main Disconnect Panel (MDP) (Continued)

The Main Disconnect Panel (MDP) is lockable to meet OSHA requirements for power Lockout/Tagout requirements. The MDP provides for the disconnection of the facility power to the PDU, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet. Individual branch circuits for the PDU, Magnet Monitor equipment, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet consisting of lockable GE Spectra circuit breakers. The MDP also has electrical contacts for an interlock to the air-conditioning units in the Equipment Room. Check local and national codes to determine if an interlock to the air-conditioning unit in the Computer/Equipment Room is required in the protective disconnect set-up.

The GE available Main Disconnect Panel is UL labeled in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110-2. If the GE MDP is not utilized then the customer supplied MDP must be listed and labeled by a Nationally Recognized Testing Lab (NRTL) such as Underwriters Laboratory (UL) in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110-2.

Note

The maximum conductor the MDP can accept is #3/0 AWG (83 mm²). For feeders larger than 3/0 AWG (83 mm²) the wires must be reduced (ie. splice, junction box, etc.) to 3/0 AWG (83 mm²) within 10 feet (3 meters) of MDP. It is important to note the maximum cable wire from the MDP to the PDU must not be larger than 2/0 AWG (70 mm²).

5-3-1 Main Disconnect Panel (MDP) (Continued)

- NOTE:**
- RUNS 296 AND 297, & POWER CORDS FOR SCC & MAGNET MONITOR EQUIPMENT (MAGNET MONITOR, UPS INPUT & OUTPUT, MODEM, OPTIONAL MULTIPLEXER) ARE GE SUPPLIED CABLES. **ALL OTHER WIRING IS CUSTOMER SUPPLIED.**
 - TWO REMOTE EMERGENCY "OFF" BUTTONS ARE SUPPLIED WITH SYSTEM MDP.
 - CIRCUIT BREAKERS ARE PROVIDED FOR PDU, TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, MAGNET MONITOR EQUIPMENT CIRCUITS.
 - ALL BRANCH CIRCUITS DROP OUT ON LOSS OF POWER. TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, & MAGNET MONITOR EQUIPMENT AUTOMATICALLY RESTART AFTER 3 SEC TIME DELAY UPON RESTORATION OF POWER. EMERGENCY OFF LOCKS OUT ALL CONTACTORS.
 - IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT USED THEN NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE POWER CABINET
 - SUPERVISORY CIRCUIT FOR HVAC INTERLOCK CONTACTS OPEN ON LOSS OF DC POWER OR EMERGENCY OFF OPERATION.

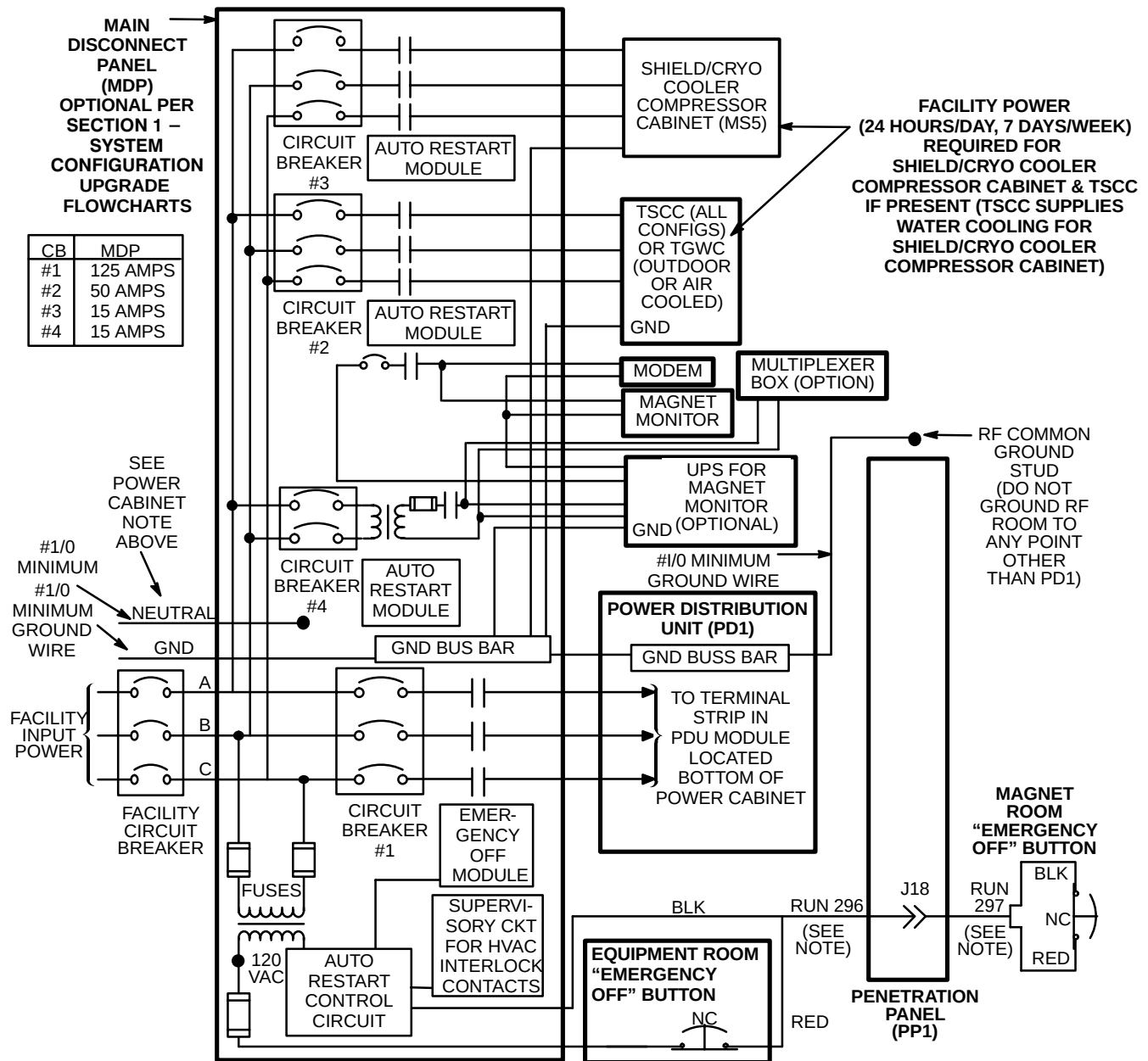


ILLUSTRATION 5-1

5-3-2 System Power Distribution Unit

The PDU Module in the lower portion of ACGD/PDU Cabinet has an integrated filter for a level of power conditioning. The largest allowable phase conductor the PDU will accept is 2/0 AWG (83 mm²). Larger feeder wires can be connected to the MDP with 2/0 AWG (83 mm²) between the MDP and PDU.

Note

The ground conductor shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.

Note

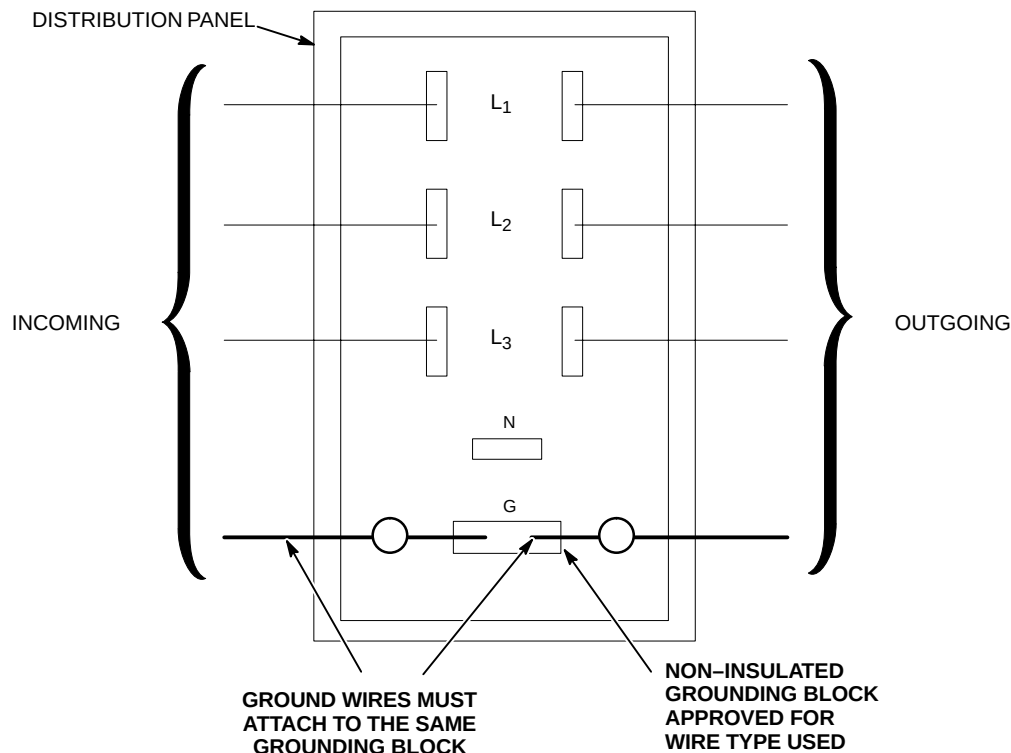
Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of ACGD/PDU Cabinet (MR3).

5-4 GROUNDING

5-4-1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into a facility, and be continuous to the MR system power disconnect in the room. This ground can be spliced with "High Compression Fittings" and should be terminated at each distribution panel it passes through. When it is broken for a connection to a panel, it shall be connected into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the block. See Illustration 5-2.

The connection at the power source shall be at the grounding point of the "Neutral – Ground" if a "Wye" transformer is used, or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.



M4301A

ILLUSTRATION 5-2

Ground Wire

The ground wire shall be copper wire with a minimum of AWG 1/0 or the same size as the power feeders whichever is larger. This means that if there is a primary feeder to a distribution panel of 500 MCM with a secondary feeder to the MR system of AWG 2 wire, the ground to the distribution panel shall be 500 MCM with an AWG 1/0 to the MR system. The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 – 1982 which can be accomplished using 3-point Fall Of Potential (3 point measurement) method or Clamp-On Ground Resistance measurement which requires a ground measurement device such as AEMC 3730.

5-4-2 System Ground

The MR system is designed with minimum ground loops to prevent noise currents and natural disturbances from flowing through the low-level signal reference path.

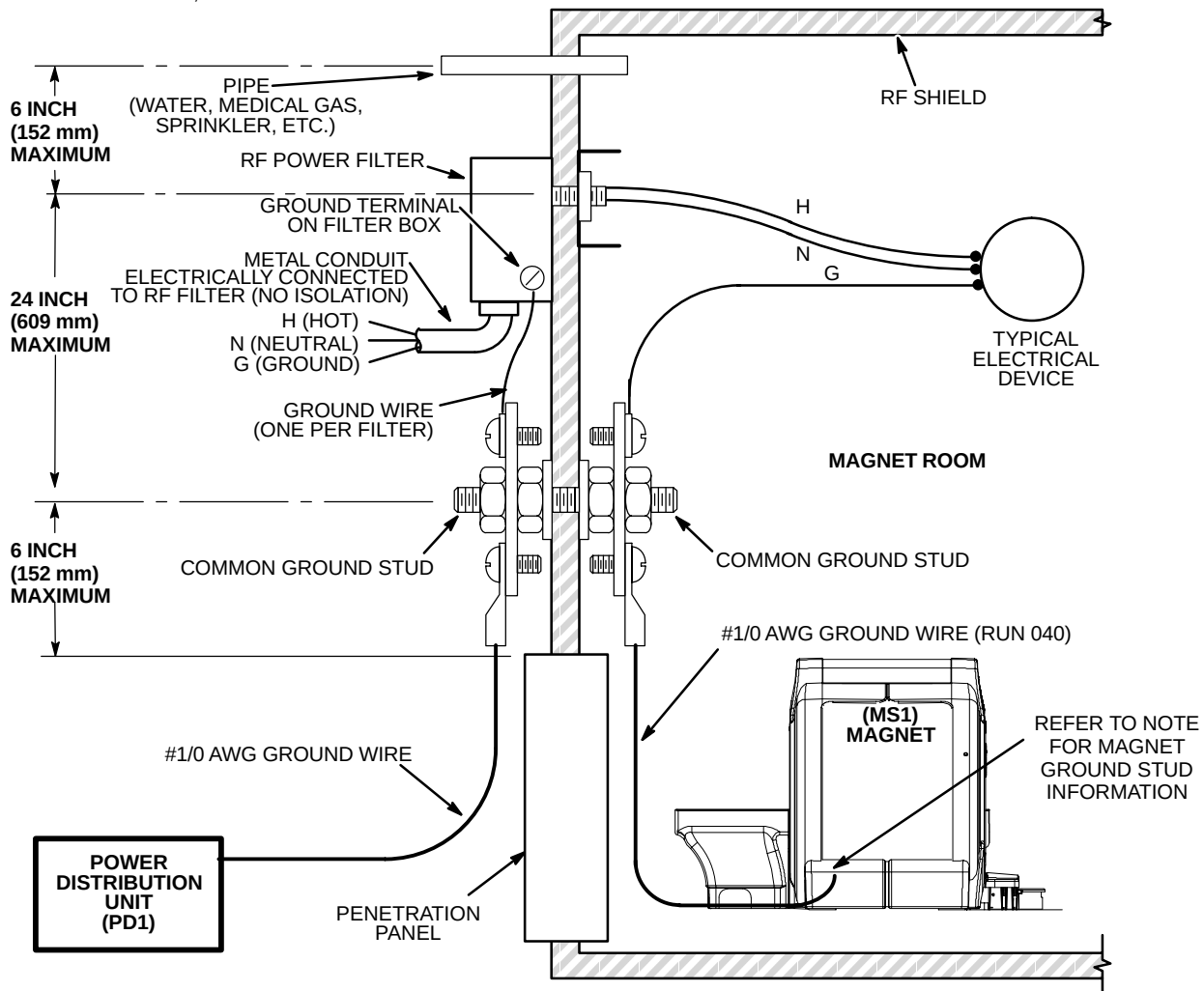
The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF shielded room common ground point. **This RF shielded room common ground point is to be located within 6 in. (152 mm) of the GE supplied Penetration Panel.** Refer to Section 7, RF SHIELDED ROOM, for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in Illustration 5-3 must be met when running power lines into the Magnet Room.

Any modifications or non-MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

5-4-2 System Ground (Continued)

- NOTE:**
- ALL ITEMS SHOWN ARE CUSTOMER SUPPLIED EXCEPT POWER DISTRIBUTION UNIT, MAGNET, AND TWO #1/0 AWG GROUND WIRES BETWEEN MAGNET GROUND STUD AND RF COMMON GROUND POINT.
 - **RESISTANCE BETWEEN ANY TWO GROUNDED DEVICES MUST NOT EXCEED 0.1 OHM TO ENSURE EQUAL POTENTIAL GROUND SYSTEM WITHIN MAGNET ROOM.**
 - **LOCATE FILTERS WITHIN 2 FEET (600 mm) OF RF COMMON GROUND STUD WHICH MUST BE LOCATED WITHIN 6 INCHES (152 mm) OF PENETRATION PANEL.**
 - ALL EXTERNAL CONDUIT MUST BE **METAL AND ELECTRICALLY CONNECTED** TO THE RF POWER FILTERS (IE. NO ISOLATION) UNLESS THE FILTERS ARE LOW VOLTAGE (<30 VOLTS).
 - RF POWER FILTERS OF 30 VOLTS OR LESS MAY BE LOCATED ANYWHERE ON THE RF SHIELD **PROVIDED THE INCOMING CONDUIT IS NON-METALLIC OR IS DIELECTRICALLY ISOLATED AND WITH NO GROUND WIRE. IF THE INCOMING CONDUIT IS METALLIC AND/OR HAVE A GROUND WIRE, THESE FILTERS MUST ALSO BE LOCATED WITHIN 24 INCHES (609 mm) OF THE RF COMMON GROUND STUD.**
 - ALL CONDUITS IN THE RF ROOM MUST BE **METAL**. STEEL IS ACCEPTABLE PROVIDED IT IS ADEQUATELY ANCHORED.
 - ALL ELECTRICAL DEVICES (IE. OUTLETS, LIGHT FIXTURES, ETC.) MUST HAVE A GROUND WIRE FROM ITS POWER SOURCE AND BE GROUNDED TO RF ROOM SHIELD AT THE RF COMMON GROUND STUD AS SHOWN BELOW.
 - ALL METALLIC PIPES ENTERING THE RF ROOM, EXCLUDING CRYOGENIC VENT AND FLOOR DRAINS, MUST BE LOCATED WITHIN 30 INCHES (762 mm) OF THE RF COMMON GROUND.
 - LCC MAGNET HAS 2 GROUND STUDS, ONE ON FRONT FOOT AND ONE ON REAR FOOT (COLDHEAD SIDE OF MAGNET). HOWEVER, THERE IS ONLY ONE #1/0 AWG GROUND WIRE TO BE CONNECTED TO ONLY ONE OF THE GROUND STUDS.



MR MAGNET ROOM GROUNDING REQUIREMENTS AND TYPICAL DIAGRAM

ILLUSTRATION 5-3

5-5 GROUND FAULT PROTECTION

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2002 Article 517.2 definitions or NEC1999 Article 517-3 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC 2002 Article 517.17 or NEC 1999 Article 517-17. NEC 2002 Article 517.17(A) or NEC 1999 Article 517-17(a) states "Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load."

NEC 2002 Article 230.95 or 215.10 or NEC 1999 Article 230-95 or 215-10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2002 Article 517.17(B) or NEC 1999 Article 517-17(b).

Check national and local electrical codes.

SECTION 6 – INTERCONNECT DATA

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6-1 INTRODUCTION

Section 6, INTERCONNECT DATA, addresses system interconnections which are impacted by the Signa 1.5T TwinSpeed upgrade catalogs. The subsections are broken down as follows:

- 6-1 INTRODUCTION
 - general upgrade information, component designations
- 6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES
 - cable interconnects for the system provided by Horizon LX with ACGD Upgrade catalogs
- 6-3 ADVANTAGE WORKSTATION INTERCONNECTS
 - Refer to Direction 2261309-100 *Advantage Workstation Ultra60 Pre-Installation* for AW interconnect information.
- 6-4 INTERCONNECTS RE-ROUTED BY ACGD UPGRADES
 - existing system cable interconnects re-labeled and re-routed during ACGD Upgrades if not present prior to Signa 1.5T TwinSpeed upgrade.
- 6-5 INTERCONNECTS REMOVED BY SIGNA 1.5T TWIN SPEED WITH EXCITE TECHNOLOGY UPGRADES
 - existing system cable interconnects removed during Signa 1.5T TwinSpeed With EXCITE Technology upgrade.
- 6-6 INTERCONNECTS REMOVED BY SIGNA 1.5T TWIN SPEED UPGRADES
 - existing system cable interconnects removed during Signa 1.5T TwinSpeed upgrade.
- 6-7 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
 - need to plan Rear Pedestal cable access and routing due to increase of cabling and hoses
- 6-8 SPECTROSCOPY OPTION INTERCONNECTS
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for Signa 1.5T TwinSpeed Upgrades.

6-1-1 Component Designators

The Component Designator System is used as the means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. (For example, the System Cabinet is referred to as MR2.) Refer to Table 6-1 for component designators impacted by Signa 1.5T TwinSpeed or TwinSpeed With EXCITE Technology upgrade.

TABLE 6-1
COMPONENT DESIGNATION

APPLICABLE SYSTEM	COMPONENT DESIGNATOR	DESCRIPTION
Signa 1.5T TwinSpeed System	EO1/EO2	Emergency Off Buttons
	MDP	Main Disconnect Panel
	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	SRF Cabinet
	MR2	System Cabinet for TwinSpeed or TwinSpeed With EXCITE Technology
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS1	Superconducting Magnet
	MS4	Magnet Rundown Unit
	MS5	Shield/Cryo Cooler Compressor Cabinet
	MSM1	Magnet Monitor
	MSM3	Modem for Magnet Monitor
	OW1	Operator Workspace
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PP1	Penetration Panel
	PT1	Patient Transport Table
	RCP	Remote Control Box for Outdoor TSCC, Outdoor TGWC, & Water Cooled TGWC
	TAC	Twin Accessory Cabinet
	TSCC	Twin System Cooling Cabinet (provides water cooling for Shield/Cryo Cooler Compressor & Gradient Coil)
	TGWC	Twin Gradient Water Cooler (provides water cooling for Gradient Coil ONLY; Customer supplied water cooling is required for Shield/Cryo Cooler Compressor Cabinet)
System Options	MNS	MNS Cabinet
	AW	Advantage Workstation
	MSM4	UPS for Magnet Monitor
	MUX	Phone Line Multiplexer for Magnet Monitor
Signa LX with ACGD or Signa Horizon LX Obsolete Equipment	HE1	Heat Exchanger
	MR1	RF/PDU Cabinet RF/Penetration Cabinet RF Cabinet
	MR3	SGD Base Gradient Cabinet for Base System or SGD Hi-Slew Gradient Cabinet for HiSpeed or EchoSpeed 8645 Gradient Cabinet Gradient Amp Cabinet
	PD1	Power Distribution Unit (PDU) is a module in lower portion of RF/PDU Cabinet Full Size (150 KVA or 75 KVA) or Compact Power Distribution Unit
	PP2	Penetration Cabinet
	WC1	Gradient Coil Water Chiller

6-1-2 Definition of Terms

The definition of terms used in following sections tables are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in interconnect diagrams
- Between Units (From/To): component designators as found in tables
- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE-supplied cable

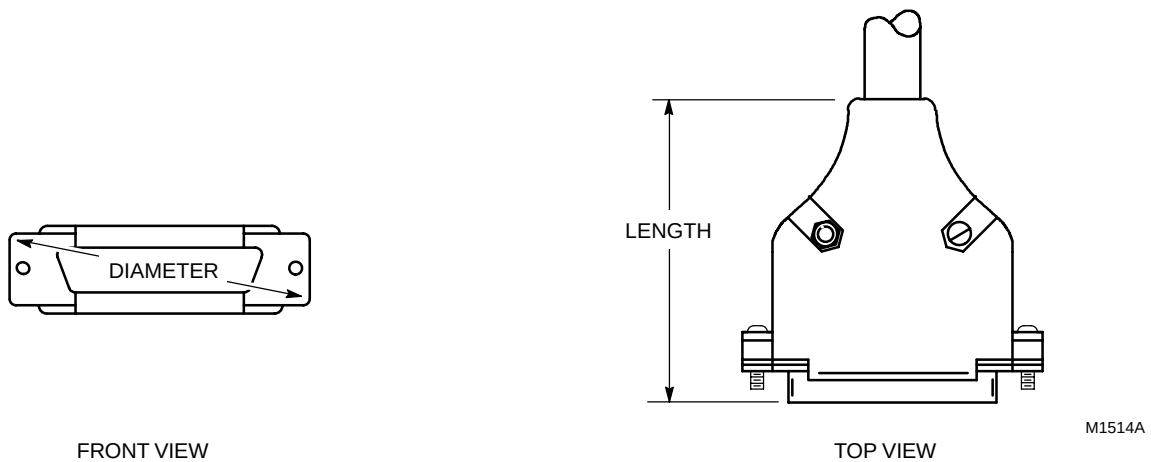
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 -x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



SUBMINIATURE-D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6-1

6-2 SYSTEM INTERCONNECTS ADDED BY ACGD UPGRADES

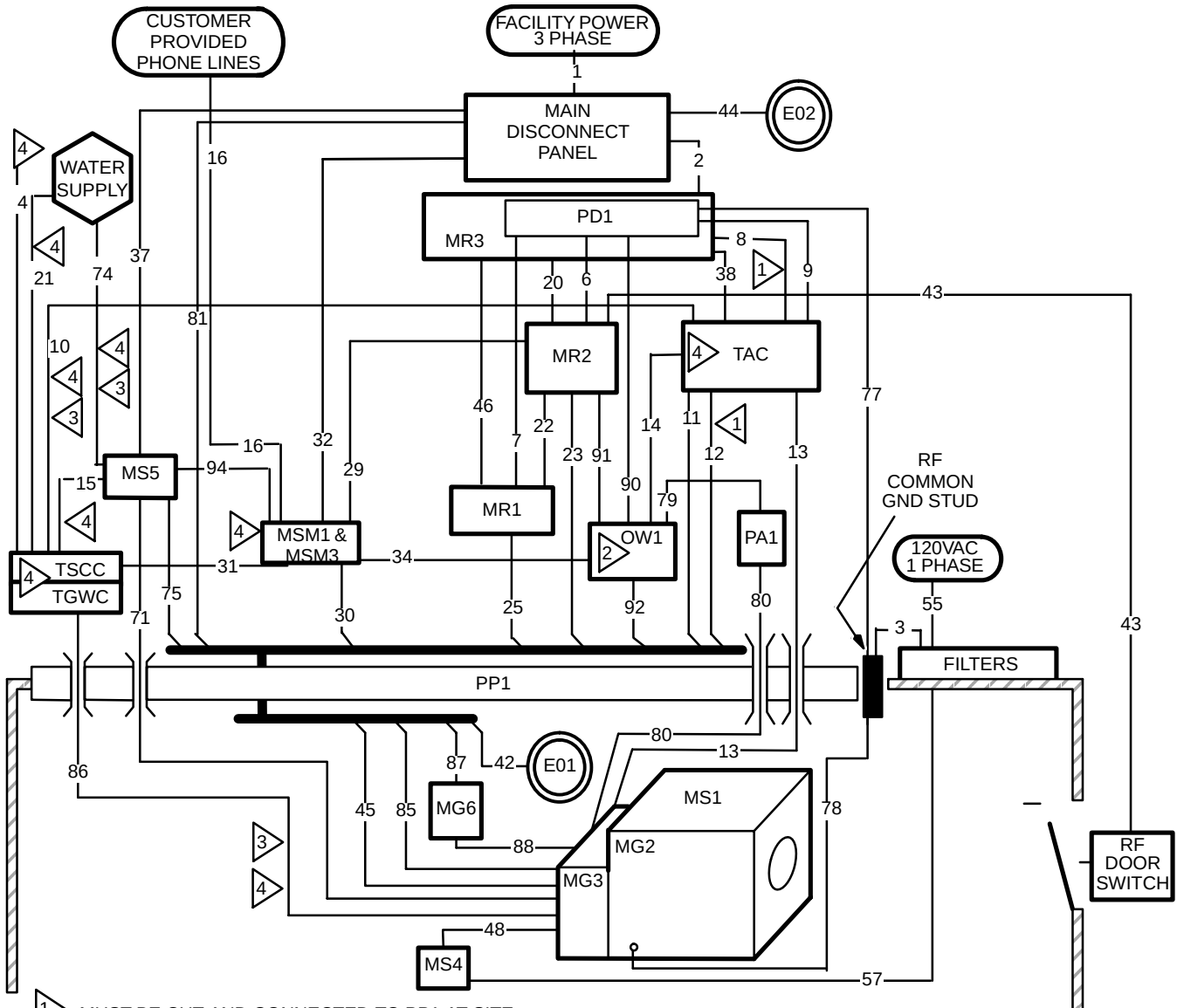
6-2-1 Introduction

Illustration 6-2 interconnect diagrams shows the groups to which cables are added by the various Signa 1.5T TwinSpeed or TwinSpeed With EXCITE Technology upgrade catalogs. The following sub-sections list the cables provided by the various upgrade catalogs. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in tables in this section are supplied by GE.

Note

Need to verify customer supplied Group1 (Facility Power to Main Disconnect Control) and Group 2 (Main Disconnect Control) are properly sized for specifications in Section 5, Power Requirements and in accordance with local and national codes.

6-2-1 Introduction (Continued)



- 1 MUST BE CUT AND CONNECTED TO PP1 AT SITE.
NOTE: IMPEDANCE IS NOT CRITICAL SO EXCESS CABLE SHOULD BE CUT OFF.
- 2 OPERATOR WORKSPACE (OW1) SUBSYSTEM EQUIPMENT IS PROVIDED WITH MAXIMUM LENGTH CABLES POSSIBLE. SEVERAL OW ASSEMBLIES ARE MOUNTED TO OW TABLE & OW INTERCONNECTS ARE ROUTED THROUGH TABLE CABLE TRAY. FOR REFERENCE USE ONLY THE OW1 RUN NUMBERS ARE 049, 792, 793, 794, 795, 796, 797, 798, 799, 806, & 807.
- 3 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- 4 FOR GROUPS 4, 10, 15, 21, 31, 49, 50, 51, 53, 74, & 86 REFER TO APPROPRIATE SECTION FOR TSCC OR TGWC INTERCONNECTS (SECTION 6-2-13 through 6-2-18) FOR ADDITIONAL AND CONFIGURATION APPROPRIATE INTERCONNECTS.

SIGNA 1.5T TWINSPEED SYSTEM UPGRADE INTERCONNECT DIAGRAM

ILLUSTRATION 6-2

6-2-2 M1033K – Indigo to Octane Computer Upgrade

M1033K provides interconnects for an Indigo to Octane Computer Upgrade listed in Table 6-2.

TABLE 6-2
CABLES PROVIDED BY M1033K

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 1	818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
Note 1 M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								

6-2-3 M1033L – Indigo to Octane II with EXCITE Upgrade

M1033L provides interconnects for an Indigo to Octane II with EXCITE Upgrade listed in Table 6-3.

TABLE 6-3
CABLES PROVIDED BY M1033L

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 1	1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
			1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
Note 1 M3044TC can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								

6-2-4 M1060JW – Shield/Cryo Cooler Compressor

M1060JW provides interconnects for Shield/Cryo Cooler Compressor for the LCC Magnet listed in Table 6-4.

TABLE 6-4
CABLES PROVIDED BY M1060JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
37	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
45	PP1	MG3	30 (9.15)	624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
71	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
75	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array

M3043TA provides standard length system interconnects for fixed site listed in Table 6-5. There are also requirements for customer supplied interconnects as indicated in Table 6-7 .

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
3	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
6	PD1 See Note 1	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
				972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
7	PD1 See Note 1	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	PD1 See Note 1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	RCP, TSCC, or TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine FROM connection.
11	TAC	PP1	21 (6.40)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	33 (10.0)	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
			33 (10.0)	980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988	0.26 (6.6)	Ring Terminal	Ring Terminal	
Ground wire								
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	14 (4.27)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	31 (9.45)	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
			See Note 2	233/ 499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
			488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)		
Note 2 M3044TA can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).								
(Continued)								

ARCHITECTURAL/INSTALLATION PLANNING USE ONLY

GE MEDICAL SYSTEMS

SIGNA 1.5T TWINSPEED UPGRADE PRE-INSTALLATION

REV 1

DIRECTION 2304880

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40) See Note 3	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable
			235	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Has a minimum bend radius of 10 in. (254 mm).	
			236	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Has a minimum bend radius of 5 in. (127 mm).	
			486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals		
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
Note 3 M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may need Special Order Inquiry (SOI) to provide longer length.								
(Continued)								

ARCHITECTURAL/INSTALLATION PLANNING USE ONLY

GE MEDICAL SYSTEMS

SIGNA 1.5T TWINSPEED UPGRADE PRE-INSTALLATION

REV 1

DIRECTION 2304880

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
43	MR2	RF Door Switch	67 (20.4) minus takeup at RF Door Switch See Note 4	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	MDP	EO2	—	—	—	—	—	Customer Furnished.
45	PP1	MG3	30 (9.15)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Has a minimum bend radius of 10 in. (254 mm).
				258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
Note 4 M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) minus takeup at RF Door Switch.								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (contin- ue)	PP1	MG3	30 (9.15)	727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
46	MR1	MR3	14 (4.27) See Note 5	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud See Note 6	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
79	OW1	PA1	5 (1.5) minus takeup at PA1	—	0.13 (3.2)	3.00x3.00 (76.2x76.2)	0.38x1.75 (9.6x44.5)	
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
81	MDP	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
Note 5 M3044TA can be ordered to increase usable length of Group 46 to 59 ft (18.0 m).								
6 M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) minus takeup at RF Common GND Stud.								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 See Note 7	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine FROM connection.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes & Note 8	784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 being mounted on floor and floor routing of cables.
88	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
90	PD1 See Note 1	OW1	51 (15.5) See Note 9	048	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				969	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1). 7 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 8 M3044TA can be ordered to increase usable length of Group 87 to 71 ft (21.6 m). 9 M3044TA can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 10	789	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8). Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
			791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes		
			836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)		
92	PP1	OW1	53 (16.1) See Note 11	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note 10 M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m). 11 M3044TA can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).								

6-2-6 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector

M3043TC provides standard length system interconnects for fixed site listed in Table 6-6. There are also requirements for customer supplied interconnects as indicated in Table 6-6.

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
3	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
6	PD1 See Note 1	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
7	PD1 See Note 1	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	PD1 See Note 1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	RCP, TSCC, or TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
11	TAC	PP1	21 (6.40)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	33 (10.0)	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
			33 (10.0)	980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988 X 3	0.26 X 3 (6.6 X 3)	Ring Terminal	Ring Terminal	A Ground wire for each of Runs 979, 980, & 981
13	TAC	MG2/3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

(Continued)

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	14 (4.27)	229	0.23 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 1033 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
				1033	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
23	MR2	PP1	31 (9.45) See Note 2	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
				233/ 499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	

Note 2 M3044TC can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).

(Continued)

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40) See Note 3	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable Has a minimum bend radius of 10 in. (254 mm). Has a minimum bend radius of 5 in. (127 mm).
			235	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)		
			236	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)		
			726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals		
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
Note 3 M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may need Special Order Inquiry (SOI) to provide longer length.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
43	MR2	RF Door Switch	67 (20.4) minus takeup at RF Door Switch See Note 4	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	MDP	EO2	—	—	—	—	—	Customer Furnished.
45	PP1	MG2/3	30 (9.15)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Has a minimum bend radius of 10 in. (254 mm).
				258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
Note 4 M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) minus takeup at RF Door Switch.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (contin- ue)	PP1	MG2/3	30 (9.15)	781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				1036	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud See Note 5	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
81	MDP	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
Note 5 M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) minus takeup at RF Common GND Stud.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 See Note 6	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes & Note 7	045	0.195 (4.95)	Ring Terminal	Ring Terminal	The usable length is based on MG6 being mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 being mounted on floor and floor routing of cables.
88	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
90	PD1 See Note 1	OW1	51 (15.5) See Note 8	048	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				969	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

6 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

7 M3044TC can be ordered to increase usable length of Group 87 to 71 ft (21.6 m).

8 M3044TC can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 9	789	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
			1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
			1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
92	PP1	OW1	53 (16.1) See Note 10	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note 9 M3044TC can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								
Note 10 M3044TC can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).								

6-2-7 M3044TA – Site Collector TwinSpeed Fixed Site With Phased Array Long Cables

M3044TA provides longer length system interconnects for fixed site as indicated in Table 6-5 notes.

6-2-8 M3044TC – EXCITE TwinSpeed Fixed Site Cable Collector Long Cables

M3044TC provides longer length system interconnects for fixed site as indicated in Table 6-6 notes.

6-2-9 M3060TA – 1.5T Magnet With TwinSpeed Enclosure

M3060TA provides 1.5T Magnet system interconnects listed in Table 6-7. There are also requirements for customer supplied interconnect as indicated in Table 6-7.

TABLE 6-7
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
16	Phone Line Connection	MSM1	—	—	—	—	—	Refer to Section 2-6-6 Telephone Lines Requirements for additional customer phone line information.
29	MSM1	MR2	54 (16.4) minus takeup at MSM1	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
30	MSM1	PP1	54 (16.4) minus takeup at MSM1	824	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				825	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				914	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
31	MSM1	TSCC/ TGWC	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
32	MDP	MSM1 & MSM3	6 (1.8) minus takeup at each end	—	0.375 (9.5)	1.2x3.00 (30.5x76.2)	1.25x2.5 (31.8x63.5)	MSM1 power cable
				—	0.25 (6.35)	2.00x2.50 (50.8x63.5)	2.00x2.50 (50.8x63.5)	Modem for MSM1 power cable. Modem located on top of MSM1. Refer to Section 2-6-5
34	MSM1	OW1	70 (21.3) minus takeup at MSM1	942	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
45	PP1	MG3	30 (9.15)	828	0.31 (7.75)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
48	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
55	Facility Power	Filter	—	—	—	—	—	Customer Furnished Magnet Room power (refer to Sections 5-1 and 7-4).
57	Filter	MS4	—	—	—	—	—	Customer Furnished (refer to Sections 5-1).

6-2-9 M3060TA – 1.5T Magnet With TwinSpeed Enclosure (Continued)TABLE 6-7 (Continued)
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
77	PD1 See Note 1	RF COM- MON GND STUD	—	—	0.584 (14.8)	Ring Terminal	Ring Terminal	Customer supplied ground cable, refer to Section 5-4-2.
94	MSM1	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

6-2-10 M3063TA – Site Collector TwinSpeed Transportable/Relocatable With Phased Array

M3063TA provides Transportable/Relocatable appropriate lengths of system interconnects similar to M3043TA. Upgrade plans are available from the van manufacturer.

6-2-11 M3063TC – EXCITE TwinSpeed T/R Cable Collector

M3063TC provides Transportable/Relocatable appropriate lengths of system interconnects similar to M3043TC. Upgrade plans are available from the van manufacturer.

6-2-12 M3080TA – 2nd Order Resistive Shim Option

M3080TA provides system interconnects for the 2nd Order Resistive Shim Option listed in Table 6-8.

TABLE 6-8
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
11	TAC	PP1	21 (6.40)	611	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 611 for system with 2nd Order Resistive Shim Option
14	TAC	OW1	25 (7.62)	568	0.36 (9.14)	1.30x2.00 (33.5x50.8)	2.30x2.10 (58.5x53.5)	Used for 2nd Order Resistive Shim Option
45	PP1	MG3	30 (9.15)	612	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 612 for system with 2nd Order Resistive Shim Option

6-2-13 M3088TA – Outdoor TSCC System Interconnects**Note**

Refer to **Section 2-5-1 TSCC Configurations Siting Considerations** Table 2-4 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 470 Pre installation Installation Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for the Outdoor TSCC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Outdoor TSCC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements

Customer must provide water supply and return lines between the Outdoor TSCC and the GE supplied hoses for the Shield/Cryo Cooler Compressor.

- Ellis And Watts provides two 0.75 in. (19 mm) FNPT fittings at the Outdoor TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-10 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 0.75 in. (19 mm) MNPT connections and 0.75 in. (19 mm) ID copper lines.
- Ellis And Watts provides two 0.5 inch (12.7 mm) ID rubber hoses of 60 ft (18.3 m) total length [usable length of 55 ft (16.7 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Shield/Cryo Cooler Compressor Cabinet. Site layout for the Shield/Cryo Cooler Compressor water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TSCC connection to the Shield/Cryo Cooler Compressor connection **MUST NOT EXCEED** 160 ft (48.8 m). Ellis And Watts supplied rubber hoses for Shield/Cryo Cooler Compressor is a maximum length of 60 ft (18.3 m) total length [55 ft (16.8 m) usable length].
 - If the 0.75 in. (19 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
 - Copper lines must be insulated outdoors.

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)**Gradient Coil Water Cooling Lines & Hoses Requirements**

- Ellis And Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-10 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TSCC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TSCC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
 - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).

RCP Data Cables Requirements

- Ellis & Watts provides three data cables that connect between the RCP and the Outdoor TSCC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TSCC and RCP placement [total length minus height of connection at Outdoor TSCC and height of connection at RCP, no cable takeup inside of TSCC or RCP].

Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

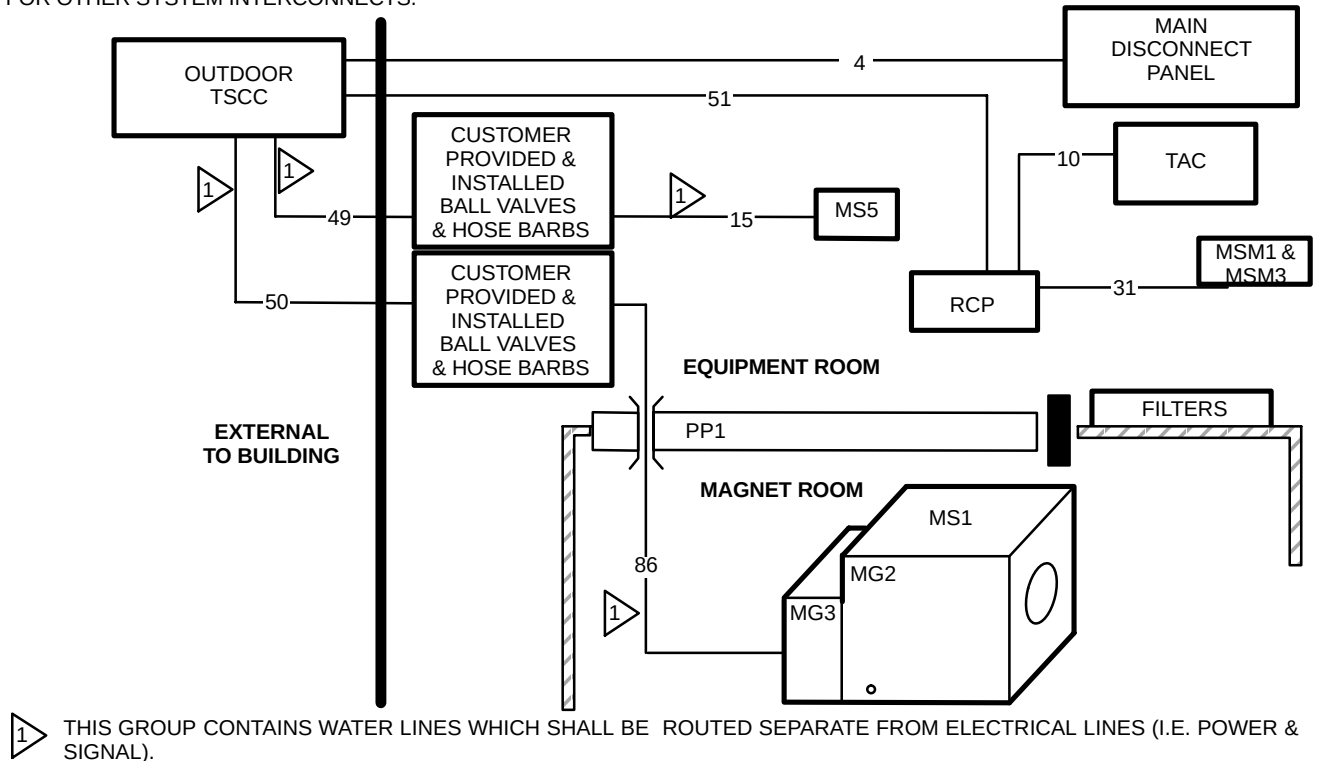
System Additional Interconnects

Illustration 6-3 shows the Outdoor TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-9 for specifics.

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



OUTDOOR TSCC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM

ILLUSTRATION 6-3

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

TABLE 6-9
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out-door TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	RCP	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	Customer provided ball valves & hose barbs for Shield/ Cryo Cooler	MS5	55 (16.7) minus takeup at hose barbs See Notes	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	Group 15 & 49 MUST NOT EXCEED limitations specified under Sec 6-2-13 sub-heading Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements
31	0.10 (66)	MSM1	RCP	50 (15.24) minus takeup at both ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
49 SEE NOTE 1	—	Out-door TSCC	Customer ball valves & hose barbs for Shield/ Cryo Cooler	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 0.75 (19 mm) ID copper lines for Shield/Cryo Cooler Compressor water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 0.75 in. (19 mm) ball valve & 0.5 in. (12.7 mm) hose barb. • Groups 49 & 15 MUST NOT EXCEED limitations specified under Sec 6-2-13 sub-heading Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).									
(Continued)									

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

TABLE 6-9 (Continued)
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
50 SEE NOTE 1	—	Out- door TSCC	Cus- tomer ball valves & hose barbs for Gradi- ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	● Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. ● Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. ● Group 50 and 86 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements
51	1.097 (613)	Out- door TSCC	RCP	See NOTES	— — —	0.63 (16.0) 0.65 (16.5) 0.76 (15.9)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TSCC and height of connection at RCP. No cable takeup inside of TSCC or RCP.
86 SEE NOTE 1 & 2	1.666 (1078)	Cus- tomer pro- vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	● Group 86 and 50 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements ● These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.									

6-2-14 M3088TB – Indoor Water Cooled TSCC System Interconnects

Note

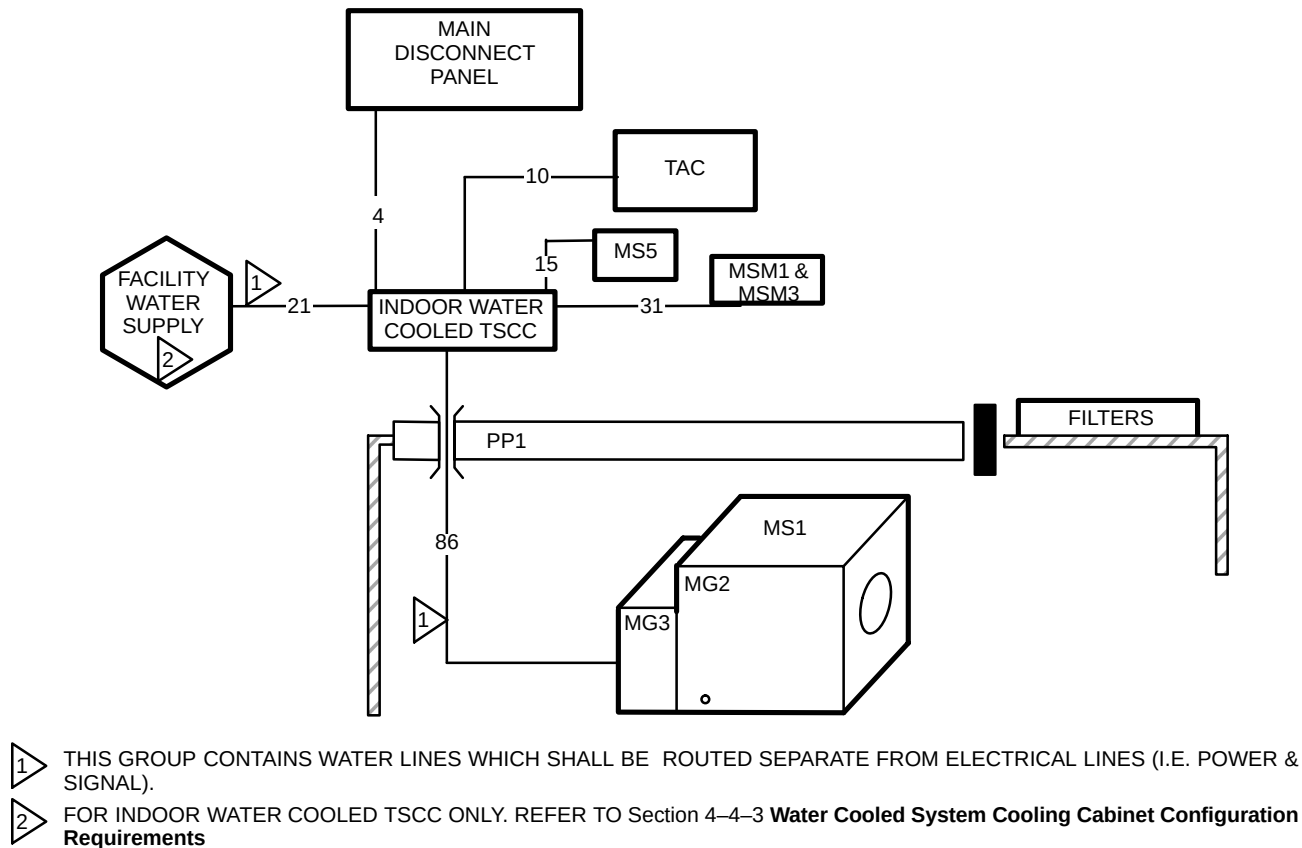
Refer to **Section 2-5-1 TSCC Configurations Siting Considerations** Table 2-4 listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 471 Pre installation Installation Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TSCC must be in accordance with local and national codes.

Illustration 6-4 shows the Indoor Water Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-10 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



INDOOR WATER COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM

ILLUSTRATION 6-4

6-2-14 M3088TB – Indoor Water Cooled TSCC (Continued)

TABLE 6-10
INDOOR WATER COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	rubber hose rubber hose	rubber hose rubber hose	See Notes 3 & 4 for vertical separation limits & TSCC hoses/connections.
21 SEE NOTE 1	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC ONLY: Customer furnished, refer to Section 4-4-3 Water Cooled System Cooling Cabinet Configuration Requirements & refer to Note 6
31	0.10 (66)	MSM1	TSCC	38 (11.6) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1 & 2	1.666 (1078)	TSCC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 5 for vertical separation limits & TSCC hoses/connections.

- Note**
- 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).
 - 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.
 - 3 The Indoor Water Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.
 - 4 The Indoor Water Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-13 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.
 - 5 The Indoor Water Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-13 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.
 - 6 Facility chilled water connections at TSCC are 1.5 inch (38.1 mm) FNPT. Customer to connect with 1.5 inch (38.1 mm) MNPT.

6-2-15 M3088TC – Indoor Air Cooled TSCC System Interconnects (Continued)

TABLE 6-11
INDOOR AIR COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	See Notes 3 & 4 for vertical separation limits & TSCC hoses/connections.
31	0.10 (66)	MSM1	TSCC	40 (12.2) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1 & 2	1.666 (1078)	TSCC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 5 for vertical separation limits & TSCC hoses/connections.
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA. 3 The Indoor Air Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. 4 The Indoor Air Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-15 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor. 5 The Indoor Air Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-15 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.									

6-2-16 M3088TE – Indoor Water Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2-5-2 TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 469 Pre installation Installation Operating LTL – 10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TGWC must be in accordance with local and national codes.

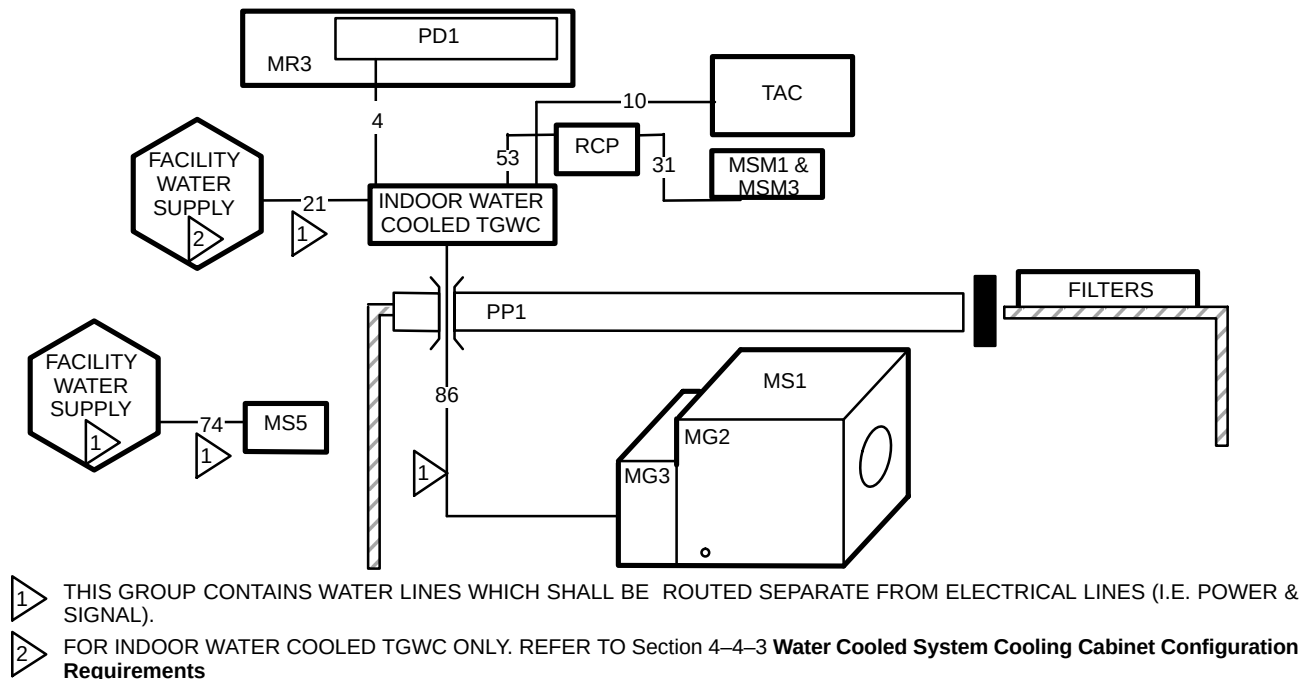
Illustration 6-6 shows the Indoor Water Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-12 for specifics.

Note

The Indoor Water Cooled TGWC and Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) of water lines. The TGWC vertical separation must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



INDOOR WATER COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-6

6-2-16 M3088TE – Indoor Water Cooled TGWC (Continued)

TABLE 6-12
INDOOR WATER COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	0.159 (103)	PD1 See Note 3	TGWC	23 (7.0)	—	0.45 (11.43)	Hard Wired	1.53x2.78 (38.9x70.6)	Ellis & Watts provided cable
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
21 SEE NOTE 1	—	Facility Water Supply	TGWC	—	—	—	—	—	For Indoor Water Cooled TGWC ONLY: Customer furnished, refer to Section 4-4-5 Water Cooled Twin Gradient Water Cooler (TGWC) Requirements
31	0.10 (66)	MSM1	RCP	50 (15.24) minus takeup at booth ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
53	0.157 (103)	Water Cooled TGWC	RCP	28 (8.53) minus takeup at RCP See Notes	—	0.258 (6.6) 0.258 (6.6) 0.258 (6.6)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	No cable takeup inside of RCP.
74	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4-4-4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	TGWC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 4 & 5 for vertical separation limits & TGWC hoses/connections.

- Note**
- 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).
 - 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.
 - 3 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).
 - 4 Indoor Water Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.
 - 5 The Indoor Water Cooled TGWC has two 0.75 in. (19 mm) SnapTite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-20 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6-2-17 M3088TF – Outdoor TGWC System Interconnects**Note**

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2-5-2 TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 474 Pre installation Installation Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for Outdoor TGWC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Single Loop Outdoor TGWC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Dual Loop Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Gradient Coil.

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis & Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-10 for location of fittings on TGWC. The Outdoor TGWC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TGWC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TGWC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
 - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduces 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
 - Copper lines must be insulated outdoors.

RCP Data Cables Requirements

- Ellis & Watts provides two data cables that connect between the RCP and the Outdoor TGWC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TGWC and RCP placement [total length minus height of connection at Outdoor TGWC and height of connection at RCP, no cable takeup inside of TGWC or RCP].

Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

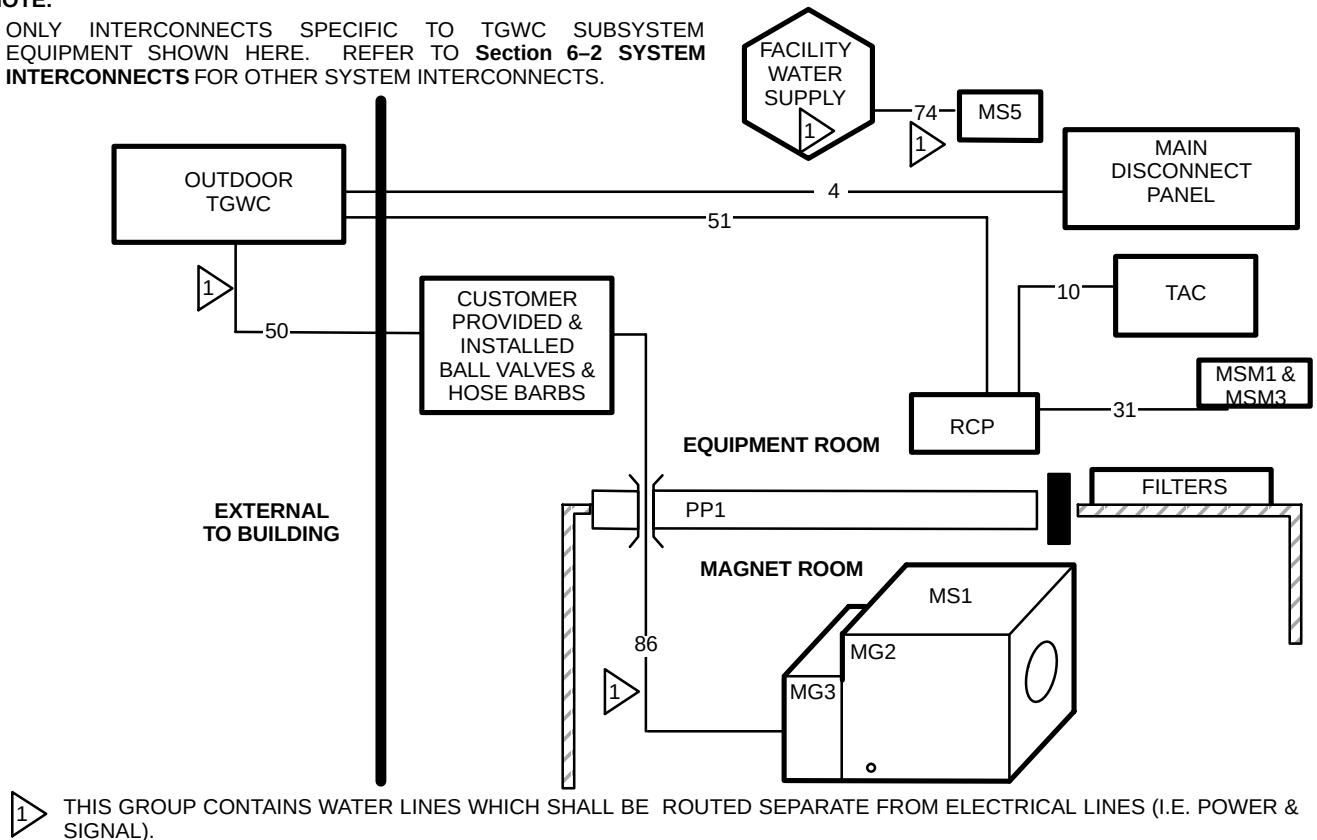
6-2-17 M3088TF – Outdoor TGWC System Interconnects (Continued)

System Additional Interconnects

Illustration 6-7 shows the Outdoor TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-13 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



OUTDOOR TGWC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM

ILLUSTRATION 6-7

6-2-17 M3088TF – Outdoor TGWC System Interconnects (Continued)

TABLE 6-13
OUTDOOR TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out- door TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.10 (66)	MSM1	RCP	60 (18.3) minus takeup at both ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
50 SEE NOTE 1	—	Out- door TGWC	Cus- tomer ball valves & hose barbs for Gradi- ent Coil	See Notes	—	—	copper supply line	copper supply line	● Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. ● Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. ● Group 50 and 86 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements
					—	—	copper return line	copper return line	
51	4.791 (3091)	Out- door TGWC	RCP	See NOTES	—	1.59 (40.4)	1.97x2.25 (50.0x57.2)	1.97x2.25 (50.0x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TGWC and height of connection at RCP. No cable takeup inside of TGWC or RCP.
					—	1.89 (48.0)	1.75x2.25 (44.5x57.2)	1.75x2.25 (44.5x57.2)	
74 SEE NOTE 1	—	Water	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4–4–4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	Cus- tomer pro- vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	—	1.03 (26.2)	Flexible Tubing	Flexible Tubing	● Group 86 and 50 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements ● These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).
					—	1.03 (26.2)	Flexible Tubing	Flexible Tubing	
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).									
2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.									

6-2-18 M3088TG – Indoor Air Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

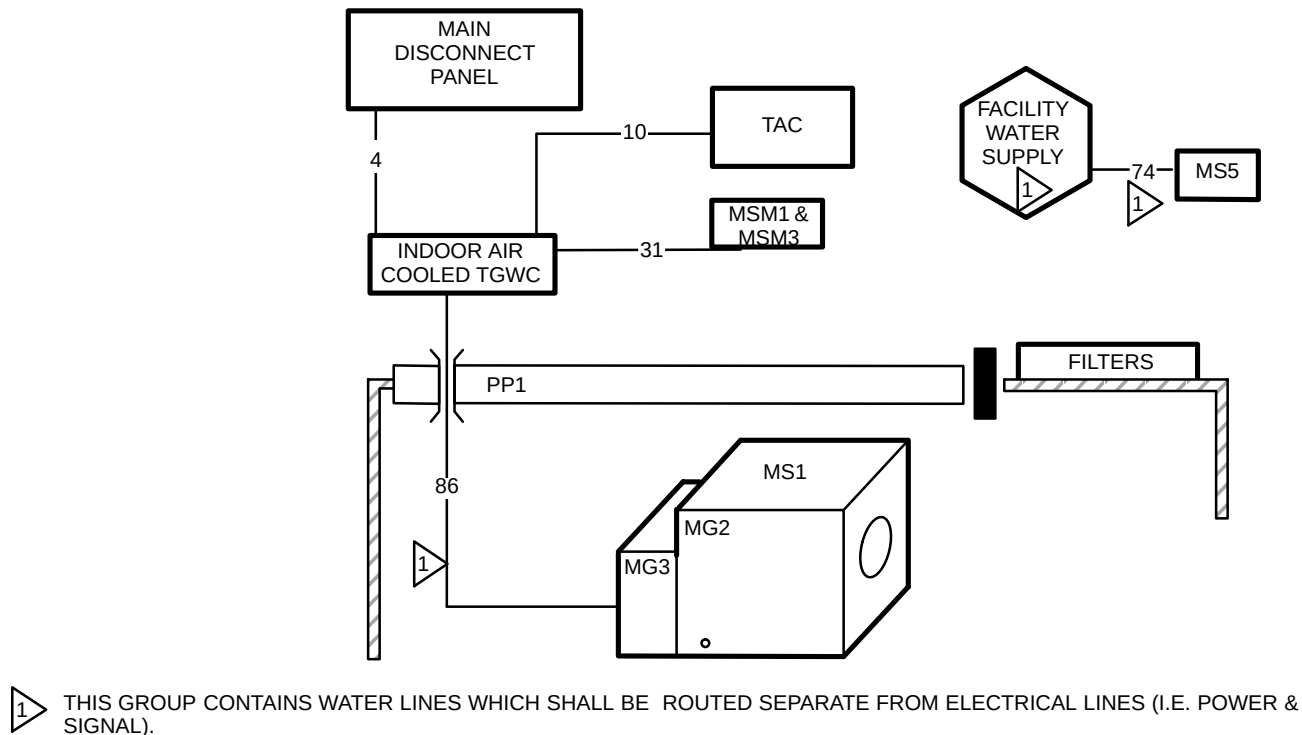
Refer to **Section 2-5-2 Fixed Site System TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 473 Pre installation Installation Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TGWC must be in accordance with local and national codes.

Illustration 6-8 shows the Indoor Air Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-14 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



INDOOR AIR COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-8

6-2-18 M3088TG – Indoor Air Cooled TGWC (Continued)

TABLE 6-14
INDOOR AIR COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.16 (103)	MSM1	TGWC	48 (14.6) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
74	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4-4-4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	TGWC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 4 for vertical separation limits & TGWC hoses/connections.
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA. 3 Indoor Air Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil. 4 The Indoor Air Cooled TGWC has two 0.75 in. (19 mm) Snpdite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-22 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.									

6-2-19 M3088TH – GE Main Disconnect Panel

Table M3080TA system interconnects for the GE Main Disconnect Panel (MDP) are listed in Table 6-15. If customer provided MDP has been selected based on meeting **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements, refer to **SECTION 5 – POWER REQUIREMENTS**.

TABLE 6-15
CABLES GE PROVIDED & CUSTOMER SUPPLIED FOR M3088TH

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	MDP	—	—	—	—	—	Customer Furnished. See Note 1.
2	MDP	PD1 See Note 2	—	—	—	—	—	Customer Furnished.
4	MDP/ PDU See Notes	TSCC/ TGWC See Notes	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
Note 1 If low Voltage Step-Up Transformer Option (R4500AW or R4500BE) is used then customer furnished interconnects are required between facility power, transformer and MDP. 2 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).								

6-2-20 M3090AH – Horizon to TwinSpeed Upgrade System Electronics

M3090AH provides system interconnects for several assemblies mounted to the OW1 Table and routed internal to the OW1 Table cable tray.

6-2-21 M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet

Catalog M3090BB provides ACGD upgrade power and ground cables. Refer to Table 6-16 for cables details.

TABLE 6-16
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-2-22 M3090EB – EXCITE UPGRADE KIT

M3090EB provides standard length system interconnects for fixed site listed in Table 6-17.

TABLE 6-17
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 or 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
45	PP1	MG2/3	30 (9.15)	1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-2-22 M3090EB – EXCITE UPGRADE KIT (Continued)

TABLE 6-17 (Continued)
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5)	1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
				1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	

6-2-23 M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Cables provided by catalogs M3090BC are the same as catalogs M3090BB. Refer to Table 6-16 for cables details.

6-2-24 M3090TA – Twin Upgrade For LCC Horizon Fixed & T/R

Catalog M3090TA provides fixed site and Transportable/Relocatable system interconnect cables for Horizon Magnet Enclosure configuration to Signa 1.5T TwinSpeed. Refer to Table 6-18 for cables details.

TABLE 6-18
CABLES PROVIDED BY CATALOG M3090TA & M3090TE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	MR3	TAC	SEE ROW NOTES	976	1.21 (30.7)	Hard Wired	Hard Wired	Runs 976, 977, & 978 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
				977	1.21 (30.7)	Hard Wired	Hard Wired	
				978	1.21 (30.7)	Hard Wired	Hard Wired	
9	PD1 SEE NOTE1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
11	TAC	PP1	41 (12.5)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	SEE ROW NOTES	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
				980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988	0.26 (6.6)	Ring Terminal	Ring Terminal	Ground wire, Ground wire, 65 ft (19.8 m) actual length.
13	TAC	MG3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.
31	MSM1	RCP/ TSCC	SEE ROW NOTES	941	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine TO connection.
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
45	PP1	MG3	30 (9.15)	300	0.44 (11.18)	2.30x2.00 (57.2x50.8)	Ring Terminals	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).								
(Continued)								

6-2-24 M3090TA – Twin Upgrade For LCC Horizon Fixed & T/R (Continued)

TABLE 6-18 (Continued)
CABLES PROVIDED BY CATALOG M3090TA & M3090TE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 SEE NOTE 2	TSCC/ TGWC	MG2	SEE ROW NOTES	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine TO connection.
Note 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).								

6-2-25 M3090TE – Twin Upgrade Covers From WideOpen For Fixed & T/R Sites

Cables provided by catalogs M3090TE are the same as catalogs M3090TA. Refer to Table 6-18 for details.

6-3 ADVANTAGE WORKSTATION INTERCONNECTS

Refer to Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation for AW interconnect information.

6-4 INTERCONNECTS RE-ROUTED BY ACGD UPGRADES

ACGD Gradients upgrades replace the existing system PDU (Full-size, Compact, or PDU in RF/PDU Cabinet) with the PDU located in the lower portion of the ACGD/PDU Cabinet. Therefore, all connections to the PDU may need to be re-routed depending on site layout of the upgraded system.

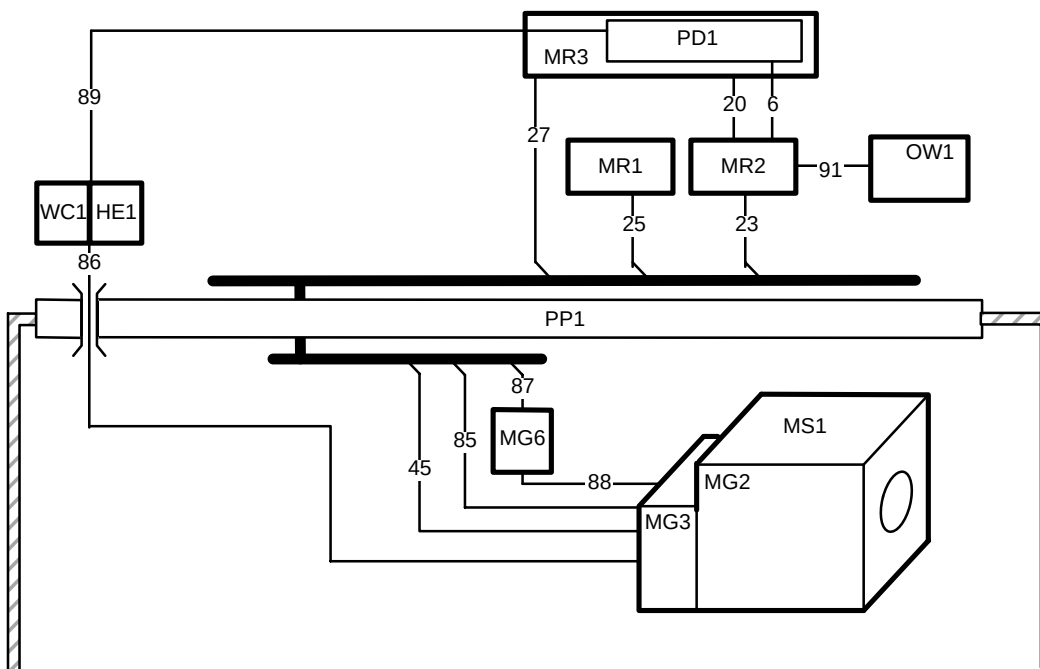
Note

GE Installation Specialist & site Field Engineer need to consider impact of upgrade site layout on re-routed cables especially if any cables (e.g. power cables) were cut to length for current installed system.

6-5 INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY UPGRADES

6-5-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets

Illustration 6-9 and Table 6-19 lists the existing Signa MR/i, CV/i, NV/i & Signa LX, CVMR system with ACGD/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



SIGNA MR/i, CV/i, NV/i OR SIGNA LX WITH ACGD/PDU CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-9

**6-5-1 Signa MR*li*, CV*li*, NV*li* Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19
SIGNA MR*li*, CV*li*, NV*li* OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
6	MR3/ PD1 See Note 1	MR2	22 (6.70)	972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40) See Note 2	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

**6-5-1 Signa MRIi, CVIi, NVIi Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19 (Continued)
SIGNA MRIi, CVIi, NVIi OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
(Continued)								

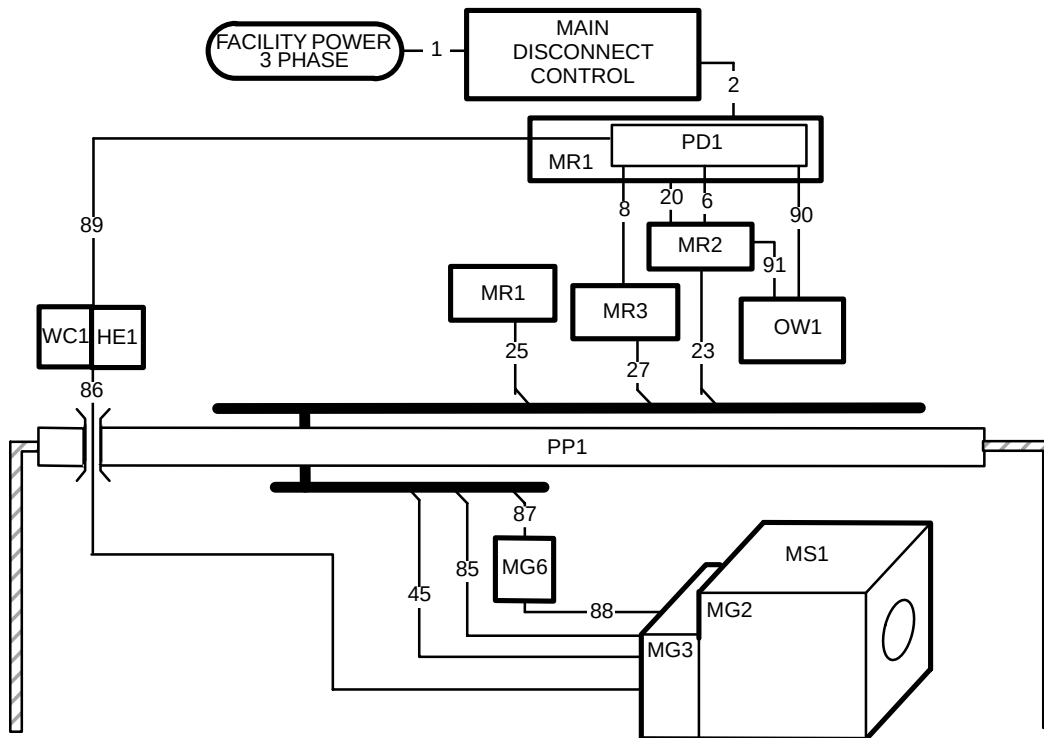
**6-5-1 Signa MR*i*, CV*i*, NV*i* Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19 (Continued)
SIGNA MR*i*, CV*i*, NV*i* OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR <i>i</i> , CV <i>i</i> , & NV <i>i</i>	MR3/ PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	MR3/ PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1). 2 M3044TA may have been ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may have been Special Order Inquiry (SOI) to provide longer length.								

6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets

Illustration 6-10 and Table 6-20 lists the existing Signa MR/i, CV/i, NV/i or LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



**SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-10**

**TABLE 6-20
SIGNA MR/II, CV/II, NV/II OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE**

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030 972	0.92 (23.4) 0.36 (9.14)	Ring Terminals 1.30x2.00 (33.5x50.8)	Ring Terminals 1.30x2.00 (33.5x50.8)	Power cable, #8 AWG 5 wires
(Continued)								

**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR/II, CV/II, NV/II OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	

(Continued)

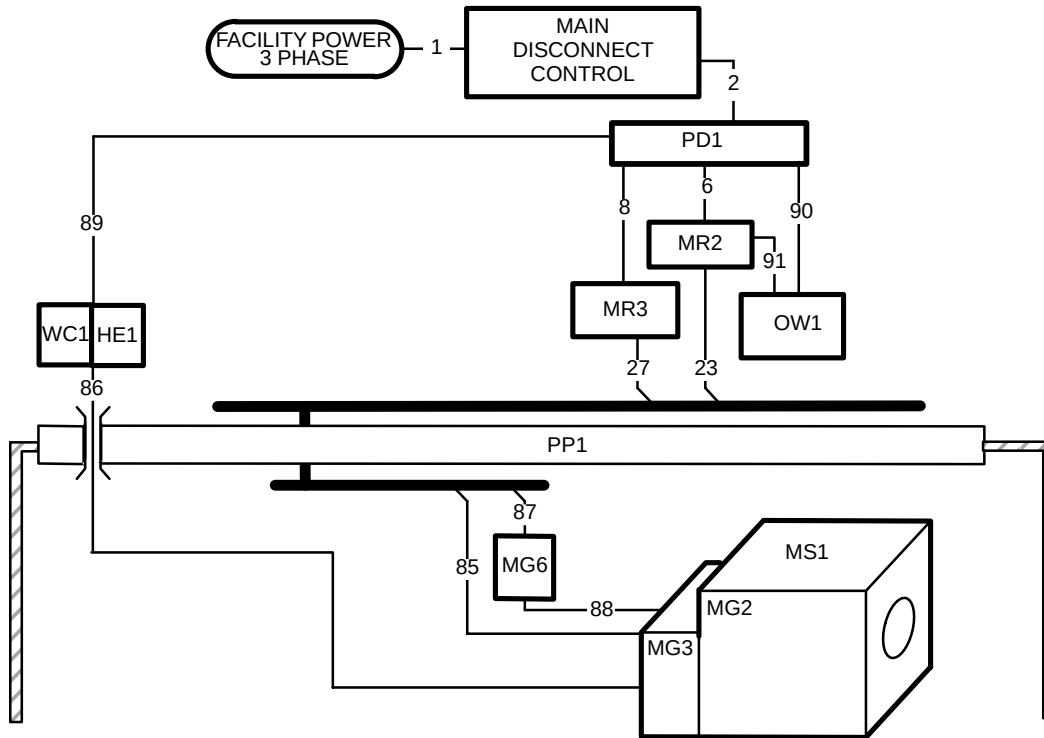
**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).

6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets

Illustration 6-11 and Table 6-21 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



**SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-11**

**TABLE 6-21
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE**

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
				972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
(Continued)								

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
(Continued)								

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).

6-5-4 Signa LX & CVMR Without LCC Magnet Installed**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa LX or CVMR (with ACGD or SGD) systems which currently do NOT have an LCC Magnet both the magnet and system electronics upgrades, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-5-5 Signa Horizon 5.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa Horizon will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-5-6 Signa Advantage 5.X & 4.X System**Note**

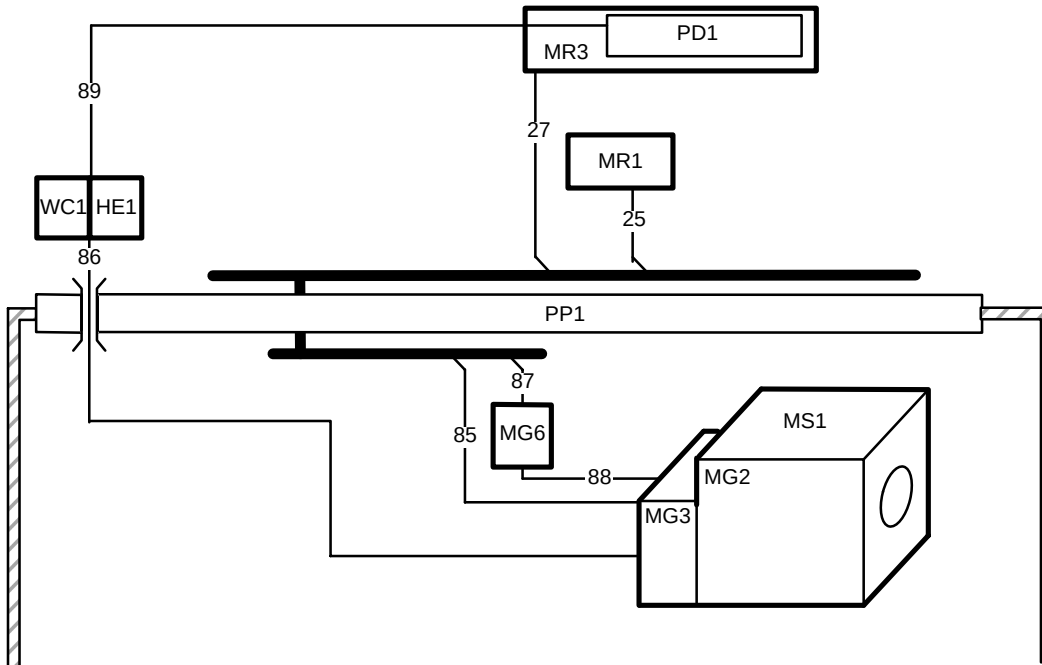
Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa Advantage 5.X and 4.X will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6 INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED UPGRADES

6-6-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets

Illustration 6-12 and Table 6-22 lists the existing Signa MR/i, CV/i, NV/i & Signa LX, CVMR system with ACGD/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA MR/i, CV/i, NV/i OR SIGNA LX WITH ACGD/PDU CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-12

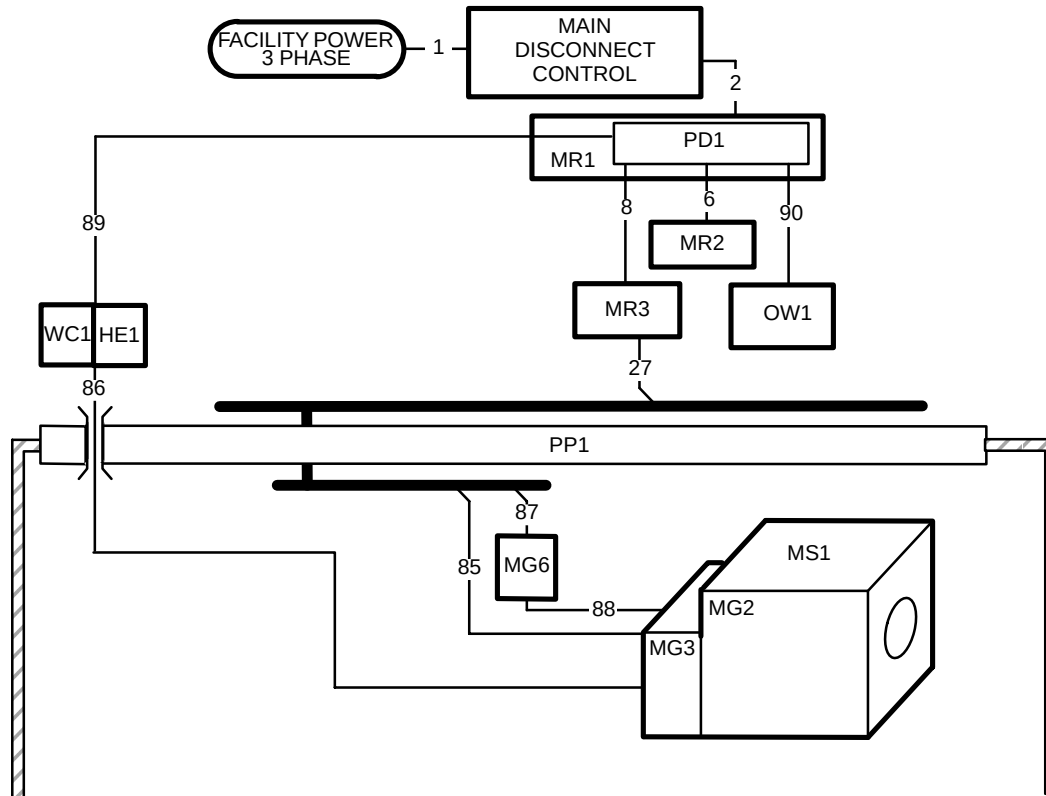
**6-6-1 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-22
SIGNA MR//, CV//, NV// OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water

6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets

Illustration 6-13 and Table 6-23 lists the existing Signa MR/i, CV/i, NV/i or LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-13

TABLE 6-23

SIGNA MR/i, CV/i, NV/i OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
(Continued)								

ARCHITECTURAL/INSTALLATION PLANNING USE ONLY

GE MEDICAL SYSTEMS

SIGNA 1.5T TWINSPEED UPGRADE PRE-INSTALLATION

REV 1

DIRECTION 2304880

**6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-23 (Continued)

SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

(Continued)

**6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

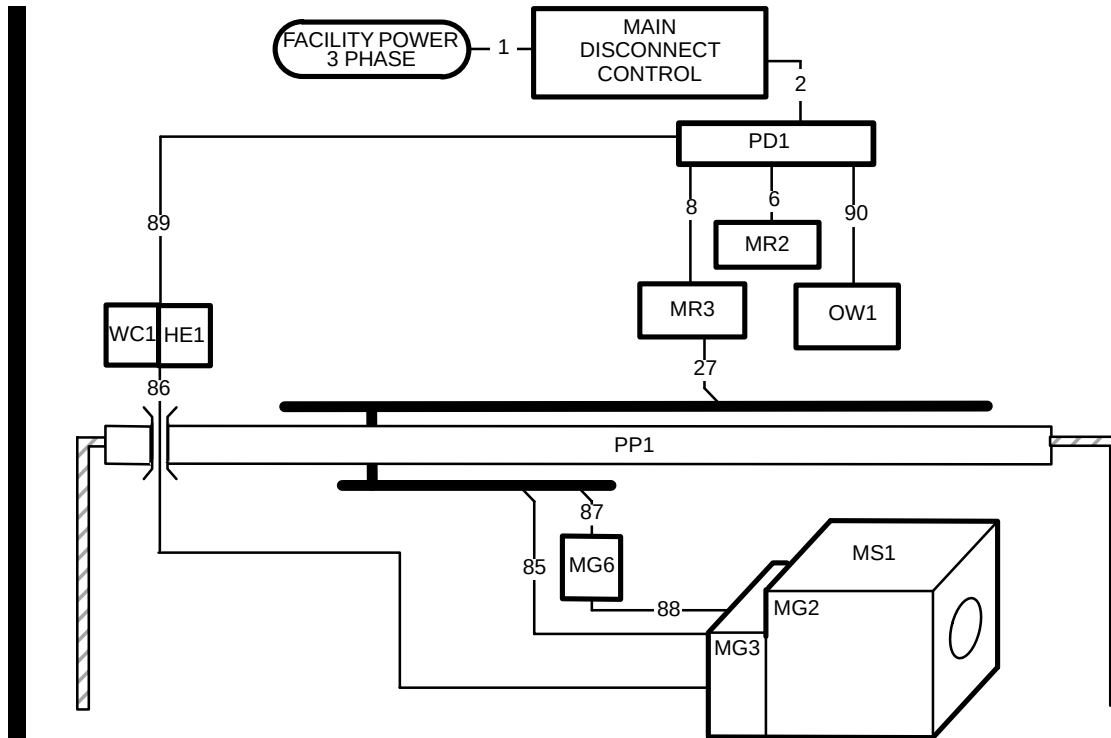
TABLE 6-23 (Continued)

SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets

Illustration 6-14 and Table 6-24 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-14

TABLE 6-24

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
(Continued)								

**6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-24 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

(Continued)

**6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-24 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-4 Signa LX & CVMR Without LCC Magnet Installed**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed upgrade for Signa LX or CVMR (with ACGD or SGD) systems which currently do NOT have an LCC Magnet both the magnet and system electronics upgrades, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6-5 Signa Horizon 5.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed upgrade for Signa Horizon will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6-6 Signa Advantage 5.X & 4.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

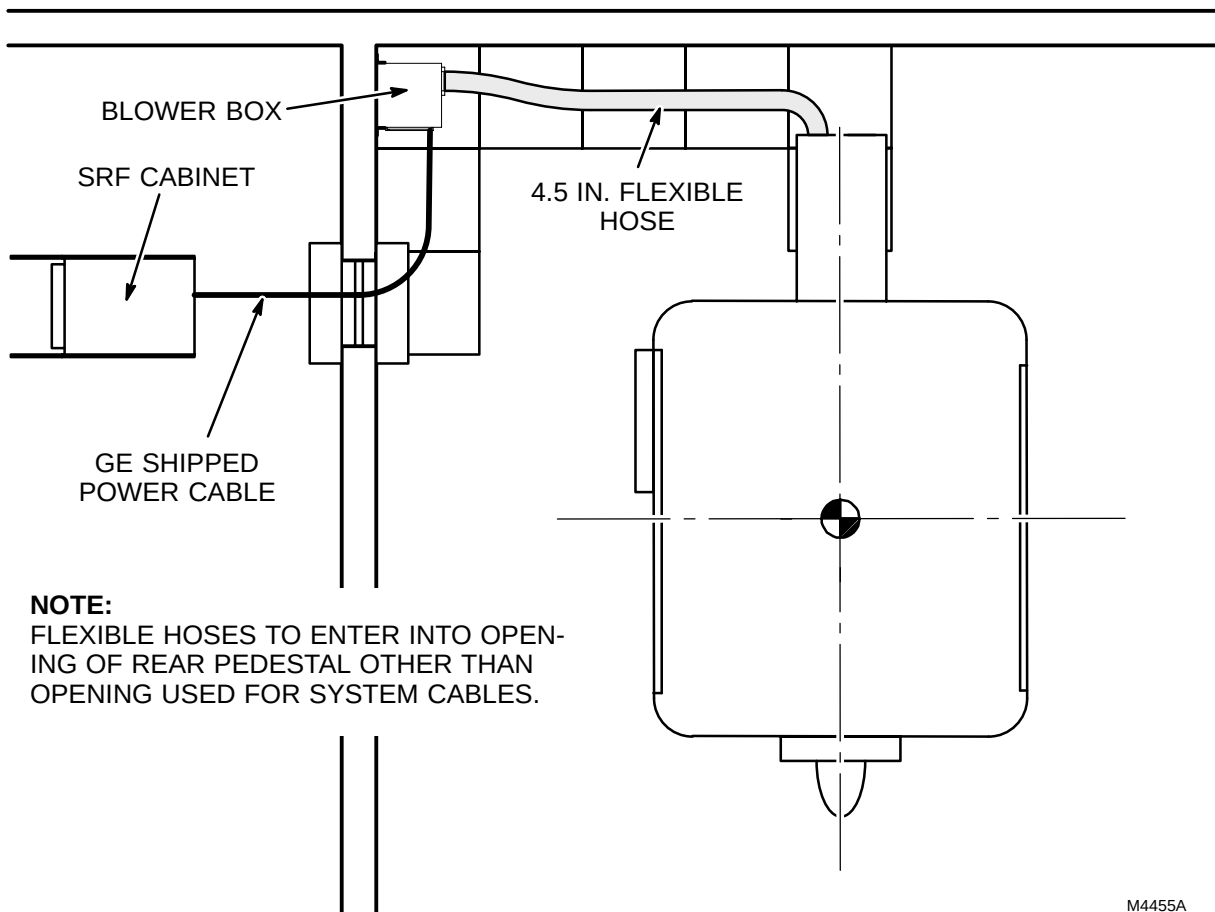
The Signa 1.5T TwinSpeed upgrade for Signa Advantage 5.X and 4.X will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-7 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-9, COMPONENT DIMENSIONS. Signa 1.5T TwinSpeed several changes to the cabling and hoses routed through the Rear Pedestal cable access areas of the Magnet Enclosure. These changes include the

- elimination of 1 Blower Box hose (6.5 inch (165 mm) OD hose)
- installation of 3 additional Gradient cables
- addition of a vacuum hose and a vacuum sensor cable for the Quiet Technology Enclosure
- 2 existing water hoses for Body Coil are replaced with 2 new larger water hoses.

The remainder of the Magnet Enclosure system interconnect cables are also still routed through the Rear Pedestal cable access. The existing cable access area utilized to site cable routing should be sufficient for the upgrade system cabling. Illustration 6-15 shows one example of Blower Box set-up with flexible hoses route separate from other interconnects.

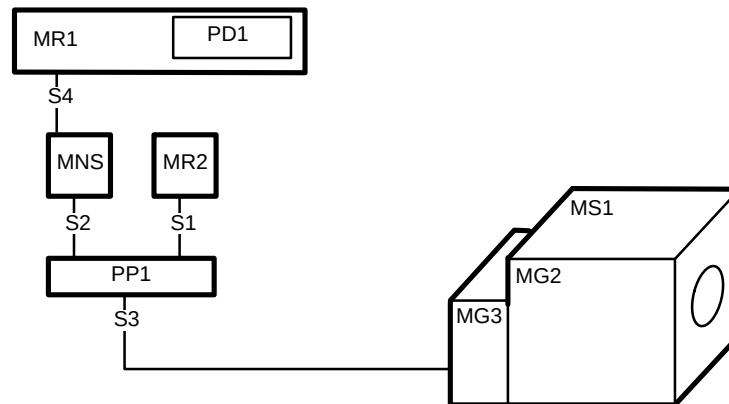


EXAMPLE OF BLOWER BOX SET-UP (PLAN VIEW)

ILLUSTRATION 6-15

6-8 SPECTROSCOPY OPTION INTERCONNECTS**6-8-1 M1033MD – MNS Kit For Signa LX Systems**

Illustration 6-16 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-25 contains the cable data for these interconnects.



CATALOG M1033MD SPECTROSCOPY OPTION SYSTEM INTERCONNECTS

ILLUSTRATION 6-16

6-8-1 M1033MD – MNS Kit For Signa LX Systems (Continued)

TABLE 6-25
CABLES PROVIDED BY CATALOG M1033MD

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-8-2 M3090DB – 5.5 MNS Upgrade Kit For ACGD Site

Illustration 6-17 shows the system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-26 contains the cable data for these interconnects.

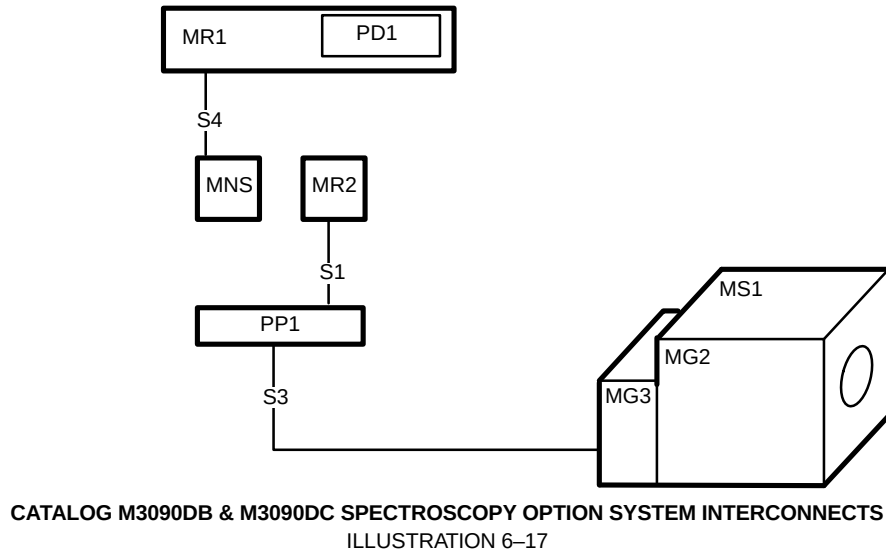


TABLE 6-26
CABLES PROVIDED BY CATALOG M3090DB & M3090DC

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-8-3 M3090DC – LX MNS Upgrade Kit For ACGD Site

Cables provided by catalog M3090DC are the same as catalog M3090DB. Refer to Illustration 6-17 and Table 6-26 cables details.

SECTION 7 – RF SHIELDED ROOM

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7-2	VENTS	7-3
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7-7	ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM	7-5
	7-7-1 Magnet Mounting	7-5
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	7-8-2 Emergency Off Buttons	7-6
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7-1 RF SHIELDED ROOM SPECIFICATION

The shielding effectiveness of the RF Shielded Room impacts the imaging ability of the system. Therefore, the RF Shielded Room needs to meet the attenuation specification before the upgrade. The RF Shielded Room RF attenuation specification with the GE system connected is 85 dB planewave (10MHz – 100MHz). If the RF Shield is not at this specification then RF vendor maintenance may be necessary.

If the magnet is upgraded to an LCC magnet with the Signa TwinSpeed system upgrade and the RF Shield is opened, modified, upgraded, etc. then the RF Shielded Room integrity **MUST** be revisited (tested) to meet or exceed the new system RF Shield Room specifications. Refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-1 RF SHIELDED ROOM SPECIFICATION.***

7-2 VENTS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-2 VENTS*** for cryogenic & HVAC vents requirements and recommendations.

7-3 PLUMBING

The Signa TwinSpeed system performance requires adherence to the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-3 PLUMBING*** for pipes, water, medical gasses, and sprinklers.

7-4 ELECTRICAL

The Signa TwinSpeed system performance requires adherence to the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-4 ELECTRICAL*** for filter attenuation, power line grounding, lighting, and common ground stud proximity.

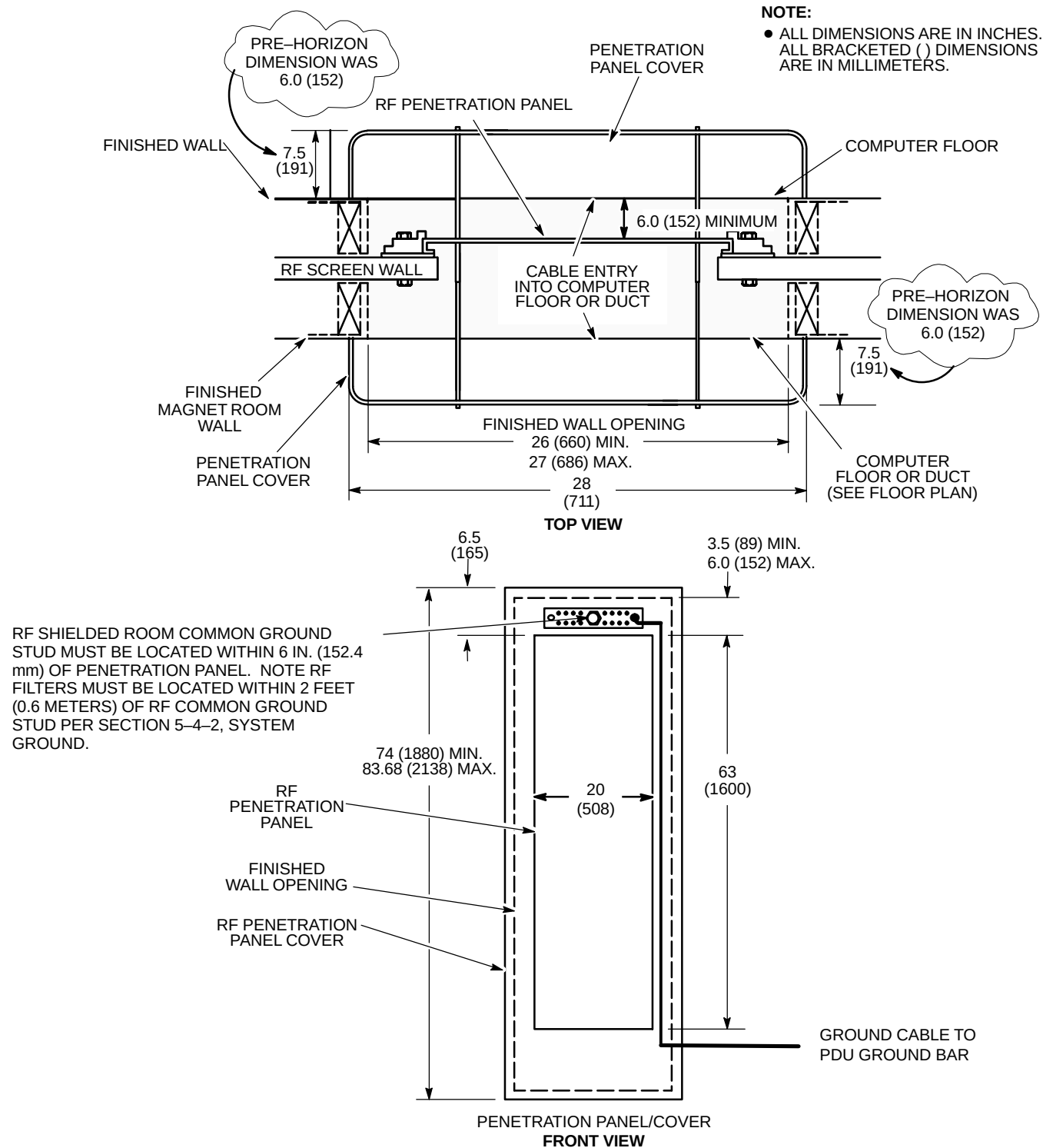
7-5 RF PENETRATION PANEL

The Penetration Panel is modified for Signa MR/i, CV/i, CV/i and LX upgrades. The Penetration Panel is replaced during Signa Horizon and prior system configurations upgrades as shown in Section 1 – SYSTEM CONFIGURATION. Current panel mounting hardware will be used to mount new Penetration Panel. The new Penetration Panel is deeper therefore new deeper Penetration Panel Covers are provided. The mounting and clearance dimensions for the Penetration Panel covering are shown in Illustration 7-1. Refer to Section 2-9, COMPONENT DIMENSIONS, for specific upgrade catalogs provide new Penetration Panel and Covers.

Note

The Penetration Panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer furnished covers or enclosures with key or tool are required for opening to limit access to the panel.

7-5 RF PENETRATION PANEL



PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS

ILLUSTRATION 7-1

7-6 PHYSICAL CONSIDERATIONS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-6 PHYSICAL CONSIDERATIONS** for door and other openings, ceiling height, walls, and floors requirements and recommendations.

7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM

7-7-1 Magnet Mounting

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Magnet and Dock Assembly requirements.

7-7-2 Blower Box

The Blower Box (MG6) provides cooling air for the Patient Comfort Module in the Magnet Enclosure for Signa TwinSpeed system.

The existing system Blower Box Panel is modified (cover plate installed) for upgrades to Signa MR/i, CV/i, CV/i, LX and CVMR with LCC Magnet installed so the existing box mounting will be used.

A new Blower Box is installed by Signa LX and CVMR without LCC Magnet installed, Signa Horizon and prior system configurations upgrades. The Blower Box will be mounted within the RF Shielded Room and connect to the Patient Comfort Module by 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted on the floor of the Magnet Room, a raised support shelf, or the Magnet Room wall or ceiling with support provided under box.** RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room. The flexible vinyl air duct route is routed from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connect to the Gradient Coil. Refer to Illustrations 2-33 for Blower Box and mounting dimensions. Refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Blower Box mounting RF Shield Integrity and electrical isolation requirements.

7-8 MAGNET ROOM EQUIPMENT MOUNTING**7-8-1 Magnet Rundown Unit (MS4)**

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then the magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the Magnet Room but outside the 200 gauss zone.

7-8-2 Emergency Off Buttons

Emergency off buttons to be located near the exit in the magnet and equipment rooms. These buttons must be clearly labeled, "Emergency Off". Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7-9 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.

SECTION 8 – SHIPPING AND DELIVERY DATA

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8-2	UPGRADE CATALOGS SHIPPING INFORMATION	8-5

8-1 INTRODUCTION

Domestic transportation for the Signa Horizon upgrade will be via an air-ride moving van. Export transportation for overseas will be via air shipment in a pressurized cargo hold.

The tables in Section 8-2 UPGRADE CATALOGS SHIPPING INFORMATION (arranged by catalog numbers) provide estimates of shipping weights and dimensions for upgrade catalog major components. Refer to the Table 8-1 for page number location for a specific catalog.

Note

The weight and dimension information is estimate only, actual shipment packaging configurations for a catalog may vary.

TABLE 8-1
LIST OF CATALOG TABLES AND PAGE NUMBER LOCATION

TABLE	TITLE	PAGE
8-2	CATALOG M1000MW, TABLE FOR USE WITH ALL MONITORS	8-5
8-3	CATALOG M1000NT, LCD COLOR MONITOR	8-5
8-4	CATALOG M1033K, INDIGO TO OCTANE UPGRADE	8-5
8-5	CATALOG M1033MD, LX MNS KIT	8-5
8-6	CATALOG M1060JW, SHIELD/CRYO COOLER COMPRESSOR	8-6
8-7	CATALOG M1090RW, 512 MBYTE BAM	8-6
8-8	CATALOG M1090TL, PATIENT TABLE	8-6
8-9	CATALOG M1090WG, FAST RECEIVER KIT	8-6
8-10	CATALOG M1090WJ, RECEIVER "0" KIT	8-6
8-11	CATALOG M1090WL, LOW PROFILE CARRIAGE COVER KIT	8-6
8-12	CATALOG M3000RA, REFLEX 100 WITH 512 BAM	8-7
8-13	CATALOG M3000TA, TWINSPEED UPGRADE SYSTEM ELECTRONICS	8-7
8-14	CATALOG M3043TA, SITE COLLECTOR KIT	8-8
8-15	CATALOG M3044TA, SITE COLLECTOR KIT	8-9
8-16	CATALOG M3060TA, 1.5T MAGNET WITH TWINSPEED ENCLOSURE	8-10
8-17	CATALOG M3063TA, SITE COLLECTOR KIT	8-10
8-18	CATALOG M3080TA, 2ND ORDER RESISTIVE SHIM KIT	8-10
8-19	CATALOG M3088TA, OUTDOOR AIR COOLED (DUAL LOOP) TSCC	8-11
8-20	CATALOG M3088TE, INDOOR WATER COOLED (SINGLE LOOP) TGWC	8-11
(Continued)		

8-1 INTRODUCTION (Continued)

TABLE 8-1 (Continued)
LIST OF CATALOG TABLES AND PAGE NUMBER LOCATION

TABLE	TITLE	PAGE
8-21	CATALOG M3088TF, OUTDOOR AIR COOLED (SINGLE LOOP) TGWC	8-11
8-22	CATALOG M3088TG, INDOOR AIR COOLED (SINGLE LOOP) TGWC	8-11
8-23	CATALOG M3090AH, HORIZON TO TWINSPEED UPGRADE SYSTEM ELECTRONICS	8-12
8-24	CATALOG M3090BB, ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PDU CABINET	8-12
8-25	CATALOG M3090BC, 1.5T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET	8-12
8-26	CATALOG M3090BE, ECHOSPEED PLUS PERFORMANCE OPTION	8-13
8-27	CATALOG M3090DB, 5.5 MNS UPGRADE KIT	8-13
8-28	CATALOG M3090DC, LXSPECTRO PRE-RF/PDU TO AGCD MNS UPGRADE KIT	8-13
8-29	CATALOG M3090JA, IPG2 BOARD	8-13
8-23	CATALOG M3090AH, HORIZON TO TWINSPEED UPGRADE SYSTEM ELECTRONICS	8-12
8-30	CATALOG M3090TA, TWIN UPGRADE FOR LCC HORIZON FIXED & T/R	8-14
8-31	CATALOG M1085BG, 1.5T BODY COIL FOR SIGNA HORIZON	8-14

8-2 UPGRADE CATALOGS SHIPPING INFORMATION

TABLE 8-2
CATALOG M1000MW, TABLE FOR USE WITH ALL MONITORS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Operator Workspace Table for Color Monitor	45 x 54 x 37 (1143 x 1372 x 940)	180 (82)	box

TABLE 8-3
CATALOG M1000NT, LCD COLOR MONITOR

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Operator Workspace Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid

TABLE 8-4
CATALOG M1033K, INDIGO TO OCTANE UPGRADE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Upgrade Kit	20 x 20 x 20 (508 x 508 x 508)	37 (17)	box
System Computer	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Octane Computer	22 x 30 x 24 (559 x 762 x 610)	50 (23)	box on skid

TABLE 8-5
CATALOG M1033MD, LX MNS KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key for 8X/LX	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
LX MNSpectro Magnet Enclosure Kit for HI/LO Carriage Covers	24 x 24 x 24 (610 x 610 x 610)	20 (9)	box
UCERD to TNS and I/F Panel Cable	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
Heliac Cable	36 x 36 x 4 (914 x 914 x 102)	20 (9)	box
MNS Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	430 (195)	on cabinet casters, wrapped with plastic
MNS Interconnect Cables	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-6
CATALOG M1060JW, SHIELD/CRYO COOLER COMPRESSOR

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Shield/Cryo Cooler Compressor Cabinet	26 x 28 x 42 (660 x 711 x 1067)	240 (109)	skid with box cover

TABLE 8-7
CATALOG M1090RW, 512 MBYTE BAM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Memory Board	16 x 16 x 5 (406 x 406 x 127)	5 (2)	box

TABLE 8-8
CATALOG M1090TL, PATIENT TABLE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Patient Table	95 x 36 x 50 (2413 x 914 x 1270)	670 (305)	pallet

TABLE 8-9
CATALOG M1090WG, FAST RECEIVER KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Fast Receiver Kit	26 x 26 x 12 (660 x 660 x 305)	35 (16)	box
Technical Publication & Software	12 x 12 x 4.5 (305 x 305 x 114)	10 (5)	box

TABLE 8-10
CATALOG M1090WJ, RECEIVER "0" KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Receiver 0 Kit	26 x 26 x 4 (660 x 660 x 102)	25 (11)	box

TABLE 8-11
CATALOG M1090WL, LOW PROFILE CARRIAGE COVER KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Low Profile Carriage Cover	18 x 30 x 14 (457 x 762 x 356)	50 (23)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-12
CATALOG M3000RA, REFLEX 100 WITH 512 BAM

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
ReFlex 100 with 512 BAM	16 x 16 x 5 (406 x 406 x 127)	5 (2)	box

TABLE 8-13
CATALOG M3000TA, TWINSPEED UPGRADE SYSTEM ELECTRONICS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Patient Table	95 x 36 x 50 (2413 x 914 x 1270)	670 (305)	pallet
Ship Collector	18 x 30 x 24 (457 x 762 x 610)	50 (23)	box
Non-magnetic chair	36 x 24 x 38 (914 x 610 x 965)	50 (23)	box
Octane Computer	24 x 36 x 23 (610 x 914 x 584)	30 (14)	box
Head Coil Assembly	24 x 24 x 24 (610 x 610 x 610)	35 (16)	box
Phantom Set	16 x 16 x 16 (406 x 406 x 406)	20 (9)	box
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 x 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Twin Accessory Cabinet	23 x 39 x 50 (584 x 991 x 1270)	600 (272)	on cabinet casters, wrapped with plastic

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-14
CATALOG M3043TA, SITE COLLECTOR KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Head & Body Transmit Cable Kit	36 x 36 x 6 (914 x 914 x 152)	20 (9)	box
Power Cables Kit	17 x 26 x 18 (432 x 660 x 457)	100 (45)	box
Signal Cables Kit	40 x 40 x 46 (1016 x 1016 x 1168)	100 (45)	4 boxes on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Gradient Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid
Gradient Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Vacuum Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Cover Kit	30 x 66 x 7 (762 x 1676 x 178)	50 (23)	box
Cryo Monitor Support Bracket Kit	30 x 30 x 5 (762 x 762 x 127)	20 (9)	box
Left & Right Trim Cover Assembly	37 x 52 x 24 (940 x 1321 x 610)	125 (57)	box on skid
Blower Assembly	21 x 15 x 27 (533 x 381 x 686)	50 (23)	box
Pedstal Cover Kit	30 x 45 x 12 (762 x 1143 x 305)	50 (23)	box
Top Front & Top Rear Cover Kits	55 x 45 x 40 (1397 x 1143 x 1016)	150 (68)	2 boxes strapped to skid
Blower Hose Kit	44 x 44 x 10 (1118 x 1118 x 254)	25 (11)	box
Rear Pedestal Assembly	23 x 49 x 36 (584 x 1245 x 914)	100 (45)	box on skid
PAC/SRI Platform Assembly, PAC I/F Assembly	18 x 45 x 12 (457 x 1143 x 305)	40 (18)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-15
CATALOG M3044TA, SITE COLLECTOR KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Head & Body Transmit Cable Kit	36 x 36 x 6 (914 x 914 x 152)	20 (9)	box
Power Cables Kit	17 x 26 x 18 (432 x 660 x 457)	100 (45)	box
Signal Cables Kit	40 x 40 x 46 (1016 x 1016 x 1168)	100 (45)	4 boxes on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Gradient Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid
Gradient Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Vacuum Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Cover Kit	30 x 66 x 7 (762 x 1676 x 178)	50 (23)	box
Cryo Monitor Support Bracket Kit	30 x 30 x 5 (762 x 762 x 127)	20 (9)	box
Left & Right Trim Cover Assembly	37 x 52 x 24 (940 x 1321 x 610)	125 (57)	box on skid
Blower Assembly	21 x 15 x 27 (533 x 381 x 686)	50 (23)	box
Pedstal Cover Kit	30 x 45 x 12 (762 x 1143 x 305)	50 (23)	box
Top Front & Top Rear Cover Kits	55 x 45 x 40 (1397 x 1143 x 1016)	150 (68)	2 boxes strapped to skid
Blower Hose Kit	44 x 44 x 10 (1118 x 1118 x 254)	25 (11)	box
Rear Pedestal Assembly	23 x 49 x 36 (584 x 1245 x 914)	100 (45)	box on skid
PAC/SRI Platform Assembly, PAC I/F Assembly	18 x 45 x 12 (457 x 1143 x 305)	40 (18)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-16
CATALOG M3060TA, 1.5T MAGNET WITH TWINSPEED ENCLOSURE

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
LCC Magnet with cryogenics, partial Quiet Technology Enclosure installed	93 x 144 x 107 (2362 x 3658 x 2718)	See Note 1	Domestic – Tarped International – crate/pallet
Magnet Accessory Equipment	48 x 48 x 28 (1219 x 1219 x 711)	400 (182)	crate
1 Approximate magnet shipping weight (includes packaging material) and weight of magnet with cryogenics, TRM Gradient & RF coils, Enclosure parts installed on magnet, lifting beams, crate, and skid (i.e. minus packaging material): 12,195 lbs (5532 kg). International shipments must add shipping crate/pallet of 2,200 lbs (998 kg).			

TABLE 8-17
CATALOG M3063TA, T/R SITE COLLECTOR KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 20 x 12 (305 x 508 x 305)	40 (18)	box
Penetration Panel	22 x 40 x 25 (559 x 1016 x 635)	70 (32)	box
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Head & Body Transmit Cable Kit	36 x 36 x 6 (914 x 914 x 152)	20 (9)	box
Power Cables Kit	17 x 26 x 18 (432 x 660 x 457)	100 (45)	box
Signal Cables Kit	40 x 40 x 46 (1016 x 1016 x 1168)	100 (45)	4 boxes on skid
Fiber Optic Cables Kit	27.5 x 31 x 33 (699 x 787 x 838)	50 (23)	box
Gradient Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid
Gradient Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Vacuum Hose Assembly	24 x 12 x 12 (610 x 305 x 305)	30 (14)	box
Cover Kit	30 x 66 x 7 (762 x 1676 x 178)	50 (23)	box
Cryo Monitor Support Bracket Kit	30 x 30 x 5 (762 x 762 x 127)	20 (9)	box
Left & Right Trim Cover Assembly	37 x 52 x 24 (940 x 1321 x 610)	125 (57)	box on skid
Blower Assembly	21 x 15 x 27 (533 x 381 x 686)	50 (23)	box
Pedestal Cover Kit	30 x 45 x 12 (762 x 1143 x 305)	50 (23)	box
Top Front & Top Rear Cover Kits	55 x 45 x 40 (1397 x 1143 x 1016)	150 (68)	2 boxes strapped to skid
Blower Hose Kit	44 x 44 x 10 (1118 x 1118 x 254)	25 (11)	box
Rear Pedestal Assembly	23 x 49 x 36 (584 x 1245 x 914)	100 (45)	box on skid
PAC/SRI Platform Assembly, PAC I/F Assembly	18 x 45 x 12 (457 x 1143 x 305)	40 (18)	box

TABLE 8-18
CATALOG M3080TA, 2ND ORDER RESISTIVE SHIM KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
2nd Order Resistive Shim Kit	36 x 24 x 18 (914 x 610 x 457)	50 (23)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-19
CATALOG M3088TA, OUTDOOR AIR COOLED (DUAL LOOP) TSCC

MR COMPONENT	APPROXIMATE W x D x H	APPROXIMATE WEIGHT	METHOD OF SHIPMENT
TSCC unit	52 x 82 x 75 (1321 x 2083 x 1905)	1320 (599)	crate/skid
Remote Control Panel	20 x 10 x 20 (51 x 25 x 51)	33 (15)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	4 boxes
Ship loose items	32 x 32 x 32 (813 x 813 x 813)	50 (23)	box

TABLE 8-20
CATALOG M3088TE, INDOOR WATER COOLED (SINGLE LOOP) TGWC

MR COMPONENT	APPROXIMATE W x D x H	APPROXIMATE WEIGHT	METHOD OF SHIPMENT
TGWC Indoor Water Cooled	38 x 24 x 27 (97 x 61 x 69)	200 (91)	box on pallet
Remote Control Panel	12 x 12 x 12 (305 x 305 x 305)	20 (9)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	2 boxes
Ship loose items	12 x 12 x 12 (305 x 305 x 305)	30 (14)	box
Ship loose items	26 x 6 x 6 (660 x 152 x 152)	15 (7)	box

TABLE 8-21
CATALOG M3088TF, OUTDOOR AIR COOLED (SINGLE LOOP) TGWC

MR COMPONENT	APPROXIMATE W x D x H	APPROXIMATE WEIGHT	METHOD OF SHIPMENT
TGWC unit	55 x 40 x 82 (141 x 102 x 208)	918 (417)	crate/skid
Remote Control Panel	13 x 7 x 15 (330 x 16 x 37)	20 (9)	box
DowFrost HD glycol & ship loose items	24 x 24 x 18 (610 x 610 x 457)	50 (23)	2 boxes

TABLE 8-22
CATALOG M3088TG, INDOOR AIR COOLED (SINGLE LOOP) TGWC

MR COMPONENT	APPROXIMATE W x D x H	APPROXIMATE WEIGHT	METHOD OF SHIPMENT
TGWC Indoor Air Cooled	34 x 40 x 76 (59 x 76 x 186)	918 (417)	crate/skid
DowFrost HD glycol & ship loose items	24 x 24 x 12 (610 x 610 x 305)	50 (23)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-23
CATALOG M3090AH, HORIZON TO TWINSPEED UPGRADE SYSTEM ELECTRONICS

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Ship Collector	18 x 30 x 24 (457 x 762 x 610)	50 (23)	box
Non-magnetic chair	36 x 24 x 38 (914 x 610 x 965)	50 (23)	box
Octane Computer	24 x 36 x 23 (610 x 914 x 584)	30 (14)	box
System Control Cabinet	24 x 33 x 78 (610 x 838 x 1981)	494 (225)	on cabinet casters, wrapped with plastic
Operator Workspace Equipment	24 x 36 x 23 (610 x 914 x 584)	60 (27)	box
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-24
CATALOG M3090BB, ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PDU CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Power Cables Kit	24 x 24 x 18 (610 x 610 x 457)	50 (23)	box
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic

TABLE 8-25
CATALOG M3090BC, 1.5T ACGD UPGRADE FOR SYSTEM WITH SGD GRADIENT CABINET & RF/PENETRATION CABINET

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Power Cables Kit	24 x 24 x 18 (610 x 610 x 457)	50 (23)	box
Dual Sensor Upgrade Kit	14 x 10 x 10 (356 x 254 x 254)	20 (9)	box
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	1810 (823)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
SRF Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	505 (230)	on cabinet casters, wrapped with plastic
Gradient Cables	40 x 40 x 18 (1016 x 1016 x 457)	620 (282)	box on skid

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-26
CATALOG M3090BE, ECHOSPEED PLUS PERFORMANCE OPTION

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Software & Install document	9 x 12 x 1 (229 x 305 x 25)	2 (1)	envelope

TABLE 8-27
CATALOG M3090DB, 5.5 MNS UPGRADE KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Multi Nuclear Spectro MOD Option Key	12 x 12 x 4 (305 x 305 x 102)	3 (1.5)	box
1.5T RF/Pen Style Cabinet MNSpectro Upgrade Kit	24 x 24 x 18 (610 x 610 x 457)	30 (14)	box
1.5T MNS Phosphorus Flex Coil Kit	18 x 24 x 6 (457 x 610 x 152)	5 (2.5)	box
MRS Phantom	16 x 16 x 16 (406 x 406 x 406)	10 (5)	box
UCERD to TNS and I/F Panel Cable	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box
MNS Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	430 (195)	on cabinet casters, wrapped with plastic
MNS Interconnect Cables	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box

TABLE 8-28
CATALOG M3090DC, LXSPECTRO PRE-RF/PDU TO AGCD MNS UPGRADE KIT

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
MNS Cabinet	24 x 42 x 71 (610 x 1067 x 1803)	430 (195)	on cabinet casters, wrapped with plastic
MNS Interconnect Cables	16 x 16 x 6 (406 x 406 x 152)	8 (4)	box

TABLE 8-29
CATALOG M3090JA, IPG2 BOARD

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
IPG2 Board	16 x 16 x 5 (406 x 406 x 127)	5 (2)	box

8-2 UPGRADE CATALOGS SHIPPING INFORMATION (Continued)

TABLE 8-30
CATALOG M3090TA, TWIN UPGRADE FOR LCC HORIZON FIXED & T/R

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Technical Publications	12 x 21 x 12 (305 x 533 x 305)	20 (9)	box
Left/right Trim Cover Assembly	37 x 52 x 22 (940 x 1321 x 102)	125 (57)	box on skid
Rear Pedestal with Split Bridge Rear Upgrade Assembly	23 x 49 x 32 (584 x 1245 x 813)	65 (30)	box
Pedstal Cover Kit	30 x 45 x 12 (762 x 1143 x 305)	50 (23)	box
Bridge	24 x 161 x 10 (610 x 4090 x 254)	125 (57)	box
Dock Control Box, Bore Light Assembly, Cable	18 x 45 x 11 (457 x 1143 x 279)	30 (14)	box
Top Front & Top Rear Cover Kits	55 x 45 x 40 (1397 x 1143 x 1016)	150 (68)	2 boxes strapped to skid
Dock Cover Kit	30 x 14 x 8 (762 x 356 x 203)	50 (23)	box
PAC/SRI Platform Assembly, PAC I/F Assembly	18 x 45 x 12 (457 x 1143 x 305)	40 (18)	box
Water Chiller power cable/circuit breaker	16 x 16 x 16 (406 x 406 x 406)	50 (23)	box
Front Cover Cart Assembly for return of old cover/endbell	36 x 106 x 57 (914 x 2693 x 1448)	100 (45)	cart
Rear Cover Cart Assembly for return of old cover/endbell	36 x 106 x 57 (914 x 2693 x 1448)	100 (45)	cart
Container for draining coil coolant	18x 18 x 24 (457 x 457 x 610)	60 (27)	box

TABLE 8-31
CATALOG M3085TA, 1.5T TRM BODY COIL

MR COMPONENT	APPROXIMATE W x D x H in. (mm)	APPROXIMATE WEIGHT lbs (kg)	METHOD OF SHIPMENT
Body Coil Assembly	35 x 94.5 x 55.88 (889 x 2400 x 1420)	3280 (1491)	Coil on cradle mated to installation cart (See Note 1)
Note: 1 Systems With LCC Magnet Currently Installed Systems with an LCC Magnet currently installed will require the Body Coil Insertion Tool to upgrade from BRM or CRM to the new TRM coil. The Body Coil Insertion Tool requires the TRM coil to be installed on the aluminum cradle and be mated onto the aluminum coil cart. The new TRM Coil will be shipped on a wooden disposable cradle, see Illustration 2-1. The TRM Coil must be transferred to an aluminum cradle utilizing a GE Service provided coil lifting bracket and a customer rigger provided forklift or crane/hoist rated for 4,000 lbs (1814 kg), refer to Section 10-3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT for additional rigger/customer supplied equipment. Systems Installing LCC Magnet Upgrade Concurrent With TwinSpeed Upgrade Systems installing an LCC Magnet Upgrade at the same time as the TwinSpeed Upgrade Systems will receive (M3060TA) LCC Magnet with TwinSpeed Enclosure with the TRM Assembly (M3085TA) installed inside the magnet bore and enclosure.			

SECTION 9 – PREINSTALLATION CHECKLIST

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9-1 INTRODUCTION

“Preinstallation” refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work.

The purpose of this section is to outline key areas of concern in the preparation of a customer's site for Signa Horizon upgrade. It is intended as a guide for GE's Installation Specialists and/or Installation Leaders when making on-site inspections during the installation planning phase of the upgrade. Note site inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE's Installation Specialists and/or Installation Leaders may observe that the Requested Arrival Date (RAD) for the upgrade equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the RAD accordingly.

All work must be in compliance with national and local safety codes.

9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY

The following items must be completed by the Installation Specialist or Site Field Engineer prior to upgrade equipment delivery. An on-site inspection by a GE Representative to confirm site readiness must be completed **at least 8 weeks prior to the Scheduled Arrival Date (SAD)** of the upgrade equipment.

- ☐ 1. Does the system to be upgraded have an LCC Magnet installed?
If yes, the magnet does not need to be upgraded.
If no, has site been planned to meet the specifications for an LCC Magnet, refer to preceding sections of this document?
- ☐ 2. Does system being upgraded have a Full-size PDU and have arrangements been made for old PDU removal? Refer to **Section 2-6-1 Power Distribution Unit (PDU)**.
- ☐ 3. Has a site equipment plan been completed, reviewed with the customer, and finalized?
- ☐ 4. Has the vibration tests/study been completed per **Section 4-13, VIBRATION**? It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits.
- ☐ 5. If there were vibration issues; has local GE Field Service verified, with the vibration consultant present, the elimination/reduction of all identified sources?
- ☐ 6. Has a cable routing plan been completed? Are all cable lengths sufficient? If lengths are not sufficient, a Special Order Inquiry (SOI) will be required. The GE Installation Specialist or Site Field Engineer must submit a SOI Request Form at least 8 weeks prior to SAD to allow for proper processing of the request.
The SOI Request Form is available on the Web at:
 <http://medquickplace02.ge.com/QuickPlace/mr/PageLibrary862569F30075E524.nsf?OpenDatabase>
- ☐ 7. Does incoming power available at PDU meet the requirements in **Section 5 – POWER REQUIREMENTS**, of this document? This includes correctly sized facility transformer, feeder wire size, Main Disconnect Panel (MDP) wire sizes and circuit breaker rating, and system ground requirements.

9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY (Continued)

- ☐ 8. If GE TwinSpeed Main Disconnect Panel (MDP) is not being used then are plans in place for customer provided MDP to be installed and operational?
- ☐ 9. Have local and national codes been check to determine if an electrician will be required to provide and/or install new feeder wires to the PDU if ACGD Upgrade is planned? If required then have arrangements been made for an electrician?
- ☐ 10. Does the site's HVAC (heating, ventilation, and air conditioning) system meet requirements in Section **4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS** and Section **4-3 AIR COOLING REQUIREMENTS** of this document? If these specifications are not strictly adhered to, failure of the ACGD modules in the ACGD/PDU Cabinet may occur.
- ☐ 11. Has the site's vibration environment been assessed including system stability tests performed as defined in Section **4-13 VIBRATION**?
- ☐ 12. Has delivery route been defined for equipment especially enclosure covers(no magnet upgrade) or LCC Magnet with covers if magnet is being upgraded (refer to **Section 2-3, MINIMUM DOOR SIZES & DELIVERY ROUTE WEIGHT CAPACITY**) including plans for protecting flooring for heavy equipment & carts (ie. Gradient Body Coil & installation cart)?
- ☐ 13. Have all upgrade exceptions, such as special cabling and options unique to the site including customer equipment, been identified and clearly tagged **DO NOT REMOVE** ?
- ☐ 14. For system with Spectroscopy option: Has the site Field Engineer removed the Spectral Module from the Exciter Board in the Systems Cabinet and is the module available at site for re-installation during upgrade.
- ☐ 15. Has a holding area for old equipment and installation debris been agreed upon with the customer? Making provision for a holding area will allow for efficient unloading and reloading of the delivery vehicle.
- ☐ 16. Has RF Shielded Room been checked to meet specifications in **Section 7 – RF SHIELDED ROOM** ?
- ☐ 17. If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade and the RF Shield is opened, modified, upgraded, etc. then are there plans to recertify (test) the RF Shielded Room integrity after magnet installation to meet or exceed the new system RF Shielded Room specifications. Refer to **Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-1 RF SHIELDED ROOM SPECIFICATION**.
- ☐ 18. If the upgrade requires installation of the TRM coil into the magnet at site then have arrangements been made with riggers and lifting equipment equipment to transfer TRM from the shipping cradle to the installation cradle/cart? Refer to **Section 2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements**.
- ☐ 19. If the upgrade requires installation of the TRM coil into the magnet at site then have arrangements been made for delivery of the following tools?
 - Empty cart/cradle (2134810 / 2134810-2)
 - BRM Insertion Tool (2164744-3)
 - Ring Alignment Tool (2299496)
 - Coil lift brackets (TBD)
- ☐ 20. If an upgrade to an LCC Magnet is being performed then the cryogenic venting must comply with the specifications in **Direction 2294003 Signa 1.5T TwinSpeed Pre-Installation Section 4-9 CRYOGENIC VENTING**. The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customer's record file stating compliance to this section.

9-3 GENERAL PREINSTALLATION REMINDERS

- ☐ 1. Has the customer's architect/contractor fully reviewed final site installation drawings using the guidelines provided by MR Siting and Shielding (MRSS)?
- ☐ 2. Have vehicle parking arrangements been made for installation personnel?
- ☐ 3. Is temporary storage space available for use during installation?
- ☐ 4. What is hospital smoking policy?
- ☐ 5. Are a first aid kit and non-ferrous fire extinguishers available at site?
- ☐ 6. Have facility arrangements been made for refuse disposal during installation?

9-4 SAFETY**WARNING!**

THE FOLLOWING SAFETY ITEMS MUST BE ADHERED TO PRIOR TO MOVING THE MAGNET FROM THE TRUCK INTO THE ROOM. THE MAGNET WILL BE EXHAUSTING HELIUM GAS DURING THIS TIME. GASEOUS HELIUM IS AN INVISIBLE, ODORLESS GAS THAT CAN CAUSE ASPHYXIATION WHEN DEPLETING THE AMBIENT OXYGEN SUPPLY. HELIUM GAS IS LIGHTER THAN AIR AND WILL RISE TO THE CEILING.

- ☐ 1. Have all of the required room ventilation items been installed and tested to make sure sufficient ventilation is available? Is exhaust fan and fan control installed and functional? The customer is to supply a copy of the Exhaust Fan Test Report to GE Medical Systems for the customers record file. Refer to **Section 4-8 ROOM VENTILATION** for specific requirements.
- ☐ 2. Are a first aid kit and non-ferrous fire extinguishers available at site?
- ☐ 3. Are removable chain or strap available for securing cryogen gas cylinders in an upright position to prevent the cylinders from falling which may cause injury or damage?
- ☐ 4. Are functioning telephones as defined in **Section 2-6-6 System Monitoring & Support Connectivity Requirements** available at site for duration of installation? Phone lines are needed to dial out in case of an emergency.
- ☐ 5. Has the cryogenic vent been installed complete from the Magnet Room RF shield penetration to the final exit and been inspected to final exit outside of the building? The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customers record file. Refer to **Section 4-9, CRYOGENIC VENTING**.
- ☐ 6. Have plans been made to connect the cryogenic vent to the magnet after magnet installation in the room? Refer to **Section 4-9, CRYOGENIC VENTING**.
- ☐ 7. Is there adequate ventilation for helium dewar storage area. Refer to **Section 4-8, ROOM VENTILATION**.

9-5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM

The following items must be completed prior to magnet delivery. A site inspection by GE Service Representative must be completed prior to magnet delivery to ensure site readiness.

- ☐ 1. Has Main Disconnect Panel been installed, electrician wiring complete, MDP operational with power available? Refer to **Table 5-1 REQUIRED CUSTOMER POWER**.
- ☐ 2. **Is 24 hour/day, 7 days/week power and air cooling available for Main Disconnect Panel (MDP) for powering the TSCC or customer water cooling equipment for Shield /Cryo Cooler (if TGWC to be used), Shield/Cryo Cooler Compressor cabinet, and Magnet Monitor equipment?**
- ☐ 3. **If MR system configuration is utilizing water cooled TSCC then is 24 hour/day, 7 days/week water cooling available?**
- ☐ 4. Are all walls and ceiling in magnet room essentially complete except for removable section?
- ☐ 5. Has a clear route to the magnet room been defined for magnet installation (refer to **Section 2-4, MINIMUM DOOR SIZES**)?
- ☐ 6. Is a secure space available to store equipment on site?
- ☐ 7. Have arrangements been made for the use of special rigging equipment for moving the magnet into the magnet room?
- ☐ 8. Has the raised floor been installed in the equipment/computer room in preparation for the superconducting and resistive power supplies?
- ☐ 9. Has work in the magnet room been completed or suspended and the magnet room closed off to provide a dust-free, closed environment?
- ☐ 10. Has all equipment been removed from the magnet room to allow space for the magnet with rigging equipment and cryogen dewars?
- ☐ 11. Are equipment anchors installed in floor as required for magnet room equipment? Refer to **Section 7-7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**.
- ☐ 12. For all magnet room anchors, has pull test been performed and installation torque value defined? Test results recorded by RF Shield Room Vendor and information forwarded to GE Medical Systems for the customers record file? Refer to **Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-7-1 Clamping Force And Pull Test**.
- ☐ 13. For all magnet room anchors, has ground impedance test been performed? Test results recorded by RF Shield Room Vendor and documentation given to GE Medical Systems for the customers record file? Refer to **Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-7-3 Electrical Isolation**.

9-6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL

- ☐ 1. Have arrangements been made for cryogen delivery and storage?
- ☐ 2. Does a clear path exist to magnet room for delivery of cryogens?
- ☐ 3. Is the magnet room adequately vented to allow exhaust of cryogen boil-off?
- ☐ 4. Is the magnet room clear to provide room for cryogen dewars during filling?

9-7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET RAMP-UP

- ☐ 1. Is the cryogenic vent connected to the magnet?
- ☐ 2. Has the magnet room been completely closed (removable section closed up and sealed)?
- ☐ 3. Has the magnet room been tested to ensure that the RF shielding meets the electrical isolation and the 100dB requirement for electric/plane waves and the 90dB requirement for magnetic waves refer to **Section 7-1 RF SHIELDED ROOM INTRODUCTION**? The customer is to supply a copy of RF shielded room vendor test reports.
- ☐ 4. Have all ferrous metal objects been removed from the magnet room?
- ☐ 3. Have adequate signs (Safety and Exclusion Zones) been posted to warn personnel about dangers of magnetic field?
- ☐ 4. Have hospital personnel been informed of magnet safety precautions and procedures?
- ☐ 7. Has power been connected from the Main Disconnect Panel to Power Distribution Unit?
- ☐ 8. Has the penetration panel been installed?
- ☐ 9. Is all contractor construction work completed?
- ☐ 10. Has all contractor equipment that could affect shimming been removed from within the three gauss zone?
- ☐ 11. Have precautions been taken to prevent movement of large metal objects within the three gauss zone?
- ☐ 12. Has outside emergency personnel (e.g. fire department and other outside ambulance or emergency personnel been informed of unique characteristics (e.g. strong magnetic field, cryogenics, etc.) of magnet and correct precautions to take in event of emergencies?

SECTION 10 – TOOLS AND TEST EQUIPMENT

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10-1 INTRODUCTION

The following tables lists the tools and test equipment needed to install a Signa 1.5T TwinSpeed and calibrate the system.

10-2 MOVING METAL MEASUREMENT EQUIPMENT

TABLE 10-1
MOVING METAL MEASUREMENT EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Magnetometer sensitive to 50 μ gauss Flux gate or equivalent Example: Barrington Instruments Ltd. Model Mag-03MC 500 μ T
2	—	Spectrum analyzer capable of moving metal measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer.
3	—	Accelerometer capable of measuring from 0.2 Hz beyond 250 Hz.

10-3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT**Note**

Table 10-2 contains an overview of equipment for initial planning purposes, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation Manual* for details of equipment and requirements.

TABLE 10-2
RIGGER/CUSTOMER SUPPLIED EQUIPMENT

SUPPLIED BY	ITEM	DESCRIPTION
Rigger	1	Crane for removing magnet crate from delivery truck: 12 ton (10,909 kg)
	2	Motorized tow vehicle for pushing/pulling magnet when it is on dollies (e.g. an electric fork lift).
	3	Steel floor plates (8) to cover floors while transporting magnet, 36 in. x 12 in. x 0.25 in. (914 mm x 305 mm x 6.4 mm).
	4	Wood blocks, assorted sizes.
	5	Jacks and accessories for lifting magnet: 20,000 lbs (9091 kg) total capacity
	6	Spreader Beam
	7	For LCC magnet: Lifting straps which meet specifications in <i>Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery/Installation Manual</i>
	8	For TRM Installation: The following equipment to meet specifications in <i>Direction 2304881 Signa TwinSpeed Upgrade Pre-Install</i> - Forklift or crane/hoist rated for 4,000 lbs (1814 kg) - spreader beam - lifting straps
Customer	9	Equipment for off loading electronics and other miscellaneous components. (e.g. fork lift, hand trucks, straps, etc.)
	10	Panel lifters for computer flooring: Smooth floor Standard carpeted floor Level loop carpet floor Indicon Industries PL2DC; Indicon Industries PL30P; Indicon Industries SCLV1.

10-4 CRYOGENIC EQUIPMENT

TABLE 10-3
CRYOGENIC EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	46-306734G1	Low Pressure Regulator Kit (Non-magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2)
2	—	Liquid helium in non-magnetic dewars is needed for refilling the magnet (See Note 2). Refer to appropriate magnet manual for helium volume.
3	46-306717G1	Non-magnetic gaseous helium cylinder cart (See Note 2)
4	46-294705G1	Universal Fill Line Kit (See Note 2)
5	46-294511P1	250 Liter Dewar Stinger (See Note 2)
6	46-294511P2	500 Liter Dewar Stinger (See Note 2)
7	46-294512P1 (12 ft) 46-294512P2 (8 ft)	Transfer Line (See Note 2)
8	46-282336P2	Dewar stinger assembly (High Efficiency) (See Note 2)
9	46-306812G1	Dewar tube "Thumper" tool (See Note 2)
10	2253802	Burst Disk and Gasket Assembly for LCC Magnet Helium Vessel (See Note 1)
11	46-271136G1	Dewar Adapting Kit including O-ring Kit for Dewar Adapting kit (46-271135P9) (See Note 2)
12	46-271137G1	Cryogen Safety Shield Kit (Includes gloves, face shield, and safety glasses.) (See Note 2)
13	46-294804G1	Non-magnetic Aeroquip wrench Set (See Note 2) includes the following: • 46-294800G1 case for bronze wrench kit • 46-294805P1 bronze 1 ⁵/₈ in. open end wrench • 46-294805P2 bronze 1 ³/₈ in. open end wrench • 46-294805P3 bronze 1 ³/₁₆ in. open end wrench • 46-294805P4 bronze 1 ¹/₈ in. open end wrench • 46-294805P5 bronze 1 in. open end wrench
14	46-281088G3	Shield Cooler Installation / Maintenance Kit (See Note 2)
15	46-265273G1	Liquid Helium Level Meter Kit (See Note 2)
16	46-306781G1	Helium Mechanical Gas Flowmeter (See Note 2)
17	46-301477G1	Cryogenic Thermometer, Lakeshore 208 Cryogenic Thermometer Kit or alternate Diode Temperature current source 46-317543G1 (See Note 2)
18	46-252210P1	3 Inch Valve Operator (vacuum break tool) (See Note 1)
19	46-294872G2	SAV-CON / Instrumentation Lead Service Kit (See Note 2)
20	46-318784G2	Shield Cooler Test Kit (See Note 2)
21	46-318696G1	Water Tee for Shield Cooler Water Samples access (See Note 2)
22	46-294052G1	Water Flow Meter Kit (for checking flow of water to Shield Cooler Compressor) (See Note 2)
23	2171219	RUO Temperature Monitor for LCC 1.5T & 1.0T Magnet (See Note 2)
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

10-5 INSTALLATION EQUIPMENT

TABLE 10-4
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	46-294060G4	Mapping Fixture for field plotting equipment for Cx/LCC magnet Typically not needed to map Cx/LCC Magnet depending on proximity of steel at site, refer to Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-14-1 Floors
6	2163971	Field Mapping Fixture Upgrade Kit to make kit compatible with Cx/LCC Magnets Used to upgrade 46-294060G3 to 46-294060G4
7	—	Magnet Log Book
8	—	Installation log book
9	—	4 foot or equivalent carpenter level
10	2230707	For LCC Magnet: Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
11	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586-2 Stripping tool replacement blade • 2134587 Cable slicer • 46-282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839-2 3/8 inch terminals
12	46-301450G1	Fiber optic connector repair kit (See Note 2)
13	46-260776G3	Magnet Service Tool 750 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
14	46-260776G4	Magnet Service Tool 1000 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
15	46-260777G3	Shim Service Tool Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 290 lbs (132 kg).
16	46-306763G1	Subminiature-D Connector Removal/Re-termination Repair Kit for Robinson-Nugent Sub-D connectors
17	46-251865G2	Field Plotting Kit – Metro Lab Teslameter
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		
(Continued)		

10-5 INSTALLATION EQUIPMENT (Continued)

TABLE 10-4 (Continued)
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
18	46-260703G3	Magnet Ramping Equipment Kit
19	46-318833G1	Ramp Cable Holder
20	46-294998G1	Ramping Supply and Equipment kit includes the following: • 46-260703G3 Magnet Ramping Equipment Kit (see Item 18) • 46-260776G3 Magnet Service Tool Power Supply Cabinet (see Item 13)
21	46-320273G3 or G4	Non-Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non-Magnetic Tool Kits needed. May substitute both of the following kits: • 46-320273G1 Non-Magnetic Tool Kit – Metric • 46-320273G2 Non-Magnetic Tool Kit – Inch
22	2164744-4	BRM Insertion Tool includes Coil Lift Kit 2326753
23	2144093	Standard Long Room coil cart with cradle for use in 20.44 ft (6.23 m) or longer Magnet Room
24	2144093-2	Short Room coil cart with cradle for use in <20.44 to 19.6 ft (<6.23 to 5.97 m) long Magnet Room
25	2299496	Ring Alignment Tool
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

10-6 RF SURVEY APPARATUS

TABLE 10-5
RF SURVEY APPARATUS

ITEM	GE PART NUMBER	DESCRIPTION
26	46-265169P1	Tuned dipole antenna (two required)
27	46-265169P2	Tripod for dipole antenna (two required)
28	—	0.305 m diameter loop antenna
29	46-255835P2	Spectrum analyzer
30	46-265171P1	RF Preamplifier
31	46-265170P1	RF Linear Power Amplifier

10-7 TEST EQUIPMENT

TABLE 10-6
TEST EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	2284754	Textronic TDS3012 digital scope
2	2284763	Fluke FLK196M scopemeter for medical applications
3	46-194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Textronic 2232
4	46-194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Textronic 2465
5	46-194427P284	Battery operated digital multi-meter, Fluke 87, 4.5 digits with frequency counter and capacitance
6	46-208572P9	Clamp-on Ammeter, 1-200A, AC/DC
7	46-317724P14	Attenuator/ load, 200 Watt, 30 dB attenuator
8	46-317724G2	RF Power Measurement Kit with Attenuator/ load, 200 Watt, 30 dB attenuator 46-317724P14
9	2218826	RPM 1650 Power Analyzer Kit
10	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
11	Catalog E6320DA or 46-194427P144	Densitometer
12	46-306801G1	Bell Gauss Meter Kit
13	46-194427P248	Microguard or ECOS leakage tester
14	46-306797G1	Fogg (ECG) Simulator and Memory Module
15	46-320433P1	Infrared Scanner 0 – 100° C
16	46-317830G1	Fiber Optic Light Meter Kit
17	46-294047G1	Shield Cooler Vacuum Pump Kit
18	46-251867G1	Main Vacuum Pump Down Kit
19	46-252805P2 (10 ft) 46-252805P3 (15 ft)	Nitrogen Transfill line (only needed if magnet is warmer than nitrogen temperture)

10-8 CALIBRATION TOOLS/FIXTURES

TABLE 10-7
CALIBRATION TOOLS/FIXTURES

ITEM	GE PART NUMBER	DESCRIPTION
1	46-307500G1	Longitudinal Drive Force Gauge
2	2133388-6	System Performance Test (SPT) Phantom Set for Systems with Cx/LCC (Active Shield) systems (See Note 1) consisting of the following: • 2125247-10 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027-2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
3	46-287900G3	Head TLT Sphere (46-265826G6), one 100 mm Nickel Chloride sphere (46-317586G1), and Loader (46-287899G1) See Note 1
4	46-255816G1	RF Test Cables Kit
5	46-265434G1	Magnet Rundown Unit Test Box
6	46-307164G4	1.5T / 1.0T GRAFIDY II Kit (See Note 4)
7	2234621	ECMT Kit (See Note 4)
8	46-328021G1	Enmet Oxygen Monitor Calibration consisting of the following: • 1800 PSI cylinder of 20.9% Oxygen in Nitrogen • 1800 PSI cylinder of 17% Oxygen in Nitrogen • Regulators with calibration adapter For use with portable or permanent Oxygen Monitor.
9	2106236	Portable Oxygen Monitor (Connecticut Analytical)
10	2107184	Permanent Oxygen Monitor (Enmet)
11	46-301549G1	TPS RF Test Cable/Adapter Kit
12	46-301927G1	TPS RF Service Interface Kit (See Note 2)
13	46-306712G1	Torque driver kit
14	46-260201P1	N2 Precool Syphon (only needed if magnet is warmer than nitrogen temperture)
15	46-306864G1	Magnet Helium Resistance Box Kit
16	2101360	Power Supply Calibration Kit
17	46-265826G3	Head Signal-to-noise (SNR) sphere (See Note 3)
18	46-265635G4	Body Signal-to-noise (SNR) sphere (See Note 3)
19	46-265635G2	Body T2 sphere (See Note 3)
20	46-328480G1	100 mm Nickel Chloride sphere (46-317586G1) and Universal Phantom Holder (46-328383P1) See Note 2
Note	1	Supplied as part of Signa.
	2	Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract.
	3	Customer may purchase these items.
	4	Either the ECMT Kit or the 1.5T/1.0T GRAFIDY II Kit can be used to calibrate the system.

10-9 TOOL KIT

TABLE 10-8
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Extension cords, with ground conductor
2	—	Power strip, grounded type, with minimum of five outlets
3	—	Soldering iron, pencil type with solder
4	—	Solder sucker
5	—	Assortment of Brady Quick labels
6	—	Micro clip leads
7	—	14 pin and 16 pin DIP clips
8	46-258218P3	Vinyl electrical tape
9	46-258218P4	Copper tape, 3 in. wide
10	46-258218P5	Copper tape, 2 in. wide
11	46-258218P6	Copper tape, 1 in. wide
12	—	Alcohol cleaning solution
13	—	Plastic or aluminum flashlight
14	—	Plastic or aluminum pen light
15	AMP No. 458994-1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183-R	Pin extractor, M-series
17	46-237072P1	Pin extractor, Sub-D
18	46-307307G1	Crimping tool for coax cable and BNC connectors Inserts for RG8, 9, 11, 214 46-255841P103 RG58, 223 46-255841P100 RG59 46-255841P101 RG174 46-255841P102 Die Removal Tool 46-255841P201
19	—	Assorted crimp tools.
20	—	Non-magnetic level
21	—	Non-magnetic tape rule, 12 ft
22	—	Assorted drill bits
23	—	Inspection mirror
24	—	Tap set, standard, and tap handle, T-type
25	—	6 in. rule, standard and metric markings
26	—	Alignment tool (tweaker)
27	—	Hex/alignment tool
28	—	Hemostat, 5 in., curved
29	46-198094P1	Wrist grounding strap
(Continued)		

10-9 TOOL KIT (Continued)

TABLE 10-7 (Continued)
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
30	Xcelite 110CG	Diagonal Cutting Pliers, 4-1/2 in.
31	—	Screw Starter, aluminum or plastic shaft
32	—	Hobby and utility knives
33	—	Spring scales, 0-10 lbs and 0-50 lbs
34	46-313413P1	Extractor for 20 - 100 pin PLCC Chips

APPENDIX A –MR SITE VIBRATION TEST GUIDELINES

A-1 TEST MEASUREMENTS

- Vibration measurements are in the range of 10^{-6} g. Test equipment must have the required sensitivity to these levels.
- All analyses are to be narrowband Fast Fourier Transforms (FFT's) over the frequency bands listed in Table A-1.

TABLE A-1
FREQUENCY BANDS FOR FFT'S

FREQUENCY BAND	FREQUENCY RESOLUTION
0.2 to 50 Hz	$\Delta f = 0.125$ Hz

- Time histories of the vibration must be recorded (i.e. acceleration levels vs. time). The resolution of the time history must be adjusted to clearly capture the transient events. The analyzer set-up will be site dependent.

A-2 EQUIPMENT (SPECTRAL ANALYZER) SET-UP

- Frequency average a minimum of 20 linear averages (do not use peak hold or 1/3 octave analysis)
- Hanning window must be applied to the entire spectra

Spectrum analyzers capable of these measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer. Accelerometers must have the capability to measure from 0.2 Hz beyond 50 Hz. Time histories can be recorded using any of the analyzers mentioned above. Good quality strip chart recorders may also be sufficient. Please note that the equipment mentioned are for example only. It is the responsibility of the Engineering test firm to provide equipment that will follow the test requirements.

A-3 DATA COLLECTION

A-3-1 Ambient Baseline Condition:

All of the measurements defined in Section A-1 and A-2 must be made in a 'quiet' environment. That is, in areas where excessive traffic, subway trains, etc. exists, a vibration measurement must also be made during periods without traffic or during periods of light traffic. Measurements must define the lowest levels of vibration possible at the site.

The source of any steady state vibration whose levels exceed the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-15-3 Specifications For 1.5T LCC (Active Shield) Magnets*** must be identified as to the source of the vibration disturbance. A second measurement should be made with all of the identified contributors powered down if possible. In situations where it is not possible to power down equipment, vibration data must be collected to identify specific source of the vibration concern. The majority of steady state vibration problems can be negated by isolating the vibration source.

A-3-2 Normal (Operating) Condition

All of the measurements listed above must be repeated during periods of 'normal' environmental conditions including the FFT's and time histories. The transient measurements must be provided to define the dynamic disturbances the MR system might be exposed to. This transient disturbance is required for a true assessment of the site.

Transient vibration is difficult to assess and to eliminate. Environments exceeding the 500 micro-g, zero to peak trigger level, requires analysis of the building's dynamic response to the transient event. The building response must remain below the vibration specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-15-3 Specifications For 1.5T LCC (Active Shield) Magnets***. The dynamic response following the transient event must exceed the specifications for sustained periods of time, i.e. for random dynamic response of duration less than a few seconds will have less impact on system performance than response of longer duration.

Building structures excited by transient events exceeding the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-15-3 Specifications For 1.5T LCC (Active Shield) Magnets*** either provide recommendations to reduce levels to meet specifications or an alternative location should be identified. A second vibration measurement should be made to help identify a more stable location.

A-4 PRESENTATION/INTERPRETATION OF RESULTS

The recommended format for site vibration data collection, presentation, and analysis is illustrated in the examples shown in Illustrations A-1 through A-2. Presentation of the data in any other format (linear units only) may result in an incorrect interpretation and diagnosis of the site. Additional data collection or presentation methods is at the option of the vibration testing service.

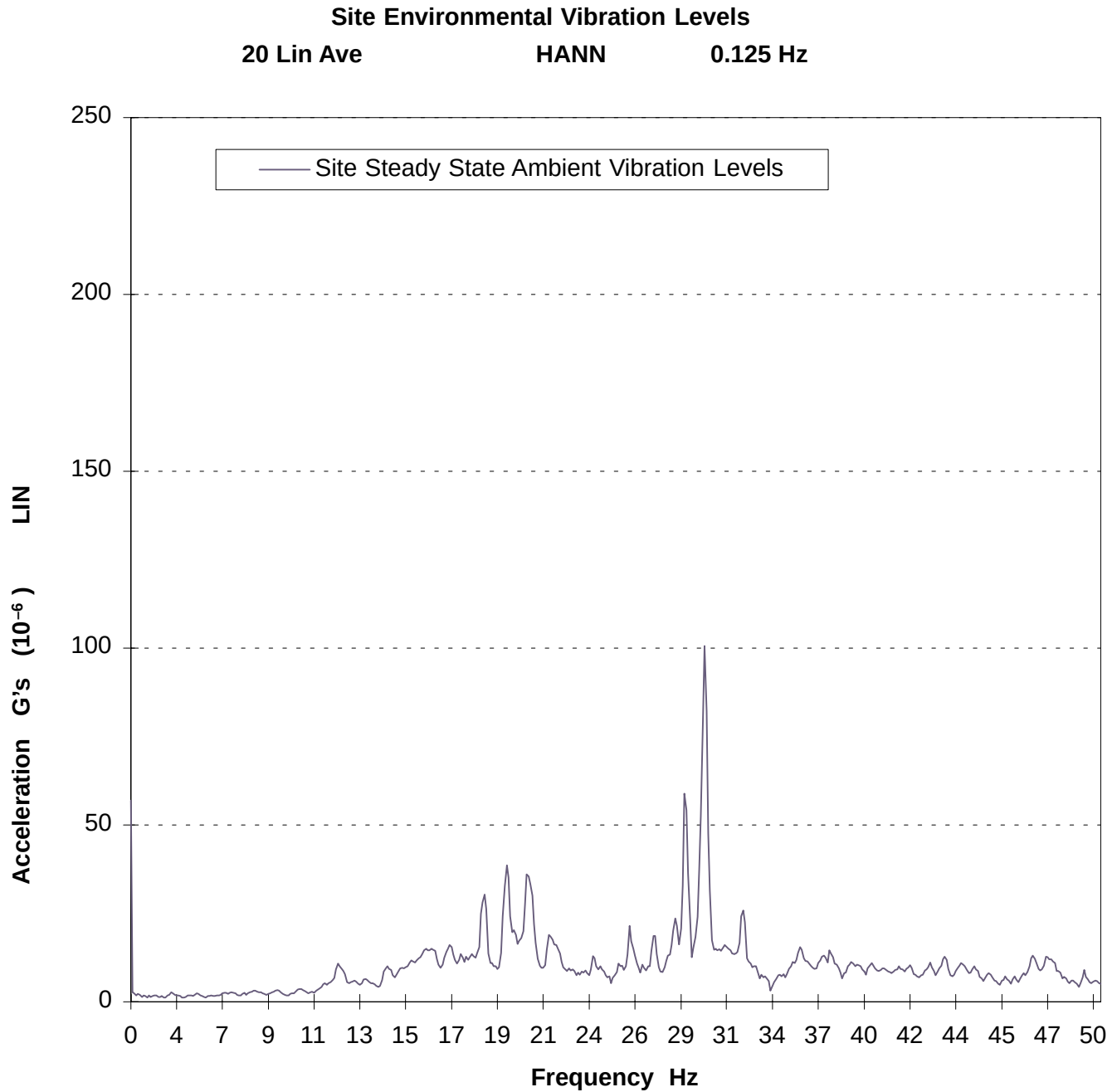
It is the responsibility of the customer's vibration testing service to interpret the results and determine if that site meets GE's specifications. Illustrations A-1 and A-2 are examples provided to assist a test consultant in the use of GE Steady State specifications (vibration specifications above ambient baseline). If the vibration levels are too high, additional data acquisition may be necessary to:

- determine the source of the vibration
- propose a solution to the problem
- find an alternate site location.

Illustrations A-3 and A-4 are examples provided to assist a test consultant in the use of GE Transient specifications. The 500 micro-g, zero to peak trigger level identifies data collection to begin assessment of the site vibration analysis. The response of the transient must be assessed relative to the Steady State vibration specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-15-3 Specifications For 1.5T LCC (Active Shield)***.

Any questions regarding test equipment requirements, test parameters, or general questions should be discussed with your GE Installation Specialist.

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



EXAMPLE SITE ENVIRONMENTAL VIBRATION
ILLUSTRATION A-1

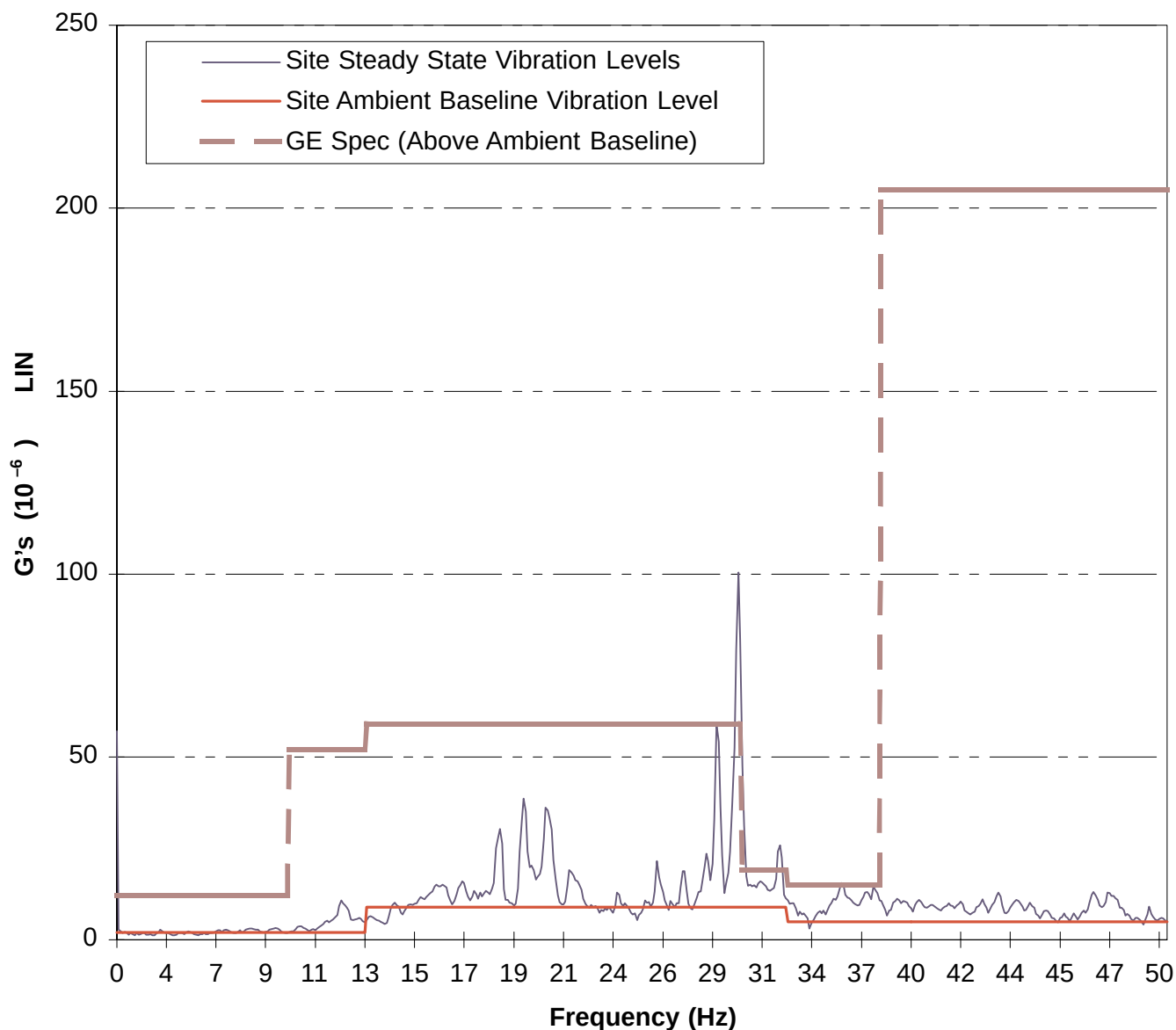
A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)

EXAMPLE: Site Environmental Vibration vs. GE Spec. for 1.5T K4 Magnet

20 Lin Ave

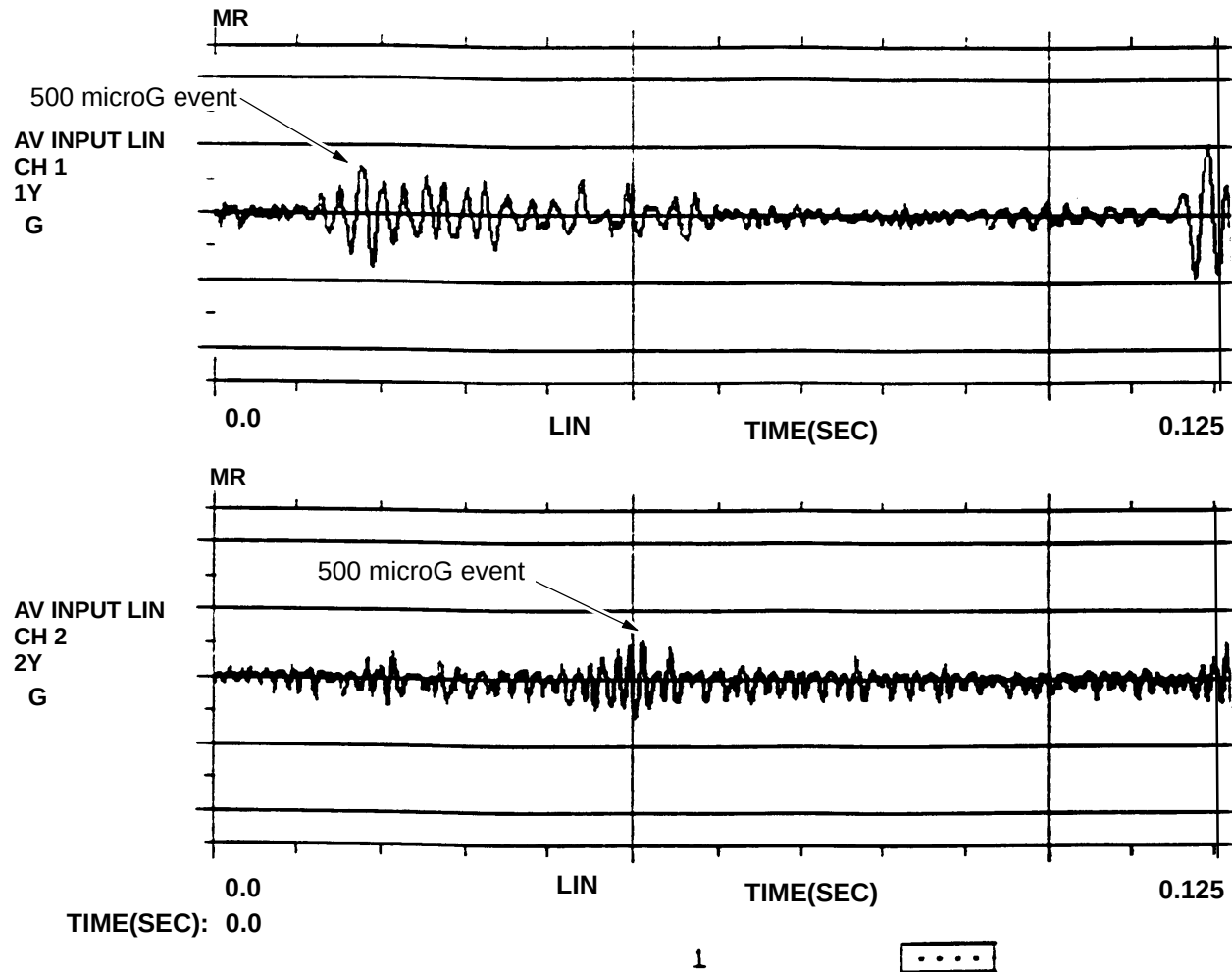
Hann

0.125Hz



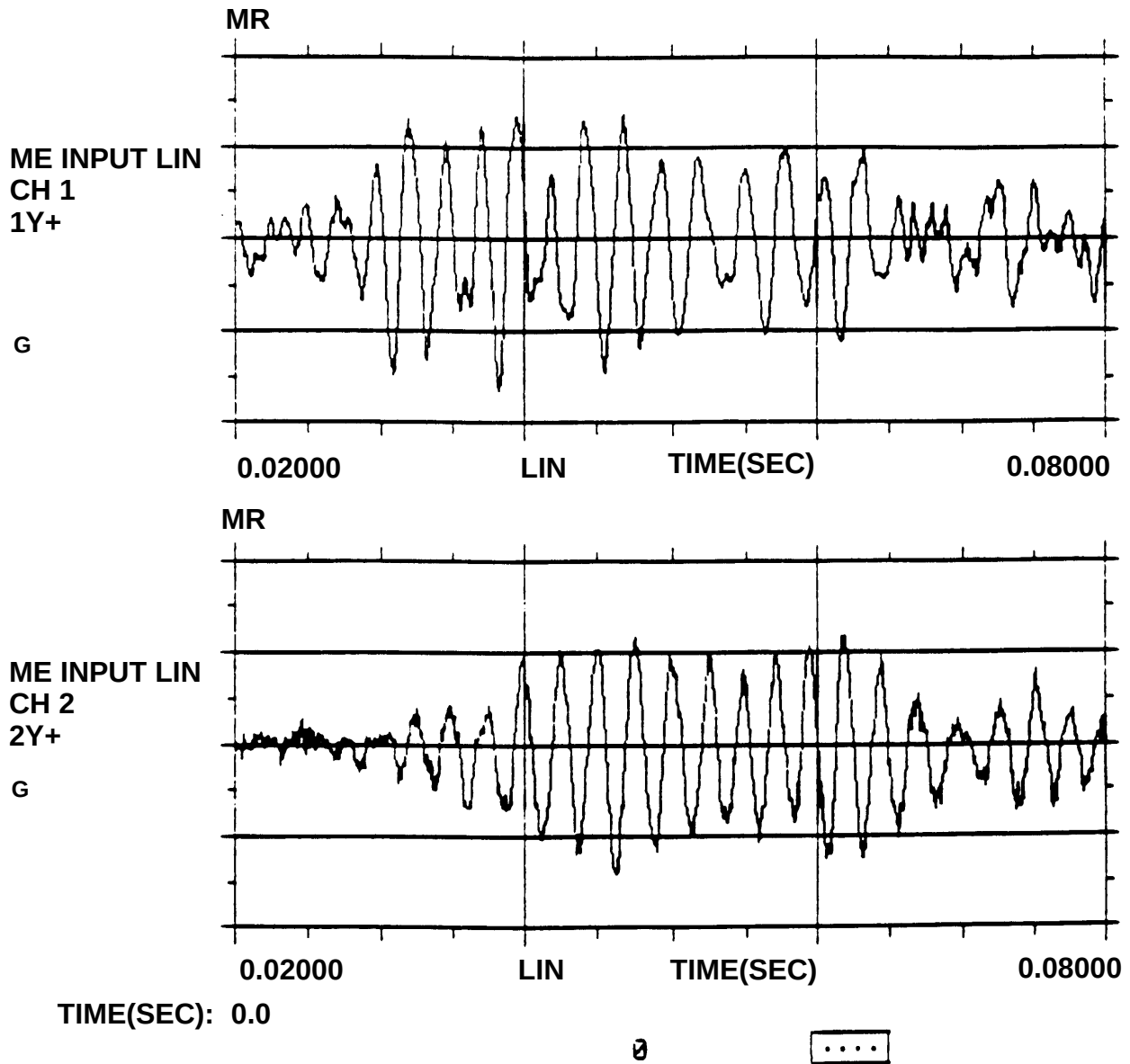
EXAMPLE SITE ENVIRONMENTAL VIBRATION VS. GE SPECIFICATION FOR 1.5T LCC MAGNET WITHOUT GRADIENT ISOLATION
ILLUSTRATION A-2

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY
ILLUSTRATION A-3

A-4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY (ZOOM IN ON 500 microG TRANSIENT EVENT)
ILLUSTRATION A-4

Prerequisites for Scheduled Performance Verification of Chillers For Signa *TwinSpeed*

Revision 02/14/02

The following list indicates items that must be completed before contacting Ellis and Watts to schedule the Factory Authorized Performance Verification of the pertinent *Twin Speed* Cooling Cabinet. When this step is completed, then the PM schedule will be established for the year. The LTL 10 (M3088TE) is started up and serviced by GE, therefore, performance verification by an Ellis and Watts Service Provider is not required.

1. Appropriate line power is connected to the Chiller and correct phasing established.
2. Chiller is permanently mounted as required by GE and / or state / local code requirements.
3. **For units with Water Cooled condensers;** (LC 18MT/WC, See Ellis and Watts Tech. Pub. 471)
 - Piping has been flushed out to remove any debris before being attached to Chiller
 - Water supply to Chiller condenser has an inline filter or strainer.
 - Full Gradient and Shield Cooler (Usually this happens about a week before turnover to customer.)
4. **For Indoor Air-Cooled Condenser units;** (LC 20MT, See Ellis and Watts Tech. Pub. 472)
(LC 10MT, See Ellis and Watts Tech. Pub. 473)
 - Ambient temperature and humidity are controlled to meet the equipment room environmental requirements.
 - Full Gradient and Shield Cooler (For Dual Loop) or Full Gradient (For Single Loop) loading is available.(Usually this happens about a week before turnover to customer.)
5. **For Outdoor Condenser unit;** (LC 20MT/O See Ellis and Watts Tech. Pub. 470)
(LC 10MT/O See Ellis and Watts Tech Pub 474.)
 - All coolant piping connections are complete and leak tested (**Do not use tap water**).

NOTE: It is recommend that distilled water or nitrogen gas be used to pressurize and leak test. Full Gradient and Shield Cooler (For Dual Loop) or Full Gradient (For Single Loop) loading is available.(Usually this happens about a week before turnover to customer.)

Performance Verification is usually completed just before the system is turned over to the Customer. Ellis and Watts service department needs a two week notification to schedule the Local Service Provider. Please provide the name and address of the Imaging Center, FE contact name with the best method of contact, requested date of service, and Cryo System ID #. Contact information is listed below.

Ellis and Watts International Inc. is determined to be as helpful as possible with the installation of the chiller equipment that we provide to support the General Electric Medical Systems equipment, but, Ellis and Watts and our Approved Service Providers are not contracted to perform **any** of the installation of this equipment. If you desire for E&W or any of our ASP's to assist with the actual installation of the chiller equipment, then, you will need to contact the Ellis and Watts Service Department at 513-752-9000 x749 or E-mail at service@elliswatts.com to discuss scheduling and payment for this work, or start-up scheduling. Information is free so call any time to discuss questions about installation or operation of our chiller equipment.

Ellis and Watts International Inc.
4400 Glenwillow Lake Lane
Batavia, Oh. 45103